

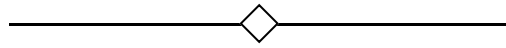
AN INTRODUCTION TO

FINANCE

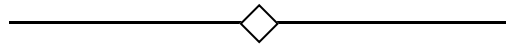
by

KUTUSH CATUSH

AN INTRODUCTION TO



FINANCE



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An Introduction to Finance

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Sundial Bakery, its people, and every date and figure given for them are invented for the purpose of teaching. Any resemblance to a real business is accidental.

First edition, 2026.

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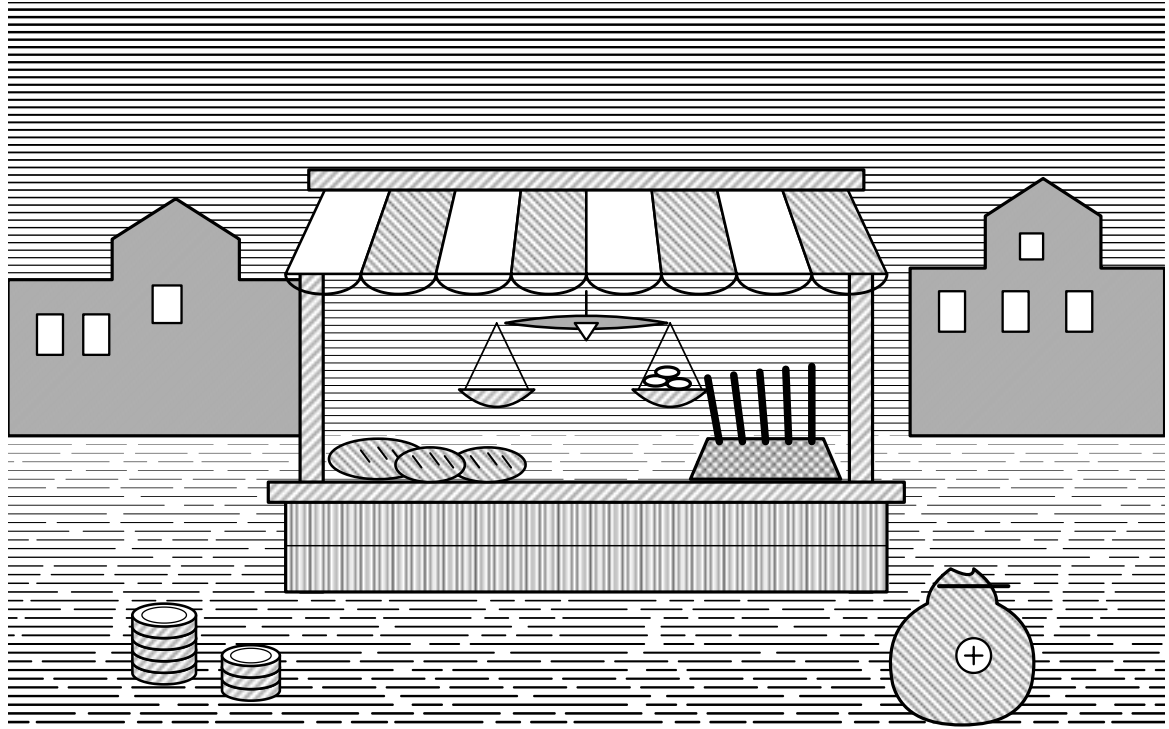


Plate I. The Market Stall.

CHAPTER I

Money, Business and the Idea of Finance

Before a firm can spend a single taka, it must find that taka somewhere.



This chapter gives you the everyday money words, introduces the small bakery that we follow through the whole book, and shows what finance is and why every business needs it. It is the slowest chapter of the book, and it teaches you how to read the rest.

A baker has a plan and some savings, but the savings are not enough.

The third question begins on the first morning: how should the money be used?

These three questions together are the subject of finance, which Section 1.4 defines.

The bakery in this book is invented, so that every idea can have exact numbers.

The book has a few regular features, and you should learn to use them now.

We begin with money as every person meets it: coming in, going out, kept, or borrowed.

GO SLOWLY HERE. THE WORDS ARE EVERYDAY WORDS, BUT EACH RECEIVES A PRECISE MEANING.

Money is whatever people accept in payment.

Money. What people accept in payment for goods and services.

Income is money coming in, and expense is money going out.

Income. Money coming in.

Expense. Money going out.

THE OPENING PUZZLE

A baker in Dhaka wishes to open a shop that sells bread, buns and cakes. The shop needs ovens, tables, flour, sugar, rent and workers. The baker has saved a large sum over many years, but the sum is smaller than the cost of everything on the list. Two questions follow at once. Where can the rest of the money come from? And what must the baker offer to the people who provide it?

A third question appears on the first morning. Once the money is in hand, how should it be used? A shop that buys too many ovens has no money left for flour. A shop that buys too much flour may be unable to pay the rent. Every taka can be used in only one way at a time, so each choice closes the other choices. Cooking skill cannot solve this problem; it is a problem about money.

Finding money, using money, and making sure that it lasts: these three tasks are the heart of *finance*. Section 1.4 gives the exact definition. This book explains how a business performs these tasks well. We begin in an ordinary place, with the money in a single person's life, because every business idea in this chapter is a larger copy of a personal one.

Throughout the book we follow one illustrative business, *Sundial Bakery* of Dhaka. It is not a real firm. Its dates and figures are invented so that each idea can be shown with exact numbers. Its records appear in paragraphs headed *Case file, illustrative*, and the figures in them are carried forward from chapter to chapter.

A few features of the book will help you from the first page. A short note in the side column repeats the main point of each paragraph. A word in bold type is a new term, and it is bold only once, where it is defined. A small raised number points to a footnote at the foot of the page. At the end of a section you will find *Before moving on*, a recap in two or three lines. In this chapter the pace is slow, and a note in the side column will tell you when to go especially slowly.

1.1 MONEY IN DAILY LIFE

Think of a student who receives Tk 5,000 from the family each month. The student buys food, pays for travel and buys books. Before we say anything about business, we need exact words for what happens to this money. Each word below will be used in the same way throughout the book.

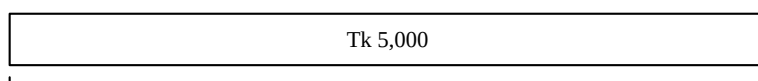
We begin with **money**. Money means what people accept in payment for goods and services. *Goods* are things that can be touched, such as bread. *Services* are work done for others, such as cutting hair. In Bangladesh the money is the taka, written Tk. A note, a coin and a balance in a bank account all count as money, because a shop will accept each of them when you pay.

Money moves in two directions. **Income** means money coming in. The student's Tk 5,000 each month is income. **Expense** means money going out. When the student pays Tk 800 for a bus pass, the Tk 800 is an expense. Notice the point of view. The same payment of Tk 800 is an expense for the student and income for the bus company.

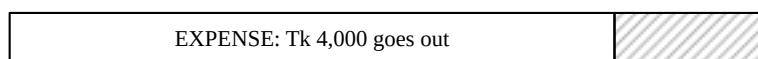
Suppose that the student spends Tk 4,000 this month. Then Tk 1,000 of the income remains. If the student keeps it, we say that the student makes a **saving**. Saving means keeping money for later use. To find the saving, subtract the expense from the income. Here the saving is Tk 5,000 less Tk 4,000, which is Tk 1,000.

Fig. 1 shows this month. This is the first figure of the book, so we shall read it slowly. The upper bar is the income, Tk 5,000. The lower bar has the same total length, but it is cut into two parts. The long plain part is the expense, Tk 4,000. The short shaded part is the saving, Tk 1,000. The length of each part matches its amount, so a part that looks four times longer is four times larger.

INCOME: Tk 5,000 comes in during the month



USE OF THE INCOME



SAVING: Tk 1,000 kept

Income less expense leaves the saving: 5,000 less 4,000 is 1,000.

FIG. 1. One student's month: the income is divided between expense and saving.

The table below states the same facts as numbers. Each row is one fact about the month, and the last column shows how the facts connect.

ONE STUDENT'S MONTH, IN TAKA

ITEM	AMOUNT (TK)	HOW IT IS FOUND
Income	5,000	Given
Expense	4,000	Given
Saving	1,000	Income less expense: 5,000 less 4,000

Saving belongs to every household, and later to every firm. A family that earns Tk 30,000 in a month and spends Tk 27,000 saves Tk 3,000. A family that spends Tk 32,000 in the same month has an expense larger than its income. It cannot save. To pay its bills, it must use earlier savings, or it must borrow.

Now suppose that the student wishes to buy a laptop for Tk 40,000. At Tk 1,000 a month, the saving would take forty months. There is another way. The student can ask someone to provide the Tk 40,000 now and agree to return it later.

This way is called **borrowing**. Borrowing means taking money now and agreeing to repay it. The money itself is called a **loan** when we think of it from the lender's side, and a loan means a sum of money lent for repayment. The lender takes a risk, because the borrower may fail to repay. The lender also loses the use of that money during the loan. For both reasons, lenders usually ask for something extra.

A **bank** is an institution that keeps people's money safely and lends money to others. Most people meet money's price at a bank. When a person deposits savings, the bank pays a small reward. When a person borrows, the bank charges a larger fee. The difference pays the bank's work and its risk.

Saving is the income that is not spent and is kept for later.

Saving. Keeping money for later use.

Fig. 1 shows the month as bars: the money in, and where it went.

The table states the same facts as numbers.

Saving is not only for students; every household and every firm can do it.

A person who wants something costly may keep saving, or may borrow.

Borrowing is taking money now and agreeing to repay it later.

Borrowing. Taking money now and agreeing to repay it.

A bank is the most familiar place that lends and keeps money.

Interest is the fee for using another person's money.

Interest. A fee for using someone's money. In daily life, interest also means curiosity.

In everyday speech, interest can also mean curiosity, but not in this book.

This first worked example finds the interest on a clean sum.

The data given and the question to answer.

The rule says that interest is the sum times the yearly rate.

The answer, and a check that it is sensible.

A second example uses a less round sum.

Common slips with interest come from the rate and the time.

Before moving on: income comes in, expense goes out, saving is kept, and a borrower repays the loan plus interest.

The one sentence for the reader to test is stated here.

The pause question checks that the reader can separate income from a loan.

A student's money is personal; a business is a different kind of money user.

A business is an activity that sells goods or services to earn money.

Business. An activity that sells goods or services to earn money.

That extra payment is called **interest**. Interest means a fee for using someone's money. Suppose that a friend lends you Tk 1,000 and asks for Tk 1,100 after one year. The interest is Tk 100. We state interest as a rate: Tk 100 for each Tk 1,000 is 10 taka for each 100 taka, which we call 10 per cent a year.¹

In everyday speech, "interest" can also mean curiosity, as in "an interest in music". In this book the word always means the fee for using money. It will appear in every chapter after this one.

Let us calculate interest in the standard order that the whole book will use. We list what is given, state what we must find, write the rule, perform the steps, state the answer and check it.

Given. A sum of Tk 10,000 is borrowed for one year at 10 per cent a year. *Find.* The interest for the year.

Rule. In words: interest equals the sum borrowed multiplied by the rate for the year. In symbols, $I = P \times r$. Here I stands for interest, P stands for the sum borrowed, and r is the rate written as a decimal. *Steps.* Ten per cent is 10 for each 100, which is 0.10. So $I = 10,000 \times 0.10 = \text{Tk } 1,000$.

Answer. The interest for the year is Tk 1,000. The borrower repays Tk 11,000 in all. *Check.* Tk 10,000 contains 100 hundreds. Each hundred pays Tk 10. So $100 \times 10 = \text{Tk } 1,000$. The two methods agree.

A second example. Given: Tk 25,000 is borrowed for one year at 12 per cent. Rate as a decimal: 0.12. Interest: $25,000 \times 0.12 = \text{Tk } 3,000$. The borrower repays Tk 28,000. Check: Tk 25,000 holds 250 hundreds, and $250 \times 12 = 3,000$.

Common slips. A learner may forget that the rate is for one year, and then use it for six months or five years. Another may write 12 per cent as 12 instead of 0.12, and so arrive at an interest larger than the loan. Always write the rate as a decimal before multiplying, and always check whether the answer is sensible. Interest of Tk 3,00,000 on a loan of Tk 25,000 is not.

Before moving on. Income is money coming in, and expense is money going out. What remains and is kept is saving. A borrower repays the loan and also pays interest. These ideas return in larger form in every business.

You should now be able to say: "Interest is what I pay for using another person's money."

Pause and check. A student receives Tk 5,000 from home and also borrows Tk 2,000 from a friend. Is the Tk 2,000 income? (The answer is at the back of the book.)

1.2 OWNERS, LENDERS AND THE FIRM

A student's money is personal. A business uses money in a larger and more complicated way, and it has special words of its own. We meet them one at a time, using the bakery.

A **business** means an activity that sells goods or services to earn money. A tea stall is a business. A large factory is a business. In both cases, someone must pay for the equipment and materials before the first sale.

¹ The words *per cent* mean "for each hundred". Chapter 3 treats percentages fully. Until then, we use only rates for which the sum is easy to follow.

When we wish to look at a business as one unit that collects and spends money, we call it a **firm**. A firm means a business seen as one money-using unit. The words *business* and *firm* name nearly the same thing. We use *business* when we think about the activity and *firm* when we think about its money.

Every firm has at least one **owner**. An owner means the person whose money and risk the business carries. If the firm earns well, the owner gains. If the firm loses, the owner is the first to lose. The owner of Sundial Bakery is one person who has chosen to use personal savings in the firm.

The money that the owner invests in the firm is called **capital**. Capital means money put in to start or run a business. In this book it does not mean a city. Capital is the first thing that a firm needs, and it is the thing the owner may not have enough of.

Large firms often have many owners. Such a firm is usually organised as a **company**. A company means a business that the law treats as a separate unit, owned through shares. A **share** means one equal part of the ownership of a company. If a company has 1,000 shares and you hold 10 of them, you own one hundredth of it. Chapter 2 returns to this idea.

An **investor** means a person who puts money into a firm, hoping to earn more later. Owners are investors. There is a second kind. A **lender** means a person or an institution that lends money for repayment plus interest. A bank that lends to the bakery is a lender. We now have two ways to bring money into a firm. They differ in an important way, which we meet next.

Case file, illustrative. On 2 January 2026 Sundial Bakery opens in Dhaka. The owner contributes Tk 6,00,000 of personal savings.² The bank lends Tk 4,00,000 at 12 per cent a year. The firm buys ovens and equipment for Tk 7,00,000. The cash left in the firm is Tk 3,00,000.

Examine what the bakery now has and what it owes. The ovens and the cash are **assets**. An asset means something the business owns that has value. The bank loan is a **liability**. A liability means an amount the business owes. The firm owns Tk 10,00,000 of assets and owes Tk 4,00,000.

Another word for the bank's loan is **debt**. Debt means money owed to a lender. The owner's part is called **equity**. Equity means the owners' share of what the business is worth. We find it by taking the assets and subtracting the debts. Here it is Tk 10,00,000 less Tk 4,00,000, which is Tk 6,00,000. This equals what the owner contributed.

Lenders and owners both hold a **claim** on the firm. A claim means a right to receive money from the firm. The two claims are very different, and the table below shows how. Read each row from left to right.

When we think of a business as one money-using unit, we call it a firm.

Firm. A business seen as one money-using unit.

The owner is the person whose money and risk the business carries.

Owner. The person whose money and risk the business carries.

Money put in to start or run a business is called capital.

Capital. Money put in to start or run a business. In daily life, a capital is also a country's chief city.

Large firms have many owners, who hold shares in a company.

Company. A business that the law treats as a separate unit, owned through shares.

Share. One equal part of the ownership of a company.

An investor puts money in to earn more; a lender is one kind of investor.

Investor. A person who puts money in, hoping to earn more.

Lender. One who lends money for repayment plus interest.

Sundial Bakery opens with money from two sources, the owner and a bank.

What the firm owns is an asset, and what it owes is a liability.

Asset. Something the business owns that has value.

Liability. An amount the business owes.

Debt is money owed to a lender; equity is what remains for the owner.

Debt. Money owed to a lender.

Equity. The owners' share of what the business is worth.

A claim is a right to receive money from the firm, and the two kinds of investor hold different claims.

Claim. A right to receive money from the firm.

² Amounts follow the South Asian custom of grouping: Tk 6,00,000 means six lakh, or six hundred thousand. One lakh is 1,00,000.

TWO KINDS OF CLAIM ON A FIRM

FEATURE	LENDER (DEBT)	OWNER (EQUITY)
What is promised	Repayment and interest	A share of what remains
Is the amount fixed?	Yes, as agreed	No, it rises and falls
Who is paid first?	The lender	The owner, after the lender
Who bears more risk?	Less	More
Who directs the firm?	Usually not	Yes

A short example shows how the lender is paid first.

The row "Who is paid first?" needs an example. Suppose that, after a bad year, Sundial Bakery must close, and it sells all its assets. The money received is shared in a fixed order. The bank, as lender, receives its Tk 4,00,000 first. The owner receives what remains. To keep the example simple, we ignore interest.

WHAT EACH INVESTOR RECEIVES IF SUNDIAL CLOSES AND SELLS ITS ASSETS

ASSETS ARE SOLD FOR	BANK RECEIVES	OWNER RECEIVES
Tk 10,00,000	Tk 4,00,000	Tk 6,00,000
Tk 8,00,000	Tk 4,00,000	Tk 4,00,000
Tk 3,00,000	Tk 3,00,000	Nothing

The bank's amount is steady while the owner's amount changes with the sale.

Read down the table. The bank receives Tk 4,00,000 in the first two rows, whatever the sale brings, while the owner's share falls from Tk 6,00,000 to Tk 4,00,000. In the last row the assets bring only Tk 3,00,000. The bank takes all of it and loses Tk 1,00,000, and the owner receives nothing. The lender is protected first, but not always completely.

Because the lender is paid first, the lender earns less and the owner has the right to decide.

This order explains the whole table. The lender is repaid first and receives a fixed amount, so the lender's risk is smaller, and the reward is smaller too. The owner receives what remains. That may be much more than the bank receives, or nothing. Because the owner carries more risk, the owner also holds the right to direct the firm.

Each person looks at a firm's money from a different seat.

From three seats. The owner hopes the firm will grow in value. The lender wishes only for repayment and interest and will refuse to lend if the risk seems too large. An employee wants wages paid on time and does not mind whether the owner's equity is large, as long as the firm continues. A good decision about money cannot please all three completely, but it must keep the firm able to meet all three.

Before moving on: assets are paid for by debt and equity, and the lender is paid first.

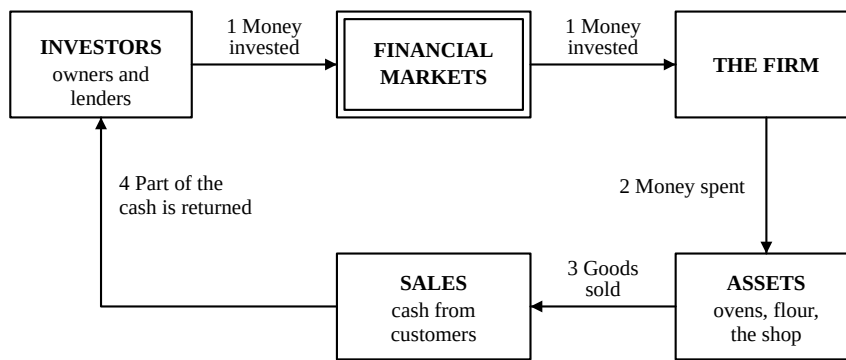
Before moving on. A business earns money by selling goods or services, and a firm is that business seen as a money-using unit. The money comes from owners (equity) and lenders (debt). What the firm owns are its assets, and what it owes are its liabilities. Assets less debt leaves equity.

The pause question tests the vocabulary of the section.

Pause and check. The firm buys a van with Tk 2,00,000 of the bank's money. Is the van an asset or a liability? Is the new loan an asset or a liability?

1.3 WHERE A FIRM'S MONEY COMES FROM AND GOES

A firm's money does not rest. It comes in, it is spent, it returns, and it goes out again. Fig. 2 shows this loop. Before reading any word in it, notice the shape: five boxes joined by arrows, and the arrows form a circle. A box is a party that holds or uses money. An arrow shows the direction in which the money moves, and the number on it shows the order.



The rest of the cash stays in the firm for the next round.

FIG. 2. The loop of a firm's money: from investors, through the firm, to its assets, to sales, and back.

Begin at the top left, with the investors. The owners and lenders have money, and they wish to place it in a firm, hoping to earn more. Arrow 1 shows the money moving toward the firm.

Between the investors and the firm stands a **financial market**. A financial market means a place or system where money is lent, borrowed or invested. A bank is the simplest example, because it brings savers and borrowers together. Larger markets bring many investors and many firms together. Chapters 11 and 14 describe them.

Arrow 2 moves downward. The firm spends the money it received on assets: ovens, flour, a shop. After this step the firm holds things instead of money. This explains why a firm that has plenty of equipment may have almost no cash. Here *cash* means money that can be spent at once, whether in notes, in coins or in a bank account.

Arrow 3 moves to the left. The assets are used to make goods and services, and the firm sells them. The money received from customers is the firm's **sales**. Sales means money earned from selling. Other books say *revenue* or *turnover* for the same thing. This book always says *sales*, so that one idea has one word.

Making goods is not free. Flour, sugar, fuel and wages must be paid. Such a payment is a **cost**. A cost means what is paid to get or make something. In everyday speech a person may say that a laptop "costs" Tk 40,000, using the word for a price. In this book, cost means what the firm pays in order to make or provide what it sells.

What remains of the sales after the costs are paid is **profit**. Profit means what is left of sales after costs. If the sales are smaller than the costs, the result is negative, and we call it a loss. The rule is: profit equals sales less costs.

A firm's money moves in a loop, and a picture shows the loop.

GO SLOWLY HERE. THIS PICTURE IS THE WHOLE BOOK IN MINIATURE.

Step 1 is money moving from investors toward the firm.

Between investors and firms stands the financial market.

Financial market. A place or system where money is lent, borrowed or invested.

Step 2 is the firm spending the money on assets.

Step 3 is the sale of goods and services, which brings money back.

Sales. Money earned from selling.

Making the goods costs money, and the word cost has an everyday meaning too.

Cost. What is paid to get or make something. In daily life, a cost can mean a price.

Profit is what remains of sales after the costs are paid.

Profit. What is left of sales after costs.

A tea stall gives a worked example of profit.

The rule and the arithmetic, step by step.

The answer, and a check that it is sensible.

A second example uses a day with a loss.

A cash flow is money actually moving in or out.

Cash flow. Money actually moving in or out.

Before moving on: money goes from investors to assets, becomes sales, and part returns to the investors.

The test sentence repeats the idea of the loop.

The bakery's first day raises a question that no recipe can answer.

Finance is the study of how money is raised and used wisely.

Finance. How money is raised and used wisely.

Raising money and using money are two sides of finance.

Finance can be organised around three questions, each answered in later chapters.

Given. A tea stall sells 100 cups in a day at Tk 10 each. The tea, milk, sugar and gas for the day cost Tk 600. *Find.* The profit for the day.

Rule. Profit = sales less costs, and sales = number of cups x price per cup. *Steps.* Sales = 100 x Tk 10 = Tk 1,000. Profit = Tk 1,000 less Tk 600 = Tk 400.

Answer. The profit for the day is Tk 400. *Check.* Each cup earns Tk 10 and costs Tk 6, since Tk 600 divided by 100 cups is Tk 6. So each cup leaves Tk 4, and 100 cups leave Tk 400. The answers agree.

A second day. The stall sells only 50 cups, but the gas, milk and tea still cost Tk 520. Sales are 50 x Tk 10 = Tk 500. Profit is Tk 500 less Tk 520, which is a loss of Tk 20. A day of fewer sales can turn a profit into a loss.

Finally, arrow 4 goes back to the investors. A part of the cash returns to them: interest to the lender and payments to the owner. The rest stays in the firm for the next round. Each movement of money into or out of the firm is a **cash flow**. A cash flow means money actually moving in or out. Cash flow is not the same as profit, as Section 1.4 will show.

Before moving on. Money flows from investors through the firm to its assets. Selling goods turns the assets back into cash. A part of that cash returns to the investors, and the rest begins the loop again. Profit is sales less costs.

You should now be able to say: "A firm's money moves in a loop, and finance is about every arrow in it."

1.4 WHAT FINANCE IS

At the end of its first day, Sundial Bakery has Tk 3,00,000 in cash. The owner sees three possible uses. The money may be kept for flour and wages. It may be used to buy a second oven. It may be used to repay a part of the bank loan and so reduce the interest. Each use has a different result, and no recipe can decide among them. A choice about money needs its own way of thinking.

We can now state the idea precisely. **Finance** means how money is raised and used wisely. The word *raised* covers every way of bringing money in: the owner's savings, a bank loan, or shares sold to other people. The word *used* covers every way of spending it: ovens, flour, wages, or repaying a loan. The word *wisely* means that each choice is judged against a purpose. Chapter 2 names the purpose.

RAISING MONEY AND USING MONEY

Finance has two sides, and the loop of Fig. 2 shows both. On the *raising* side, arrows 1 and 4 matter: money comes in from investors, and part of it returns to them. On the *using* side, arrows 2 and 3 matter: money buys assets, and the assets earn sales. A firm that raises money with great skill but uses it badly will fail. A firm that uses money well but cannot raise enough will never begin.

THREE QUESTIONS

Finance can be organised around three questions. First, where should the firm find its money? Second, which long-lasting things should the firm buy with it? Third, how can the firm be sure that it can pay its bills every day? Chapter 2 calls these the financial manager's decisions. Later chapters give each question a detailed answer.

PRICE AND RISK

Two facts make these questions hard. The first is that money has a price. A firm that borrows pays interest, and a firm that uses the owner's savings gives up what the owner might have earned elsewhere. The second is that money carries risk, which means the chance that results will differ from what was hoped. Almost every idea in this book is a way of handling the price of money, its risk, or both.

Money has a price, which is interest, and a risk, which is the chance of loss.

A FIRST DECISION, WITH NUMBERS

Return to the three uses of the bakery's Tk 3,00,000. We cannot judge them all yet, but one can be measured. Suppose the owner repays Tk 3,00,000 of the Tk 4,00,000 loan on the first day. How much yearly interest would the firm save?

A small calculation shows how one choice of the three would save interest.

Given. Repayment Tk 3,00,000; rate of interest 12 per cent a year. *Find.* The interest saved in one year.

The data given and the question to answer.

Steps. Interest saved = $3,00,000 \times 0.12 = \text{Tk } 36,000$.

The rule and the arithmetic, step by step.

Answer. The firm saves Tk 36,000 of interest in a year. *Check.* Tk 3,00,000 holds 3,000 hundreds, and $3,000 \times 12 = 36,000$.

The answer, and a check that it is sensible.

The saving is real, but it has a price. After the repayment, the firm has no cash for flour or wages. Finance does not give the answer by arithmetic alone; it also asks whether the firm will survive the next month. Keeping some cash may be worth more than the interest saved.

But the saving costs the firm all its cash, so the choice is not yet settled.

PROFIT IS NOT CASH

Here is the most important warning of the chapter: profit and cash are not the same. Suppose a tea stall sells 1,000 cups at Tk 10 each to a nearby office during a month. The office pays at the end of the month. During the month, the stall spends Tk 6,000 on tea, milk, sugar and gas.

Profit and cash are different, and a firm needs both.

Given. Sales Tk 10,000, to be paid at month end; costs Tk 6,000, paid during the month. *Find.* The profit, and the cash in hand before the office pays.

GO SLOWLY HERE. MANY STUDENTS FIND THIS STEP CONFUSING, BECAUSE THE TWO WORDS FEEL ALIKE.

The data given and the question to answer.

Steps. Profit = Tk 10,000 less Tk 6,000 = Tk 4,000. But the stall has paid out Tk 6,000 and received no money. Before the office pays, the cash of the stall is lower by Tk 6,000.

The profit is Tk 4,000 on paper, yet the stall has spent Tk 6,000 and received nothing.

Answer. The profit is Tk 4,000, but the cash movement so far is a payment of Tk 6,000. *Check.* After the office pays Tk 10,000, the stall's cash is higher by Tk 10,000 less Tk 6,000, which is Tk 4,000, and now the cash equals the profit.

The answer, and a check that it is sensible.

If the stall cannot pay the milk supplier on the fifth day, the supplier stops delivery, and the stall closes with a profit on paper and no tea to sell. This is why finance watches cash flows and not only profit.

A firm can show a profit on paper and still lack the cash to pay its bills.

Before moving on. Finance studies how money is raised and used wisely. It asks where money comes from, what it buys, and whether bills can be paid. Money has a price, which is interest, and a risk. Profit is not cash.

Before moving on: finance studies how money is raised and used, under price and risk, and profit is not cash.

You should now be able to say: "A firm can have a profit and still run out of cash."

The test sentence states the central warning.

Finance has limits: it cannot create money, and it cannot make a poor product sell.

The pause question asks for the difference between profit and cash.

Several subjects study the firm, and finance works with the others.

Management directs; accounting records; economics explains choice.

Management. Directing people and resources to reach a goal.

Accounting. Recording and reporting what has happened to a firm's money.

Economics. The study of how people and societies choose when resources are limited.

Accounting records the past; finance uses the record to decide the future.

Finance is part of management, because every decision about a firm has a result in money.

One decision, the second shop, shows all four subjects at work together.

WHAT FINANCE CANNOT DO

Finance has limits. It cannot create money from nothing, and it cannot make a poor product sell. A firm that makes goods nobody wants will fail even if its finance is excellent. Finance helps a good business to survive and grow; it does not replace the work of making and selling things that people want.

Pause and check. In the tea stall example, how does the Tk 4,000 profit differ from Tk 4,000 in the till?

1.5 FINANCE BESIDE ACCOUNTING, ECONOMICS AND MANAGEMENT

A firm is studied by several subjects. Three of them are close neighbours of finance, and we must separate them from the start so that they are not confused.

Management means directing people and resources to reach a goal. **Accounting** means recording and reporting what has happened to a firm's money. **Economics** means the study of how people and societies choose when resources are limited. Each describes a different job. The table below places finance beside them.

FOUR SUBJECTS THAT STUDY THE FIRM

SUBJECT	THE QUESTION IT ASKS	A BAKERY EXAMPLE
Management	What should we do, and who will do it?	Plan the week's baking and the staff
Accounting	What has happened to our money?	Record Tk 3,00,000 in cash and Tk 4,00,000 in debt
Economics	How do buyers, sellers and prices behave?	Why customers buy fewer buns when the price rises
Finance	How do we raise money and use it wisely?	Whether to buy a second oven, and how to pay for it

Accounting and finance are the closest pair. The accountant records what has happened and reports it. The finance worker uses those records to decide what should happen next. Without accounting, finance has no facts. Without finance, accounting records facts that nobody uses.

Finance is also closely linked to management. Every management decision has a result in money. To open a second shop, the manager must find money, buy equipment, hire staff and pay for goods until the first sales arrive. Each of these tasks is a financial question. Finance is therefore part of good management, and the person who answers these questions is a member of the management team.

Consider one decision seen from all four seats. The manager plans the work of a second shop. The economist asks whether enough customers live nearby and how much they will pay. The accountant records what the shop earns and spends. The finance worker asks how much money the shop needs, where it will come from, and whether the firm can bear the risk. No single subject can answer the whole question.

Before moving on. Management directs, accounting records, economics explains how people choose, and finance decides how money is raised and used. The four work together in every firm.

Before moving on: management directs, accounting records, economics explains, and finance decides about money.

A WORKED CASE: THE OPENING OF SUNDIAL BAKERY

Case file, illustrative. On 2 January 2026 the owner of Sundial Bakery invests Tk 6,00,000. The bank lends Tk 4,00,000 at 12 per cent a year. The firm buys ovens and equipment for Tk 7,00,000 and keeps the rest as cash.

This case applies the chapter's words to the bakery's first day.

Question. How much cash does the firm hold? What are its assets, its debt and its equity? How much interest does it owe for one year?

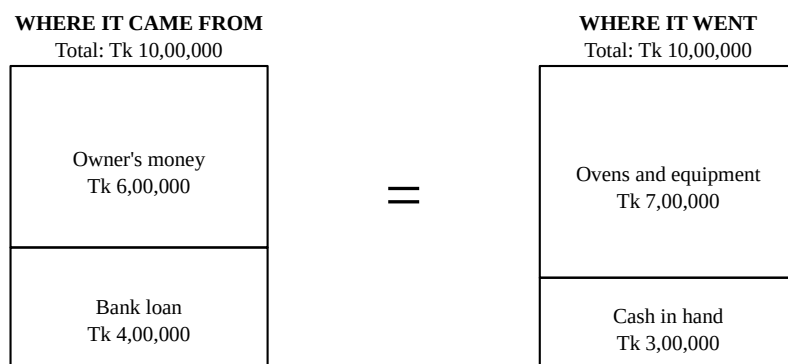
The question asks for the cash, assets, debt, equity and one year's interest.

Reasoning. The money received is Tk 6,00,000 plus Tk 4,00,000, which is Tk 10,00,000. The ovens and equipment cost Tk 7,00,000, so the cash is Tk 10,00,000 less Tk 7,00,000, which is Tk 3,00,000. The assets are equipment Tk 7,00,000 plus cash Tk 3,00,000, which is Tk 10,00,000. The debt is the bank loan, Tk 4,00,000. The equity is assets less debt: Tk 10,00,000 less Tk 4,00,000, which is Tk 6,00,000.

The cash and assets come first; then the debt and equity follow.

For the interest, Tk 4,00,000 holds 4,000 hundreds. Each hundred earns Tk 12 in a year, so the interest is $4,000 \times \text{Tk } 12$, which is Tk 48,000. Interest for one month is Tk 48,000 divided by 12, which is Tk 4,000.

The interest for a year is 12 taka for each 100 taka borrowed.



Each side adds to the same total.

FIG. 3. The opening position of Sundial Bakery: the money that came in equals the money that went out.

Fig. 3 shows the result. The left side shows where the money came from, and the right side shows where it went. Both sides total Tk 10,00,000.

Fig. 3 shows two sides of the same total.

Conclusion. The firm holds Tk 3,00,000 in cash and has assets of Tk 10,00,000, debt of Tk 4,00,000 and equity of Tk 6,00,000. It owes Tk 48,000 in interest for the first year.

The conclusion states the five answers.

Check. Debt plus equity is Tk 4,00,000 plus Tk 6,00,000, which is Tk 10,00,000, and this equals the assets. If the two sides differ, a figure has been copied wrongly.

The check shows that debt plus equity equals assets.

A second case, with new figures (not part of the bakery's record). Brightway Tailors begins with an owner's investment of Tk 4,00,000 and a bank loan of Tk 2,00,000 at 9 per cent a year. It buys machines for Tk 5,00,000. Cash: Tk 6,00,000 less Tk 5,00,000 is Tk 1,00,000. Assets: Tk 5,00,000 plus Tk 1,00,000 is Tk 6,00,000. Equity: Tk 6,00,000 less Tk 2,00,000 is Tk 4,00,000. Interest for a year: $2,000 \text{ hundreds} \times \text{Tk } 9 = \text{Tk } 18,000$.

A second case, with new figures, tests the same steps.

The summary restates the chapter in a few lines.

This problem asks the reader to apply the chapter to a new firm.

Four common mistakes are stated here and corrected.

The first mistake mixes up profit and cash.

The second mistake treats a loan as income.

The third mistake mixes up equity and cash.

The fourth mistake treats finance and accounting as one subject.

A real case shows that a very small loan can begin a large idea.

The experiment grew into a bank in 1983.

SUMMARY

Money comes in as income and goes out as expense; what is kept is saving. Borrowed money must be repaid with interest. A business is an activity that earns money, and a firm is a business seen as one money-using unit. A firm's money comes from owners (equity) and lenders (debt) and is spent on assets. Sales bring cash back, and profit is sales less costs. Finance is the study of how money is raised and used wisely. It works beside accounting, economics and management, and it must watch cash as well as profit.

THINK IT THROUGH

A new firm, Greenleaf Stationers, begins with the owner's Tk 2,00,000 and a bank loan of Tk 3,00,000. It buys shelves and a counter for Tk 4,00,000 and keeps the rest as cash. (a) What are its assets, its debt and its equity? (b) In its first month, sales are Tk 90,000 and costs paid are Tk 60,000, but customers have paid only Tk 40,000. What is the profit, and what is the cash received less the cash paid? (c) Explain why the two answers differ.

WHERE PEOPLE GO WRONG

"Profit is the money in the bank." Profit is sales less costs, while cash is money actually held. A firm may show a profit and hold no cash, as the tea stall showed.

"A loan is income." A loan brings cash in, but the firm must repay it. Only money earned from selling is sales. Borrowed money is debt.

"Equity is the cash the firm holds." Equity is the owners' share of what the firm is worth, which is assets less debt. Sundial's equity is Tk 6,00,000, while its cash is Tk 3,00,000.

"Finance and accounting are the same." Accounting records what happened. Finance decides what to do about money next.

IN HISTORY: A LOAN OF TWENTY-SEVEN DOLLARS

In 1976 Muhammad Yunus, a professor at the University of Chittagong, visited the village of Jobra, near the university.³ He found that women who made bamboo furniture borrowed money at very high interest and gave most of their profit to the lenders. From his own pocket he lent a total of US\$27 to 42 villagers. The experience convinced him that very small loans could change the work and the lives of poor people.

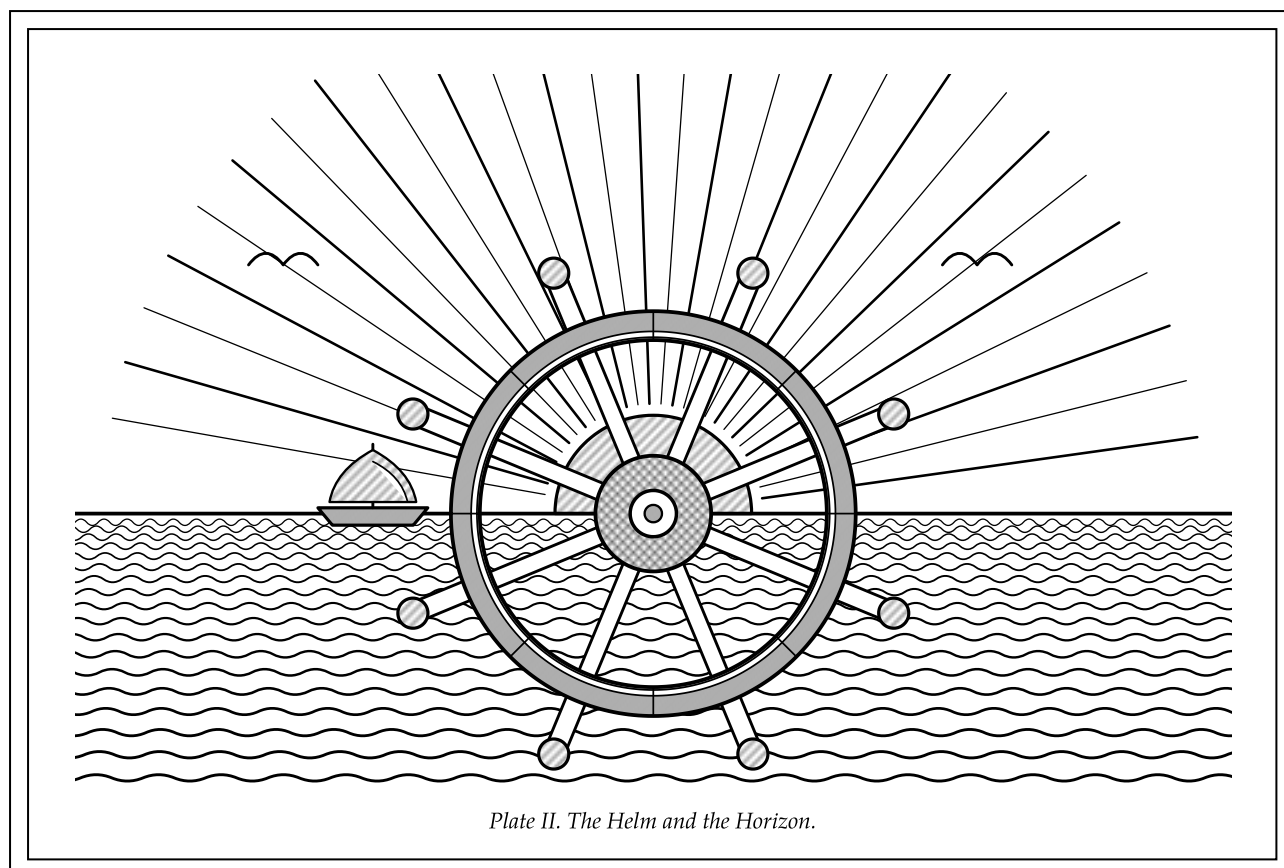
Yunus then arranged a credit line with Janata Bank, an established bank, in December 1976, and he signed as guarantor of the loans. In October 1983 the project became an independent bank, Grameen Bank, whose name means "village bank". In 2006 Yunus and the bank shared the Nobel Peace Prize. Every idea in this chapter appears in the story: borrowing, interest, risk, the lender's claim, and the way that finance can change what a person is able to do.

³ The facts in this paragraph follow published accounts of Grameen Bank; the sums are given in US dollars, as those accounts give them.

REVIEW QUESTIONS

1. Define income, expense and saving, and show how they are related. 2. What is interest? Calculate the interest for one year on Tk 20,000 borrowed at 8 per cent a year. 3. Distinguish a business from a firm. 4. Explain the difference between an owner's claim and a lender's claim. 5. Define asset, liability and equity. How is equity found? 6. Describe the loop of a firm's money, using Fig. 2. 7. A shop sells goods for Tk 50,000 and pays costs of Tk 35,000. What is the profit? Why may the shop still lack cash? 8. Explain how finance is related to accounting, economics and management.

These questions test the chapter's main ideas.



CHAPTER II

The Financial Manager and the Goal of the Firm

A firm that does not know what it is trying to achieve cannot know whether it has succeeded.



This chapter shows what the people who manage a firm's money decide, why those decisions decide whether a firm survives, and what the firm is trying to increase. It ends with the problem that arises when owners and managers want different things.

This chapter relies only on the money words of Chapter 1.

The bakery has Tk 3,00,000 and a chance to buy a second oven, but who decides, and by what test?

Behind the oven lie two harder questions, one about who decides and one about the aim.

A firm's money problems fall into four kinds of decision.

GO SLOWLY HERE. FOUR NEW TERMS ARRIVE TOGETHER, AND EACH ANSWERS A DIFFERENT QUESTION.

The first decision is what long-lasting things the firm should buy.

Investment decision. Choosing what long-lasting things to buy.

You will need: from Chapter 1, the words *owner, lender, asset, debt, equity, sales, cost* and *profit*, and the picture of the loop of money in Fig. 2. The pace is still slow, because new ideas arrive here.

THE OPENING PUZZLE

In February 2026 the owner of Sundial Bakery sees a chance to grow. A second oven would cost Tk 3,00,000, which is exactly the cash that the firm holds. With two ovens, the bakery could bake twice as much bread in the same hours. But if the whole sum is spent on the oven, nothing remains for flour and wages. If the firm borrows more money to buy the oven, it must pay more interest. If it waits, a competitor may take the customers.

Two harder questions stand behind this choice. The first is *who* should decide such things, and what skills the decider needs. The second is *by what test* a decision is called good. A bigger bakery is not always a better one, and a larger profit this year is not always a better result. Before a firm can choose among its options, it must know what it is trying to achieve. This chapter answers both questions.

2.1 THE DECISIONS THAT FINANCE MAKES

In Chapter 1 we followed a loop of money from investors, through the firm and its assets, to sales and back. Every arrow in that loop hides a choice. Experience shows that the choices fall into four groups. Fig. 4 shows them around the firm. Examine the boxes first, then the direction of each arrow.

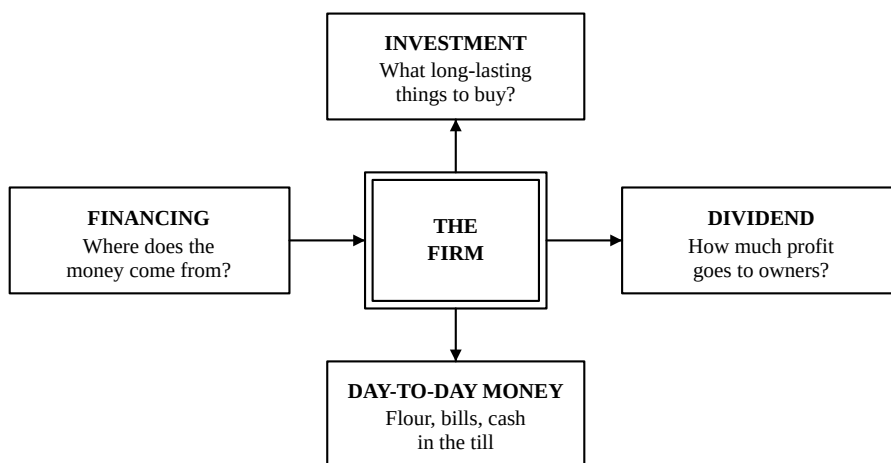


FIG. 4. The four decisions about a firm's money: three about money in motion, and one about profit.

The first group is the **investment decision**. An investment decision means choosing what long-lasting things to buy. A second oven is a clear example. The oven will bake for many years, so the choice affects the firm for a long time. A wrong choice is expensive, because a bought oven cannot easily be returned. Chapters 6 and 7 teach how to judge such choices.

The second group is the **financing decision**. A financing decision means choosing where the money comes from. The oven can be paid for from the owner's further savings, from a bank loan, or by inviting a new owner to invest. Each source has a different price and a different risk, as Chapter 1 showed. Chapters 13 and 14 describe the sources in detail.

The third group is about the firm's daily life. A bakery must always hold flour, owe money to its supplier, be owed money by customers who buy on credit, and keep some cash for wages. The money tied up in all this is the firm's **working capital**. Working capital means money tied up in daily running, such as goods waiting to be used or sold, bills not yet paid, and cash. Deciding how much of it to hold is the third decision. Chapter 12 treats it fully.

The fourth group concerns profit. When a firm earns a profit, someone must decide how much to pay to the owners and how much to keep in the firm for later use. The part paid to the owners is called a **dividend**. A dividend means the part of profit paid to the owners. Keeping more profit in the firm leaves more money for growth but gives the owners less now. Chapter 14 returns to this choice.

The table below gathers the four decisions. Read each row across: the name, the question, the bakery's own case, and the chapter where it is taught.

THE FOUR DECISIONS OF FINANCE

DECISION	THE QUESTION	THE BAKERY'S CASE	CHAPTER
Investment	What long-lasting things to buy?	The second oven	6 and 7
Financing	Where does the money come from?	Savings, a loan, or a new owner	13 and 14
Working capital	How much money for daily running?	Flour, bills, cash in the till	12
Dividend	How much profit goes to the owners?	Take it out, or keep it in the firm	14

The four decisions are not separate. Suppose the owner decides to buy the oven using all the cash. That is an investment decision, and it also answers the financing question: the money comes from the firm's own cash. But it damages the third decision, because no cash is left for flour. A good manager therefore never makes one decision alone. Each choice is tested against the other three.

Before moving on. Finance makes four kinds of decision: what to buy (investment), where the money comes from (financing), how to run daily money (working capital), and how much profit to pay out (dividend). They are linked, so a manager judges them together.

You should now be able to say: "Finance decides what to buy, where the money comes from, how much to hold for daily use, and how much profit to pay out."

Pause and check. A firm decides to borrow Tk 5,00,000 from a bank and use it to buy a delivery van. Which two of the four decisions are involved?

The second decision is where the money comes from.

Financing decision. Choosing where the money comes from.

The third decision concerns the money tied up in daily running.

Working capital. Money tied up in daily running, such as flour, unpaid bills and cash.

The fourth decision is how much profit to give to the owners.

Dividend. The part of profit paid to the owners.

The table lists the four decisions in one place, with the bakery's example of each.

The four decisions are linked, because a choice in one changes the others.

Before moving on: finance has four decisions, and each one affects the others.

The test sentence states the four decisions.

The pause question applies the four groups to a new case.

2.2 THE FINANCIAL MANAGER AND THE WORK

A person must take these decisions, and that person is the financial manager.

Financial manager. The person who decides how a firm raises and uses money.

The financial manager is not the shop manager, and the two should not be confused.

The work has five tasks: plan, raise, allocate, control and report.

Each task serves a different group of people.

The financial manager works closely with the accountant but has a different task.

From three seats, the financial manager has three different responsibilities.

Before moving on: the financial manager plans, raises, allocates, controls and reports.

Someone must make the four decisions, and that person is the **financial manager**. A financial manager means the person who decides how a firm raises and uses money. In a small firm, such as Sundial Bakery, the owner is the financial manager. In a large firm there is a team, led by a senior manager. The title changes with the size of the firm, but the work is the same.

The financial manager is not the same person as the shop manager. The *shop manager* directs the daily work of the shop: baking, serving and staff. The *financial manager* directs the firm's money. In March 2026 the owner of Sundial Bakery hires a shop manager, but the owner remains the financial manager. In a larger firm these are always different people.

The work of a financial manager can be divided into five tasks. The table below states each task in plain words and gives the bakery's example. Read it row by row.

THE FIVE TASKS OF THE FINANCIAL MANAGER

TASK	IN PLAIN WORDS	A BAKERY EXAMPLE
Plan	Look ahead at how much money will be needed and when	Estimate the money needed for flour in the next three months
Raise	Find the money from investors and lenders	Ask a bank for a loan
Allocate	Choose what to spend the money on	Decide between a second oven and a delivery van
Control	Watch cash and costs so that bills can be paid	Check the till and the supplier's bills each week
Report	Tell owners and lenders how the firm is doing	Show the bank the year's results

Each task serves a different group of people. Planning and control serve the firm's own managers, who need to know that the firm can pay its bills. Raising money serves the firm's future growth. Reporting serves the owners and the lenders, who have lent or invested their money and have a right to know what has happened to it.

The financial manager works closely with the accountant. Chapter 1 showed that the accountant records what has happened and reports it, while finance decides what to do next. The financial manager uses the accountant's records as facts, and the accountant records the effect of the financial manager's decisions.

From three seats. To the owner, the financial manager is a guardian of the owner's savings. To the lender, the financial manager is a person who promises that the loan will be repaid and keeps that promise. To the employees, the financial manager is the person who ensures that wages are paid on time. A good financial manager keeps all three promises together.

Before moving on. The financial manager makes the four decisions. The work has five tasks: plan, raise, allocate, control and report. In a small firm the owner does this work, and the shop manager is a different person.

2.3 HOW FINANCE SHAPES EFFICIENCY AND SURVIVAL

Why does the quality of financial decisions matter so much? There are two answers, and both concern the survival of the firm. The first is about how well the firm uses its money. The second is about whether the firm can continue at all.

The first answer is **efficiency**. Efficiency means getting the most result from the money and effort used. A firm uses money efficiently when little of it is idle or wasted. Consider cash that lies in the till for a whole year while the firm pays interest on a bank loan. The cash earns nothing, and the loan costs interest. Both are signs of poor use.

Given. Sundial Bakery holds Tk 1,00,000 of cash that it does not need for the coming year. The bank loan costs 12 per cent a year. *Find.* The interest that the firm would save in a year if it used this cash to repay part of the loan.

Rule. Interest saved = sum repaid $\times$ rate, as in Chapter 1. *Steps.* Interest saved = $1,00,000 \times 0.12 = \text{Tk } 12,000$.

Answer. The idle cash costs the firm Tk 12,000 a year in interest that could have been saved. *Check.* Tk 1,00,000 holds 1,000 hundreds, and $1,000 \times 12 = 12,000$.

The Tk 12,000 is not a payment that anyone sends to the bank for the idle cash. It is a saving that the firm failed to make. The point for finance is that money which does no work still has a price.

The second answer is **survival**. A firm may sell good products and still fail, if it cannot pay its bills on the dates they fall due. Wages are due every month, the supplier's invoice is due every week, and the bank's interest is due on fixed days. If cash is not available on those days, the supplier stops delivering, the workers leave, and the bank may demand repayment. Chapter 1 showed that profit is not cash. This is the practical result of that warning.

A firm can also fail because it has borrowed too much. Interest is a fixed payment, and it falls due in good months and bad months alike. A firm with large debt must find that payment whatever its sales are. Chapter 9 shows how such fixed payments increase a firm's risk.

Finance therefore shapes growth as well as survival. A firm with too little money cannot grow, because it cannot buy the ovens or the stock. A firm with too much debt may grow quickly but is in danger if sales fall. The financial manager's work is to find the path between the two.

Before moving on. Finance matters because it decides how efficiently a firm uses money and whether the firm survives. Idle money has a price, cash must be available on the days bills fall due, and heavy debt adds risk.

Finance decides whether a firm does much with its money or little, and whether it lasts.

Efficiency means getting the most result from the money and effort used.

Efficiency. Getting the most result from the money and effort used.

A worked example shows that idle cash has a price.

The rule and the arithmetic, step by step.

The answer, and a check that it is sensible.

The result shows that idle money is not free; it has a price in missed saving.

The second answer is survival: a firm that cannot pay its bills cannot continue.

Too much debt is a second danger, because interest must be paid in every month, good or bad.

Finance also shapes growth: too little money prevents it, and too much debt makes it dangerous.

Before moving on: good finance uses money efficiently and keeps the firm able to pay its bills.

2.4 THE GOAL: PROFIT OR WEALTH?

Return to the question left open in the opening puzzle: by what test is a decision called good? A manager who must choose among ovens, loans and cash needs one aim to measure them all. The first answer that most people give is profit. A firm exists to earn money, they say, so the best decision is the one that brings the largest profit.

The firm needs a single aim, and the obvious first answer is profit.

GO SLOWLY HERE. THIS IS THE CENTRAL IDEA OF THE CHAPTER, AND IT WILL GUIDE EVERY DECISION IN THE BOOK.

Profit maximisation means aiming for the biggest profit.

Profit maximisation. Aiming for the biggest profit.

The aim ignores when the profit arrives.

This aim has a name. **Profit maximisation** means aiming for the biggest profit. For a first answer it is reasonable, because profit is what remains for the owners after all costs. But it fails in three ways, and each failure teaches something important about finance.

FIRST WEAKNESS: TIMING

The first weakness is *timing*. Profit maximisation does not say when the profit arrives. Suppose that a firm may receive Tk 1,00,000 today or Tk 1,00,000 in three years. The two choices have the same profit, so profit maximisation cannot separate them. But nearly everyone prefers the money now. Money received today can be used today, to repay a loan or to buy goods, and it earns interest while the other waits.

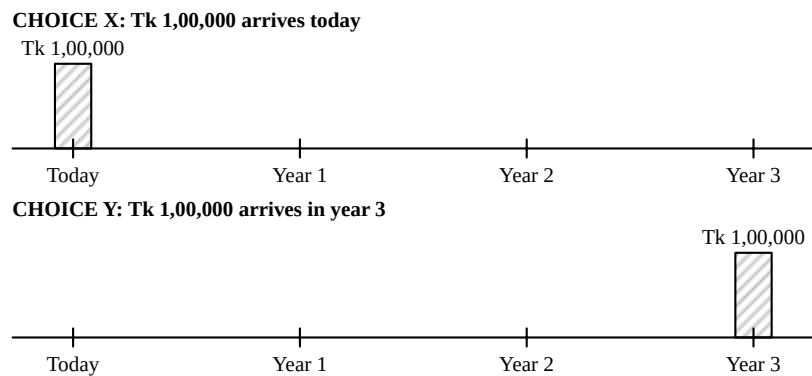


FIG. 5. Two choices with the same amount of money arriving at different times.

Fig. 5 shows the same amount arriving at two different times.

Fig. 5 shows the two choices. Each line is a timeline, with time moving from the left to the right. The tall bar on the first line is the money arriving today. The bar on the second line is the same money arriving in year 3. The sums are equal, but the choices are not. Chapter 4 teaches how to measure the difference with numbers.

SECOND WEAKNESS: RISK

The second weakness is *risk*. Suppose a firm may choose between two plans. Plan P gives a sure profit of Tk 50,000. Plan Q gives Tk 1,00,000 if sales are good and nothing if they are not, and the two outcomes are equally likely. On average, plan Q gives Tk 50,000 as well. Yet most people prefer the sure sum, because they prefer to avoid the chance of nothing. Profit maximisation does not notice the difference.

THIRD WEAKNESS: CASH

The third weakness is *cash*. Chapter 1 showed that profit and cash are different. A firm that counts profit when it makes a sale may have no cash until its customers pay. A goal built on profit alone can therefore lead a firm to approve plans that earn on paper but leave the firm unable to pay its bills.

The three weaknesses point to a better aim. First we need a word. **Wealth** means the total value of what a person owns less what the person owes. For the owner of a firm, wealth in the firm is the equity of Chapter 1, valued as fairly as possible.

The aim also ignores how sure the profit is.

The aim counts profit, but a firm pays its bills with cash.

Wealth is the total value of what the owners own after debts are paid.

Wealth. The total value of what a person owns less what the person owes.

The better aim is **wealth maximisation**. Wealth maximisation means aiming to raise the owners' total value over time. This aim answers all three weaknesses. It looks at *when* money arrives, because money later is worth less than money now. It looks at *risk*, because a sure sum is worth more than an uncertain sum of the same average size. And it looks at *cash*, because only cash can be spent or paid to the owners.

TWO POSSIBLE AIMS OF THE FIRM

QUESTION	PROFIT MAXIMISATION	WEALTH MAXIMISATION
Does it consider timing?	No	Yes
Does it consider risk?	No	Yes
Does it use cash?	No, it uses profit	Yes
Does it look beyond this year?	Not necessarily	Yes, over time

How can anyone see whether wealth is rising? In a large company, a **shareholder** means the owner of a share of the company, and many shares are bought and sold every day. The **share price** means the price at which one share can be bought or sold. When buyers expect the firm to bring in more money, with less risk and sooner, they are willing to pay more. A rising share price is therefore a scoreboard for the owners' wealth.

The scoreboard has limits. A share price may rise or fall because of events outside the firm, such as news about the whole country. A manager who tries to push the price up this month, by hiding bad news or by cutting useful spending, may reduce the firm's real value in the years that follow. For this reason wealth is judged over years, not days. A firm whose shares are not traded, such as Sundial Bakery, has no price. Its owner must estimate the firm's value, and the methods of Chapters 7 and 11 are the tools.

Finally, a word of caution about the aim itself. Wealth maximisation does not permit a firm to cheat its customers, to deceive its lenders, or to harm its workers. A firm that does these things soon loses its customers, its credit and its good name, and its wealth falls. The aim works best within the law and fair dealing, and lasting wealth usually grows from them.

Before moving on. Profit maximisation ignores timing, risk and cash. Wealth maximisation aims to raise the owners' total value over time, and in a company the share price is its scoreboard. Both aims must be pursued within the law and with fair dealing.

You should now be able to say: "A firm aims to raise the owners' wealth over time, because profit alone ignores when, how surely and in what form the money arrives."

Pause and check. A firm may choose a sure gain of Tk 60,000 or an equal chance of Tk 1,20,000 and nothing. Why can profit maximisation not tell the two apart?

Wealth maximisation aims to raise the owners' total value over time.

Wealth maximisation. Aiming to raise the owners' total value over time.

The table sets the two aims side by side.

Wealth is hard to see inside a firm, so a share price is used as a scoreboard.

Shareholder. The owner of a share of a company.

Share price. The price at which one share can be bought or sold.

The scoreboard has limits, and the manager should not chase the price day by day.

Wealth maximisation does not mean seeking riches by any means.

Before moving on: profit alone fails on timing, risk and cash, while wealth maximisation meets all three.

The test sentence states the goal.

The pause question applies the idea of risk.

2.5 AGENCY: WHEN OWNERS AND MANAGERS DIFFER

A hired manager may want something different from the owner, and the bakery's bonus rule shows how.

The hiring terms are recorded in the case file.

The owner is the principal and the manager is the agent.

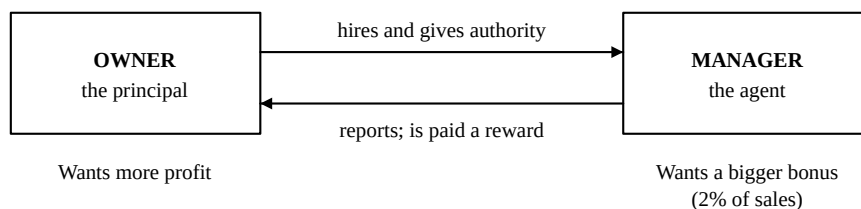
Principal. The one who hires another to act for them.

Agent. The one hired to act for another.

When the owner of a small bakery also runs the shop, the owner's wishes and the manager's wishes are the same, because they are one person. The position changes when the owner hires someone else. On 1 March 2026 Sundial Bakery hires a shop manager. The terms are in the case file.

Case file, illustrative. On 1 March 2026 Sundial Bakery hires a shop manager at Tk 30,000 a month plus a bonus of 2 per cent of the sales of the month. The bonus depends on sales, not on profit.

Two words describe the arrangement. The **principal** means the one who hires another to act for them. The **agent** means the one hired to act for another. The owner of Sundial Bakery is the principal, and the shop manager is the agent. Fig. 6 shows the arrangement.



When the two wants differ, an agency problem can arise.

FIG. 6. The owner (principal) and the manager (agent), and what each wants.

The two want different things, and that is the source of trouble.

A worked example shows how the bonus rule can lead the manager to lower prices.

Agency problem. A situation in which the agent's aims differ from the principal's.

The data given and the question to answer.

The rule and the arithmetic, step by step.

The answer, and a check that it is sensible.

Read Fig. 6 from the left. The owner hires the manager and gives authority, shown by the upper arrow. The manager reports and is paid a reward, shown by the lower arrow. Under the boxes are the two wants. The owner wants more profit. The manager, whose bonus is 2 per cent of sales, wants more sales. These two wants are not the same.

When the agent's aims differ from the principal's, an **agency problem** arises. An agency problem means a situation in which the agent's aims differ from the principal's. An example will show how this happens. Suppose that in one month the bakery sells 7,500 boxes of buns at Tk 40, and each box costs Tk 24 to make. In a second version of the same month, the manager lowers the price to Tk 34 and sells 9,000 boxes.

Given. Normal month: price Tk 40, 7,500 boxes. Lower-price month: price Tk 34, 9,000 boxes. Cost to make each box: Tk 24. The bonus is 2 per cent of sales. *Find.* In each month, the sales, the manager's bonus, and the money left to the owner after the cost of making the boxes.

Rule. Sales = price $\times$ boxes. Bonus = $0.02 \times$ sales. Money left = (price less cost per box) $\times$ boxes. *Steps (normal).* Sales = $40 \times 7,500 =$ Tk 3,00,000. Bonus = $0.02 \times 3,00,000 =$ Tk 6,000. Money left = $(40 \text{ less } 24) \times 7,500 = 16 \times 7,500 =$ Tk 1,20,000. *Steps (lower price).* Sales = $34 \times 9,000 =$ Tk 3,06,000. Bonus = $0.02 \times 3,06,000 =$ Tk 6,120. Money left = $(34 \text{ less } 24) \times 9,000 = 10 \times 9,000 =$ Tk 90,000.

Answer. The manager's bonus rises from Tk 6,000 to Tk 6,120, while the money left to the owner falls from Tk 1,20,000 to Tk 90,000, a loss of Tk 30,000. *Check.* The lower price loses Tk 6 on each of the 7,500 boxes

that would have sold anyway: $7,500 \times 6 = \text{Tk } 45,000$. The extra 1,500 boxes gain $1,500 \times 10 = \text{Tk } 15,000$. The net change is 15,000 less 45,000, a loss of Tk 30,000, and this agrees with the answer.

THE SAME MONTH WITH TWO PRICES

ITEM	NORMAL MONTH	LOWER-PRICE MONTH
Price of a box	Tk 40	Tk 34
Boxes sold	7,500	9,000
Sales	Tk 3,00,000	Tk 3,06,000
Manager's bonus (2% of sales)	Tk 6,000	Tk 6,120
Money left to the owner after the cost of the boxes	Tk 1,20,000	Tk 90,000

The table puts the two months side by side.

The manager gains a little and the owner loses a great deal. The manager may not intend harm. The bonus rule rewards whatever raises sales, and the manager responds to the reward. The fault lies in the rule, which pays for the wrong result.

The manager gains slightly and the owner loses much, though the manager may not intend harm.

The owner has two kinds of loss here. One is the loss of Tk 30,000 caused by the conflict. The other is the effort and money spent on preventing it, such as checking the manager's prices. Together they are **agency cost**, which means the cost of the conflict or of preventing it. Agency costs are a real cost of hiring someone to act for you.

The cost of such a conflict, and of preventing it, is called agency cost.

Agency cost. The cost of the conflict, or of preventing it.

The owner can reduce the problem with an **incentive**, which means a reward that guides behaviour. A good incentive rewards the result that the owner wants. Suppose the bonus were 5 per cent of the money left to the owner, not 2 per cent of sales. In the normal month the bonus is $0.05 \times 1,20,000 = \text{Tk } 6,000$, the same as before. In the lower-price month it is $0.05 \times 90,000 = \text{Tk } 4,500$, which is less. Now the manager has no reason to lower the price in this way.

An incentive is a reward that guides behaviour, and a better rule can align the two aims.

Incentive. A reward that guides behaviour.

There are other ways to reduce an agency problem. The owner may check the reports and prices each week, which takes time. The owner may set a rule, for example that no price may be lowered without permission. Or the owner may share ownership with the manager, so that the manager gains when the firm's value rises. Each method has its own cost, and no method removes the problem completely.

Other ways to reduce the problem are checking, rules and ownership.

In a large company the problem is larger. The shareholders may number in the thousands and cannot watch the managers daily. The managers, as agents, may prefer a larger firm, more comfort for themselves, or safer choices than the shareholders would want. The ideas of this section explain why companies use bonus rules linked to shareholder wealth, and why they require careful reports.

In a large company the problem is larger, because the owners are many and far from the firm.

Before moving on. The owner who hires a manager is the principal, and the manager is the agent. If the manager's aims differ from the owner's, an agency problem arises. The conflict and the cost of preventing it are agency cost. A good incentive rewards what the owner truly wants.

Before moving on: when owners hire managers, their aims may differ, and the cost of this is an agency cost.

You should now be able to say: "An agency problem arises when the one who acts wants something different from the one who hires."

The test sentence states the main point.

A WORKED CASE: THE SECOND OVEN AND THE BONUS RULE

This case applies the four decisions, the manager's pay and the incentive to the bakery's February and March.

The questions ask for the decisions, the manager's pay and the effect on the owner.

The oven touches all four decisions, and the pay follows from the rule.

Part (c) needs the sales, the bonus and the money left, before and after.

The conclusion states the three answers.

The check confirms the differences and the lesson.

The summary restates the chapter in a few lines.

This problem applies the four decisions and the idea of agency to a new firm.

Case file, illustrative. In February 2026 Sundial Bakery considers a second oven, with an outlay of Tk 3,00,000. On 1 March 2026 it hires a shop manager at Tk 30,000 a month plus 2 per cent of the month's sales as a bonus. In a later month the manager lowers the price of a box from Tk 40 to Tk 38 and sells 8,500 boxes instead of 7,500. Each box costs Tk 24 to make.

Question. (a) Which of the four decisions does the oven involve? (b) What is the manager's pay in a month with sales of Tk 3,00,000? (c) What happens to the manager's bonus and to the owner's money when the price falls to Tk 38?

Reasoning, part (a). Choosing to buy the oven is an investment decision. Choosing how to pay for it is a financing decision. Making sure that flour and wages can still be paid is a working capital decision. Taking money out of the firm, or leaving it in, is the dividend decision. The oven touches all four. *Reasoning, part (b).* Pay = Tk 30,000 + 0.02 × 3,00,000 = Tk 30,000 + Tk 6,000 = Tk 36,000.

Reasoning, part (c). Before: sales Tk 3,00,000, bonus Tk 6,000, money left (40 less 24) × 7,500 = Tk 1,20,000. After: sales 38 × 8,500 = Tk 3,23,000, bonus 0.02 × 3,23,000 = Tk 6,460, money left (38 less 24) × 8,500 = 14 × 8,500 = Tk 1,19,000.

Conclusion. (a) The oven involves all four decisions. (b) The manager's pay is Tk 36,000. (c) The bonus rises by Tk 460, from Tk 6,000 to Tk 6,460, while the owner's money left falls by Tk 1,000, from Tk 1,20,000 to Tk 1,19,000.

Check. The bonus change is 6,460 less 6,000, which is 460. The owner's change is 1,19,000 less 1,20,000, which is a fall of 1,000. The manager gains while the owner loses, though the loss here is small. The bonus on sales rewards a result that the owner does not fully want.

SUMMARY

A firm's money decisions fall into four groups: investment, financing, working capital and dividend. The financial manager plans, raises, allocates, controls and reports. Finance decides how efficiently a firm uses its money and whether it survives. Profit maximisation ignores timing, risk and cash, so the better aim is wealth maximisation: to raise the owners' total value over time, for which the share price is a scoreboard. When owners hire managers, an agency problem can arise, and incentives, checking and rules can reduce it at some cost.

THINK IT THROUGH

Greenleaf Stationers (the firm of Chapter 1) hires a shop manager at Tk 20,000 a month plus 3 per cent of sales. In a normal month it sells 2,000 notebooks at Tk 50, and each notebook costs Tk 30. The manager offers a discount that lowers the price to Tk 45 and raises sales to 2,400 notebooks. (a) Find the bonus in each case. (b) Find the money left to the owner after the cost of the notebooks in each case. (c) Does the bonus rule serve the owner? Suggest a change to the rule.

WHERE PEOPLE GO WRONG

"The financial manager runs the shop." The shop manager directs the daily work. The financial manager directs the firm's money. In a small firm one person may do both, but the tasks are different.

"The firm's aim is the biggest profit." Profit ignores when it arrives, how sure it is, and whether it is cash. The better aim is the owners' wealth over time.

"Wealth maximisation means making money at any cost." A firm that cheats customers or harms its workers loses its good name and its customers, and so loses wealth.

"A manager always acts for the owner." A manager acts for the owner when the rewards and the checks make this the manager's own interest. Where they do not, an agency problem arises.

Four common mistakes are stated here and corrected.

The first mistake mixes up the shop manager and the financial manager.

The second mistake thinks profit is the same as the owners' wealth.

The third mistake treats wealth maximisation as permission to do anything.

The fourth mistake thinks that managers always act for the owners.

IN HISTORY: WHEN OWNERSHIP AND CONTROL PARTED

In 1932 two American scholars, Adolf Berle and Gardiner Means, published *The Modern Corporation and Private Property*. They studied large American companies. They argued that the ownership of such companies was spread among many shareholders, and that the control of the companies had passed into the hands of salaried managers. The owners provided the money, but they no longer directed the firm.

This separation of ownership and control raised a question that remains important: how can many scattered owners be sure that managers act in their interest? The ideas of principal and agent, of incentives and of agency cost are later answers to that question. Section 2.5 applied them to a single bakery, but they apply with greater force to companies with thousands of shareholders.

A real study showed that, in large companies, owners and managers had become different people.

The argument raised the question this chapter asks: who watches the managers?

SOURCES

Berle, A. and Means, G. (1932), *The Modern Corporation and Private Property*, discussed above, first described the separation of ownership and control in large companies. Jensen, M. and Meckling, W. (1976), "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure", in the *Journal of Financial Economics*, gave the term *agency cost* a precise meaning and developed the theory of the conflict between owners and managers.

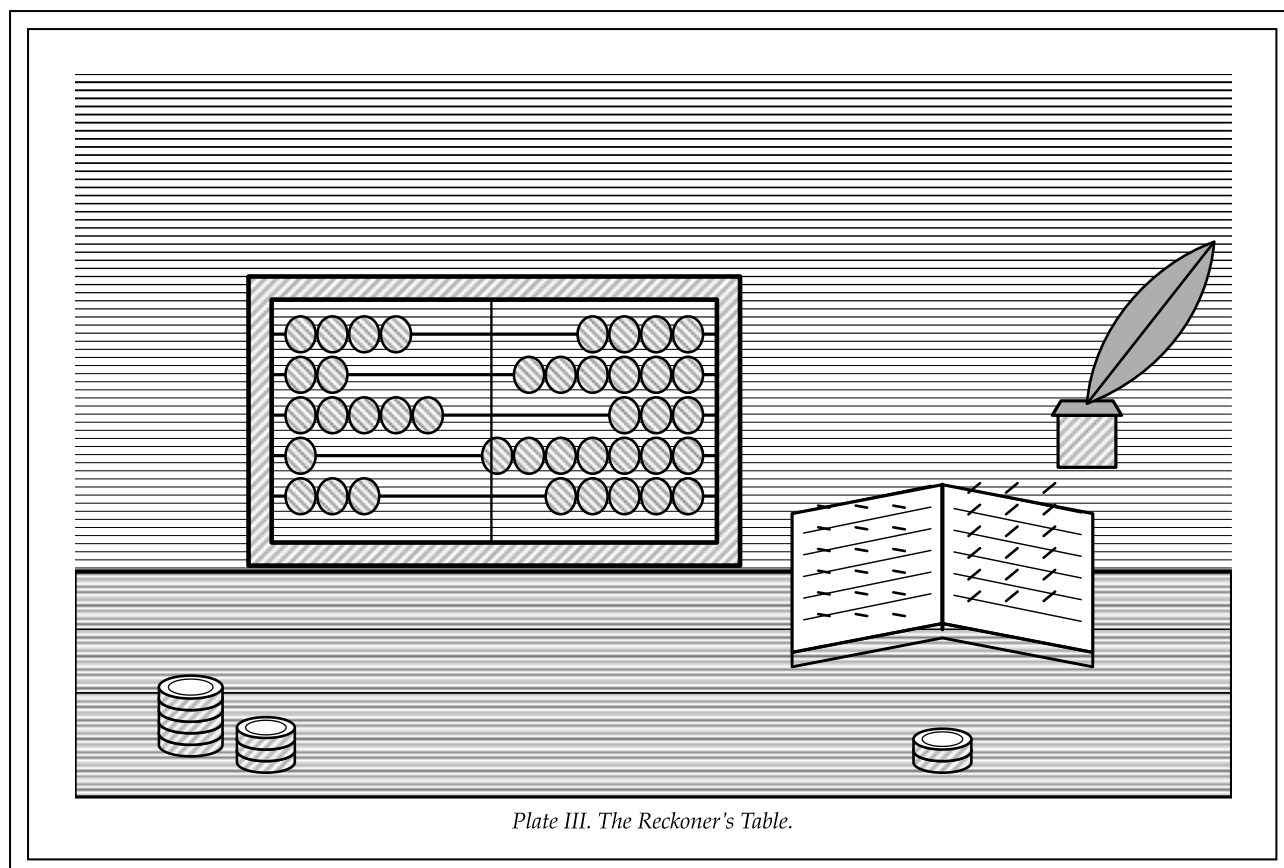
Two works stand behind the ideas of this chapter.

REVIEW QUESTIONS

1. Name the four decisions of finance and give an example of each. 2. Who is the financial manager? How does this person differ from a shop manager? 3. List and explain the five tasks of the financial manager. 4. Give two reasons why good finance matters to the survival of a firm. 5. A firm holds Tk 2,00,000 of idle cash while it pays 10 per cent a year on a bank loan. Find the yearly interest it could save by repaying part of the loan. 6. Explain three weaknesses of profit maximisation. 7. What is wealth maximisation? Why is the share price called a scoreboard, and

These questions test the chapter's main ideas.

what are its limits? 8. Define principal, agent, agency problem and agency cost. Show with an example how a bonus rule can create an agency problem, and how it can be changed.



CHAPTER III

Numbers You Will Need

How much, how many, and how to be sure



This chapter gathers the small tools of arithmetic that every later chapter uses: percentages, powers, tables, and the layout that carries a firm's sales down to its profit. Each tool is taught slowly, with the bakery as our guide.

A shirt, a rise of 20%, and a fall of 20%: where does the price end?

Work it with a pencil, and the answer changes.

A second puzzle: two true statements about the same profit.

Finance runs on a few small tools of arithmetic, which this chapter teaches slowly.

You will need only the words from Chapter I.

Slow passages are flagged, and each section ends with a short recap.

We begin in the market, where a percentage is easy to feel.

Percentage. A part out of 100. So 25% means 25 out of every 100.

A percentage is a part out of 100.

Three different questions use percentages, and beginners often mix them.

A shopkeeper in a Dhaka market sells a shirt for Tk 100. In the festival season she raises the price by 20%. When the festival ends, she lowers the new price by 20%. A customer watches both changes and says, "Good, the shirt is back at Tk 100." The shopkeeper smiles and says nothing. Is the customer right?

Try it step by step. A rise of 20% on Tk 100 adds Tk 20, so the price becomes Tk 120. A fall of 20% now takes away one fifth of Tk 120, which is Tk 24. The new price is Tk 96. The customer is wrong by four taka. The reason hides in two small words: a percentage is always a percentage of *something*, and the something changed.

A second puzzle lives in the bakery's books. The owner says, "Our profit rose by 20% this quarter." The bank manager says, "Your profit rose from 10 to 12 per cent of sales, which is a rise of two." Both speak the truth, yet their numbers differ. They are measuring the same event with two different tools. By the end of this chapter you will be able to say exactly which tool each person used.

Finance is the study of money, and money is counted with numbers. Later chapters will use a small set of tools again and again: percentages, powers, tables, and one standard layout that carries sales down to profit. None of these is difficult. Each is easy to misuse when we hurry. So this chapter moves slowly, one tool at a time, so that later chapters can move faster.

You will need: the words sales, cost and profit from Chapter I, and the picture of the bakery that began there. The arithmetic itself is arithmetic you already know. We arrange it with care and give it names.

Two signs will guide you. A note in the side column that says *Go slowly here* marks a passage that deserves a second reading. A short recap called *Before moving on* ends every section. Questions called *Pause and check* appear after the harder passages; their answers are at the back of the book.

3.1 PERCENTAGES, RATIOS AND ROUNDING

A woman buys a shirt marked Tk 800. A sign says "10% off". She wants to know what she will pay. She thinks: ten out of every hundred taka are taken away. There are eight hundreds in 800, so Tk 80 is taken away. She pays Tk 720. This small sum already contains the whole of the first idea in this chapter.

A **percentage** means a part out of 100. The sign % is read "per cent", which means "for each hundred". So 25% means 25 out of every 100, or one quarter. If 50 of your 200 customers buy cake, then 25 of every 100 customers buy cake, and we say that 25% of your customers buy cake. A percentage lets us compare parts of wholes of different sizes.

Percentages answer three different questions. The first asks: what is 25% of a given amount? The second asks: this part is what percentage of that whole? The third asks: by what percentage did a number change? Each question has its own rule. We take them in turn. In this book every calculation is shown in the same order: *Given, Find, Formula, Steps, Answer, Check*.

FINDING A PERCENTAGE OF AN AMOUNT

The rule in words: the part equals the whole multiplied by the percentage, divided by 100. In symbols, $\text{part} = \text{whole} \times p \div 100$, where p stands for the percentage. The word *whole* means the full amount that the percentage is taken of. In the shirt, the whole was Tk 800 and p was 10.

Given. On 2 January 2026 the bank lent the bakery Tk 4,00,000 at 12% a year. *Find.* The interest for one year. *Formula.* $\text{Interest} = \text{whole} \times p \div 100$, with the whole equal to the loan and p equal to 12.

Steps. First, $4,00,000 \times 12 = 48,00,000$. Second, $48,00,000 \div 100 = 48,000$. *Answer.* The interest for one year is Tk 48,000. *Check.* Ten per cent of Tk 4,00,000 is Tk 40,000, and 2% is Tk 8,000. Together they make Tk 48,000, so the answer is sensible.

Given. A loan of Tk 3,75,000 carries interest at 9.5% a year. *Find.* One year's interest. *Steps.* $3,75,000 \times 9.5 = 35,62,500$, and $35,62,500 \div 100 = 35,625$. *Answer.* Tk 35,625. *Check.* Ten per cent of Tk 3,75,000 is Tk 37,500, and half of 1% is Tk 1,875. Then $37,500 - 1,875 = \text{Tk } 35,625$. A decimal in the rate changes nothing about the method.

Pause and check. Use the same three steps to find 15% of Tk 2,40,000. Write the given, the formula and the check. The answer is at the back of the book.

FINDING WHAT PERCENTAGE A PART IS OF A WHOLE

Now we turn the question around. We know the part and the whole, and we want the percentage. The rule in words: the percentage equals the part divided by the whole, multiplied by 100. In symbols, $p = \text{part} \div \text{whole} \times 100$. This rule is the first rule run backwards, and the check is to run it forwards again.

Given. In the first three months of 2026, the bakery's sales were Tk 9,00,000. The cost of flour, sugar and other items that grow with each box baked was Tk 5,40,000. *Find.* What percentage of sales this cost is. *Formula.* $p = \text{part} \div \text{whole} \times 100$.

Steps. First, $5,40,000 \div 9,00,000 = 0.6$. Second, $0.6 \times 100 = 60$. *Answer.* The cost is 60% of sales. *Check.* Run the first rule forwards: 60% of Tk 9,00,000 is $9,00,000 \times 60 \div 100 = \text{Tk } 5,40,000$, which is the part we began with. The answer is correct. We will meet this 60% again in section 3.4.

FINDING A PERCENTAGE CHANGE

Here is the passage that settles the shirt puzzle, so read it twice. A **percentage change** means the change in a number, shown as a percentage of the old number. A bus fare rises from Tk 20 to Tk 25. The rise is Tk 5. Compare Tk 5 with the *old* fare of Tk 20: it is one quarter, so the fare rose by 25%. The old value is the whole in this rule.

The rule in words: first find the change, which is the new value less the old value. Then divide the change by the old value, and multiply by 100. In symbols, $\text{percentage change} = (\text{new} - \text{old}) \div \text{old} \times 100$. A positive answer is a rise. A negative answer is a fall.

The rule does not always give a round answer. Later we will see that schools bought Tk 2,44,000 of the bakery's Tk 9,00,000 sales in the first quarter. Then $p = 2,44,000 \div 9,00,000 \times 100 = 27.11\%$ to two decimal places. *Check:* 27.11% of 9,00,000 is 2,43,990, which is close to 2,44,000.

To find a percentage of an amount, multiply the amount by the percentage and divide by 100.

First worked example: the bank's interest on the opening loan.

The steps, the answer and a check by a different route.

A second, messier example uses a rate with a decimal part.

Pause and check: find 15% of Tk 2,40,000.

To find what percentage a part is of a whole, divide the part by the whole and multiply by 100.

Second worked example: how large is the bakery's variable cost compared with its sales?

The steps, the answer and the check by running the rule forwards.

GO SLOWLY HERE.

Percentage change. The change in a number, shown as a percentage of the old number.

A percentage change compares the change with the old value, not the new one.

The rule in words and symbols.

A second example of the part-to-whole rule, with an awkward answer.

Third worked example: the price of flour goes up, and then comes back down.

The steps and the answers show that the two changes are not equal.

The check, and the reason: the old value changed.

Fig. 1 shows the shirt puzzle as three bars.

The small gap of Tk 10 comes from rounding the percentage, and this is normal.

Given. The bakery's flour cost Tk 50 for one kilogram. A month later it cost Tk 56. *Find.* The percentage change from Tk 50 to Tk 56, and then the percentage change from Tk 56 back to Tk 50. *Formula.* Percentage change = $(\text{new} - \text{old}) \div \text{old} \times 100$.

Steps for the rise. The change is $56 - 50 = 6$. Then $6 \div 50 = 0.12$, and $0.12 \times 100 = 12$. *Steps for the fall.* The change is $50 - 56 = -6$. Then $-6 \div 56 = -0.107142\dots$, and $\times 100$ gives -10.71% to two decimal places. *Answer.* The price rose by 12%, and the return trip is a fall of 10.71%, not 12%.

Check. Increase Tk 50 by 12%: $50 + 6 = 56$. Decrease Tk 56 by 10.71%: 10.71% of 56 is about 6.00, so $56 - 6 = 50$. Both work. The two percentages differ because the old value in the first case was Tk 50 and in the second case was Tk 56. A rise and a fall of the same size, measured against different old values, give different percentages.

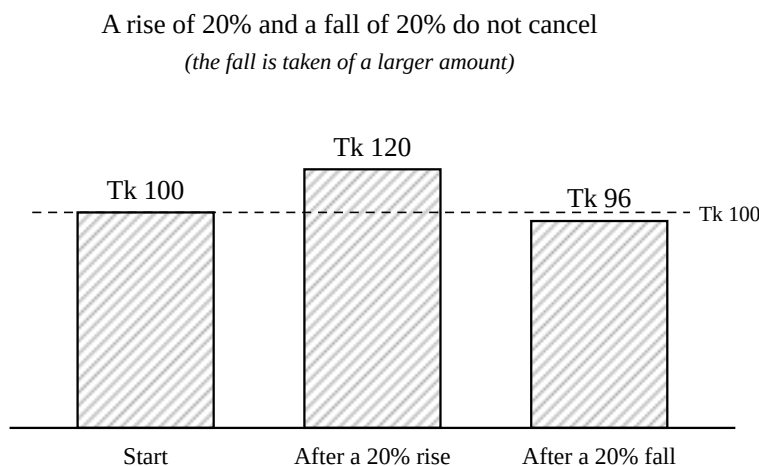


FIG. 7. The shirt puzzle. After a 20% rise and a 20% fall the price is Tk 96, not Tk 100.

The shirt puzzle is solved: the second 20% was taken of a larger amount.

A fall can be measured in the same way, as a negative change.

Percentage point. The gap between two percentages. From 10% to 12% is two percentage points.

Percentage points measure the gap between two percentages, which is not the same as percentage change.

Pause and check: a loan rate moves from 12% to 14%.

Now we can settle the shirt. The rise of 20% was taken of Tk 100 and added Tk 20. The fall of 20% was taken of Tk 120 and removed Tk 24. Because the second percentage was taken of a larger whole, it removed more than the first added. To return from Tk 120 to Tk 100, the price must fall by $20 \div 120 \times 100 = 16.67\%$, not 20%.

A percentage change can also be applied to a rate. Suppose the bank cuts its yearly rate from 12% to 10.5%. The change is $10.5 - 12 = -1.5$. Divide by the old rate: $-1.5 \div 12 = -0.125$, so the rate fell by 12.5%. The gap is 1.5 percentage points, and the percentage change is 12.5%. Both statements are correct and they answer different questions.

The second puzzle needs one more term. A **percentage point** means the plain gap between two percentages. When the bakery's profit share of sales moves from 10% to 12%, the gap is two percentage points. That is the bank manager's number. But 12 is 20% more than 10, because $2 \div 10 = 0.20$. That is the owner's number. Later chapters compare interest rates, so remember both tools: points for the gap, percent for the relative change.

Pause and check. A loan rate moves from 12% to 14%. State the change in percentage points. Then state the change as a percentage of

the old rate. Compare your two answers with those at the back of the book.

You should now be able to say: "A percentage change is the change divided by the old value, times 100, and a rise and a fall of equal percentages do not cancel."

RATIOS

A **ratio** means one number compared with another. Ratios are written with a colon, as in 3 : 2, and read "3 to 2". At the start of the bakery, the owner put in Tk 6,00,000 and the bank lent Tk 4,00,000. The ratio of the owner's money to the bank's money is 6,00,000 : 4,00,000. Divide both numbers by 2,00,000 and the ratio becomes 3 : 2, which is the same comparison in smaller numbers.

A ratio can be changed into a single number by dividing: $6,00,000 \div 4,00,000 = 1.5$, so the owner put in 1.5 times as much as the bank. It can also be changed into percentages of the total. The total money was Tk 10,00,000. The owner's share is $6,00,000 \div 10,00,000 \times 100 = 60\%$, and the bank's share is 40%. The three forms carry the same information. Later chapters will switch between them without warning, so practise all three.

ROUNDING

Many divisions never end. Dividing 1 by 7 gives 0.142857142857..., and no one wants to write it all. **Rounding** means replacing a long number by a nearby shorter one. We choose how many decimal places to keep. Look at the first digit we would drop. If it is 5 or more, increase the last kept digit by one. If it is 4 or less, leave the last kept digit as it is. So 7.1428 becomes 7.14 to two decimal places, and 7.1463 becomes 7.15.

In this book, money is rounded to the nearest whole taka unless the amount is small, when we keep paisa. Percentages and ratios are rounded to two decimal places. A word of warning: round only at the end. If you round in the middle of a calculation, the small errors travel into the answer and grow. Section 3.6 shows this with real numbers.

Before moving on. A percentage is a part out of 100. Three questions use it: the part of a whole, the percentage a part makes, and the change from an old value. A ratio compares two numbers. Round at the end, to two decimal places for rates.

3.2 POWERS AND ROOTS; USING A CALCULATOR

Suppose a rumour starts with one person. That person tells two friends, and each of those tells two more. After one round, two people know. After two rounds, four know. After three rounds, eight know. Each round multiplies the number by 2. We multiplied 2 by itself three times, and the result is 8. Finance meets this pattern whenever money grows by the same percentage each year. Chapter IV builds on it, so we meet it now in its simplest form.

You should now be able to say how a percentage change is found.

Ratio. One number compared with another, such as 3 to 2.

A ratio compares two numbers by showing how many times one contains the other.

A ratio can also be written as one number, or as percentages of the total.

Rounding. Replacing a number by a nearby shorter one, such as 7.1428 by 7.14.

Rounding shortens a number, and a fixed rule keeps everyone's answers equal.

This book's rounding habits, so that your answers match ours.

A recap of the section.

GO SLOWLY HERE.

Repeated growth is repeated multiplication, which is why powers matter to finance.

Power. A number multiplied by itself a stated number of times. (In daily life, strength or authority.)

A power is a number multiplied by itself a stated number of times.

Exponent. The small raised figure that says how many times a number is multiplied by itself.

The exponent counts the multiplications, not the number itself.

First worked example: a price that rises 10% each year for three years.

The steps, the answer and a check.

Second worked example: a messier number, 1.08 to the power 5.

The calculator does the same work in one step.

Order of operations. The fixed order in which a calculation is worked: brackets, powers, times and divide, add and subtract.

A calculation has a fixed order of work, and ignoring it gives wrong answers.

Typing a formula in the wrong order is a very common slip.

Root. The number that, raised to a stated power, gives a given number. (In daily life, part of a plant.)

A root undoes a power, and finance uses roots to find growth rates.

A **power** means a number multiplied by itself a stated number of times. We write the number, and a small raised figure beside it. So 2^3 means $2 \times 2 \times 2$, and the answer is 8. We say "2 to the power 3". The raised figure is called the **exponent**. It tells us how many times the number appears in the multiplication. The exponent 2 is also called "squared", and the exponent 3 is also called "cubed".

Notice that the exponent counts the *copies* of the number, not the number of times we multiply. In 2^3 there are three copies of 2 but only two multiplication signs. A common slip is to confuse 2^3 with 2×3 . The first is 8 and the second is 6. In finance the number is usually close to 1, such as 1.08, and then the powers grow slowly and steadily.

Given. A price of Tk 1 rises by 10% each year for three years. After one year, it is multiplied by 1.10. *Find.* The number that Tk 1 is multiplied by after three years. *Formula.* Multiplier = $1.10^3 = 1.10 \times 1.10 \times 1.10$.

Steps. First, $1.10 \times 1.10 = 1.21$. Second, $1.21 \times 1.10 = 1.331$. *Answer.* After three years, Tk 1 becomes Tk 1.331. *Check.* If each year added a fixed 10 paise, Tk 1 would reach Tk 1.30. The true answer is a little larger, because later rises are taken of larger amounts, as in the shirt puzzle. So 1.331 is sensible.

Given. The number 1.08 raised to the power 5. *Find.* The value of 1.08^5 . *Formula.* $1.08^5 = 1.08 \times 1.08 \times 1.08 \times 1.08 \times 1.08$. *Steps.* $1.08^2 = 1.1664$. Then $1.1664 \times 1.08 = 1.259712$. Then $1.259712 \times 1.08 = 1.36048896$. Then $1.36048896 \times 1.08 = 1.4693280768$. *Answer.* $1.08^5 = 1.4693$ to four decimal places.

The calculator key for powers is usually marked $\wedge$ or x^y or y^x . Type 1.08, press the power key, type 5, and press equals. The display shows 1.469328077, which agrees with our hand work. *Check.* The answer must be a little above $1 + 5 \times 0.08 = 1.40$, because growth on growth adds a little more. It is. Learn where the power key sits on your own calculator before you need it in a hurry.

A formula often mixes brackets, powers, multiplication and addition. The **order of operations** is the fixed order in which we work them: brackets first, then powers, then multiplication and division from left to right, then addition and subtraction from left to right. Consider $1,00,000 \times (1 + 0.08)^3$. The bracket gives 1.08. The power gives $1.08^3 = 1.259712$. The multiplication gives Tk 1,25,971.

Compare two wrong ways. If you type $1,00,000 \times 1 + 0.08^3$, the calculator works the power first and then multiplies, and your answer is Tk 1,00,000.0005, which shows almost no growth at all. If you type 1.08×3 , you have multiplied instead of raising to a power and your answer is 3.24. Always write the bracket, or press the bracket keys, and then check that the size of your answer is sensible.

A power takes a number and builds a larger one. A **root** undoes this, and returns to the number we started with. Since $12 \times 12 = 144$, the square root of 144 is 12. Since $1.08^5 = 1.469328$, the fifth root of 1.469328 is 1.08. Roots matter to us because the growth rate of money, such as 8% a year, is hidden inside a power. A root uncovers it. Chapter IV does this, and Chapter VIII uses square roots again.

Given. The number 8. *Find.* The square root of 8. *Formula.* Find the number that, multiplied by itself, gives 8. *Steps.* Press the square root key ($\sqrt{}$) on your calculator, then type 8. The display shows 2.828427... *Answer.* The square root of 8 is 2.83 to two decimal places. *Check.* $2.83 \times 2.83 = 8.0089$, which is very near 8. We will use this exact number in Chapter VIII.

For roots other than the square root, many calculators have a root key. Where this is absent, raise the number to the power of one divided by the root. The fifth root is found by raising to the power $1 \div 5 = 0.2$. Type 1.469328, press the power key, type 0.2, and press equals. The display shows 1.08. This is the method we will use in Chapter IV to find a rate.

There is one more form. A power with a minus sign in the exponent means one divided by the ordinary power. So 1.08^{-5} means $1 \div 1.08^5 = 1 \div 1.469328 = 0.680583$. Many calculators have a key marked $1/x$ for the division. Chapter IV needs this form to bring money back from the future. For now, note only that 1.08^5 makes a number larger and 1.08^{-5} makes it smaller.

Given. The number 1.10 with the exponent -3 . *Find.* The value of 1.10^{-3} . *Formula.* $1.10^{-3} = 1 \div 1.10^3$. *Steps.* We found in the first example that $1.10^3 = 1.331$. Then $1 \div 1.331 = 0.7513$ to four decimal places. *Answer.* $1.10^{-3} = 0.7513$. *Check.* Multiply back: $0.7513 \times 1.331 = 1.0000$ to four decimals. A number and its reciprocal always multiply to 1.

Pause and check. Calculate 1.06^4 by repeated multiplication, and then with the power key. Then find the square root of 50, to two decimal places. The answers are at the back of the book.

Before moving on. A power is a number multiplied by itself, and the exponent counts the copies. Follow the order of operations. A root undoes a power, and a minus sign in the exponent means one divided by the power. You should now be able to say: " 1.08^5 means 1.08 multiplied by itself five times, and a root takes me back."

Third worked example: a square root.

Other roots are found with a fractional power.

A power with a minus sign means one divided by the power.

A worked example of a minus sign in the exponent.

Pause and check: powers and roots.

A recap of the section.

3.3 READING AND USING A TABLE

Think of a bus timetable. Down the left side are the stops. Across the top are the times of the buses. To find when the 9 o'clock bus reaches your stop, you run one finger along the row for your stop and another down the column for 9 o'clock, and read the number where they meet. That small habit is all that reading a table requires. Finance uses tables constantly, and Chapter IV introduces tables of numbers that save long calculations.

A **table** means numbers arranged in rows and columns so that each can be found quickly. A **row** runs across the page. A **column** runs down the page. The place where a row and a column meet is called a *cell*, and it holds one number. In this book, every table has a caption above it in small capitals, a heading row, and ruled lines only at the top, under the headings, and at the bottom.

The bakery sold the same number of boxes of buns in each of the first three months of 2026, but the boxes went to different buyers. The shop

A table is an everyday tool, and the bakery's records use it too.

Table. Numbers arranged in rows and columns so that each can be found quickly. (In daily life, a piece of furniture.)

Row. One line of a table that runs across the page.

Column. One line of a table that runs down the page.

A table arranges numbers in rows and columns, and each number sits in a cell.

The bakery's boxes of buns are best shown in a table, which we first introduce in words.

counter sold to walk-in customers. Some boxes went to schools and some to offices. The table below shows boxes sold, with the three channels in rows and the three months in columns. A final row and a final column hold totals.

BOXES OF BUNS SOLD, BY CHANNEL AND MONTH, JANUARY TO MARCH 2026

CHANNEL	JANUARY	FEBRUARY	MARCH	QUARTER
Shop counter	4,000	3,800	3,700	11,500
Schools	2,000	2,100	2,000	6,100
Offices	1,500	1,600	1,800	4,900
All channels	7,500	7,500	7,500	22,500

To read a single cell, find its row, find its column, and read where they meet.

Totals give a check: the row totals and the column totals must agree.

A table can be built into a second table by using percentages.

Let us read the table. How many boxes did the schools take in February? Find the row marked Schools and the column marked February. Where they meet, the cell holds 2,100. So the schools took 2,100 boxes in February. To read a row total, look at the last column: the shop counter sold 11,500 boxes in the quarter. To read a column total, look at the last row: in March, all channels together sold 7,500 boxes.

A table with totals carries its own check. Add the three row totals: $11,500 + 6,100 + 4,900 = 22,500$. Add the three monthly totals: $7,500 + 7,500 + 7,500 = 22,500$. Both sums give the same grand total, and the grand total sits in the bottom right cell. If the two sums disagree, a number is wrong. This check costs ten seconds and saves many mistakes.

Tables become useful when we ask questions of them. What share of all boxes did each channel take? Divide each channel's quarter total by the grand total and multiply by 100, as in section 3.1. For the shop counter, $11,500 \div 22,500 \times 100 = 51.11\%$. The next table gives all three shares, and the sales in taka, because each box sold for Tk 40.

SHARE OF BOXES AND SALES, BY CHANNEL, FIRST QUARTER 2026

CHANNEL	BOXES	SHARE OF BOXES	SALES (TK)
Shop counter	11,500	51.11%	4,60,000
Schools	6,100	27.11%	2,44,000
Offices	4,900	21.78%	1,96,000
All channels	22,500	100.00%	9,00,000

The shares must sum to 100%, and the sales must agree with the quarter's total.

Another common table gives a number to use according to which band a case falls in.

Always read the title, the headings and the units of a table before reading a number.

Check the table. The three shares add to $51.11 + 27.11 + 21.78 = 100.00$. The three sales amounts add to Tk 9,00,000, which is the quarter's sales in the profit layout of section 3.5. And $22,500 \text{ boxes} \times \text{Tk } 40 = \text{Tk } 9,00,000$. Two different routes reach the same total. When several routes agree, we trust the numbers.

A second kind of table gives a number to use. Imagine that the bakery's van charges a delivery fee by distance: Tk 20 for 0 to 5 kilometres, Tk 35 for 6 to 10 kilometres, and Tk 60 for 11 to 20 kilometres. A school is 8 kilometres away. Eight lies in the band 6 to 10, so the fee is Tk 35. Chapter IV uses tables in this way, to find a number that we then multiply.

A habit worth forming: read the caption and the headings before reading any number. The caption of the second table says that sales are in taka and that the period is the first quarter. Without this, a cell such as

4,60,000 has no meaning. Also check whether a table shows amounts, percentages or both, because a column of shares looks much like a column of boxes.

Pause and check. From the first table, find the percentage change in the boxes sold to offices from January to March. Then find the share of all boxes that schools took in February. The answers are at the back of the book.

Before moving on. A table has rows, columns and cells, and you read a cell where a row and a column meet. Totals give a check. A table can be turned into shares with the rules of section 3.1. You should now be able to say: "I can find any number in a table and check the totals."

Pause and check: reading the channel table.

A recap of the section.

3.4 FIXED AND VARIABLE COSTS

A woman runs a tea stall. She pays Tk 6,000 each month for the stall, whether she sells 100 cups or 3,000. She also pays Tk 4 for the tea leaves, milk and sugar in each cup. In a slow month she sells 1,000 cups, and in a busy month she sells 2,000. Her stall cost does not change. Her cost for ingredients changes with every cup. Two kinds of cost live side by side in her little business.

A **fixed cost** means a cost that stays the same however much the firm makes or sells. The stall rent of Tk 6,000 is a fixed cost. In a bakery, the shop rent and the monthly wages of permanent staff are fixed costs. They must be paid in a quiet month and in a busy one.

A **variable cost** means a cost that rises and falls with the amount the firm makes or sells. The tea leaves, milk and sugar are variable costs. We measure the amount in **units**: a unit is one item of what the firm sells, such as one cup of tea or one box of buns. If each unit carries a variable cost of Tk 4, then 1,000 units carry Tk 4,000 and 2,000 units carry Tk 8,000.

The rule in words: total cost equals the fixed cost plus the variable cost of each unit multiplied by the number of units. In symbols, total cost = $F + v \times q$, where F is the fixed cost, v is the variable cost per unit, and q is the number of units. The tea stall with 1,000 cups: $6,000 + 4 \times 1,000 = \text{Tk } 10,000$. With 2,000 cups: $6,000 + 4 \times 2,000 = \text{Tk } 14,000$.

Look at the cost of one cup. At 1,000 cups, $\text{Tk } 10,000 \div 1,000 = \text{Tk } 10$ per cup. At 2,000 cups, $\text{Tk } 14,000 \div 2,000 = \text{Tk } 7$ per cup. Doubling the cups did not double the cost, because the rent was shared among more cups. This fact will matter greatly in Chapter IX, where we ask why a small rise in sales can bring a large rise in profit.

Now the bakery. In the first quarter of 2026 the bakery's fixed cost was Tk 2,40,000, which is Tk 80,000 each month. Each box of buns cost Tk 24 in ingredients and other variable items, and each box sold for Tk 40. We will build a table of total cost at three levels of monthly output.

GO SLOWLY HERE.

A tea seller shows the idea in daily life before we use it in the bakery.

Fixed cost. A cost that stays the same however much the firm makes or sells.

A fixed cost stays the same as the amount sold changes.

Variable cost. A cost that rises and falls with the amount the firm makes or sells.

Unit. One item of what the firm sells, such as one box of buns.

A variable cost changes in step with the number of units made.

Total cost is the fixed cost plus the variable cost per unit times the number of units.

Cost for each unit falls as more units are made, because the fixed cost is shared.

The bakery's own numbers: a fixed cost of Tk 80,000 a month and a variable cost of Tk 24 a box.

MONTHLY COST OF MAKING BOXES OF BUNS

BOXES MADE	FIXED COST (TK)	VARIABLE COST (TK)	TOTAL COST (TK)	COST PER BOX (TK)
5,000	80,000	1,20,000	2,00,000	40.00
7,500	80,000	1,80,000	2,60,000	34.67
10,000	80,000	2,40,000	3,20,000	32.00

How to read the cost table.

Read the middle row. At 7,500 boxes, the variable cost is $24 \times 7,500 = \text{Tk } 1,80,000$. Add the fixed cost of Tk 80,000 and the total cost is Tk 2,60,000. Divide by 7,500 boxes and each box carries Tk 34.67. Look down the last column: the cost per box falls as the boxes rise, from Tk 40.00 to Tk 32.00. The fixed cost is the reason.

Fig. 2 shows the same costs as lines.

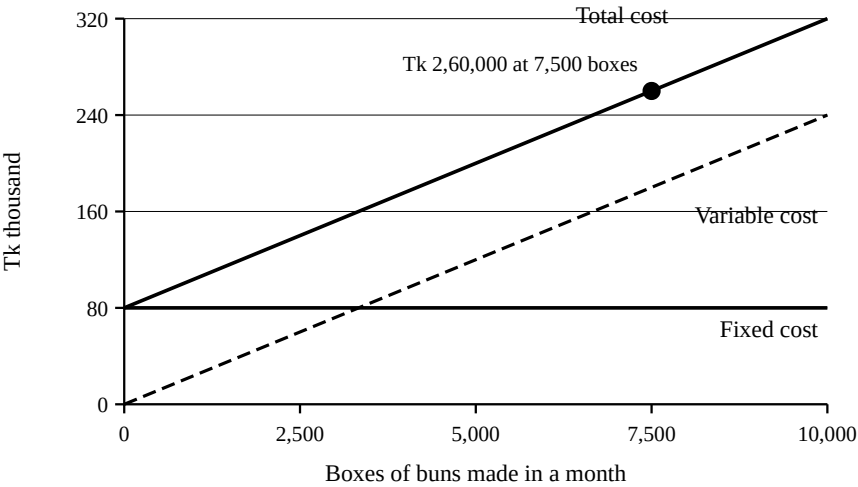


FIG. 8. The bakery's monthly costs. The fixed cost is level; the variable cost rises steadily; the total cost is the two together.

A figure shows the three costs, and how to read it.

To read Fig. 2, look along the bottom for the boxes made and up the left side for the cost in thousands of taka. The level line is the fixed cost, Tk 80,000, which does not move. The dashed line is the variable cost, which starts at zero and rises with every box. The top line is their sum. The dot marks the total cost of Tk 2,60,000 at 7,500 boxes, the same as in the table.

The two kinds of cost are compared in a short table, which we first introduce in words.

The two kinds of cost are easiest to remember side by side. The table below compares them on four points. Read each row from left to right and ask which cost it describes.

FIXED COST AND VARIABLE COST COMPARED

POINT	FIXED COST	VARIABLE COST
When boxes made rise	Stays the same	Rises in step
A bakery example	Shop rent, permanent wages	Ingredients, packaging
Cost of each box	Falls as boxes rise	Stays the same
Paid when nothing is sold	Yes	No

Contribution means the selling price of one unit less its variable cost. A box sells for Tk 40 and carries Tk 24 of variable cost, so each box has a contribution of $40 - 24 = \text{Tk } 16$. Each box sold adds Tk 16 toward paying the fixed cost. Once the fixed cost is paid, further contribution is profit before interest and tax. At 7,500 boxes the total contribution is $7,500 \times 16 = \text{Tk } 1,20,000$. Subtract the fixed cost of Tk 80,000 and Tk 40,000 remains each month.

Given. A cake sells for Tk 500 and carries Tk 320 of variable cost. *Find.* The contribution of one cake, and its share of the price. *Steps.* $500 - 320 = \text{Tk } 180$. Then $180 \div 500 \times 100 = 36\%$. *Answer.* Each cake contributes Tk 180, which is 36% of its price. *Check.* Variable cost is $320 \div 500 = 64\%$ of the price, and $100\% - 64\% = 36\%$.

Here are the exceptions, and they matter. First, no cost is fixed for ever. If the bakery needs a second oven (Chapter II), the rent of extra space and the wages of extra staff would rise, so the fixed cost climbs in steps. We say a cost is fixed *within a range of output and for a period of time*. Second, some costs are mixed. An electricity bill may have a fixed monthly charge plus a charge for each unit of power used. We separate such a cost into its fixed part and its variable part.

Viewpoint helps. The owner sees the fixed cost as a promise that must be paid even when no one buys. The shop manager sees the variable cost as the part that moves with her work: each box she sells brings a cost of Tk 24 but also adds Tk 16 of contribution. Both views are correct, and both use the same two labels.

Pause and check. The bakery pays shop rent, the cost of flour, the wages of two permanent bakers, and the packaging for each box. Classify each as fixed or variable. The answer is at the back of the book.

Before moving on. A fixed cost stays the same as sales change, and a variable cost rises with each unit. Total cost is F plus v times q . Contribution is selling price less variable cost for each unit. You should now be able to say: "Fixed cost is shared among more units as output rises, so the cost of each unit falls."

Contribution is what each unit sold adds toward the fixed costs and the profit.

Contribution. The selling price of one unit less its variable cost.

A second example of contribution, with a larger price.

A cost is not fixed or variable by its name, but by how it behaves.

The same cost can look different from the owner's and the manager's seats.

Pause and check: sort the costs.

A recap of the section, with the sentence to test yourself.

3.5 THE PROFIT LAYOUT: SALES TO NET PROFIT

Every firm in this book reports its profit in the same order. We call this order the *profit layout*. It starts with sales at the top. At each step, one kind of cost is taken away, and a new subtotal appears. The last subtotal is the profit that belongs to the owners. Learning the ladder once will save you from confusion in Chapters VI, IX, XIII, XIV and XV.

The first step takes the variable cost from sales. The result is the total contribution, as in section 3.4. The second step takes the fixed cost from the contribution. We give the result a name: **EBIT**, which means profit before interest and tax. The letters stand for Earnings Before Interest and Tax. EBIT is the profit of the firm's work itself, before we count what it pays to lenders and to the government.

The profit layout is a fixed ladder that carries sales down to profit, one step at a time.

The first steps: from sales, take variable cost to find contribution, then fixed cost to find EBIT.

EBIT. Profit before interest and tax: sales less variable cost and fixed cost.

Interest expense is the price of borrowed money, and it is taken next.

Interest expense. The money a firm pays lenders for using their money.

Profit before tax. Profit left after interest expense and before tax is paid.

Tax is the government's share of profit, and it is taken fourth.

Tax. Money a firm must pay the government on its profit.

Net profit. The profit left after interest and tax have been paid.

Net profit is the last line before payments to special owners.

Preferred dividend. A fixed payment to a special kind of owner, taught in Chapter XIV.

First, a small clean example of the ladder, with round numbers.

A table lays the whole ladder out for the bakery's first quarter of 2026.

The third step takes the interest. **Interest expense** means the money a firm pays to its lenders for using their money. The bakery's bank loan was Tk 4,00,000 at 12% a year, and interest for one year is Tk 48,000 (section 3.1). The first quarter is one quarter of a year, so interest expense is $\text{Tk } 48,000 \times 3 \div 12 = \text{Tk } 12,000$. What remains is **profit before tax**, the profit that the government will tax.

The fourth step takes the tax. **Tax** means money that a firm must pay to the government on its profit. The rate depends on the law and on the kind of company, and it changes from time to time.¹ In our illustrations we use a rate of 25% of profit before tax. For the bakery, 25% of Tk 1,08,000 is Tk 27,000. What remains is **net profit**, the profit left after interest and tax.

The profit layout has one more line, which we only preview. Some firms have owners of a special kind who are promised a fixed payment before the ordinary owners receive anything. That payment is called a **preferred dividend**. Chapter XIV explains this kind of owner. The bakery has none, so the line is zero. After this line, what remains belongs to the ordinary owners.

Before the bakery, take a small stationery shop with round numbers. *Given.* Sales Tk 1,00,000; variable cost Tk 60,000; fixed cost Tk 20,000; interest expense Tk 5,000; tax at 25%. *Steps.* Contribution = $1,00,000 - 60,000 = 40,000$. EBIT = $40,000 - 20,000 = 20,000$. Profit before tax = $20,000 - 5,000 = 15,000$. Tax = $15,000 \times 25\% = 3,750$. Net profit = $15,000 - 3,750 = \text{Tk } 11,250$. *Check.* Net profit is below EBIT by interest and tax together: $5,000 + 3,750 = 8,750$, and $20,000 - 8,750 = 11,250$.

The table below follows the ladder with the bakery's figures. The first column names the line, the second shows the working, and the third shows the amount in taka. Read it from top to bottom, and notice that each subtotal is the previous subtotal less one cost.

THE BAKERY'S PROFIT LAYOUT, JANUARY TO MARCH 2026

LINE	WORKING	TK
Sales	22,500 boxes $\times$ Tk 40	9,00,000
Less variable cost	22,500 $\times$ Tk 24	5,40,000
Contribution	9,00,000 $-$ 5,40,000	3,60,000
Less fixed cost	3 months $\times$ Tk 80,000	2,40,000
EBIT	3,60,000 $-$ 2,40,000	1,20,000
Less interest expense	4,00,000 $\times$ 12% $\times$ 3 $\div$ 12	12,000
Profit before tax	1,20,000 $-$ 12,000	1,08,000
Less tax at 25%	1,08,000 $\times$ 25%	27,000
Net profit	1,08,000 $-$ 27,000	81,000
Less preferred dividend	None for the bakery	0
Profit for ordinary owners	81,000 $-$ 0	81,000

¹ The tax rate depends on the kind of company and on the law in force. Twenty-five per cent is used here only for illustration.

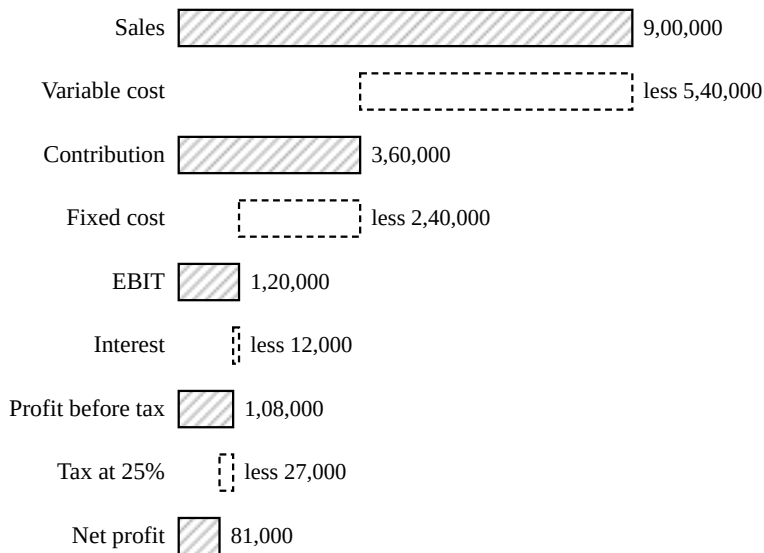


Fig. 3 draws the same ladder, and how to read it.

FIG. 9. From sales to net profit, first quarter 2026. Shaded bars are subtotals; dashed bars are the amounts taken away.

To read Fig. 3, begin at the top. The long shaded bar is sales of Tk 9,00,000. The dashed bar beneath it shows the Tk 5,40,000 taken away for variable cost, and it sits at the right end of the sales bar. The next shaded bar is what remains, the contribution. The pattern repeats down the page, and each shaded bar is shorter than the last. The final shaded bar is net profit, Tk 81,000.

We can describe the ladder in percentages of sales. Variable cost is 60%. Contribution is 40%. EBIT is $1,20,000 \div 9,00,000 \times 100 = 13.33\%$. Net profit is $81,000 \div 9,00,000 \times 100 = 9.00\%$. Each line can be compared with sales in this way, and the same method will compare one firm with another. The owner and the bank manager of our opening puzzle used exactly such percentages.

Two warnings. First, EBIT and net profit are different numbers. EBIT is Tk 1,20,000 and net profit is Tk 81,000. The gap is the interest and the tax. A firm that borrows heavily can have a high EBIT and a small net profit. Second, tax is taken from profit *after* interest, because the law lets a firm deduct interest before it calculates tax. Reversing the order would give a wrong tax.

Before moving on. The ladder runs: sales, less variable cost, contribution, less fixed cost, EBIT, less interest, profit before tax, less tax, net profit, less preferred dividend. You should now be able to say: "EBIT is the profit before interest and tax, and net profit is what remains after both."

The figure shows the ladder as bars that step down from sales to net profit.

Percentages of sales show how each cost compares with sales, using section 3.1.

Two common confusions: EBIT is not net profit, and the order cannot change.

A recap of the section.

3.6 DECIMALS, UNITS AND SENSIBLE CHECKING

Bangladesh groups large numbers in lakhs and crores, and so does this book. After the first three digits from the right, the commas come every two digits. So Tk 6,00,000 is six lakh, which is 600,000 in the international style. One lakh is 1,00,000, and one crore is 100 lakh, written 1,00,00,000. When you read a number with commas, count the digits carefully, because one misplaced comma changes the amount tenfold.

Numbers in this book follow the lakh style of grouping.

A percentage, a decimal and a fraction are three ways to write one number.

Every number needs its label, and every answer needs a size check.

Rounding early lets small errors grow, as this example shows.

A short list of three checking habits.

A recap of the section.

The case file for the first quarter, and the owner's question.

The situation, the question and the plan for the case.

Step one: the new boxes and the new sales.

Step two: the fixed cost and interest stay the same; tax and net profit follow.

A percentage can be written as a decimal or as a fraction, and later chapters use all three. To change a percentage to a decimal, divide by 100: $12\% = 0.12$ and $8.5\% = 0.085$. To change it to a fraction, write it over 100: $12\% = 12/100 = 3/25$. To go back, multiply the decimal by 100. Formulas in this book use the decimal form, so a rate of 8% enters a formula as 0.08.

Always write what a number measures: Tk, boxes, per cent, years. A bare 7,500 might be boxes or taka. Mixing labels is a common cause of wrong answers. For example, a percentage must be changed to a decimal before it is multiplied: 12% becomes 0.12. A good habit is to estimate before you calculate. If you expect an answer near 1,26,000 and the calculator shows 12,600, a decimal point or a comma has gone astray.

Now we show why rounding should come last. *Given.* We want $1,00,000 \times 1.08^5$, and the value of 1.08^5 is 1.469328. *Wrong route.* Round 1.469328 to 1.47 and multiply: $1,00,000 \times 1.47 = \text{Tk } 1,47,000$. *Right route.* Keep all the digits: $1,00,000 \times 1.469328 = \text{Tk } 1,46,933$. The early rounding caused an error of Tk 67 on a sum of one lakh. On a larger sum, or in a longer calculation, the error would be larger.

Three habits make you reliable. First, estimate the size of the answer before calculating. Second, check the answer by a different route, such as running a rule backwards or comparing two totals. Third, ask whether the answer is sensible: a profit larger than the sales, or a negative cost, is a signal to stop and look again. These habits cost very little time and they will be used in every chapter that follows.

Before moving on. Lakh grouping puts commas every two digits after the first three. Label every number. Estimate first, keep all digits until the end, and check by another route.

THE WORKED CASE: THE BAKERY'S FIRST QUARTER

Case file, illustrative. Sundial Bakery, Dhaka. Quarter from 1 January to 31 March 2026. Sales Tk 9,00,000 (22,500 boxes at Tk 40). Variable cost Tk 5,40,000 (Tk 24 a box). Fixed cost Tk 2,40,000. Interest Tk 12,000 (loan Tk 4,00,000 at 12% a year). Illustrative tax rate 25%. The owner asks what net profit would be if sales rose by 10% with the same unit prices and the same costs per box.

Situation. The owner has seen the profit layout and wants to plan the next quarter. *Question.* If the number of boxes sold rises by 10%, by what percentage does net profit change? *Reasoning plan.* We will rebuild the layout with the new number of boxes. The fixed cost and the interest expense do not change when boxes rise, because they are fixed. Variable cost rises with the boxes. We will then compare the net profit with the old one using the percentage change rule.

Given. Boxes rise by 10%: $22,500 \times 1.10 = 24,750$ boxes. *Steps.* Sales = $24,750 \times 40 = \text{Tk } 9,90,000$. Variable cost = $24,750 \times 24 = \text{Tk } 5,94,000$. Contribution = $9,90,000 - 5,94,000 = \text{Tk } 3,96,000$. Check: contribution per box is Tk 16, and $24,750 \times 16 = \text{Tk } 3,96,000$.

Steps. The fixed cost remains Tk 2,40,000. EBIT = $3,96,000 - 2,40,000 = \text{Tk } 1,56,000$. Interest expense remains Tk 12,000. Profit before tax = $1,56,000 - 12,000 = \text{Tk } 1,44,000$. Tax at 25% = Tk 36,000. Net profit = $1,44,000 - 36,000 = \text{Tk } 1,08,000$.

Steps. The old net profit was Tk 81,000. The change is $1,08,000 - 81,000 = \text{Tk } 27,000$. Then $27,000 \div 81,000 = 0.3333$, and $\times 100$ gives 33.33%. *Answer.* A rise of 10% in boxes sold raises net profit by 33.33%. *Check.* EBIT rose from Tk 1,20,000 to Tk 1,56,000, which is a rise of 30%. Net profit rose by even more, because the interest of Tk 12,000 did not rise.

Conclusion. Sales rose by 10% but net profit rose by 33.33%. The reason lies in section 3.4: the fixed cost and the interest were shared over more boxes, so each extra box brought its full Tk 16 of contribution into profit. The same effect works downwards, and a fall of 10% would cut profit sharply. Chapter IX explains this magnifying effect and measures it. For now, notice only that the percentage change in profit can differ greatly from the percentage change in sales.

Step three: compare the two net profits with the percentage change rule.

The conclusion and the lesson for later chapters.

SUMMARY

A percentage is a part out of 100. It answers three questions: the part of a whole, the percentage a part makes, and the change from an old value. A rise and a fall of equal percentages do not cancel, because each is taken of a different whole. A ratio compares two numbers. A power is a number multiplied by itself, a root undoes it, and a minus sign in the exponent means one divided by the power. Round only at the end.

A table is read where a row and a column meet, and totals give a check. A fixed cost stays the same as sales change, and a variable cost changes with each unit. Contribution is price less variable cost for each unit. The profit layout runs from sales, through contribution, EBIT, profit before tax and net profit. Fixed costs and interest make profit change by a larger percentage than sales.

A summary of the chapter's tools.

A summary of the table, cost and profit ideas.

THINK IT THROUGH

A café has sales of Tk 6,00,000 in a quarter. Its variable cost is 50% of sales, its fixed cost is Tk 1,80,000, its interest expense is Tk 20,000, and the illustrative tax rate is 25%. Build its profit layout. Then suppose sales fall by 10%, with variable cost still at 50% of sales and the other costs unchanged. Build the layout again. By what percentage does net profit fall? Explain in two sentences why this percentage is larger than 10%.

A café, and a fall of 10% in sales.

WHERE PEOPLE GO WRONG

Taking the percentage of the new value. A fall of 20% is taken of the value before the fall. Many students divide by the new value, and so get the wrong percentage. Always ask: what is the old value?

Believing that +20% and -20% cancel. They do not. After a rise of 20% and a fall of 20%, Tk 100 becomes Tk 96. The second percentage is taken of a different amount.

Mixing percentage points with per cent. A rate that moves from 10% to 12% has risen by two percentage points and by 20%. Say which you mean.

Rounding too early. Keep all digits until the final line. Rounding 1.469328 to 1.47 changed an answer by Tk 67 on one lakh.

Treating all costs as variable. If every cost rose with sales, profit would rise by the same percentage as sales. Fixed costs make it rise faster.

Mistake one: taking a percentage of the wrong whole.

Mistake two: believing a rise and a fall of equal percentages cancel.

Mistake three: mixing percentage points and percent.

Mistake four: rounding in the middle of a calculation.

Mistake five: confusing EBIT with net profit, and fixed with variable costs.

Mistake six: typing a formula in the wrong order.

Treating EBIT as the last line. Interest and tax still have to be paid after EBIT.

Ignoring the order of operations. Brackets come first, then powers. A formula such as $1,00,000 \times (1.08)^3$ must be typed with its bracket, and the result must look sensible.

Question one: a percentage of an amount.

Question two: the percentage a part makes.

Question three: a percentage change in both directions.

Question four: a ratio.

Question five: powers and roots.

Question six: reading the channel table.

Question seven: fixed and variable costs.

Question eight: the profit layout from raw figures.

Question nine: explain in your own words.

REVIEW QUESTIONS

1. A manufacturer's variable cost is 18% of a sales amount of Tk 2,50,000. Find the variable cost in taka. Show the given, the formula, the steps and a check.

2. A café sells Tk 1,20,000 in a month, and Tk 36,000 of this is from cakes. What percentage of the month's sales came from cakes?

3. The price of sugar rises from Tk 90 to Tk 108 for one kilogram. Find the percentage rise. Then find the percentage fall that would bring the price back to Tk 90, and explain why the two percentages differ.

4. Two partners put Tk 3,00,000 and Tk 5,00,000 into a business. State the ratio of their contributions in lowest terms, and the percentage of the total that each put in.

5. Calculate 1.06^4 to four decimal places. Then find the fifth root of 1.3382, and say what percentage it suggests for the yearly growth of money.

6. Use the table of boxes sold. What share of the boxes sold in January went to the shop counter? By what percentage did the boxes sold to the shop counter change from January to March?

7. A firm sells a unit for Tk 60. The variable cost is Tk 36 a unit, and the fixed cost is Tk 90,000 a month. Find the contribution of each unit, the total cost at 5,000 units, and the cost per unit at 5,000 units.

8. A firm has sales of Tk 12,00,000, variable cost of Tk 7,20,000, fixed cost of Tk 3,00,000 and interest expense of Tk 30,000. Using a tax rate of 25%, build the profit layout down to net profit.

9. In two or three sentences, explain the difference between EBIT and net profit, and say why a firm's net profit can change by a larger percentage than its sales.

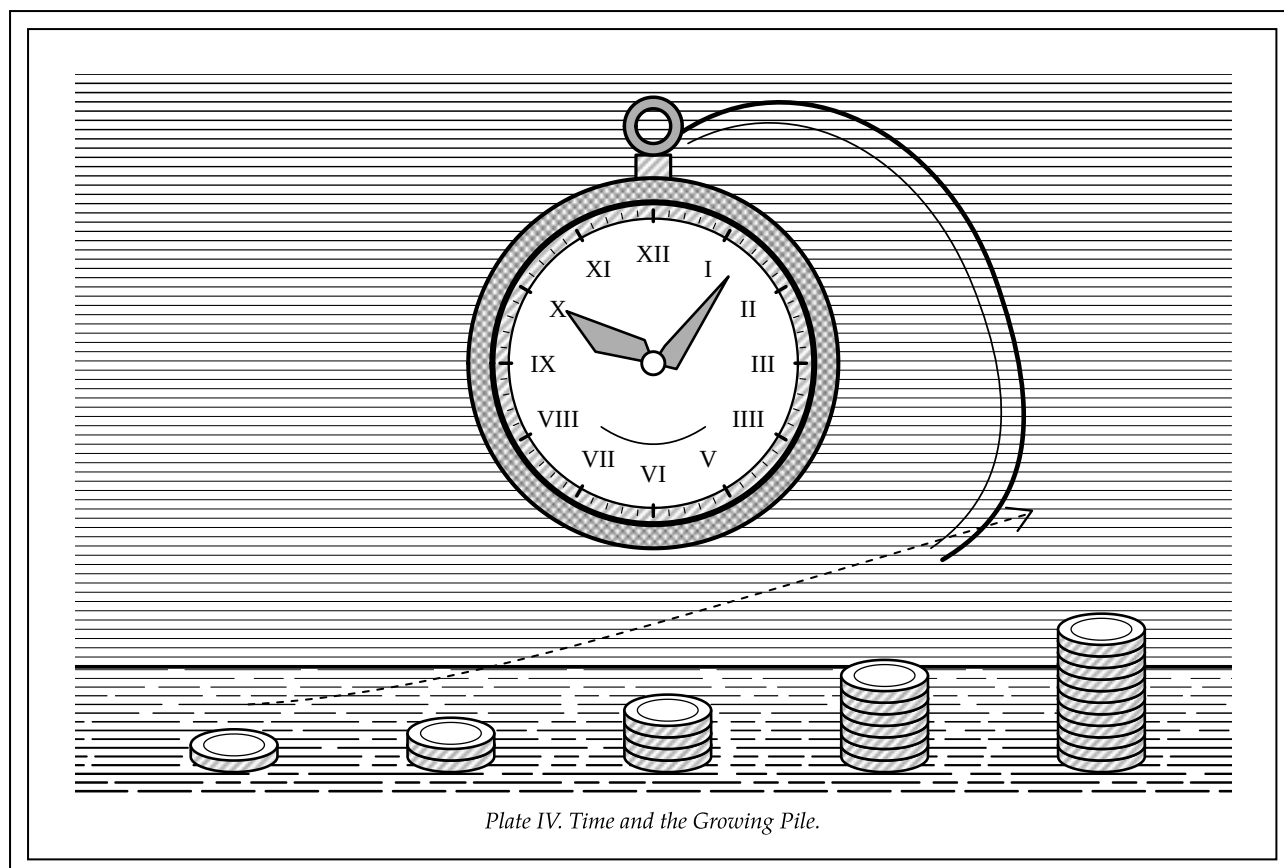


Plate IV. Time and the Growing Pile.

CHAPTER IV

Money Over Time: Single Sums

Is a taka today worth the same as a taka next year?



This chapter teaches that a taka now and a taka later are different things, and shows how to move one sum of money forward and backward through time. We learn future value, present value, and the tables that make the work quick.

A friend offers Tk 1,00,000 today or Tk 1,00,000 in a year.

Now the later offer is larger, and the choice is no longer easy.

The bakery faces such questions on 1 April 2026.

Sums at different dates must be brought to one date before they are compared.

You will need the arithmetic of Chapter III.

GO SLOWLY HERE.

Everyday bridge: a neighbour owes you Tk 1,000 and offers two ways to pay.

First reason: money in hand can earn interest.

Second reason: prices usually rise, so money buys less later.

Third reason: a promise may be broken.

A friend offers you a choice. You may take Tk 1,00,000 today, or you may take Tk 1,00,000 in exactly one year. The friend is honest and will certainly pay. Almost everyone chooses today. Why? The two amounts are equal, and yet no sensible person treats the two offers as equal.

Make the offer harder. You may take Tk 1,00,000 today, or Tk 1,10,000 in one year. The later sum is larger by Tk 10,000. Is it better to wait? The answer depends on something we have not yet named: what the Tk 1,00,000 could earn in the meantime. Different people, with different opportunities, will answer differently, and both can be right.

The bakery faces such questions on 1 April 2026. After its first quarter, the owner has Tk 1,00,000 of spare cash, and a bank offers 8% a year. The owner also knows that equipment will need replacing in five years, at a cost of Tk 2,00,000. How much will the spare cash be worth in three years? How much must be set aside today to meet the replacement cost? Each question asks us to move money through time.

Money at different dates cannot be compared directly, as kilograms cannot be added to litres. We must first bring both sums to the same date. This chapter teaches how: moving one sum forward in time, moving one sum backward, and doing both quickly with tables. The idea behind it has a name, which we meet in section 4.1. It is the most important idea in finance, and the rest of the book stands on it. So we shall move slowly.

You will need: percentages, powers and roots, the order of operations, and the reading of tables, all from Chapter III. If any of these feels uncertain, read sections 3.1 to 3.3 again before going on. The words interest, saving and borrowing come from Chapter I.

4.1 WHY TIME MATTERS

Imagine that a neighbour owes you Tk 1,000. She offers to pay today. Or she says, "I will pay you next year, and I promise." Which do you prefer? Most people answer, "Today." Ask them to explain, and three reasons usually appear. We take them one at a time, because each reason teaches something.

The first reason is that money received today can be put to work. You could place Tk 1,000 in a bank. After a year the bank would return your Tk 1,000 and add interest. Recall from Chapter I that interest is a fee for using someone's money. If you wait a year for your Tk 1,000, you lose the interest that you could have earned. Waiting has a cost.

The second reason is that prices usually rise. A box of buns that costs Tk 40 today may cost Tk 42 next year. Then Tk 1,000 next year buys fewer boxes than Tk 1,000 today. The money is the same, but what it can buy is smaller.

The third reason is uncertainty. A promise about next year may fail, because the neighbour may be unable to pay. Money in hand is certain, and money promised is not. Chapter VIII studies this reason. In this chapter we remove it by assuming that every future payment is certain. Then we can study the first reason, which is the heart of finance.

The three reasons together give an idea its name. The **time value of money** means that a taka now is worth more than a taka later. Notice what the idea does not say. It does not say that a later taka is worthless. It says that a later taka is worth *less than a taka today*, and that we can state by how much.

Interest is the tool for measuring this difference. The **interest rate** means the interest for one year, shown as a percentage of the sum lent. If the rate is 10% a year, then Tk 100 lent today earns Tk 10 in a year. So Tk 100 today and Tk 110 in a year are *equal in value* to a person who can earn 10%. That person has no reason to prefer one to the other. This small comparison is the seed of the whole chapter.

Given. A sum of Tk 100 today, and an interest rate of 10% a year. *Find.* The sum, one year from now, that is worth the same as Tk 100 today. *Formula.* Sum after one year = sum today + interest, and interest = sum today $\times$ rate. *Steps.* Interest = $100 \times 10\% = \text{Tk } 10$. Sum after one year = $100 + 10 = \text{Tk } 110$. *Answer.* Tk 100 today is worth the same as Tk 110 in one year. *Check.* Lend Tk 100 and, one year later, you hold Tk 110. The two sums are two names for one value at two dates.

To follow money through time we draw a **timeline**. A timeline means a line that shows the dates at which money moves. We mark today as Year 0. One year from today is Year 1, two years from today is Year 2, and so on. A **single sum** means one payment made at one date. Every sum in this chapter is a single sum: one amount at one point on the line.

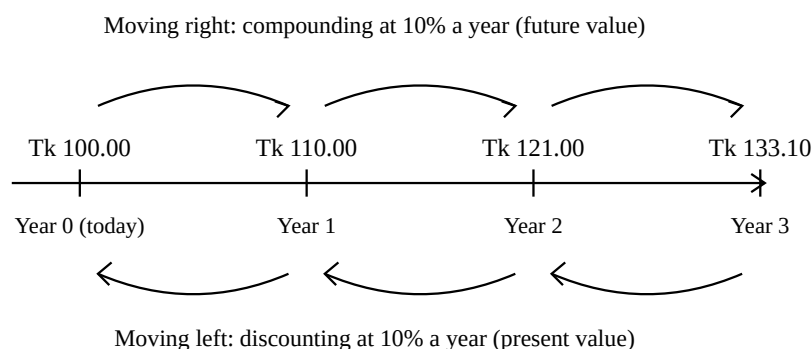


FIG. 10. A timeline for Tk 100 at 10% a year. Moving right, the sum grows; moving left, it is brought back.

To read Fig. 4, begin at the left with Year 0 and Tk 100. Each step to the right is one year, and the sum grows by 10% at each step: Tk 110.00, Tk 121.00, and Tk 133.10. We call the move to the right *compounding*. Each step to the left brings a sum back by one year. We call the move to the left *discounting*. Sections 4.2 and 4.3 teach each move in turn. Notice that the same rate, 10%, is used in both directions.

Different seats give different views of the same rate. To the lender, interest is income for waiting. To the borrower, it is the cost of having money early. To the owner of the bakery it is both: a price paid on the bank's loan, and an income when spare cash is placed in a deposit. In each case the rate is the exchange price between today's money and next year's money.

Time value of money. The idea that a taka now is worth more than a taka later.

These reasons together give the idea its name: the time value of money.

Interest rate. The interest for one year, shown as a percentage of the sum lent.

The interest rate is the tool that measures the difference between now and later.

First worked example, with the smallest numbers: Tk 100 at 10%.

Timeline. A line that shows the dates at which money moves.

Single sum. One payment made at one date.

To follow money through time, we draw a timeline.

Fig. 4 shows a timeline and the two directions of movement.

How to read the timeline figure.

Viewpoint: lender, borrower and owner each see interest differently.

Two limits: the equal-value statement holds only at a stated rate, and our payments are certain.

Pause and check: lending Tk 200 at 10%.

You should now be able to say why a taka now beats a taka later.

A recap of the section.

We begin with the bakery's deposit and ask what it becomes.

Simple interest. Interest worked on the first sum only.

Compound interest. Interest worked on the first sum and on the interest already added.

Simple interest is worked on the first sum only; compound interest on the growing sum.

Compounding. The process of earning interest on interest already added.

The extra Tk 1,000 is interest earned on interest.

Future value. What a sum today grows to at a later date.

Present value. What a later sum is worth today. (In daily life, a present can mean a gift.)

Two names are needed: the sum today is the present value, and the grown sum is the future value.

The year-by-year pattern: each year the sum is multiplied by one plus the rate.

The rule for future value, in words and in symbols.

Two limits matter. First, the equality of Tk 100 today and Tk 110 in a year holds only if 10% is the right rate. At 5%, Tk 100 would grow to Tk 105 only, and a promise of Tk 110 in a year would then be worth *more* than Tk 100 today. The rate we choose decides the answer. Second, we assume that interest is paid at the end of each year and that all future sums are certain. Later chapters relax these assumptions one at a time.

Pause and check. Tk 200 is lent today at 10% a year. How much does the lender hold at Year 1? What sum today is worth the same as Tk 220 at Year 1? The answers are at the back of the book.

You should now be able to say: "A taka now is worth more than a taka later, because money can earn interest, and the interest rate tells me by how much."

Before moving on. The time value of money says that a taka now is worth more than a taka later, mainly because money can earn interest. A timeline puts dates on money. A single sum is one payment at one date. Moving right on the line is compounding, and moving left is discounting.

4.2 FUTURE VALUE OF A SINGLE SUM

On 1 April 2026 the owner places Tk 1,00,000 of spare cash in a bank at 8% a year. The owner wants to know what the deposit will be worth in three years. This is a question about moving a sum to the right on the timeline. Before we answer it, we must understand how a bank counts interest, because there are two ways.

With **simple interest**, the bank works the interest on the first sum only. With **compound interest**, the bank works the interest on the first sum *and* on the interest already added. Take Tk 1,00,000 at 10% for two years. With simple interest, each year brings Tk 10,000, so two years bring Tk 20,000, and the sum is Tk 1,20,000. With compound interest, year 1 brings Tk 10,000 and the sum becomes Tk 1,10,000. Year 2 brings 10% of Tk 1,10,000, which is Tk 11,000, and the sum becomes Tk 1,21,000.

The two methods differ by Tk 1,000. That is the interest earned in year 2 on the Tk 10,000 of interest from year 1, since 10% of Tk 10,000 is Tk 1,000. The process of earning interest on interest is called **compounding**. Almost all bank deposits and bank loans work this way, so compound interest is the method we shall use from now on unless we say otherwise.

We need two names. The **future value** means what a sum today grows to at a later date. The **present value** means what a later sum is worth today. In this section we start from a present value and find a future value. In section 4.3 we do the reverse. The word *present* here means "now", and has nothing to do with a gift.

Look at the pattern in the compound case. At the end of year 1, the sum is the first sum multiplied by 1.10. At the end of year 2, it is that sum multiplied by 1.10 again: $1,00,000 \times 1.10 \times 1.10$. Each year multiplies the sum by the same number, one plus the rate. After n years we multiply n times, which is the power of section 3.2. The exponent counts the years.

The rule in words: the future value equals the present value multiplied by one plus the interest rate, raised to the power of the number of years. In symbols, $FV = PV \times (1 + r)^n$. Here FV is the future value, PV is

the present value, r is the interest rate for one year written as a decimal (so 8% is 0.08), and n is the number of years. The number $(1 + r)^n$ is the multiplier that carries a sum n years forward.

Given. $PV = \text{Tk } 1,00,000$, $r = 10\% = 0.10$, $n = 2$ years. *Find.* FV . *Formula.* $FV = PV \times (1 + r)^n$. *Steps.* First, $1 + 0.10 = 1.10$. Second, $1.10^2 = 1.21$. Third, $FV = 1,00,000 \times 1.21 = \text{Tk } 1,21,000$. *Answer.* The future value is Tk 1,21,000. *Check.* This agrees with the year-by-year count above: Tk 1,10,000 after year 1 and Tk 1,21,000 after year 2.

Given. $PV = \text{Tk } 1,00,000$, placed on 1 April 2026; $r = 8\% = 0.08$; $n = 3$ years. *Find.* FV on 1 April 2029. *Formula.* $FV = PV \times (1 + r)^n$. *Steps.* First, $1 + 0.08 = 1.08$. Second, $1.08^2 = 1.1664$, and $1.1664 \times 1.08 = 1.259712$, so $1.08^3 = 1.259712$. Third, $FV = 1,00,000 \times 1.259712 = \text{Tk } 1,25,971.20$, which is Tk 1,25,971 to the nearest taka.

Answer. The deposit will be worth Tk 1,25,971 on 1 April 2029. *Check.* Count year by year: $\text{Tk } 1,00,000 \times 1.08 = \text{Tk } 1,08,000$; $\times 1.08 = \text{Tk } 1,16,640$; $\times 1.08 = \text{Tk } 1,25,971.20$. The answer also exceeds the simple-interest answer, $\text{Tk } 1,00,000 + 3 \times \text{Tk } 8,000 = \text{Tk } 1,24,000$, as it should.

Given. $PV = \text{Tk } 60,000$, $r = 9.5\% = 0.095$, $n = 4$ years. *Find.* FV . *Formula.* $FV = PV \times (1 + r)^n$. *Steps.* First, $1 + 0.095 = 1.095$. Second, $1.095^2 = 1.199025$. Third, $1.199025^2 = 1.437661$, so $1.095^4 = 1.437661$. Fourth, $FV = 60,000 \times 1.437661 = \text{Tk } 86,259.66$, which is Tk 86,260. *Answer.* Tk 86,260. *Check.* With simple interest the sum would be $60,000 + 4 \times 5,700 = \text{Tk } 82,800$, which is smaller than Tk 86,260, as it should be.

First worked example of the rule:
Tk 1,00,000 at 10% for two years.

Second worked example: the
bakery's spare cash at 8% for three
years.

The answer and a check by the
year-by-year route.

Third worked example: an
awkward rate, 9.5% for four years.

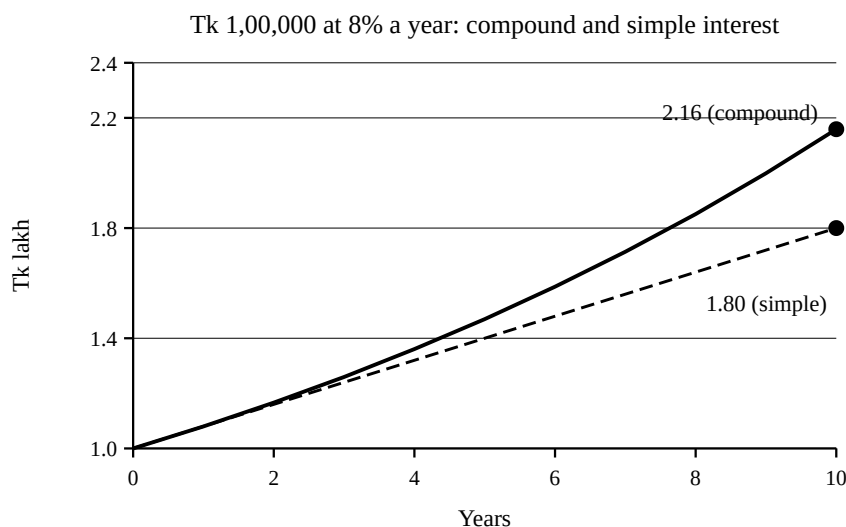


FIG. 11. Tk 1,00,000 at 8% a year for ten years. The curve is compound interest; the dashed line is simple interest.

To read Fig. 5, look along the bottom for the years and up the left side for the sum in lakhs of taka. Both lines start at 1.0, which is Tk 1,00,000. The dashed line rises by the same amount each year, because simple interest is always worked on the first sum. The curve rises faster each year, because compound interest is worked on a larger sum each time. After ten years, compounding gives Tk 2,15,892 and simple interest gives Tk 1,80,000.

The figure carries a lesson. In the first two years the two lines are almost the same. After ten years, they are Tk 35,892 apart. Compounding rewards patience, and it punishes a borrower who does not repay. This is

Fig. 5 shows how compounding
pulls away from simple interest.

How to read the growth figure and
what it teaches.

The lesson: compounding matters
most when time is long.

GO SLOWLY HERE.

Solving for the rate: the rule can be turned round to find the interest rate.

Fourth worked example: finding the rate from two sums.

Solving for the number of years: try year after year until the target is passed.

An advanced note for readers who know logarithms, which a beginner may skip.

Common slips in future value.

Pause and check: the future value of Tk 50,000.

A recap of the section.

why a small difference in the interest rate, or in the length of time, can lead to a very large difference in the sum.

Sometimes we know the present value, the future value and the years, and we want the rate. The rule can be turned round using a root from section 3.2. In words: the rate equals the root of the future value divided by the present value, less 1, where the root is the n -th root and n is the number of years. In symbols, $r = (FV \div PV)^{1/n} - 1$. The exponent $1/n$ is the instruction to take the n -th root.

Given. PV = Tk 1,00,000, FV = Tk 1,46,933, $n = 5$ years. *Find.* r . *Formula.* $r = (FV \div PV)^{1/n} - 1$. *Steps.* First, $FV \div PV = 1,46,933 \div 1,00,000 = 1.46933$. Second, raise 1.46933 to the power $1/5 = 0.2$. The calculator shows 1.08000. Third, $r = 1.08000 - 1 = 0.08000$. *Answer.* The rate is 8% a year. *Check.* Raise 1.08 to the power 5: it gives 1.469328, which is the multiplier we began with.

We can also ask how long a sum takes to reach a target. The simplest method needs no new rule. Keep multiplying by $(1 + r)$ and count the years. *Given.* Tk 1,00,000 grows at 8% a year. *Find.* The time to double. *Steps.* $1.08^8 = 1.8509$, which is below 2. $1.08^9 = 1.9990$, which is still a little below 2. $1.08^{10} = 2.1589$, which is above 2. *Answer.* The sum doubles in a little more than 9 years. *Check.* Run the calculator to nine years: $Tk\ 1,00,000 \times 1.9990 = Tk\ 1,99,900$, which is very near Tk 2,00,000.

Advanced; a beginner may skip this paragraph. Readers who know logarithms can solve for the years directly: $n = \ln(FV \div PV) \div \ln(1 + r)$. For doubling at 8%, $\ln 2 = 0.693147$ and $\ln 1.08 = 0.076961$, so $n = 9.006$ years. This agrees with the trial method. Section 4.5 shows a third method, using a table.

Four slips are common. First, writing 8 in the formula instead of 0.08. Second, multiplying by n instead of raising to the power n : $1.08 \times 3 = 3.24$ is not 1.08^3 . Third, using $n - 1$ or $n + 1$ years, which comes from miscounting the dates on the timeline: from 1 April 2026 to 1 April 2029 is three years. Fourth, mixing the units of the rate and the time. A rate per year goes with years, not months.

Pause and check. Find the future value of Tk 50,000 invested for four years at 10% a year. Show the given, the formula, the steps and a check. The answer is at the back of the book.

Before moving on. Compound interest is worked on the sum and on the earlier interest. The future value of a present value is $PV \times (1 + r)^n$. The rule can be turned round to find the rate, using a root, or to find the years, by trial. You should now be able to say: "Each year multiplies the sum by 1 plus the rate, and the exponent counts the years."

4.3 PRESENT VALUE OF A SINGLE SUM

GO SLOWLY HERE.

The reverse question: how large a deposit today will reach a target later?

Now we turn the question round. The bakery's owner needs Tk 2,00,000 in five years to replace equipment. Instead of asking what a deposit will become, the owner asks: how large a deposit today will become Tk 2,00,000? This is a move to the *left* on the timeline. Begin with small numbers. If Tk 1,21,000 is wanted in two years, and the rate is 10%, we know from section 4.2 that Tk 1,00,000 grows to Tk 1,21,000. So the answer is Tk 1,00,000.

Finding a present value is called **discounting**, and the rate we use is called the **discount rate**. It is the same yearly rate as before. In section 4.2 we called it the interest rate, because it carried a sum forward. When the same rate brings a sum back to today, we call it the discount rate. It is one rate with two names, chosen by the direction of travel. In daily life a discount is a cut in price, and the two meanings are related: a later sum is priced below its face amount today.

Discounting is compounding run backwards. Where we multiplied, we now divide. The rule in words: the present value equals the future value divided by one plus the discount rate, raised to the power of the number of years. In symbols, $PV = FV \div (1 + r)^n$. Dividing by a number is the same as multiplying by one divided by it, so we can write $PV = FV \times (1 + r)^{-n}$, the minus-sign power of section 3.2. The number $(1 + r)^{-n}$ is the multiplier that carries a sum n years back.

Given. $FV = \text{Tk } 1,21,000$ due in 2 years; $r = 10\% = 0.10$. *Find.* PV . *Formula.* $PV = FV \div (1 + r)^n$. *Steps.* First, $1 + 0.10 = 1.10$. Second, $1.10^2 = 1.21$. Third, $PV = 1,21,000 \div 1.21 = \text{Tk } 1,00,000$. *Answer.* The present value is Tk 1,00,000. *Check.* Carry it forward again with section 4.2: $1,00,000 \times 1.21 = \text{Tk } 1,21,000$, which is the sum we began with.

Given. $FV = \text{Tk } 2,00,000$, needed in 5 years, so on 1 April 2031; $r = 8\% = 0.08$; the deposit would be made on 1 April 2026. *Find.* PV . *Formula.* $PV = FV \div (1 + r)^n$. *Steps.* First, $1.08^5 = 1.469328$ (section 3.2). Second, $PV = 2,00,000 \div 1.469328 = \text{Tk } 1,36,116.64$, which is Tk 1,36,117 to the nearest taka.

Answer. The owner must set aside Tk 1,36,117 today. *Check.* Carry the sum forward: $1,36,117 \times 1.469328 = \text{Tk } 2,00,000.5$, which is Tk 2,00,000 within the rounding. The result means that Tk 1,36,117 today and Tk 2,00,000 in five years are equal in value at 8%. The difference, Tk 63,883, is the interest that the deposit will earn.

Discounting. Finding the present value of a later sum.

Discount rate. The yearly rate used to bring a later sum back to today. (In daily life, a discount is a cut in price.)

Finding a present value is called discounting, and the rate used is the discount rate.

The rule for present value: divide by the multiplier that compounding would use.

First worked example of discounting, with clean numbers.

Second worked example: the bakery's replacement fund.

The answer, a check, and what the answer means.

Fig. 6 shows how present value shrinks as time passes and as the rate rises.

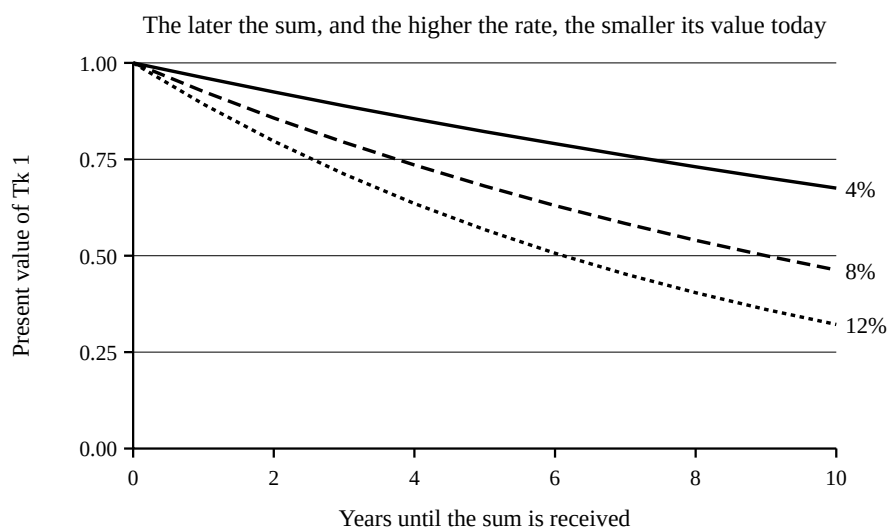


FIG. 12. The present value of Tk 1 due at the end of the year shown, at three discount rates.

To read Fig. 6, look along the bottom for the number of years until the sum is received, and up the left side for what Tk 1 is worth today. All three lines start at 1.00, because a sum received today is worth its full amount. Each line falls as the years increase. The line for 12% falls fastest.

How to read the discount figure and what it shows.

Third worked example: the opening puzzle is settled.

Check, and the lesson: the answer depends on the discount rate.

Exceptions: the discount rate is not the same for everyone, and risk raises it.

A sum at Year 0 is not discounted, because no time passes.

Common slips in present value.

Pause and check: the present value of Tk 90,000.

You should now be able to say how a later sum is brought back.

A recap of the section.

Interest may be added more often than once a year.

After ten years, Tk 1 is worth Tk 0.68 today at 4%, Tk 0.46 at 8%, and only Tk 0.32 at 12%.

Now we can settle the second offer from the start of the chapter. *Given.* Take Tk 1,00,000 today, or Tk 1,10,000 in one year. *Find.* Which is worth more today at 8%, and at 12%. *Steps at 8%.* $PV = 1,10,000 \div 1.08 = \text{Tk } 1,01,852$. *Steps at 12%.* $PV = 1,10,000 \div 1.12 = \text{Tk } 98,214$. *Answer.* At 8% the later offer is worth Tk 1,852 more than Tk 1,00,000, so it is better to wait. At 12% it is worth Tk 1,786 less, so it is better to take the money today.

Check. At 10%, $PV = 1,10,000 \div 1.10 = \text{Tk } 1,00,000$ exactly, so the two offers are equal. The owner who can earn only 8% should wait, and a person who can earn 12% elsewhere should take the money now. The lesson is important: the same two sums can rank in either order. The ranking depends on the discount rate, which is why the choice of rate matters so much in later chapters.

Two exceptions need care. First, there is no single discount rate for the whole world. Each person or firm uses the rate it could earn elsewhere, or the rate it pays to borrow. Second, a risky promise should be discounted at a higher rate than a certain one, because risk lowers what a promise is worth today. We have assumed all payments certain. Chapters VIII to X teach how a higher rate is chosen when they are not.

One small point on the timeline. A sum received at Year 0 has $n = 0$, and $(1 + r)^0 = 1$. So its present value is the sum itself. Only sums that lie to the right of Year 0 are discounted. This will matter in Chapter V, where a stream of payments often begins with a sum today.

Four slips are common. First, multiplying by $(1 + r)^n$ instead of dividing. The answer then grows when it should shrink. Second, using the wrong number of years. Third, forgetting that a higher discount rate gives a *smaller* present value. Fourth, rounding the multiplier early, as in section 3.6. A quick check never fails: a present value must be smaller than the future value it comes from, if the rate is positive.

Pause and check. Find the present value of Tk 90,000 due in three years at a discount rate of 6%. State the given, the formula, the steps and a check. The answer is at the back of the book.

You should now be able to say: "To bring a later sum to today, I divide it by 1 plus the discount rate, raised to the power of the number of years."

Before moving on. Discounting is compounding backwards. The present value is $FV \div (1 + r)^n$. The discount rate is the interest rate used in the backward direction. A higher rate or a longer time gives a smaller present value, and the same two sums can rank differently at different rates.

4.4 RATES, PERIODS AND COMPOUNDING FREQUENCY

Until now interest was added once a year. Many banks add it more often: every six months, every quarter, or every month. The bank still states one *yearly* rate, such as 8%, but it adds a share of it at each shorter period. Because each addition begins to earn interest at once, more frequent addition gives a slightly larger sum. We need a rule that handles this, and the rule has two parts.

Let m be the number of times in a year that interest is added. For half-yearly addition, $m = 2$. For quarterly, $m = 4$. For monthly, $m = 12$. Then the rate for each period is $i = r \div m$, and the number of periods is $N = n \times m$. The earlier rule keeps its shape, with the new letters: $FV = PV \times (1 + i)^N$, where N appears as the exponent. Always use i and N together, never r with N .

Given. $PV = \text{Tk } 1,00,000$; $r = 8\%$ a year; $n = 3$ years; $m = 2$. *Find.* FV . *Formula.* $i = r \div m$, $N = n \times m$, $FV = PV \times (1 + i)^N$. *Steps.* First, $i = 8\% \div 2 = 4\% = 0.04$, and $N = 3 \times 2 = 6$ periods. Second, $1.04^2 = 1.0816$ and $1.04^3 = 1.124864$. Third, $1.04^6 = 1.124864^2 = 1.265319$. Fourth, $FV = 1,00,000 \times 1.265319 = \text{Tk } 1,26,531.90$, which is $\text{Tk } 1,26,532$.

Answer. The future value is $\text{Tk } 1,26,532$. *Check.* This is slightly larger than $\text{Tk } 1,25,971$, the sum we found for yearly addition in section 4.2. The difference is $\text{Tk } 561$. Addition twice a year gave the interest of the first half-year a chance to earn interest in the second half. The table below shows the same 8% yearly rate with four ways of adding interest.

TK 1,00,000 AT 8% A YEAR FOR THREE YEARS, INTEREST ADDED AT DIFFERENT INTERVALS

INTEREST ADDED	RATE EACH PERIOD	PERIODS	SUM AFTER 3 YEARS (TK)	EFFECTIVE ANNUAL RATE
Yearly	8.00%	3	1,25,971	8.00%
Half-yearly	4.00%	6	1,26,532	8.16%
Quarterly	2.00%	12	1,26,824	8.24%
Monthly	0.67%	36	1,27,024	8.30%

The last column of the table needs a new term. The **effective annual rate** means the yearly rate that gives the same growth when interest is added once a year. In symbols, effective annual rate $= (1 + i)^m - 1$. For half-yearly addition at 8% : $1.04^2 = 1.0816$, so the effective annual rate is 0.0816 , or 8.16% . A bank that states 8% and adds half-yearly really pays 8.16% a year. Always compare rates through the effective annual rate, when they are added at different intervals.

Read down the table. Each shorter interval gives a larger sum, but the gains shrink. Moving from yearly to half-yearly adds $\text{Tk } 561$. Moving from half-yearly to quarterly adds only $\text{Tk } 292$. Moving from quarterly to monthly adds $\text{Tk } 200$. The effective rate rises from 8.00% to 8.30% but would never reach even 8.4% , however often interest is added. Frequency matters, but the stated rate matters far more.

Given. $FV = \text{Tk } 2,00,000$ due in 3 years; $r = 12\%$ a year added quarterly, so $m = 4$. *Find.* PV . *Formula.* $PV = FV \div (1 + i)^N$, with $i = r \div m$ and $N = n \times m$. *Steps.* First, $i = 12\% \div 4 = 3\% = 0.03$, and $N = 3 \times 4 = 12$. Second, $1.03^{12} = 1.425761$. Third, $PV = 2,00,000 \div 1.425761 = \text{Tk } 1,40,275.98$, which is $\text{Tk } 1,40,276$.

Answer. The present value is $\text{Tk } 1,40,276$. *Check.* Carry it forward: $1,40,276 \times 1.425761 = \text{Tk } 2,00,000.0$, as it should be. With yearly addition at 12% , the present value would be $2,00,000 \div 1.404928 = \text{Tk } 1,42,356$. The quarterly case needs a *smaller* deposit, because the deposit grows faster when interest is added more often.

Periods need not be whole years. *Given.* $PV = \text{Tk } 1,00,000$ at 8% a year, added half-yearly, for 18 months. *Find.* FV . *Steps.* $i = 4\%$, and $N = 18$

The rule: divide the yearly rate by the number of periods in a year, and multiply the years by the same number.

First worked example: 8% for three years, added half-yearly.

The answer, a check, and the gain from the shorter period.

Effective annual rate. The yearly rate that gives the same growth as the stated rate with interest added once a year.

The effective annual rate tells us what a stated rate really earns in a year.

Reading the table: more frequent addition gains, but with a limit.

Second worked example: present value with quarterly addition.

The answer, a check, and a comparison with yearly addition.

Third worked example: an awkward period of eighteen months.

months $\div$ 6 months = 3 periods. Then $1.04^3 = 1.124864$, so $FV = 1,00,000 \times 1.124864 = \text{Tk } 1,12,486$. *Answer.* Tk 1,12,486. *Check.* The result lies between the sum after one year, Tk 1,08,160 (two half-years), and after two years, Tk 1,16,986 (four half-years), so it is sensible.

Common slips with frequency.

Three slips are common. First, using the yearly rate with the number of periods: 8% for 6 periods is wrong; the rate must be 4%. Second, using the right rate and the wrong number of periods: three years of half-yearly addition is 6 periods, not 3. Third, forgetting to say how often interest is added. If the question gives no interval, assume it is added once a year.

Pause and check: Tk 75,000 at 6%, added quarterly.

Pause and check. Find the future value of Tk 75,000 invested for two years at 6% a year, with interest added every quarter. The answer is at the back of the book.

A recap of the section.

Before moving on. When interest is added m times a year, use $i = r \div m$ and $N = n \times m$. The effective annual rate, $(1 + i)^m - 1$, shows what the stated rate really earns. More frequent addition gives a larger sum, with gains that shrink. You should now be able to say: "I match the rate to the period before I raise it to a power."

4.5 READING THE FOUR FACTOR TABLES

Tables store the multipliers in advance, so that no power key is needed.

A power key makes our work quick, but not everyone has one at hand, and people prepared tables of multipliers long before calculators existed. A table stores the multiplier for each pair of rate and years. We find the multiplier and use it. Tables are still taught because they show patterns that a single calculator answer hides, and because a table lets us check a calculator result quickly.

Factor. A table number to multiply by. (In daily life, a cause or influence.)

A factor is a table number that we multiply by.

In a table of this kind, each number is called a **factor**. A factor means a table number to multiply by. The factor for a sum moving forward is $(1 + r)^n$, and the factor for a sum moving back is $(1 + r)^{-n}$. Four tables are in common use. This chapter needs the first two. We preview the other two, which belong to Chapter V.

Tables one and two are introduced in words before they are shown.

Table 1 gives the *future value factor of one taka*: what Tk 1 becomes after n years at rate r . Table 2 gives the *present value factor of one taka*: what Tk 1 due after n years is worth today. Each table has the years in rows and the rates in columns. Because the factors are for Tk 1, we multiply the factor by our own sum to find the answer.

TABLE 1. FUTURE VALUE FACTOR OF TK 1 = $(1 + r)^n$

YEARS	4%	6%	8%	10%	12%
1	1.0400	1.0600	1.0800	1.1000	1.1200
2	1.0816	1.1236	1.1664	1.2100	1.2544
3	1.1249	1.1910	1.2597	1.3310	1.4049
4	1.1699	1.2625	1.3605	1.4641	1.5735
5	1.2167	1.3382	1.4693	1.6105	1.7623
6	1.2653	1.4185	1.5869	1.7716	1.9738
7	1.3159	1.5036	1.7138	1.9487	2.2107
8	1.3686	1.5938	1.8509	2.1436	2.4760
10	1.4802	1.7908	2.1589	2.5937	3.1058

TABLE 2. PRESENT VALUE FACTOR OF TK 1 = $(1 + r)^{-n}$

YEARS	4%	6%	8%	10%	12%
1	0.9615	0.9434	0.9259	0.9091	0.8929
2	0.9246	0.8900	0.8573	0.8264	0.7972
3	0.8890	0.8396	0.7938	0.7513	0.7118
4	0.8548	0.7921	0.7350	0.6830	0.6355
5	0.8219	0.7473	0.6806	0.6209	0.5674
6	0.7903	0.7050	0.6302	0.5645	0.5066
7	0.7599	0.6651	0.5835	0.5132	0.4523
8	0.7307	0.6274	0.5403	0.4665	0.4039
10	0.6756	0.5584	0.4632	0.3855	0.3220

To read a factor, run one finger along the row for the number of years and another down the column for the rate, as we learned in section 3.3. Where they meet is the factor. In Table 1, the row for 5 years and the column for 8% meet at 1.4693. In Table 2, the same row and column meet at 0.6806. These are the two numbers we have already met by calculator in sections 4.2 and 4.3.

The tables are not separate. Each factor in Table 2 equals one divided by the matching factor in Table 1. Check with 5 years at 8%: $1 \div 1.4693 = 0.6806$. This is the same link as between compounding and discounting. It also means that a single table is enough, if we are ready to divide. The check is a quick way to see whether a number in a table has been copied wrongly.

Given. Tk 1,00,000 at 8% for 3 years. *Find.* FV by the table. *Steps.* Table 1, row 3 and column 8%, gives 1.2597. Then $1,00,000 \times 1.2597 = \text{Tk } 1,25,970$. *Answer.* Tk 1,25,970. *Check.* The calculator gave Tk 1,25,971 in section 4.2. The gap of Tk 1 comes from the table's four decimals: 1.259712 was cut to 1.2597. In this book, answers use the calculator unless a question says to use the table, and a table answer may differ by a few taka.

Given. Tk 2,00,000 due in 5 years at 8%. *Find.* PV by the table. *Steps.* Table 2, row 5 and column 8%, gives 0.6806. Then $2,00,000 \times 0.6806 = \text{Tk } 1,36,120$. *Answer.* Tk 1,36,120. *Check.* The calculator gave Tk 1,36,117. The gap of Tk 3 is again the effect of four decimals. Both are close enough to show that the method is right.

Tables also solve for an unknown rate or time. *Given.* Tk 80,000 grows to Tk 1,17,544 in 5 years. *Find.* The rate. *Steps.* The factor is $1,17,544 \div 80,000 = 1.4693$. In Table 1, read along the row for 5 years until you find 1.4693. It sits under 8%. *Answer.* The rate is 8%. *Check.* $80,000 \times 1.4693 = \text{Tk } 1,17,544$, as given.

Given. Tk 50,000 is to double to Tk 1,00,000 at 10% a year. *Find.* The years. *Steps.* The factor is 2.0000. Look down the 10% column of Table 1. Row 7 shows 1.9487, which is below 2, and row 8 shows 2.1436, which is above 2. So the answer is between 7 and 8 years. A straight-line estimate gives $7 + (2.0000 - 1.9487) \div (2.1436 - 1.9487) = 7 + 0.0513 \div 0.1949 = 7.26$ years. *Answer.* About 7.3 years. *Check.* The calculator method of section 4.2 gives 7.27 years.

To read a factor, find the row for the years and the column for the rate.

The two tables are linked: each number in Table 2 is one divided by the matching number in Table 1.

Using a table to find a future value, and why the answer differs slightly from the calculator's.

Using a table to find a present value.

Using a table backwards to find the rate.

Using a table backwards to find the years, with an estimate between two rows.

The limits of tables: only printed rates and years, and a straight-line estimate is only an estimate.

Tables three and four are previewed here, and used in Chapter V.

Tables have limits. They print only some rates and some years, so a rate such as 9.5% needs a calculator. The straight-line estimate between two rows is close, but not exact, because compounding follows a curve. A careful worker uses the table to see the pattern and the calculator for the exact answer.

Two more tables complete the set. Table 3 gives the future value of Tk 1 paid at the end of every year for n years. Table 4 gives the present value of Tk 1 paid at the end of every year for n years. Chapter V teaches how to use them, with equal payments called annuities. The table below shows all four factors side by side at 8%, so that you can see how they relate.

THE FOUR TABLES AT 8%, YEARS 1 TO 5

YEARS	TABLE 1	TABLE 2	TABLE 3	TABLE 4
1	1.0800	0.9259	1.0000	0.9259
2	1.1664	0.8573	2.0800	1.7833
3	1.2597	0.7938	3.2464	2.5771
4	1.3605	0.7350	4.5061	3.3121
5	1.4693	0.6806	5.8666	3.9927

Reading the four-table comparison.

Read the table across. Tables 1 and 2 are for a single sum, and each Table 2 factor is one divided by the Table 1 factor in the same row. Tables 3 and 4 are for equal payments each year. The first row of Table 3 is 1.0000, because one payment at the end of year 1 has not yet earned anything. The second row is $1 + 1.08 = 2.08$, which is the second payment plus the first payment grown by one year. Do not worry if this is not clear yet; Chapter V explains it.

Pause and check: using the tables.

Pause and check. Use Table 1 to find the future value of Tk 2,00,000 invested for 5 years at 6%. Use Table 2 to find the present value of Tk 3,00,000 due in 4 years at 10%. The answers are at the back of the book.

A recap of the section.

Before moving on. A factor is a table number to multiply by. Table 1 carries a sum forward and Table 2 brings it back, and each Table 2 factor is one divided by the Table 1 factor. A table answer can differ from a calculator answer by a few taka. You should now be able to say: "I can read a factor, and I can use a table to find a rate or the years."

THE WORKED CASE: THE SPARE CASH AND THE REPLACEMENT FUND

The case file for April 2026, and the owner's three questions.

Case file, illustrative. Sundial Bakery, Dhaka. On 1 April 2026 the owner places Tk 1,00,000 of spare cash in a bank deposit at 8% a year, with interest added once a year. The equipment will need replacing in 5 years, on 1 April 2031, at an expected cost of Tk 2,00,000. The owner asks three questions: what the deposit will be worth on 1 April 2029; what sum must be set aside on 1 April 2026 to meet the replacement cost; and whether adding interest half-yearly would change the sum needed.

The plan: two moves along the timeline and one change of period.

Reasoning plan. The first question moves a sum to the right on the timeline, so it asks for a future value. The second question moves a sum to the left, so it asks for a present value. The third question keeps the move to the left but changes the period from one year to half a year. We answer each in turn,

with the standard layout of Given, Find, Formula, Steps, Answer and Check.

Given. PV = Tk 1,00,000; $r = 8\%$; $n = 3$ years. *Find.* FV. *Formula.* $FV = PV \times (1 + r)^n$. *Steps.* $1.08^3 = 1.259712$, and $1,00,000 \times 1.259712 = \text{Tk } 1,25,971.20$. *Answer.* Tk 1,25,971 on 1 April 2029. *Check.* Table 1, row 3 and column 8%, gives 1.2597, so the table answer is Tk 1,25,970, within one taka of the calculator.

Given. FV = Tk 2,00,000; $r = 8\%$; $n = 5$ years. *Find.* PV. *Formula.* $PV = FV \div (1 + r)^n$. *Steps.* $1.08^5 = 1.469328$, and $2,00,000 \div 1.469328 = \text{Tk } 1,36,116.64$. *Answer.* Tk 1,36,117. *Check.* Table 2, row 5 and column 8%, gives 0.6806. Then $2,00,000 \times 0.6806 = \text{Tk } 1,36,120$, within Tk 3 of the calculator answer.

Given. FV = Tk 2,00,000; $r = 8\%$ a year, added half-yearly; $n = 5$ years. *Find.* PV. *Formula.* $PV = FV \div (1 + i)^N$, with $i = r \div 2$ and $N = n \times 2$. *Steps.* $i = 4\%$ and $N = 10$. Then $1.04^{10} = 1.480244$, and $2,00,000 \div 1.480244 = \text{Tk } 1,35,112.83$. *Answer.* Tk 1,35,113. *Check.* This is Tk 1,004 less than the yearly case, as it should be, because interest added twice a year makes a deposit grow faster.

Conclusion. The spare cash of Tk 1,00,000 will be worth Tk 1,25,971 on 1 April 2029. To meet Tk 2,00,000 on 1 April 2031, the owner needs Tk 1,36,117 today with yearly interest, or Tk 1,35,113 with half-yearly interest. The Tk 1,00,000 is therefore short by Tk 36,117 today. The same gap can be seen at the end: Tk 1,00,000 would grow to Tk 1,46,933 in five years, which is Tk 53,067 short of Tk 2,00,000, and $53,067 \div 1.469328 = \text{Tk } 36,117$. Chapter V shows how a plan of equal yearly deposits can reach the same goal.

Question one: the deposit on 1 April 2029.

Question two: the sum to set aside today.

Question three: half-yearly addition lowers the sum needed.

The conclusion: the spare cash is not yet enough, and two routes show the shortfall.

SUMMARY

A taka now is worth more than a taka later, mainly because money can earn interest. This is the time value of money. A timeline marks today as Year 0 and shows each sum at its date. Moving a sum to the right is compounding, and it gives a future value: $FV = PV \times (1 + r)^n$. Moving a sum to the left is discounting, and it gives a present value: $PV = FV \div (1 + r)^n$. The same rate is used in both directions, and it is called the discount rate when it brings a sum back.

When interest is added m times a year, use the rate $i = r \div m$ for $N = n \times m$ periods. The effective annual rate shows what a stated rate really earns. A factor is a table number to multiply by, and Tables 1 and 2 carry a sum forward and back. A higher rate or a longer time makes a present value smaller and a future value larger. All our sums were certain; risk comes in Chapter VIII.

A summary of the time value of money and the two moves.

A summary of frequency, tables and the cautions.

THINK IT THROUGH

An uncle wishes to help his niece. He offers her one of two gifts. Offer A is Tk 5,00,000 today. Offer B is Tk 3,00,000 today and a further Tk 2,50,000 in two years. First find the present value of Offer B at a discount rate of 10%, and say which offer is better. Then do the same at 20%. Finally explain, in two sentences, why the answer changed, and say what the niece should ask herself before choosing.

An uncle's two offers, compared at two different discount rates.

Mistake one: comparing sums at different dates without bringing them to one date.

Mistake two: using the percentage as a whole number in the formula.

Mistake three: multiplying by the number of years instead of using a power.

Mistake four: multiplying when the question needs division, or the reverse.

Mistake five: mismatching the rate and the period.

Mistake six: believing a higher discount rate raises present value.

Mistake seven: trusting a table answer to be exact.

Fibonacci's book of 1202 taught Italian merchants the arithmetic of trade, which included interest.

The book's later problems came from trade, and included the price of goods, profit and interest.

A short note on the person and his time, for readers unfamiliar with him.

Question one: a future value with new data.

WHERE PEOPLE GO WRONG

Comparing Tk 1,00,000 today with Tk 1,10,000 next year as if they were taken on the same day. Sums at different dates cannot be compared until they are brought to the same date. Move them both to Year 0, or both to the same later year.

Writing 8 instead of 0.08. A rate enters the formula as a decimal. With 8 in place of 0.08, the multiplier becomes 9 instead of 1.08, and the answer is absurd. Estimate first, as in section 3.6.

Multiplying instead of raising to a power. The future value after three years is $PV \times 1.08^3$, not $PV \times 1.08 \times 3$. The second gives 3.24, and the first gives 1.2597.

Multiplying when the sum lies in the future and we want today. A present value must be smaller than the later sum it comes from. If your answer is larger, you have multiplied where you should divide.

Using the yearly rate with half-yearly periods. If interest is added half-yearly, the rate in the formula is half the yearly rate, and the number of periods is twice the years.

Thinking a higher discount rate gives a larger present value. The opposite is true. A higher rate makes later sums worth less today, as Fig. 6 shows.

Expecting a table answer to match the calculator to the last taka. Table factors have four decimals. A difference of a few taka is normal and does not mean the method is wrong.

IN HISTORY

In 1202, a man from Pisa, in what is now Italy, completed a book. His name was Leonardo of Pisa, and today he is better known as Fibonacci. As a boy he had lived in Africa, where he met the Hindu-Arabic figures that we use today. His book, *Liber Abaci*, or the *Book of Calculation*, taught these figures to Italian merchants. A second edition followed in 1228.¹

The book did not stop at the figures. It gave merchants worked problems from their trade: the price of goods, the profit on a sale, the change of one currency into another, and the computation of interest. A merchant who lent money, or who waited for a payment, needed to know what a sum would become after a time. The idea of this chapter, that money has a value that changes with time, belongs to that long tradition of practical arithmetic.

For readers meeting him for the first time: Fibonacci lived in Pisa, a trading city, in the late twelfth and early thirteenth centuries. Merchants there traded with ports around the Mediterranean, and they needed methods that were quick and exact. His book gave them such methods. We carry the sums of this chapter on calculators now, but the questions are the questions of his merchants: how much will this grow, and what is that later sum worth today?

REVIEW QUESTIONS

1. A firm places Tk 2,50,000 in a deposit at 6% a year, with interest added yearly.

¹ Dates and the content of *Liber Abaci* are from standard histories of mathematics; see the verification note at the end of the chapter work.

Find its value after 4 years. Show the given, the formula, the steps and a check.

2. A firm will receive Tk 4,00,000 in 6 years. The discount rate is 12%. Find the present value. Show a check by carrying your answer forward.

3. Tk 40,000 grows to Tk 58,564 in 4 years. Find the yearly rate, and check your answer.

4. How many years does Tk 1,00,000 need to double at 12% a year? Use trial multiplication, and say between which two whole years the answer lies.

5. Find the value of Tk 3,00,000 after 2 years at 12% a year, with interest added quarterly. Compare it with the value if interest is added yearly, and state the difference.

6. A bank states 12% a year, with interest added monthly. Find the effective annual rate.

7. Tk 5,00,000 is due in 3 years. Interest is added half-yearly at 10% a year. Find the present value.

8. A school fund needs Tk 10,00,000 in 8 years. The discount rate is 9% a year, added yearly. How much must be set aside today? If only Tk 4,00,000 is available today, what extra sum is needed today?

9. Use Table 1 to find the future value of Tk 1,50,000 at 10% for 6 years. Use Table 2 to find the present value of Tk 2,00,000 due in 8 years at 12%.

10. In two or three sentences each, explain why a present value is smaller at 12% than at 8%, and why compound interest gives a larger sum than simple interest over a long time.

Question two: a present value with new data.

Question three: finding the rate.

Question four: finding the years by trial.

Question five: compounding frequency.

Question six: the effective annual rate.

Question seven: a present value with half-yearly addition.

Question eight: a goal case.

Question nine: using the tables.

Question ten: explain in your own words.

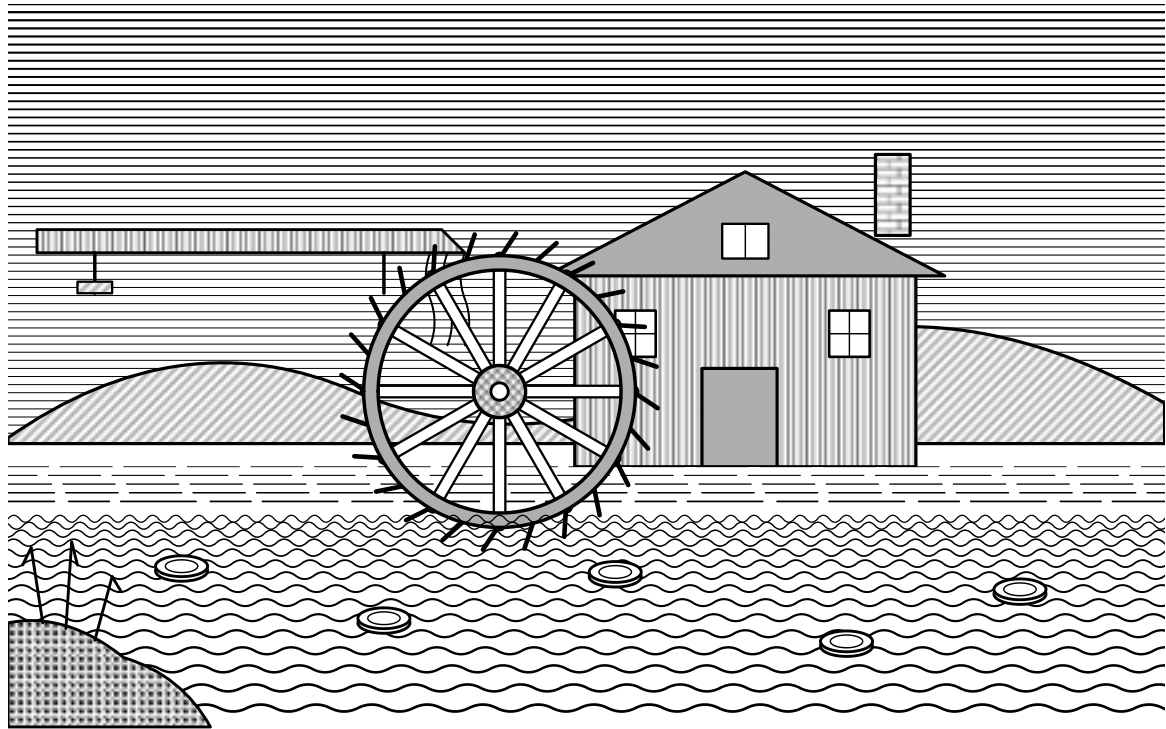


Plate V. The Mill and the Stream.

CHAPTER V

Streams of Cash: Annuities and Mixed Streams

Many payments, spread across the years, and one honest total.



Chapter 4 taught us to move one sum through time. Most business money does not travel alone: rent, loan repayments, savings and sales arrive again and again. This chapter shows how to value a whole series of payments, equal or unequal, and how to find the regular saving that reaches a goal.

The bakery needs a large sum in five years but cannot pay it today.

Each yearly deposit grows for a different length of time.

Equal payments allow a shortcut; unequal payments need a patient sum.

A stream is any series of payments that arrive over time.

Stream. A series of payments made or received over time.

A single sum is the smallest stream: one payment at one time.

Streams differ in the size of the payments and in the gaps between them.

Annuity. Equal payments made at equal gaps of time.

The word annuity once meant yearly, but now covers any equal gap.

A mixed stream has payments of different sizes.

Mixed stream. A stream whose payments have different sizes.

Sundial Bakery, in Dhaka, must replace some equipment in five years. The replacement will cost Tk 2,00,000. Chapter 4 told the owner what that future sum is worth today: Tk 1,36,117 placed at 8% a year would grow to Tk 2,00,000. The owner, however, does not hold Tk 1,36,117 in spare cash. She holds something different: a steady profit, from which she can save a smaller sum at the end of each year.

Here lies the puzzle. The first deposit will earn interest for four years. The second will earn for three years, and the last will earn for none, because the goal date arrives on the day it is paid. Chapter 4 can value each deposit, but only one at a time. Five deposits need five calculations and one addition. A savings plan of twenty years would need twenty calculations. Is there a shorter road?

There is, and it rests on one observation. When the payments are equal and regular, the same arithmetic repeats, and a single table number can do the repeating for us. When the payments are unequal, no such shortcut exists, but the method stays plain: treat every payment as a single sum from Chapter 4, and add the results. This chapter teaches both, and it closes by returning to the bakery's question.

5.1 KINDS OF CASH STREAMS

Money rarely moves once and then stops. A tenant pays rent every month. A bank customer repays a loan every year. A shop receives sales every day. We give the name **stream** to any series of payments made or received over time. A stream of five yearly deposits of Tk 10,000 is a stream; so is a list of six monthly rent payments.

Chapter 4 dealt with the simplest case, the single sum, which is one payment at one moment. We can now see it as a stream with only one member. Every longer stream is, in the end, a set of single sums placed at different moments on the timeline. This simple fact will carry us through the whole chapter.

Streams differ in two ways: the size of each payment and the length of the gap between payments. Suppose first that all the payments are equal and all the gaps are equal. We call such a stream an **annuity**. An annuity is a stream of equal payments made at equal gaps of time. A deposit of Tk 10,000 at the end of each year for three years is an annuity. So is a rent of Tk 15,000 paid every month for a year.

The word *annuity* comes from a Latin root meaning *year*, so beginners expect it to mean yearly payments only. Today it covers any equal gap: a month, a quarter or a year. In this chapter the gap is always one year, to keep the arithmetic clear. The ideas, however, work for months in the same way, as Chapter 4 showed for compounding.

Now suppose the payments are not equal. A **mixed stream** is a stream whose payments have different sizes. A catering contract that pays Tk 50,000 in the first year, Tk 60,000 in the second and Tk 40,000 in the third is a mixed stream. Most real investments produce mixed streams, because sales rise and fall from year to year. We shall meet them again in Chapters 6 and 7.

Business people also use the word **instalment**. An instalment is one of several regular part-payments that together repay a debt or buy something over time. Suppose a shop buys a refrigerator for Tk 60,000 and pays Tk 5,000 each month for twelve months. The twelve instalments form an annuity. A loan repaid in equal instalments is an annuity from the lender's point of view.

Why should a firm care about streams? Because almost every large business decision creates one. A loan creates a stream of instalments. A new machine creates a stream of extra sales over its life. A savings plan creates a stream of deposits. To judge any of these decisions, the manager must give the whole stream a single value today. Without that value, a choice between two plans would rest on guesswork.

The table below compares the three kinds. Read each row from left to right: the first column names the kind, the next two describe the payments and the gaps, and the last gives an example.

THREE KINDS OF CASH STREAM

KIND	PAYMENTS	GAPS	EXAMPLE
Single sum	One payment	Not applicable	Tk 1,00,000 received once, in 3 years
Annuity	All equal	All equal	Tk 10,000 deposited each year for 3 years
Mixed stream	Not all equal	Equal in this book	Tk 50,000, then Tk 60,000, then Tk 40,000

Now consider an exception, which catches many students. A stream of Tk 10,000, Tk 10,000 and Tk 10,500 is *not* an annuity, although it is almost equal. One different payment makes it a mixed stream. In the same way, equal payments at unequal gaps do not form an annuity. Both conditions must hold: equal payments and equal gaps.

Pause and check. Classify each stream as a single sum, an annuity or a mixed stream. (a) Tk 8,000 received at the end of each month for a year. (b) Tk 20,000, Tk 30,000 and Tk 25,000 received at the ends of three years. (c) Tk 90,000 received once in two years. The answers are at the back of the book.

Before moving on. A stream is a series of payments over time. An annuity has equal payments at equal gaps. A mixed stream has unequal payments. A single sum is one payment. You should now be able to say, in your own words, why Tk 10,000, Tk 10,000, Tk 10,500 is a mixed stream.

5.2 ORDINARY ANNUITY AND ANNUITY DUE

Two people each pay Tk 10,000 a month. The first is a worker who receives a wage at the end of each month, after the work is done. The second is a tenant who pays rent at the start of each month, before living in the house. Both streams are annuities. Yet one payment comes at the end of its month and the other at the start. Does that small difference change the value? In finance it does, and so we must name the two cases.

An instalment is one of several regular payments that clear a debt.

Instalment. One of several regular payments that together repay a debt.

Firms meet streams whenever they borrow, save, invest or sell.

A table sets the three kinds of stream side by side.

One unequal payment is enough to destroy an annuity.

Pause and check: classify each stream.

Recap: three kinds of stream, and what separates them.

Equal payments can still differ in when each period's payment is made.

GO SLOWLY HERE.

GO SLOWLY HERE.

The two timings have two names: ordinary annuity and annuity due.

Ordinary annuity. An annuity whose payments come at the end of each period.

Annuity due. An annuity whose payments come at the start of each period.

Reading the figure: arrows show when each payment arrives.

An **ordinary annuity** is an annuity whose payments are made at the end of each period. An **annuity due** is an annuity whose payments are made at the start of each period. A period here means one gap, which is one year in this chapter. The wage in our example resembles an ordinary annuity. The rent resembles an annuity due.

The figure below shows both kinds. Each line is a timeline running from left to right through time. A short tick mark is a moment, and an arrow pointing down is a payment of Tk 10,000 at that moment. In the upper timeline the payments arrive at the ends of years 1, 2 and 3. In the lower timeline they arrive today, at the end of year 1 and at the end of year 2. Every payment has moved one year earlier.

Ordinary annuity

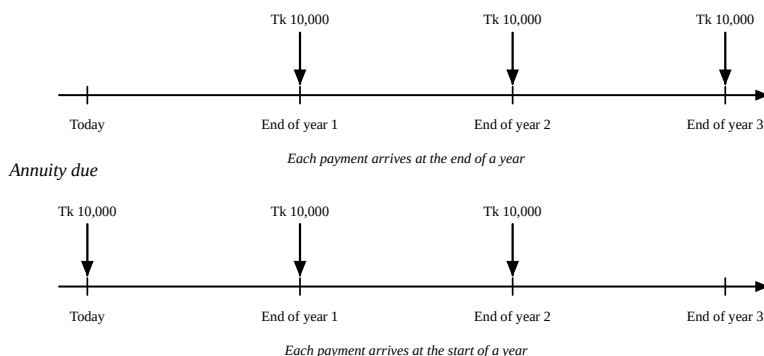


FIG. 13. Three yearly payments of Tk 10,000: ordinary annuity above, annuity due below.

The end of one year and the start of the next are one moment.

An earlier payment earns more interest, or loses less to discounting.

A second bridge: a loan repaid at the end of each year.

When a problem is silent, we treat the payments as ordinary.

Many students find this step confusing, because the lower timeline says *end of year 1* while the description says *start of year 2*. These are the same moment: the instant one year finishes is the instant the next begins. So a payment at the start of year 2 sits at the same point as a payment at the end of year 1. Keep this fact in mind, and the figure becomes easy to read.

Why does the timing change the value? A deposit made one year earlier earns one more year of interest, so a stream of early deposits grows to more. A receipt that arrives one year earlier is discounted for one year less, so its present value is higher. In both cases, earlier means worth more. An annuity due is therefore always worth more than the ordinary annuity with the same payments, at any positive discount rate.

Here is a second everyday bridge. A trader borrows money and promises to repay at the end of each year. The bank receives its money after the year has passed, so the repayments form an ordinary annuity. Compare a shop that sells goods on a plan requiring the first payment on the day of purchase. That plan resembles an annuity due. In each case, the only question to ask is: does the first payment come at the end of the first period, or at its start?

A rule of the book: unless a problem states otherwise, an annuity is an ordinary annuity, with each payment at the end of its year. Problems that intend the other case will say *at the start of each year*, or will describe a

payment made today. Always search the wording for these clues before you begin.

Pause and check. A school pays its teacher at the end of each month. Is that stream an ordinary annuity or an annuity due? A shopkeeper pays rent on the first day of each month. Which kind is the rent? The answers are at the back of the book.

Before moving on. Ordinary means end of period; due means start of period. Due is worth more, because each payment is one year earlier. You should now be able to say why a payment on the first day of a year is the same moment as a payment on the last day of the year before.

Pause and check: ordinary or due?

Recap: the two timings and why they matter.

5.3 FUTURE AND PRESENT VALUE OF ANNUITIES

We begin with the method we already own. To value a stream, treat each payment as a single sum, value each one, and add. We shall do this once, slowly, for a small annuity. Then we shall see how a table number replaces the repeated work. Throughout this section, i is the discount rate written as a decimal, so that 10% is 0.10, and n is the number of years.

The future value of an annuity is what all its payments are worth at the end of the last year, when each payment has earned interest from the day it was paid. Suppose Tk 10,000 is deposited at the end of each of three years, and the discount rate is 10%. The table below grows each deposit to the end of year 3.

The first method treats each payment as a single sum from Chapter 4.

The future value of an annuity is the sum of what each payment grows to.

GROWING THREE DEPOSITS OF TK 10,000 AT 10% TO THE END OF YEAR 3

DEPOSIT	MADE AT	YEARS OF GROWTH	GROWTH FACTOR	VALUE AT END OF YEAR 3 (TK)
First	End of year 1	2	1.2100	12,100
Second	End of year 2	1	1.1000	11,000
Third	End of year 3	0	1.0000	10,000
Total			3.3100	33,100

Look at the last row. The three growth factors add to 3.3100, and the three values add to Tk 33,100. Because every deposit is the same Tk 10,000, we may add the factors first and multiply once: $\text{Tk } 10,000 \times 3.3100 = \text{Tk } 33,100$. The number 3.3100 depends only on the discount rate and the number of years. It does not depend on the size of the deposit.

Add the growth factors once, and the result serves every deposit.

We call such a number an **annuity factor**. It is a table number that, multiplied by the regular payment, gives the value of the whole annuity. There are two: the *future value annuity factor*, which gives the future value, and the *present value annuity factor*, which gives the present value. In words, the rule for the future value is simple:

The annuity factor is the table number that serves a whole annuity.

Annuity factor. A table number that, times the payment, gives an annuity's value.

IN SHORT

Future value of an annuity = regular payment × future value annuity factor. In symbols, $FVA = PMT \times FVAF$. Here FVA is the future value of the annuity, PMT is the regular payment, and FVAF is the future value annuity factor for the given discount rate and number of years.

Where the factor comes from: a short formula, or the table.

Worked example 5.1 lays out the future value in six steps.

The factor can be found from the tables of Chapter 4, or from a short formula. In words: take 1 plus the discount rate, raise it to the power n , subtract 1, and divide by the discount rate. In symbols, $FVAF = [(1 + i) \text{ to the power } n - 1] \div i$. For $i = 0.10$ and $n = 3$, the power is 1.10 to the power 3, which is 1.3310, so $FVAF = (1.3310 - 1) \div 0.10 = 3.3100$. This matches the sum of the growth factors above, as it should.

Now the same calculation in the standard layout. Study each row; every example in this chapter follows the same order.

WORKED EXAMPLE 5.1: FUTURE VALUE OF AN ORDINARY ANNUITY

STEP	WORKING
Given	Deposit Tk 10,000 at the end of each year; 3 years; discount rate 10%
Find	The value of the deposits at the end of year 3
Formula	$FVA = PMT \times FVAF$
Step 1	FVAF for 10% and 3 years = 3.3100
Step 2	$FVA = Tk\ 10,000 \times 3.3100$
Answer	Tk 33,100
Check	Deposits total only Tk 30,000, so Tk 33,100 is plausible; the extra Tk 3,100 is interest. The table above also gave Tk 33,100

The present value of an annuity is the sum of what each payment is worth today.

The same shortcut works: add the factors once and multiply once.

The present value of an annuity is what all its payments are worth today, when each payment is discounted back to the present. The method mirrors the one above. We discount each payment with the factors of Chapter 4 and add. For the same three payments of Tk 10,000 at 10%, the payments are worth Tk 9,091, Tk 8,264 and Tk 7,513 today, which total Tk 24,868.

The three discount factors are 0.9091, 0.8264 and 0.7513, which add to 2.4868. The table gives 2.4869, because its factor is calculated exactly and then rounded once, whereas our sum rounds three times. This tiny gap of one taka is a rounding effect, not an error. In words: present value of an annuity = regular payment × present value annuity factor. In symbols, $PVA = PMT \times PVAF$. The factor can be found from $[1 - 1 \div (1 + i) \text{ to the power } n] \div i$.

WORKED EXAMPLE 5.2: PRESENT VALUE OF AN ORDINARY ANNUITY

STEP	WORKING
Given	Receive Tk 10,000 at the end of each year; 3 years; discount rate 10%
Find	The value today of the three receipts
Formula	$PVA = PMT \times PVAF$
Step 1	$PVAF$ for 10% and 3 years = 2.4869
Step 2	$PVA = Tk\ 10,000 \times 2.4869$
Answer	About Tk 24,869
Check	The receipts total Tk 30,000, and today's value must be smaller: Tk 24,869 is smaller. Also, $Tk\ 24,869 \times 1.3310 = Tk\ 33,100$, the future value of Example 5.1

The first two examples used round numbers, so that the method stood alone. Real figures are less tidy. The next example uses 8% and five years, with a payment of Tk 18,000. The factors come from the table that follows this example, which gives 5.8666 for the future value and 3.9927 for the present value. Notice that the answers are not round, and that rounding to the nearest taka is sensible.

A second, less tidy example uses an 8% rate and five years.

WORKED EXAMPLE 5.3: BOTH VALUES OF A LESS TIDY ANNUITY

STEP	WORKING
Given	A café receives Tk 18,000 at the end of each year; 5 years; discount rate 8%
Find	(a) the present value today; (b) the value at the end of year 5
Formula	$PVA = PMT \times PVAF$; $FVA = PMT \times FVAF$
Step 1	$PVAF = 3.9927$ and $FVAF = 5.8666$ for 8% and 5 years
Step 2	$PVA = Tk\ 18,000 \times 3.9927 = Tk\ 71,869$
Step 3	$FVA = Tk\ 18,000 \times 5.8666 = Tk\ 1,05,599$
Answer	Present value Tk 71,869; value at end of year 5 Tk 1,05,599
Check	The receipts total Tk 90,000. The present value is below that total, and the year-5 value is above it, as discounting and growth require. Also $Tk\ 71,869 \times 1.4693 = Tk\ 1,05,599$

The next table gives both annuity factors at 8% for one to five years, the rate used in the bakery's case. To read a row, find the number of years, then choose the column you need. For three years, 3.2464 means that three yearly deposits of Tk 1 grow to Tk 3.2464 at the end of year 3. The value 2.5771 means that three yearly receipts of Tk 1 are worth Tk 2.5771 today.

The tables below give annuity factors at 8% for the bakery's case.

ANNUITY FACTORS AT A DISCOUNT RATE OF 8%

YEARS	FUTURE VALUE ANNUITY FACTOR	PRESENT VALUE ANNUITY FACTOR
1	1.0000	0.9259
2	2.0800	1.7833
3	3.2464	2.5771
4	4.5061	3.3121
5	5.8666	3.9927

An annuity due is the ordinary annuity multiplied by one plus the rate.

Now the annuity due. Every payment arrives one year earlier, so every payment earns one more year of interest, or is discounted one year less. This is the same as multiplying the ordinary answer once by $(1 + i)$. The rule: value of an annuity due = value of the ordinary annuity $\times (1 + i)$. It holds for both future and present values.

WORKED EXAMPLE 5.4: FUTURE VALUE OF AN ANNUITY DUE

STEP	WORKING
Given	Deposit Tk 10,000 at the start of each year; 3 years; discount rate 10%
Find	The value of the deposits at the end of year 3
Formula	$FVA \text{ due} = FVA \text{ ordinary} \times (1 + i)$
Step 1	$FVA \text{ ordinary} = \text{Tk } 33,100$ (Example 5.1)
Step 2	$FVA \text{ due} = \text{Tk } 33,100 \times 1.10$
Answer	Tk 36,410
Check	Grow each deposit separately: $\text{Tk } 13,310 + \text{Tk } 12,100 + \text{Tk } 11,000 = \text{Tk } 36,410$

The present value of the same due stream is also higher.

The present value follows the same logic. If the three payments of Tk 10,000 begin today, they are worth $\text{Tk } 10,000 + \text{Tk } 9,091 + \text{Tk } 8,264 = \text{Tk } 27,355$, which is larger than the Tk 24,869 of the ordinary case. The first payment is not discounted at all, because it is already in the present.

Four slips cause most wrong answers in this section.

Beware of four common slips. The first is to use the future value factor when the problem asks for the present value, or the reverse. The second is to count the years wrongly: for three deposits, n is 3, not 2 or 4. The third is to mix months and years, using a yearly rate with monthly payments. The fourth is to forget that a table factor serves a payment of Tk 1, so it must be multiplied by the actual payment.

Pause and check: a small annuity at 10%.

Pause and check. Using the method of Example 5.1, find the future value of Tk 5,000 deposited at the end of each of two years at 10%. Then find the future value if the deposits are made at the start of each year. The answers are at the back of the book.

Recap: annuity value equals payment times factor.

Before moving on. Value of an annuity = payment $\times$ annuity factor. The factor depends on the rate and the years, not on the payment. For an annuity due, multiply the ordinary value by $(1 + i)$. You should now be able to say why a factor of 3.3100 serves any size of payment.

5.4 FINDING THE REGULAR PAYMENT FOR A GOAL

The same formula, turned around, gives the payment that reaches a target.

Until now the payment was known and the value was unknown. Often the position is reversed: the goal is known and the payment is unknown. The bakery's owner knows the goal, Tk 2,00,000 in five years. She needs the yearly deposit. Because $FVA = PMT \times FVAF$, we divide both sides by the factor, and the rule becomes: payment = goal $\div$ future value annuity factor.

A sinking fund payment is the regular saving that reaches a target.

A **sinking fund payment** is a regular saving made to reach a target sum on a fixed date. The old name *sinking fund* describes a fund that grows by regular deposits until it can pay a large cost, such as replacing a machine. In symbols, $PMT = FVA \div FVAF$.

Sinking fund payment. A regular saving that reaches a target sum.

WORKED EXAMPLE 5.5: THE BAKERY'S YEARLY DEPOSIT

STEP	WORKING
Given	Goal Tk 2,00,000 in 5 years; deposits at the end of each year; discount rate 8%
Find	The yearly deposit
Formula	$PMT = FVA \div FVAF$
Step 1	FVAF for 8% and 5 years = 5.8666 (table above)
Step 2	$PMT = \text{Tk } 2,00,000 \div 5.8666$
Answer	Tk 34,091 a year
Check	Grow each deposit to the end of year 5, as in the figure below: $\text{Tk } 46,381 + 42,945 + 39,764 + 36,819 + 34,091 = \text{Tk } 2,00,000$

The figure below is the check in a picture. It is one tall column, built from five blocks, one for each deposit, with the total at the top. Each block shows what one deposit has become by the end of year 5. The first deposit, which was in the account for four years, has grown most. The last deposit has not grown at all, because it was paid on the goal date.

Reading the figure: each deposit has a different amount of growth.

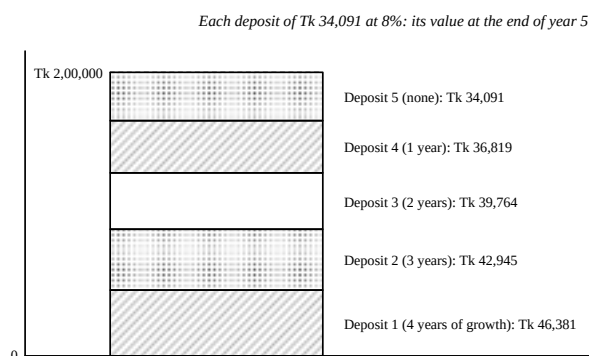


FIG. 14. Each yearly deposit of Tk 34,091 at 8%, valued at the end of year 5.

Compare this with Chapter 4. A single sum of Tk 1,36,117 placed today would reach the goal. The owner chooses instead five deposits of Tk 34,091, which total Tk 1,70,455. The interest supplies the other Tk 29,545. She pays the cost gradually, from yearly profit, and the account earns the rest.

The deposits total less than the goal; interest supplies the rest.

Suppose, instead, she deposits at the start of each year. Each deposit then earns one more year, so the yearly deposit can be smaller. The annuity due factor is $5.8666 \times 1.08 = 6.3359$, and the payment is $\text{Tk } 2,00,000 \div 6.3359 = \text{Tk } 31,566$. The saving is Tk 2,525 a year, which is the reward for paying one year earlier.

If deposits are made at the start of each year, the payment falls.

The reverse idea works for loans. A lender gives a sum today, and the borrower repays it in equal instalments. The sum borrowed is the present value of the instalments. So $\text{instalment} = \text{amount borrowed} \div \text{present value annuity factor}$. Example 5.6 shows this for a small loan.

The present value factor gives the instalment that repays a loan.

WORKED EXAMPLE 5.6: THE INSTALMENT THAT REPAYS A LOAN

STEP	WORKING
Given	Borrow Tk 1,00,000 today; repay in 4 equal instalments at the end of each year; discount rate 10%
Find	The yearly instalment
Formula	$PMT = PVA \div PVAF$
Step 1	PVAF for 10% and 4 years = 3.1699
Step 2	$PMT = Tk\ 1,00,000 \div 3.1699$
Answer	Tk 31,547 a year
Check	Four instalments total Tk 1,26,188, which is more than Tk 1,00,000, as it should be, because interest is paid. Also $Tk\ 31,547 \times 3.1699 = Tk\ 1,00,000$

Common slips: dividing by the wrong factor, and mixing the timing.

Pause and check: a goal and a loan.

Two slips are frequent here. A student may divide by the present value factor when saving for a goal, which gives the wrong answer; a goal in the future needs the *future* value factor. Another slip is to use the ordinary factor when the problem says *at the start of each year*. Choose the factor by asking two questions: is the known sum in the future or today, and do payments come at the end or the start?

Pause and check. A trader wants Tk 1,00,000 in three years and saves at the end of each year at 8%. The future value annuity factor for three years is 3.2464. Find the yearly saving. Then decide whether the same trader, paying at the start of each year, would save more or less each year, and why. The answers are at the back of the book.

5.5 MIXED STREAMS AND COMPARING STREAMS

Two contracts pay the same total in different orders: are they equal?

A caterer offers the bakery two contracts. Each pays four yearly receipts, and each pays Tk 2,20,000 in total. The only difference is the order of the payments. Is one contract worth more than the other? Our understanding of time says yes, since earlier money is worth more. This section shows how to measure the difference in taka.

A mixed stream has no shortcut; value each payment and add.

When payments are unequal, no single annuity factor exists. The method of Chapter 4 remains, and it is complete. Find the present value of each payment with its own discount factor, then add the present values. The sum is the present value of the stream. A stream's future value is found in the same way by growing each payment and adding.

Example 5.7 discounts each receipt separately, then adds.

We now value Contract A, the bakery's own four-year catering contract, one receipt at a time. The table gives the factors, the present values and the total.¹

¹ The factors in the table are rounded to four places. The present values are calculated with the more exact factors of a calculator. The four rounded present values add to Tk 1,80,941; the exact total is Tk 1,80,942.

WORKED EXAMPLE 5.7: PRESENT VALUE OF THE BAKERY'S CATERING RECEIPTS (CONTRACT A)

STEP	WORKING
Given	Receipts at the end of years 1 to 4: Tk 50,000; Tk 60,000; Tk 40,000; Tk 70,000; discount rate 8%
Find	The present value of the stream
Formula	PV of stream = sum of (each receipt $\times$ its present value factor)
Step 1	Factors at 8%: 0.9259 (year 1), 0.8573 (year 2), 0.7938 (year 3), 0.7350 (year 4)
Step 2	Present values: Tk 46,296; Tk 51,440; Tk 31,753; Tk 51,452
Step 3	Add the four present values
Answer	Tk 1,80,942
Check	The receipts total Tk 2,20,000, and the present value, Tk 1,80,942, is lower, as it must be. Each present value is below its own receipt

The figure below shows the same result as a picture. For each year there are two bars. The plain bar is the receipt, and the shaded bar beside it is its present value at 8%. The shaded bar is always shorter, and the gap grows as the year grows, because later money loses more value.

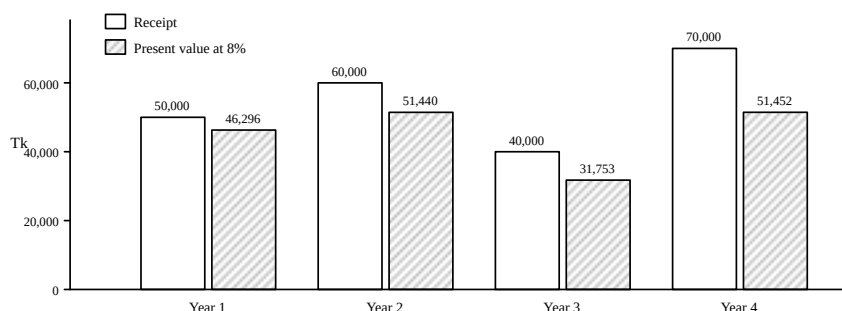


FIG. 15. The bakery's catering receipts and their present values at 8%.

Now Contract B. It pays Tk 70,000, Tk 50,000, Tk 60,000 and Tk 40,000 at the ends of years 1 to 4. The four payments are the same sums as Contract A, with the largest one moved to the first year. The table below sets both contracts side by side, using the same four discount factors.

PRESENT VALUES OF TWO CONTRACTS AT A DISCOUNT RATE OF 8%

YEAR	FACTOR	A: RECEIPT (TK)	A: PRESENT VALUE (TK)	B: RECEIPT (TK)	B: PRESENT VALUE (TK)
1	0.9259	50,000	46,296	70,000	64,815
2	0.8573	60,000	51,440	50,000	42,867
3	0.7938	40,000	31,753	60,000	47,630
4	0.7350	70,000	51,452	40,000	29,401
Total		2,20,000	1,80,942	2,20,000	1,84,713

Both contracts pay Tk 2,20,000, yet Contract B is worth Tk 1,84,713 today and Contract A is worth Tk 1,80,942. The difference is Tk 3,771. Contract B delivers its large payment early, when it loses least value. This is the lesson of the whole book so far: *compare streams by their present values, at the same discount rate and on the same date, never by their totals.*

Reading the figure: each receipt has a shorter twin, its present value.

Contract B pays the same sums in a different order.

Contract B is worth Tk 3,771 more, though the totals are equal.

A higher discount rate rewards early payments even more.

Pause and check: which stream is worth more?

Recap: how to value and compare a mixed stream.

A perpetuity is an annuity that never ends.

Perpetuity. An annuity that never ends.

A perpetuity is an idealisation, but a useful one.

The case file applies both methods to one business.

The question has two parts: a deposit and a comparison.

The reasoning chooses the factor by the kind of stream.

The conclusion gives both answers in taka.

The checks show both answers are sensible.

One exception deserves a note. The ranking of two streams can change if the discount rate changes. A higher rate lowers distant payments by more, so the stream with early payments gains. A stream that wins at 4% may lose at 15%. For this reason, a comparison must always name the rate it used.

Pause and check. Stream P pays Tk 10,000 at the end of year 1 and Tk 50,000 at the end of year 2. Stream Q pays Tk 50,000 at the end of year 1 and Tk 10,000 at the end of year 2. Both total Tk 60,000. Without calculating, which is worth more at any positive discount rate, and why? The answers are at the back of the book.

Before moving on. A mixed stream is valued by discounting each payment and adding. Two streams are compared by present value at the same rate. You should now be able to say why the order of the payments matters when the total does not change.

5.6 PERPETUITY IN BRIEF

A **perpetuity** is an annuity that never ends. Its payments continue for ever. At first this seems to give an infinite value, but later payments are discounted so heavily that the total stays finite. The present value is the payment divided by the discount rate: $PV = PMT \div i$. A payment of Tk 5,000 every year for ever, at 8%, is worth $Tk\ 5,000 \div 0.08 = Tk\ 62,500$ today. Check: Tk 62,500 earning 8% gives Tk 5,000 each year. We shall use this idea again in Chapter 11.

No real payment continues for ever, so a perpetuity is an idealisation. Even so, it helps when payments last for a very long time, because distant payments have almost no present value. The perpetuity formula then gives a good estimate with one division. Its main use for us will come in Chapter 11, where we value a share that is expected to pay a steady dividend for many years.



Case file, illustrative. *Situation.* Sundial Bakery, Dhaka, must replace equipment costing Tk 2,00,000 in five years (Chapter 4). The owner will make five equal deposits, one at the end of each year, into an account paying 8% a year. Separately, a caterer offers two four-year contracts. Contract A pays Tk 50,000, Tk 60,000, Tk 40,000 and Tk 70,000 at the ends of years 1 to 4. Contract B pays Tk 70,000, Tk 50,000, Tk 60,000 and Tk 40,000.

Question. What yearly deposit reaches Tk 2,00,000 in five years? Which contract is worth more today at 8%?

Reasoning. The deposits are equal and regular, so they form an ordinary annuity with a known future value. We divide the goal by the future value annuity factor for 8% and five years. The catering receipts are mixed streams, so we discount each receipt and add.

Conclusion. The deposit is $Tk\ 2,00,000 \div 5.8666 = Tk\ 34,091$ a year. Contract A is worth Tk 1,80,942 today. Contract B is worth Tk 1,84,713 today, which is Tk 3,771 more, although each contract pays Tk 2,20,000 in total.

Check. Growing each deposit to year 5 gives Tk 2,00,000 in total (see the figure in section 5.4). Each contract's present value is below Tk 2,20,000, as discounting requires. The

earlier-paying contract has the higher present value, as the theory predicts.

SUMMARY

A stream is a series of payments over time. An annuity has equal payments at equal gaps; a mixed stream does not. An ordinary annuity pays at the end of each period, and an annuity due pays at the start, so the due is worth more by the factor $(1 + i)$. The value of an annuity is the payment times an annuity factor, and the payment for a goal is the goal divided by the factor. A mixed stream is valued by discounting each payment and adding. Streams are compared by present value, never by total. A perpetuity is worth its payment divided by the discount rate.

The summary restates the chapter's four ideas.

THINK IT THROUGH

A school canteen will need Tk 5,00,000 in four years to build a new kitchen. The canteen can earn 10% a year. (a) Find the yearly deposit, made at the end of each year, that reaches the goal. The future value annuity factor for 10% and four years is 4.6410. (b) Find the yearly deposit if payments are made at the start of each year. (c) Explain, in two sentences, why the second answer is smaller. The answers are at the back of the book.

Apply the methods to a new school canteen case.

WHERE PEOPLE GO WRONG

Adding the payments as if time did not matter. A stream of Tk 10,000 for five years is not worth Tk 50,000 today. Each payment is discounted, so the total present value is smaller. Correction: always multiply by a factor, or discount each payment.

Error one: adding payments without discounting them.

Using the wrong factor. The future value annuity factor answers *what will my deposits become?* The present value annuity factor answers *what are my receipts worth today?* Correction: ask first whether the known figure sits in the future or in the present.

Error two: using the wrong table.

Ignoring the timing. Payments at the start of each year are not payments at the end. Correction: look for words such as *start*, *beginning*, *today* or *in advance*, which signal an annuity due, and multiply the ordinary answer by $(1 + i)$.

Error three: forgetting the timing of the payments.

Calling an uneven stream an annuity. A single different payment breaks the annuity. Correction: use the annuity factor only when every payment is equal and every gap is equal. Otherwise, discount each payment separately.

Error four: calling a nearly equal stream an annuity.

IN HISTORY

In the seventeenth century, several European governments raised money by selling life annuities. A life annuity pays a fixed sum each year for as long as the buyer lives. The price had to be fair: too low, and the state lost; too high, and no one bought. In 1671 Johan de Witt, a leading statesman of the Dutch Republic, wrote a report on the fair price of such annuities. He treated each yearly payment as a sum to be discounted, weighted by the chance that the buyer would still be alive. His method is a close relative of the present-value method of this chapter.

A seventeenth-century state priced annuities by the methods of this chapter.

REVIEW QUESTIONS

Review questions close the chapter.

REVIEW QUESTIONS FOR CHAPTER V

NO.	QUESTION
1	Define an annuity. State two ways in which a stream can fail to be an annuity.
2	Explain the difference between an ordinary annuity and an annuity due. Which has the higher future value, and why?
3	Tk 20,000 is deposited at the end of each year for four years at 10%. Find the future value. (FVAF = 4.6410)
4	Find the future value if the deposits in question 3 are made at the start of each year.
5	Find the present value of Tk 25,000 received at the end of each year for five years at 8%.
6	A loan of Tk 2,00,000 is repaid in three equal year-end instalments at 8%. Find each instalment.
7	Stream X pays Tk 30,000, Tk 30,000 and Tk 90,000 at the ends of years 1 to 3. Stream Y pays Tk 90,000, Tk 30,000 and Tk 30,000. At 10%, which is worth more today, and by how much?
8	A perpetuity pays Tk 12,000 a year. Find its present value at 6%.
9	Explain why two streams with the same total can have different present values.

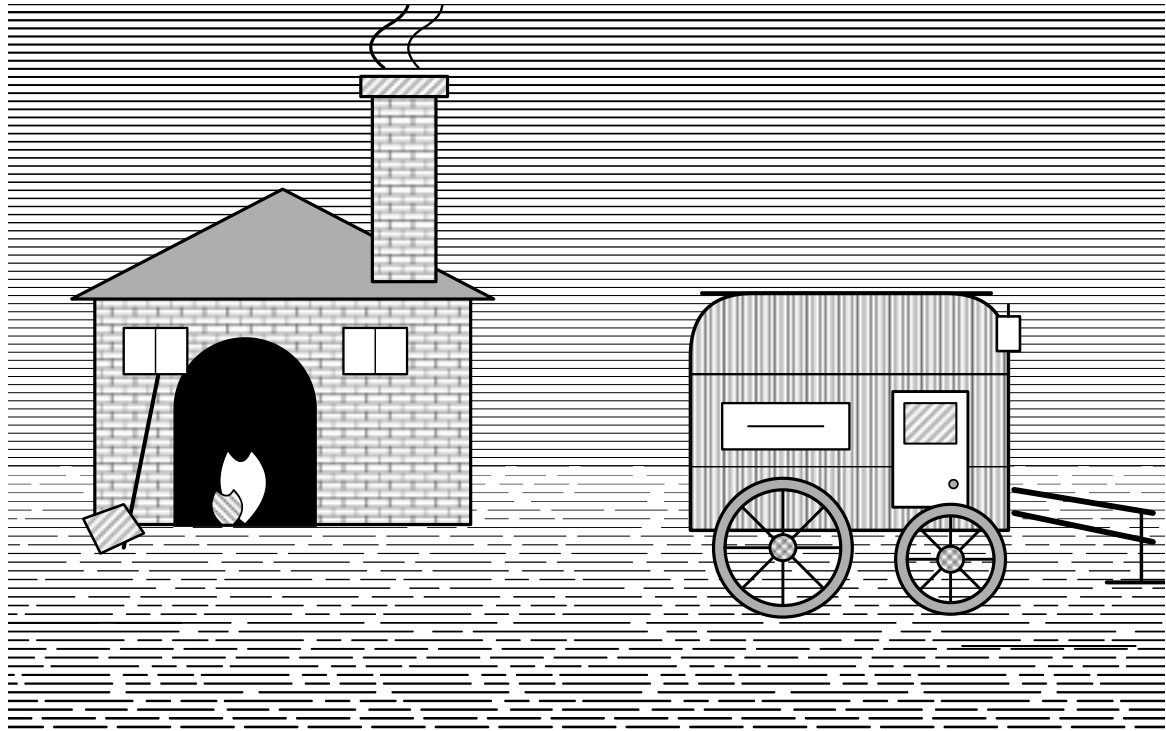


Plate VI. The Oven and the Van.

CHAPTER VI

Choosing Big Purchases: Process and Cash Flows

Before the money is spent, the plan must be drawn.



Some purchases last for years and cannot be returned. This chapter explains how a firm plans such purchases, the steps it follows, and which cash flows belong in the plan. We then draw the cash-flow timeline of two choices facing the bakery, so that Chapter 7 can judge them.

The bakery has money for one large purchase and two good options.

A wrong choice lasts for years, so the plan must come before the purchase.

This chapter prepares the numbers; Chapter 7 chooses between them.

Capital expenditure is spending on assets that serve for many years.

Capital expenditure. Money spent on assets that serve for more than a year.

A project is a single plan to make one such purchase.

Project. One investment plan, such as buying a single oven.

Capital budgeting is the planning and choosing of long-lived projects.

Capital budgeting. Planning and choosing long-lived investments.

Five features make these decisions more serious than daily spending.

The table contrasts daily spending with capital expenditure.

On 1 July 2026, Sundial Bakery has two options. The first is a second oven. The second is a delivery van. Each costs Tk 3,00,000, and each is expected to serve for five years. The bank will lend the bakery only Tk 3,00,000 for new equipment, so the owner can buy one option but not both. Both look useful. Which should she choose?

The puzzle is harder than it seems. A purchase of flour can be corrected next week, when the next order is placed. An oven cannot. It will occupy the kitchen for five years, and its cost has already left the bank account. The owner needs a way to look ahead, to write down what each option will cost and bring, and only then to decide. That way of looking ahead has a name, and this chapter teaches it.

We shall not choose between the oven and the van in this chapter. The choice needs the techniques of Chapter 7. Our task here is earlier and, in practice, harder: to decide *which* cash flows belong in the plan, and to place them on a timeline. A mistake in this step cannot be repaired by clever arithmetic later.

6.1 WHAT CAPITAL BUDGETING IS AND WHY IT MATTERS

A firm spends money in two broad ways. Some spending is used up quickly: flour, wages and electricity are consumed within days or weeks. Other spending buys something that serves for many years: an oven, a van, a building. We call this second kind **capital expenditure**. It is money spent on assets that will serve the firm for more than one year.

Each such plan has a name. A **project** is a single investment plan, such as the purchase of one oven, together with the cash it is expected to bring. The bakery's second oven is one project. The delivery van is another. A firm may have many projects in view at the same time.

Capital budgeting is the planning and choosing of long-lived investments. It answers one question for each project: should the firm spend this money now, in return for cash over the coming years? Chapter 2 named this the *investment decision*. Capital budgeting is the careful method for making it.

Why does this decision deserve its own chapter? Five features set it apart from everyday spending. First, the sums are large compared with the firm's size. Second, the effects last for years, so a poor choice is paid for repeatedly. Third, the decision is hard to reverse: a used oven sells for much less than its price. Fourth, the future is uncertain, so every forecast may prove wrong. Fifth, these decisions shape what the firm will be, because the firm becomes what it buys.

The table below contrasts the two kinds of spending. Read each row across: the first column names a feature, and the next two columns say how it appears in daily spending and in capital expenditure.

DAILY SPENDING COMPARED WITH CAPITAL EXPENDITURE

FEATURE	DAILY SPENDING (FLOUR)	CAPITAL EXPENDITURE (OVEN)
Size	Small	Large
Life of the benefit	Days or weeks	Many years
Can it be undone?	Easily	With difficulty and loss
Needs a formal plan?	Seldom	Always

There is an exception. Not every long-lived purchase needs a full plan. A new shop sign costing Tk 2,000 serves for years, yet no firm would draw a five-year timeline for it. Firms set a limit: above a certain size, a formal plan is required; below it, the manager may decide quickly. The size of the limit depends on the size of the firm.

Pause and check. Decide whether each item is daily spending or capital expenditure for the bakery. (a) Tk 12,000 of flour. (b) A new mixing machine costing Tk 1,50,000 that will serve for eight years. (c) Tk 30,000 of monthly wages. The answers are at the back of the book.

One exception: small purchases of long life need little formality.

Pause and check: spending or capital expenditure?

6.2 KINDS OF PROJECTS AND THE STEPS OF THE PROCESS

Projects rarely stand alone. Whether the firm may accept one depends on what else it wishes to do. The relations between projects matter for three ideas, named in the next three paragraphs.

An **independent project** is a project whose acceptance does not affect any other project. If the bakery wished to buy a new shop sign and also a second oven, and the two had no effect on each other, each would be judged on its own merits. The firm could accept both, one or neither.

A **mutually exclusive project** is a project that rules out another if it is chosen. A firm that buys the oven cannot also buy the van with the same loan. The two compete for the same limited money. Mutually exclusive projects must be compared with one another, and only the best may be chosen.

Capital rationing is a limit on the money available for projects. It occurs when the firm has more worthwhile projects than money. The bakery's position is an example: the bank will lend only Tk 3,00,000 for new equipment, so the oven and the van become mutually exclusive in practice, although they might be independent if money were plentiful.

The table below sets the three ideas side by side, with an example from the bakery for each.

Projects are related to one another in three important ways.

An independent project does not affect the others.

Independent project. A project whose acceptance does not affect other projects.

Mutually exclusive projects rule one another out.

Mutually exclusive project. A project that rules out another if it is chosen.

Capital rationing means the firm has too little money for all its projects.

Capital rationing. A limit on the money available for projects.

A table sets the three relations side by side.

THREE RELATIONS BETWEEN PROJECTS

IDEA	MEANING	BAKERY EXAMPLE
Independent	Choosing one does not affect another	A new shop sign and a second oven
Mutually exclusive	Choosing one rules out another	A second oven or a delivery van
Capital rationing	Money limits how many can be chosen	Tk 3,00,000 available for projects costing Tk 6,00,000

The process has five steps, from finding ideas to reviewing results.

Now the process itself. A firm that handles projects well follows five steps, shown in the figure below. Read the figure from left to right. Each box is a step, and each arrow means that the next step follows. The dashed arrow at the bottom returns from the last step to the first, to show that the process repeats. The next paragraphs describe each step.

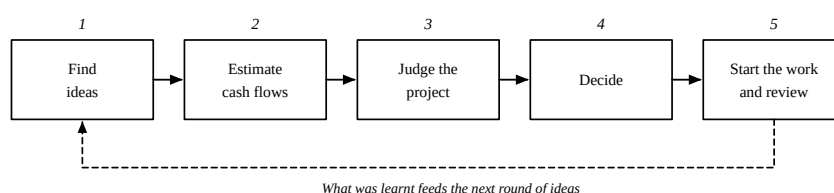


FIG. 16. The five steps of capital budgeting.

Step one: ideas come from many places.

Step 1, find ideas. Projects begin as ideas. They come from staff who see a problem, from customers who ask for something new, and from equipment that is wearing out. A firm that welcomes ideas from every level finds more good projects than a firm that waits for the owner to think of them.

Step two: estimate the cash flows each project will bring.

Step 2, estimate cash flows. For each project, someone must forecast the money it will cost and the money it will bring in each future year. This step is the hardest and the most important, because every later step relies on it. A forecast will be wrong in some degree, so the person making it should say how sure he or she is. The rest of this chapter concerns this step.

Step three: judge each project by a rule.

Step 3, judge the project. The forecast cash flows are put through a decision rule, such as the payback period or net present value. Chapter 7 teaches these rules. A rule turns a timeline of cash flows into a verdict: accept, reject, or rank.

Step four: someone with authority decides.

Step 4, decide. The verdict is only advice. A person with authority, the owner of the bakery or, in a larger firm, a committee or the board, makes the decision. Larger projects go to higher levels, because the damage of a wrong choice grows with the sum.

Step five: start the work, then compare results with the forecast.

Step 5, start the work and review. If the project is accepted, the firm buys the asset and begins. Months and years later, the firm compares the actual cash flows with the forecast. The review is not a search for blame. Its purpose is to improve the next forecast, which is why the dashed arrow in the figure leads back to step one.

6.3 RELEVANT CASH FLOWS: OUTLAY, YEARLY INFLOWS AND END OF LIFE

A project may be described in two ways: by the profit it earns or by the cash it brings. Capital budgeting uses cash. The reason is practical. A firm cannot pay wages, repay a bank or reward its owner with *profit*; it can do so only with *cash*. Chapter 1 defined a cash flow as money actually moving in or out. Profit contains items, such as depreciation, which do not move money at all, as we shall see in section 6.4.

Every project produces cash flows of three kinds, in an order that follows the life of the asset. The figure below shows them on a timeline. Read it as before: the line moves through time, an arrow pointing up is money received, and an arrow pointing down is money paid out. Today's outlay is at the left, the yearly receipts are in the middle, and the final sum is at the right.

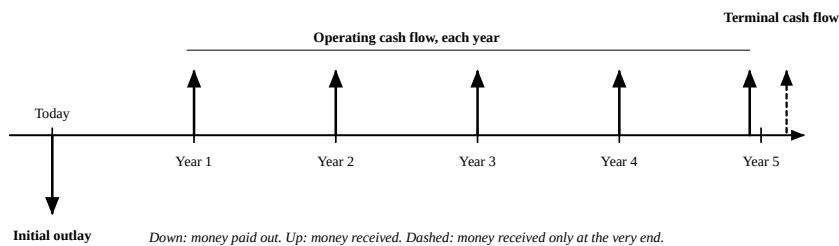


FIG. 17. The three kinds of cash flow in a project that lasts five years.

The **initial outlay** is all the money spent at the start of a project, to make the asset ready for use. It includes the price of the asset and every other cost needed to begin. Suppose an oven costs Tk 2,50,000, delivery costs Tk 5,000, and installation costs Tk 10,000. The initial outlay is Tk 2,65,000, not Tk 2,50,000. The outlay is a cash flow *out*, so we write it with a minus sign.

The **operating cash flow** is the cash a project brings in each year of its working life. It is the extra cash received from sales, less the extra cash paid for materials, wages and other running costs, and after the extra tax that the project causes. Section 6.4 shows how to find it. For the oven, it is the extra cash earned each year by baking more.

The **terminal cash flow** is cash received or paid at the end of the project's life. The most common case is the sale of the old asset. If the oven can be sold for Tk 20,000 after five years, that sum is a terminal cash flow of plus Tk 20,000, received at the end of year 5. Money tied up in extra stock may also be released at the end. To keep the arithmetic clear, this book ignores tax on such sales.

For the bakery's two options, we assume that the oven and the van have no resale value at the end of five years, and that no extra stock is needed. Their terminal cash flows are therefore zero. This assumption keeps the case small; the method is the same when a terminal sum exists.

Many students find this step confusing, because the outlay is a payment, not a profit or a cost in the ordinary sense. They forget to place it on the timeline, and the project then looks free. Rule: every taka that leaves or enters the firm because of the project appears on the timeline, with its own date and sign.

Why capital budgeting uses cash flows, not profit.

A project's cash flows fall into three kinds.

The initial outlay is all the money spent at the start.

Initial outlay. All the money spent at the start of a project.

Operating cash flow is the cash a project brings each year.

Operating cash flow. The cash a project brings in each year of its life.

Terminal cash flow is cash at the very end of the project.

Terminal cash flow. Cash received or paid at the end of a project's life.

The bakery's two options are assumed to be worth nothing at the end.

Many students forget that the outlay is a cash flow too.

6.4 SUNK COSTS, EXTRA COSTS, DEPRECIATION AND TAX EFFECTS

Only the extra cash caused by the project belongs in the plan.

Incremental cash flow. The extra cash flow caused by the project alone.

The test counts extra costs and ignores costs that do not change.

A sunk cost has already been spent and cannot return.

Sunk cost. Money already spent that cannot be recovered.

Depreciation spreads an asset's cost over its working life.

Depreciation. Spreading an asset's cost over the years of its life.

Depreciation is a charge in the profit statement, but no cash moves.

Depreciation still matters, because it lowers the tax paid.

The table shows the tax with and without the depreciation charge.

Which cash flows belong in the plan? A simple test answers the question. Compare the firm's cash *with* the project and *without* it. The difference, year by year, is the **incremental cash flow**: the extra cash flow caused by the project alone. Only incremental flows belong on the timeline.

The test works in both directions. If the oven raises the electricity bill by Tk 20,000 a year, that extra cost counts, because the project causes it. If the bakery pays Tk 30,000 a year in shop rent whether or not it buys the oven, the rent does not count, because the rent is the same with and without the project. A project that takes sales away from the firm's other products also has a cost: the lost sales are a negative incremental flow.

A **sunk cost** is money already spent that cannot be recovered. Suppose that last month the bakery paid Tk 5,000 for a survey of delivery routes for the van. That money is gone whether or not the van is bought. It is therefore not an incremental flow, and it must stay off the timeline. This rule feels strange, since the money was real. But the decision concerns only the future; the past cannot be changed by choosing either option.

Depreciation means spreading an asset's cost over the years of its life. One simple method is the *straight-line* rule: depreciation each year equals the cost, less the expected resale value, divided by the number of years. The oven costs Tk 3,00,000, has no resale value, and serves for five years, so its yearly depreciation is $\text{Tk } 3,00,000 \div 5 = \text{Tk } 60,000$.

An important fact follows. The Tk 3,00,000 left the bank on the day of purchase; that was the initial outlay. When the accountant charges Tk 60,000 of depreciation in each later year, no money leaves the bank. Depreciation is a *non-cash charge*. It reduces profit, but it does not reduce cash. For this reason, a project's operating cash flow is not the same as its profit.

Yet depreciation affects cash in one real way: it lowers taxable profit, and so it lowers the tax the firm pays. The tax saved is a true cash gain. Suppose a machine costs Tk 1,00,000, serves for five years, and has no resale value. Its yearly depreciation is Tk 20,000. Suppose the machine earns Tk 60,000 a year before depreciation, and the tax rate is 25%.¹

The table below computes the tax in two ways: first as if depreciation were not allowed, then as the law allows. The difference is the tax saving from depreciation.

¹ The tax rate of 25% is illustrative. The actual rate depends on the type of firm and on the law in force. The straight-line rule is also used here for teaching; tax law prescribes its own depreciation rates. Both should be checked against current Bangladesh rules before use.

TAX WITH AND WITHOUT THE DEPRECIATION CHARGE (TK A YEAR)

ITEM	WITHOUT DEPRECIATION	WITH DEPRECIATION
Earnings before depreciation	60,000	60,000
Less depreciation	0	20,000
Taxable profit	60,000	40,000
Tax at 25%	15,000	10,000
Cash left after tax	45,000	50,000

The tax saving is Tk 15,000 – Tk 10,000 = Tk 5,000. This equals the depreciation multiplied by the tax rate: Tk 20,000 $\times$ 25% = Tk 5,000. The cash left with depreciation allowed is Tk 50,000, which is Tk 5,000 more than without. A firm that earns no taxable profit cannot use this saving in that year, because it has no tax to reduce. This exception matters for small or loss-making firms.

There is a quick route from profit to cash. Start with the profit after tax, then add back the depreciation, because depreciation was subtracted although no cash left. In words: operating cash flow = profit after tax + depreciation. Here the project's profit after tax is Tk 40,000 – Tk 10,000 = Tk 30,000, and adding Tk 20,000 of depreciation gives Tk 50,000, the same as the last row above.

The saving equals depreciation multiplied by the tax rate.

A simple route from profit to cash: add back depreciation.

WORKED EXAMPLE 6.1: OPERATING CASH FLOW OF A MACHINE

STEP	WORKING
Given	Earnings before depreciation Tk 60,000 a year; depreciation Tk 20,000 a year; tax rate 25%
Find	The yearly operating cash flow
Formula	Operating cash flow = profit after tax + depreciation
Step 1	Profit before tax = Tk 60,000 – Tk 20,000 = Tk 40,000
Step 2	Tax = 25% $\times$ Tk 40,000 = Tk 10,000; profit after tax = Tk 30,000
Step 3	Operating cash flow = Tk 30,000 + Tk 20,000
Answer	Tk 50,000 a year
Check	Earnings Tk 60,000 less tax Tk 10,000 = Tk 50,000, the same answer by a second route

One more rule completes the section. Interest paid on a loan, and the repayment of the loan itself, are *not* part of the project's operating cash flow. They belong to the financing decision, not to the investment decision. Chapter 7 charges each project for the cost of its funds through the discount rate. If interest were also subtracted here, the project would be charged twice. In the profit layout of Chapter 3, this means starting from EBIT, which is profit before interest and tax.

Before moving on. Four tests tell us whether an item belongs on the timeline. Is it cash? Is it caused by the project? Is it in the future? Is it not interest on the loan? An item that fails any test stays off. You should now be able to say why depreciation is not a cash flow but still changes the cash flow.

Interest on a loan does not belong in the project's cash flows.

Four tests decide whether an item belongs on the timeline.

6.5 BUILDING A PROJECT'S CASH-FLOW TIMELINE

Four steps build a timeline for any project.

The oven's yearly cash flow is built from its extra profit and its depreciation.

A timeline can be built for any project in four steps. First, list the initial outlay at the start, with a minus sign. Second, estimate the operating cash flow for each year. Third, add the terminal cash flow to the last year. Fourth, draw the line with each flow at its date. We shall now follow these steps for the bakery's two options.

For the oven, the bakery's forecast is that the extra baking will add profit of Tk 53,333 a year before interest and tax. This figure is rounded to the nearest taka. Tax at 25% takes Tk 13,333, leaving Tk 40,000 of profit after tax. The oven's depreciation is Tk 60,000 a year, as found in section 6.4. Adding the depreciation back gives the operating cash flow.

WORKED EXAMPLE 6.2: THE OVEN'S YEARLY OPERATING CASH FLOW

STEP	WORKING
Given	Outlay Tk 3,00,000; life 5 years; no resale value; extra EBIT Tk 53,333 a year; tax rate 25%
Find	The operating cash flow for each year
Formula	Operating cash flow = profit after tax + depreciation
Step 1	Depreciation = Tk 3,00,000 ÷ 5 = Tk 60,000
Step 2	Tax = 25% × Tk 53,333 = Tk 13,333; profit after tax = Tk 40,000
Step 3	Operating cash flow = Tk 40,000 + Tk 60,000
Answer	Tk 1,00,000 a year, for years 1 to 5
Check	EBIT Tk 53,333 + depreciation Tk 60,000 – tax Tk 13,333 = Tk 1,00,000

The van brings uneven cash, with a loss after tax in year one.

The van is different. Its sales grow as customers learn of the delivery service, so its cash flows rise year by year. The profit after tax is a loss of Tk 10,000 in year 1, then Tk 20,000, Tk 50,000, Tk 80,000 and Tk 80,000. The van also depreciates by Tk 60,000 a year. Adding that back gives the operating cash flows in the table below.

THE VAN'S OPERATING CASH FLOWS (TK)

YEAR	PROFIT AFTER TAX	ADD BACK DEPRECIATION	OPERATING CASH FLOW
1	–10,000	60,000	50,000
2	20,000	60,000	80,000
3	50,000	60,000	1,10,000
4	80,000	60,000	1,40,000
5	80,000	60,000	1,40,000

A loss in profit can still be a gain in cash.

Look at year 1. The van's profit after tax is a loss, yet its cash flow is a gain of Tk 50,000. The reason is depreciation: it reduced the profit by Tk 60,000, but no cash left. This is a clear example of why capital budgeting follows cash, not profit.

Both timelines are now complete and can be drawn.

With both operating cash flows known, and with a terminal cash flow of zero for each, we may place everything on the timeline. The table below lists the flows year by year. Money paid out carries a minus sign; money received carries none.

CASH-FLOW TIMELINES OF THE TWO OPTIONS (TK)

YEAR	OVEN	VAN
Today (year 0)	-3,00,000	-3,00,000
1	1,00,000	50,000
2	1,00,000	80,000
3	1,00,000	1,10,000
4	1,00,000	1,40,000
5	1,00,000	1,40,000
Total received	5,00,000	5,20,000

The figure below shows the same flows as bars. Each timeline has a horizontal line for time. A shaded bar above the line is money received in that year; its height shows its size. The dotted bar below the line on the left is the outlay paid today. The oven's bars are all equal. The van's bars begin small and grow.

Reading the figure: bars above the line are receipts, the dotted bar is the outlay.

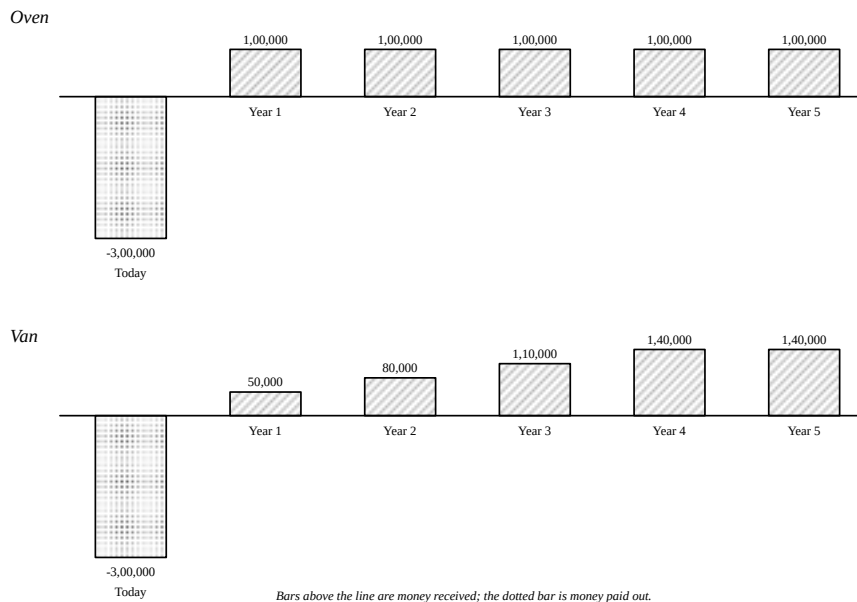


FIG. 18. The oven's and the van's cash flows, year by year (Tk).

The van returns Tk 5,20,000 in total, and the oven returns Tk 5,00,000. A hasty reader would choose the van. But Chapters 4 and 5 taught us that when money arrives matters as much as how much arrives. The van's larger sums come later, and later sums are worth less today. The totals therefore cannot settle the choice. We shall use present value in Chapter 7 to do so.

The two timelines differ in shape, and Chapter 5 gave a tool for each. The oven's five equal receipts form an ordinary annuity, so one annuity factor will value them. The van's receipts form a mixed stream, so each receipt must be discounted separately and the results added. Chapter 7 uses both tools at a provisional discount rate of 10%.

The totals favour the van, but totals alone mislead us.

The two timelines have different shapes, and Chapter 5 supplies a tool for each.



Case file, illustrative. *Situation.* On 1 July 2026, Sundial Bakery considers two options, each with an initial outlay of Tk 3,00,000 and a life of five

The case file records both options in full.

years. The bank will lend only Tk 3,00,000 for new equipment, so one option may be chosen. Neither has resale value, and neither needs extra stock. The tax rate is 25% (illustrative). Depreciation is straight-line, Tk 60,000 a year for each. The oven adds EBIT of Tk 53,333 a year (rounded). The van adds profit after tax of –Tk 10,000, Tk 20,000, Tk 50,000, Tk 80,000 and Tk 80,000.

The question asks for the timelines, not yet for a choice.

The reasoning applies the tests of section 6.4.

The conclusion lists the two timelines.

The checks confirm the arithmetic and the logic.

Question. What are the relevant cash flows of each option, year by year? Which option has the larger total?

Reasoning. The outlay of Tk 3,00,000 is a cash flow out today. The operating cash flow for each year is profit after tax plus depreciation. Interest on the bank loan is excluded, because the discount rate in Chapter 7 will charge for the funds. The survey cost of Tk 5,000 mentioned in section 6.4 is sunk and excluded. Terminal cash flows are zero.

Conclusion. The oven's flows are –Tk 3,00,000 today and Tk 1,00,000 in each of years 1 to 5. The van's flows are –Tk 3,00,000 today, then Tk 50,000; Tk 80,000; Tk 1,10,000; Tk 1,40,000; and Tk 1,40,000. The van's total is larger, by Tk 20,000, but the choice waits for Chapter 7.

Check. The oven's Tk 1,00,000 is confirmed by two routes (Example 6.2). The van's five flows add to Tk 5,20,000, and Tk 50,000 + Tk 80,000 + Tk 1,10,000 + Tk 1,40,000 + Tk 1,40,000 agrees. Each option's total received exceeds its outlay of Tk 3,00,000, as any worthwhile project would require.

SUMMARY

Capital budgeting is the planning and choosing of long-lived projects. It matters because the sums are large, the effects last, and the decision is hard to undo. Projects may be independent or mutually exclusive, and money may be rationed. The process has five steps: find ideas, estimate cash flows, judge, decide, and review. Only incremental cash flows belong in the plan: the initial outlay, the yearly operating cash flows, and any terminal cash flow. Sunk costs, unchanged costs and interest are left out. Depreciation is a non-cash charge, but it lowers tax, so operating cash flow equals profit after tax plus depreciation.

THINK IT THROUGH

A tailoring shop considers a machine that costs Tk 2,00,000 and serves for four years. It can be sold for Tk 20,000 at the end of year 4; ignore tax on that sale. The machine will add Tk 2,50,000 of sales a year. It will add Tk 1,50,000 of materials and wages and Tk 20,000 of electricity each year. The shop pays Tk 30,000 a year of rent whether or not it buys the machine, and last month it paid Tk 10,000 for a survey of the market. Depreciation is Tk 45,000 a year, and the tax rate is 25%. (a) Say which items stay off the timeline, and why. (b) Find the operating cash flow for each year. (c) Write the complete timeline, from today to year 4. The answers are at the back of the book.

WHERE PEOPLE GO WRONG

Counting money already spent. A manager who has paid for a survey feels that the money must be justified, and so adds it to the project.

Build the timeline for a new tailoring machine.

Error one: counting a sunk cost.

Correction: a sunk cost cannot change with the decision, so it never belongs on the timeline.

Using profit instead of cash. A project's accounting profit includes depreciation, which is not cash. Correction: add depreciation back to profit after tax, or start from earnings and subtract only the cash tax.

Subtracting interest as a project cost. A student sees that the oven is bought with a loan, and deducts the interest from each year's cash flow. Correction: financing is charged later, through the discount rate. Deducting interest too would charge the project twice.

Counting costs that continue anyway. Shop rent, the owner's salary and similar overheads often appear in a project's cost sheet. Correction: if the amount is the same with and without the project, it is not incremental and must be omitted.

Error two: using profit instead of cash.

Error three: subtracting interest.

Error four: counting costs that do not change.

IN HISTORY

In 1858 a company was formed in France to build a canal across the Isthmus of Suez, in Egypt, linking the Mediterranean Sea with the Red Sea. The company's leader was Ferdinand de Lesseps, a French diplomat. The company raised money by selling shares. The canal opened to ships in November 1869, more than ten years after the company was formed, and only then could the company begin to earn tolls. Until that day the investors had paid out money and received no operating cash. They were judging the project by the cash flows they expected far in the future, which is the central idea of capital budgeting.

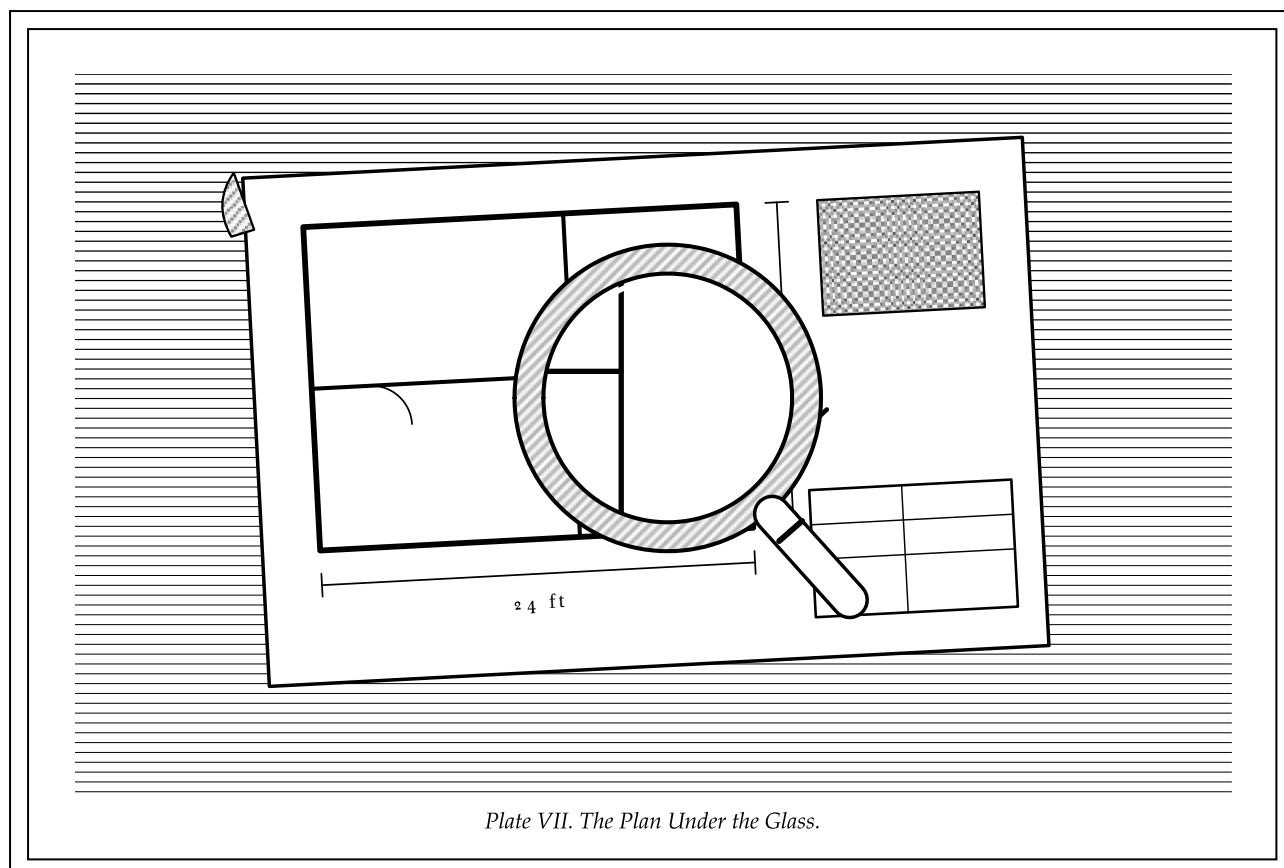
A great canal shows how long investors may wait for cash.

REVIEW QUESTIONS

REVIEW QUESTIONS FOR CHAPTER VI

NO.	QUESTION
1	Define capital budgeting and explain why a firm needs it.
2	State five features that distinguish capital expenditure from daily spending.
3	Explain the difference between an independent project and a mutually exclusive project. Give an example of each.
4	Name the five steps of capital budgeting and describe the third and fifth steps.
5	Explain why capital budgeting uses cash flows rather than profit.
6	Name the three kinds of cash flow in a project and give an example of each.
7	Define a sunk cost. A firm paid Tk 8,000 for a survey before deciding on a project. Should the Tk 8,000 count? Explain.
8	A machine costs Tk 4,00,000, serves for five years, and has no resale value. It adds EBIT of Tk 1,00,000 a year. The tax rate is 25%. Find the yearly depreciation and the yearly operating cash flow.
9	Explain why a project's profit after tax can be negative while its cash flow is positive.

Review questions close the chapter.



CHAPTER VII

Judging Projects

A list of cash flows is only a list until a rule turns it into a decision.



This chapter teaches how a firm decides whether a long-lived purchase is worth its cost. It begins with the quick test of payback, builds the trusted test of net present value, and adds two close relatives. It ends by asking what to do when the tests disagree.

The oven and the van tempt the owner with two different sets of yearly cash.

A plain addition of the inflows seems to favour the van.

The fault lies in adding sums that arrive at different times.

Five tests are taught here, from the quick and crude to the careful.

The owner's first question is how long it takes to get the money back.

Payback period. The time a project takes to bring back its initial outlay.

The payback period is the number of years needed to recover the outlay.

The rule compares the payback with a limit the owner sets in advance.

When the inflows are equal, one division gives the payback.

The first example uses a mixer that costs Tk 1,00,000.

Dividing the outlay by the yearly inflow gives two years.

The owner of Sundial Bakery has Tk 3,00,000 to spend and two ways to spend it. An oven would bring in Tk 1,00,000 in each of five years. A van would bring in a smaller sum at first and larger sums later: Tk 50,000, Tk 80,000, Tk 1,10,000, Tk 1,40,000 and Tk 1,40,000. Chapter VI showed how to find these yearly cash flows. This chapter asks what to do with them.

Begin with the plainest test: add up what each choice brings in. The oven brings in Tk 5,00,000 over five years. The van brings in Tk 5,20,000. The van gives Tk 20,000 more, so it seems to be the better choice. Yet the oven is the better choice, and we will show why before the chapter ends. Something in the plain test must be wrong.

The fault lies in the adding. Chapter IV taught that a taka received later is worth less than a taka received today. The van's larger sums come later, so the plain total flatters it. A fair test must first give each sum its true value today, and only then add.

This chapter teaches five ways of judging a project. Each turns a timeline of cash flows into a decision. The first, payback, is quick but crude. The second, net present value, is the most trusted. The next two, the internal rate of return and the profitability index, are close relatives of the second. The last section asks what to do when the tests give different answers.

PAYBACK: HOW LONG UNTIL THE MONEY RETURNS

THE QUESTION

Every investment begins with a payment. The oven costs Tk 3,00,000 at the start. A careful owner asks a simple question at once: how long will it take to get this money back? The answer to that question is a number of years, and it is the oldest test of a project.

The number of years has a name. The **payback period** is the length of time a project takes to bring back its initial outlay from its yearly cash inflows. A short payback means the money returns quickly. A long payback means the owner must wait.

The owner then needs a rule. The owner chooses a cut-off, which is the longest wait that the firm will accept. A project whose payback is equal to or shorter than the cut-off is accepted. A project whose payback is longer is rejected. If two projects compete for the same money, the one with the shorter payback is preferred.

EVEN INFLOWS

When every year brings the same inflow, the payback needs one division. We use a small example with clean numbers so that the method is the only new thing. The firm may buy a mixer for its kitchen.

Given. The mixer costs Tk 1,00,000 at the start. It brings in Tk 50,000 at the end of each of three years. *Find.* The payback period. *Formula.* Payback period equals the initial outlay divided by the yearly inflow.

Steps. Payback = Tk 1,00,000 ÷ Tk 50,000 = 2 years. *Answer.* The mixer pays back its cost in 2 years. *Check.* Two years of inflow are 2 × Tk 50,000 = Tk 1,00,000, which is exactly the outlay.

UNEVEN INFLOWS

Often the inflows are not equal, and one division no longer works. We then keep a running total of the cash received and watch the unpaid part of the outlay shrink. We take a display case as the example. It costs Tk 1,50,000 and brings in Tk 40,000, Tk 60,000 and Tk 80,000 at the ends of years 1, 2 and 3.

When the inflows differ, we keep a running total instead.

THE DISPLAY CASE: OUTLAY STILL UNPAID AT THE END OF EACH YEAR (TK)

YEAR	CASH INFLOW	TOTAL RECEIVED SO FAR	OUTLAY STILL UNPAID
0	none	0	1,50,000
1	40,000	40,000	1,10,000
2	60,000	1,00,000	50,000
3	80,000	1,80,000	none

The table shows that the outlay is repaid during year 3. After two full years Tk 50,000 is still unpaid. Year 3 brings in Tk 80,000, so the unpaid Tk 50,000 needs $50,000 \div 80,000 = 0.625$ of that year. We assume that cash arrives evenly through the year. The payback is therefore 2 years plus 0.625 of a year, which is 2.625 years, or about 2.6 years.

The outlay is fully repaid during year 3, so we need a fraction of that year.

Check. After two years the firm has received Tk 1,00,000, which is less than Tk 1,50,000. After three years it has received Tk 1,80,000, which is more. The payback must lie between two and three years, and 2.6 years does.

The check is to confirm that two full years leave less than the outlay received.

WHAT PAYBACK DOES NOT TELL US

Payback is popular because it is quick, easy to explain, and easy to compute by hand. It also tells the owner how soon cash comes back, which matters when money is short or the future is hard to foresee. But the test has three weaknesses, and a student must know all three.

Payback is quick and easy, but it has three weaknesses.

The first weakness is that payback ignores the time value of money. A taka received in year 1 counts the same as a taka received in year 2, although Chapter IV showed that they are not worth the same.¹ The second weakness is that payback ignores everything after the payback date. The third is that the cut-off is a matter of habit and not of reasoning. Nothing in the test tells the owner whether three years is the right limit or four.

The first weakness is that payback treats a taka now and a taka later alike.

The second weakness needs an example. Two projects, D and E, each cost Tk 1,00,000. Their inflows are the same for the first three years. Project E then continues to earn, and project D stops.

Two projects can have the same payback and very different value.

¹ A variation called discounted payback first reduces each inflow to its present value, and then counts the years. For the oven, discounting at 10% gives about 3.75 years instead of 3.0. It removes the first weakness, but not the second.

TWO PROJECTS WITH THE SAME PAYBACK (TK)

YEAR	PROJECT D	PROJECT E
1	40,000	40,000
2	40,000	40,000
3	20,000	20,000
4	0	80,000
5	0	80,000
Payback	3.0 years	3.0 years

Payback cannot tell D from E, though E is far the better project.

Payback gives both projects 3.0 years. Yet project E earns Tk 1,60,000 more after the payback date, and project D earns nothing. A test that cannot tell these two apart is a poor guide when it stands alone. We return to D and E after we meet a better test.

USING PAYBACK SENSIBLY

The cut-off is chosen by judgement, and it should fit the firm's own position.

How should an owner choose the cut-off? There is no formula. A firm that is short of cash may want its money back quickly, so it sets a short cut-off such as two years. A firm whose machines grow old fast may also choose a short cut-off, because the machine may be useless after a few years. A firm with a long-lived building may accept a longer wait. The cut-off is a judgement about the firm's own position, and it should never be copied from another firm without thought.

Payback is best used as a first filter and a measure of waiting, and not as the final test.

Payback therefore has a modest but real use. It is a quick filter, which removes projects that wait too long before the owner studies the others in detail. It also states the waiting time in a form that every owner understands. But it is never enough to decide a large purchase alone. An owner who relies on payback only will reject slow projects that are very valuable, and will accept fast projects that add little.

COMMON SLIPS WITH PAYBACK

Four slips are common when finding a payback period.

Four slips are common. The first is to add all the inflows and compare the total with the outlay, instead of keeping a running total year by year. The second is to forget the fraction of the last year, and to answer 3 years when the answer is 2.6. The third is to divide the unpaid amount by the wrong year's inflow, such as the first year's. The fourth is to forget the unit: 2.625 is a number of years, and 0.625 of a year is $0.625 \times 12 = 7.5$ months. A check against the running-balance table will find all four.

NET PRESENT VALUE: THE CENTRAL TEST

THE IDEA

GO SLOWLY HERE. THIS IS THE MAIN IDEA OF THE CHAPTER.

We bring every cash flow to the same date, which is today.

Discounting shrinks each later sum, and it shrinks the later ones more.

The opening puzzle showed that sums received at different dates cannot be added fairly. The remedy comes from Chapter IV. We bring every sum back to today by discounting, so that all the sums are measured on the same date. After that, adding them is fair.

Look at the oven again. It brings in Tk 1,00,000 at the end of each of five years. At a discount rate of 10% a year, the Tk 1,00,000 of year 1 is worth Tk 90,909 today. The Tk 1,00,000 of year 5 is worth only Tk 62,092 today. The later the sum, the smaller its value today. The figure below shows this shrinking.

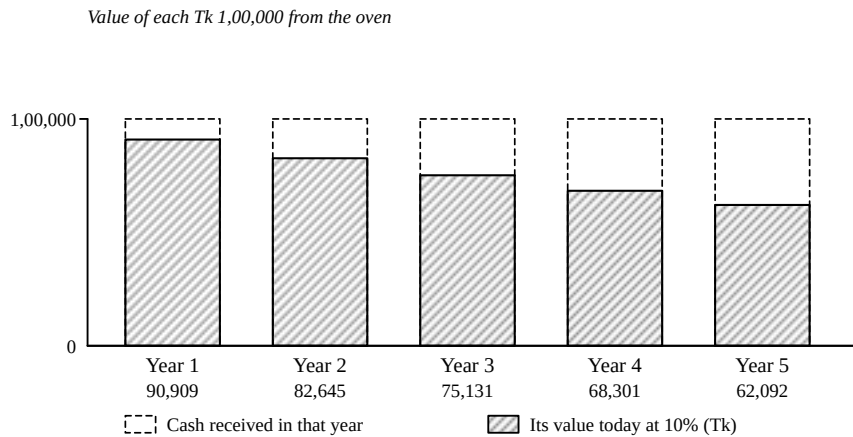


FIG. 19. The oven's yearly inflow of Tk 1,00,000, and what each inflow is worth today at a discount rate of 10%.

Adding the five values gives about Tk 3,79,079. This is the present value of the oven's inflows. It is what the oven's whole stream of cash is worth to the owner today. The oven costs Tk 3,00,000 today. The inflows are worth more than the cost, and the difference is the gain.

That gain has a name and a symbol. The **net present value**, or NPV, of a project is the present value of its cash inflows minus its initial outlay. It states how many taka today the project adds to the owner's wealth, or removes from it. Chapter II taught that a firm should aim to raise the wealth of its owners. NPV measures exactly that aim for one project.

THE FORMULA

In words: net present value equals the present value of all cash inflows, minus the initial outlay. In symbols, let CF1 be the cash inflow in year 1, CF2 the inflow in year 2, and so on up to CF_n in the last year n. Let r be the discount rate written as a decimal, so that 10% is 0.10. Then:

NPV = CF1 ÷ (1 + r) + CF2 ÷ (1 + r)² + ... + CF_n ÷ (1 + r)ⁿ – Outlay.

The outlay is paid today, at the start, so it has no discounting. Each inflow is divided by (1 + r) once for each year of waiting. Chapter V gave a shorter way for equal inflows. If the yearly inflow is the same in every year, then the present value of the inflows is the yearly inflow multiplied by the annuity factor for r and n.²

The decision follows at once. The **accept-reject rule** says: accept a project if its NPV is above zero, and reject it if its NPV is below zero. A positive NPV means that the inflows are worth more today than the outlay. The project adds wealth. A negative NPV means that the project takes wealth away.

A fixed routine keeps the working in order, and it does not change from one project to another. First, draw the timeline and list every cash flow with its year. Second, write the discount rate as a decimal. Third, find the present value of each inflow, or use the annuity factor when the inflows are equal. Fourth, add the present values. Fifth, subtract the initial outlay. Sixth, apply the

² The tables show factors to four decimals. In this chapter we multiply by the unrounded factor from a calculator and round the answer to the nearest taka, so that our answers match the case file. A student who uses the four-decimal table factor may find an answer that differs by a few taka. This is rounding and not a mistake.

Adding the values today gives the present value of all the inflows.

Net present value. The present value of a project's inflows minus its initial outlay.

The gain, measured in taka today, is the net present value.

The formula discounts each inflow and then subtracts the outlay.

The formula in symbols adds the discounted inflows and subtracts the outlay.

The outlay is not discounted, and equal inflows can use the annuity factor.

Accept-reject rule. Accept a project if its NPV is above zero; reject it if it is below zero.

The decision rule is to accept a project whose NPV is above zero.

A fixed routine of six moves keeps the working in order.

accept-reject rule and write the answer in a sentence, with its unit, which is the taka.

A FIRST EXAMPLE WITH CLEAN NUMBERS

We return to the mixer, so that the method is the only thing to learn. *Given.* The outlay is Tk 1,00,000. The inflow is Tk 50,000 at the end of each of years 1, 2 and 3. The discount rate is 10%. *Find.* The NPV, and whether to accept. *Formula.* NPV equals the yearly inflow times the annuity factor, minus the outlay.

Steps. The annuity factor for 10% and 3 years is 2.4869 in the table, and 2.486852 on a calculator. Present value of the inflows = Tk 50,000 $\times$ 2.486852 = Tk 1,24,343. Then NPV = Tk 1,24,343 – Tk 1,00,000 = Tk 24,343. *Answer.* The NPV is Tk 24,343. It is above zero, so the rule says accept.

Check. Discount each inflow separately: Tk 50,000 $\div$ 1.10 = Tk 45,455; Tk 50,000 $\div$ 1.21 = Tk 41,322; Tk 50,000 $\div$ 1.331 = Tk 37,566. These add to about Tk 1,24,343, the same present value as before.

A SECOND EXAMPLE WITH MESSIER NUMBERS

Now the display case, whose inflows are uneven. No annuity factor can be used, because the inflows differ. Each inflow must be discounted separately. *Given.* The outlay is Tk 1,50,000. The inflows are Tk 40,000, Tk 60,000 and Tk 80,000 in years 1, 2 and 3. The discount rate is 10%. *Find.* The NPV.

NET PRESENT VALUE OF THE DISPLAY CASE AT 10% (TK)

YEAR	CASH INFLOW	DIVIDE BY	PRESENT VALUE
1	40,000	1.10	36,364
2	60,000	1.21	49,587
3	80,000	1.331	60,105
Total present value of inflows			1,46,056
Less initial outlay			1,50,000
Net present value			–3,944

The present value of the inflows is Tk 1,46,056, and the outlay is Tk 1,50,000. The NPV is Tk 1,46,056 – Tk 1,50,000 = –Tk 3,944. *Answer.* The NPV is below zero, so the rule says reject. *Check.* The undiscounted inflows add to Tk 1,80,000, which is above the outlay, so a careless addition would say accept. Discounting reverses the verdict, as it should, because the largest sum comes last.

Notice what has happened. The display case paid back in 2.6 years, which would satisfy an owner with a cut-off of three years. Its NPV is negative all the same. Payback said yes, and NPV said no. The sums after the payback date were too small to cover the cost of waiting.

THE TWO PROJECTS OF THE LAST SECTION

Now go back to projects D and E, which payback could not separate. At 10%, project D has a present value of inflows of Tk 84,448 and an NPV of –Tk 15,552. Project E has a present value of inflows of Tk 1,88,763 and an NPV of Tk 88,763. The test that counts all the cash flows sees the difference at once.

We test the mixer from the payback section at a discount rate of 10%.

The factor for 10% and three years is about 2.4869, and the steps follow.

The check adds the three discounted inflows one by one.

The display case has uneven inflows, so each is discounted by itself.

The inflows are worth Tk 1,46,056 today, which is less than the cost.

This project paid back in 2.6 years, yet its NPV is negative.

NPV separates D from E, which payback could not do.

WHAT A ZERO NPV MEANS

Suppose the NPV is exactly zero. The inflows are then worth the same as the outlay, once the waiting is paid for. The project earns exactly the discount rate, and no more. It neither adds wealth nor removes it. The discount rate is the least return the owner will accept, so zero is the dividing line between accept and reject.

One warning belongs here. Every cash flow in the future is an estimate. An NPV that is only a few thousand taka above or below zero can change sign when an estimate changes a little. A large positive or negative NPV is a stronger signal. Chapter VIII begins to measure this uncertainty.

An NPV of zero means the project earns exactly the discount rate.

Estimates of cash flows are uncertain, so a small NPV needs care.

WHERE THE DISCOUNT RATE COMES FROM

Every NPV needs a discount rate, and the answer depends on it. What does the rate stand for? It stands for what the money could earn if it were not spent on this project. If the owner could place Tk 3,00,000 in a safe investment that pays 8%, then a project must at least match that return to be worth the trouble. The rate is therefore the price of waiting, as Chapter IV explained.

The discount rate is the return the owner could earn on the money elsewhere.

A risky project needs a higher rate than a safe one, because the owner asks for more reward when the result is less sure. A rate that is too low makes weak projects look good. A rate that is too high makes good projects look weak. In this chapter we are given a provisional rate of 10%, so that the method can be learned first. Chapter X shows how risk shapes the return that investors require, and Chapter XV finds one rate for the whole firm.

The rate should also allow for the risk of the project.

NET PRESENT VALUE ADDS UP

An important property of NPV is that it adds. If the firm accepts two independent projects, the wealth added by the pair is the sum of the two NPVs. The mixer adds Tk 24,343, and a second project that adds Tk 10,000 would bring the pair to Tk 34,343. The payback period and the internal rate of return do not have this property. It is one more reason why NPV matches the goal of raising the owners' wealth.

The NPV of several projects is the sum of their separate NPVs.

COMMON SLIPS WITH NPV

Four slips are common. The first is to discount for the wrong number of years, for example by dividing the year 3 inflow by 1.10 only twice. The second is to read the wrong row or column of the factor table, such as 12% in place of 10%. The third is to add the outlay instead of subtracting it. The fourth is to mix periods: if the rate is 10% a year and the cash flows are half-yearly, then the rate and the number of periods must both be changed, as Chapter IV showed. A check by discounting one year by hand will find most of these slips.

Four slips are common when finding an NPV.

IN SHORT

Net present value is the present value of a project's inflows minus its initial outlay. Accept the project if the result is above zero, and reject it if the result is below zero. The test counts every cash flow, and it gives each its true value today.

ACCEPT, REJECT AND RANK

THREE SITUATIONS

The same NPV rule is used in three different situations.

Independent projects are each accepted whenever their NPV is above zero.

Mutually exclusive projects are ranked, and the highest NPV wins.

A small example shows that the larger outlay is not always the better choice.

Chapter VI separated projects into kinds. The NPV rule works for all of them, but it is applied a little differently in each. We take the three situations in turn: projects that stand alone, projects that compete, and projects that share a limited sum of money.

If the projects are independent, the decision about one does not affect the others. The firm applies the accept-reject rule to each project, one at a time. Every project with a positive NPV is accepted, because each adds wealth. If the mixer and the display case were independent, the firm would accept the mixer and reject the display case.

If the projects are mutually exclusive, choosing one rules out the other, as with the oven and the van. Here the rule has two stages. First, discard any project with a negative NPV. Then, among those left, choose the one with the highest NPV. A higher NPV means more wealth added, so the highest NPV is the best use of the money.

An example shows the second stage at work. The firm may buy a small oven or a large oven, and it cannot buy both. The small oven costs Tk 2,00,000 and brings in Tk 90,000 at the end of each of three years. The large oven costs Tk 3,00,000 and brings in Tk 1,30,000 at the end of each of three years. The discount rate is 10%, and the annuity factor for 3 years is 2.486852.

A SMALL OVEN AND A LARGE OVEN AT 10% (TK)

	SMALL OVEN	LARGE OVEN
Initial outlay	2,00,000	3,00,000
Present value of inflows	2,23,817	3,23,291
Net present value	23,817	23,291

Both NPVs are positive, but the small oven is slightly better.

Both ovens have a positive NPV, so neither is rejected. The small oven has the higher NPV by Tk 526, so it is chosen. The large oven earns more cash, but it also costs Tk 1,00,000 more. The extra Tk 40,000 a year is worth Tk 99,474 today, which is less than the extra Tk 1,00,000 spent. A larger project is not better because it is larger. It is better only if it adds more wealth.

When money is limited, we choose the group of projects with the largest total NPV.

Capital rationing means that the firm has less money than it has good projects. The firm cannot accept every project with a positive NPV. It must choose the group of projects whose total outlay fits within the money available, and whose total NPV is the largest. We test this with three independent projects and a budget of Tk 4,00,000.

THREE PROJECTS AND A BUDGET OF TK 4,00,000 (TK)

PROJECT	OUTLAY	NET PRESENT VALUE	NPV FOR EACH TAKA OF OUTLAY
R	1,00,000	50,000	0.50
S	2,00,000	1,20,000	0.60
T	2,00,000	80,000	0.40

Only some groups fit the budget, and one group gives the largest total NPV.

The three projects together cost Tk 5,00,000, which is more than the budget. We list the groups that fit. Projects S and T cost Tk 4,00,000 and give a total NPV of Tk 2,00,000. Projects S and R cost Tk 3,00,000 and give

Tk 1,70,000. Projects R and T cost Tk 3,00,000 and give Tk 1,30,000. The best group is S and T, with a total NPV of Tk 2,00,000.

Notice that project R, taken by itself, has a higher NPV for each taka of outlay than project T. Yet the best group contains T and not R. The reason is that R leaves Tk 1,00,000 unused, and nothing else in the list can use it. The firm should judge the whole group, and not each project alone. If the unused money could be placed elsewhere to earn something, the answer might change.

A group should be judged by its total NPV, not by one project at a time.

THE INTERNAL RATE OF RETURN AND THE PROFITABILITY INDEX

THE INTERNAL RATE OF RETURN

NPV gives its answer in taka. Many owners and bankers prefer an answer in per cent, because a rate is easy to compare with the interest on a loan. The next test gives such an answer. It asks a different question: what yearly return does the project itself earn?

The **internal rate of return**, or IRR, is the discount rate at which a project's NPV is exactly zero. At the IRR, the present value of the inflows equals the outlay. The IRR is therefore the yearly return that the project earns on the money tied up in it. It is called internal because it depends only on the project's own cash flows, and not on any outside rate.

NPV gives an answer in taka, but some owners prefer an answer in per cent.

Internal rate of return. The discount rate at which a project's NPV is exactly zero.

The internal rate of return is the discount rate that makes NPV zero.

A simple case shows the idea. Suppose the firm pays Tk 1,00,000 today and receives a single sum of Tk 1,21,000 after two years. Chapter IV taught that Tk 1,00,000 growing at a rate r for two years becomes $Tk\ 1,00,000 \times (1 + r)^2$. This equals Tk 1,21,000 when r is 10%, because $1.10 \times 1.10 = 1.21$. The IRR is therefore 10%.

A simple case shows that IRR is a growth rate.

When the inflows are equal, we can find the IRR from the annuity table. For the oven, the outlay is Tk 3,00,000 and the yearly inflow is Tk 1,00,000, for five years. The needed factor is the outlay divided by the yearly inflow, which is $3,00,000 \div 1,00,000 = 3.0000$. We look along the row for 5 years and find the rate that gives this factor.

For equal inflows, the factor method finds the IRR from the table.

In the 5-year row, the factor at 19% is 3.0576 and the factor at 20% is 2.9906. Our factor of 3.0000 lies between them, so the IRR lies between 19% and 20%. It is closer to 20%. A straight-line estimate gives $(3.0576 - 3.0000) \div (3.0576 - 2.9906) = 0.0576 \div 0.0670 = 0.86$. The IRR is about 19.9%.

The table row shows that the factor 3.0 lies between 19% and 20%.

When the inflows are uneven, no table row helps. We try a rate, compute the NPV, and then try another rate. If the NPV is positive, the rate is too low. If it is negative, the rate is too high. For the van, the NPV is about Tk 183 at 18%, and -Tk 15,342 at 20%. The IRR is slightly above 18%. A calculator or spreadsheet gives 18.0%.

For uneven inflows, we use trial and error with different rates.

How does the owner use the IRR? The owner compares it with a **hurdle rate**, which is the least return a project must earn to be accepted. In this chapter the hurdle rate is the same 10% as our discount rate. The IRR rule says: accept the project if its IRR is above the hurdle rate, and reject it if the IRR is below. The oven's IRR of 19.9% is well above 10%, and so is the van's 18.0%. Both are accepted.

Hurdle rate. The least return a project must earn to be accepted.

The IRR is compared with the hurdle rate, which is the least return required.

Profitability index. The present value of a project's inflows divided by its initial outlay.

The profitability index shows the present value gained for each taka spent.

The PI rule is equivalent to the NPV rule for accepting or rejecting.

A small example shows that the PI follows from the NPV.

The IRR can mislead when projects differ in size.

The IRR can also fail when the cash flows change direction more than once.

One picture shows both projects at every discount rate.

THE PROFITABILITY INDEX

The last test is simple. The **profitability index**, or PI, is the present value of a project's inflows divided by its initial outlay. It shows how many taka of present value the project returns for each taka spent. For the oven, $PI = \text{Tk } 3,79,079 \div \text{Tk } 3,00,000 = 1.264$. For the van, $PI = \text{Tk } 3,76,766 \div \text{Tk } 3,00,000 = 1.256$.

The PI rule says: accept a project if its PI is above 1, and reject it if the PI is below 1. This is the same decision as the NPV rule. The PI is above 1 exactly when the present value of the inflows is above the outlay, and that is exactly when the NPV is positive. The PI is useful when money is limited, because it shows the return for each taka spent.

Take a counter extension that costs Tk 80,000 and whose inflows have a present value of Tk 1,00,000. Its PI is $1,00,000 \div 80,000 = 1.25$. Its NPV is $\text{Tk } 1,00,000 - \text{Tk } 80,000 = \text{Tk } 20,000$, which is positive, as the PI above 1 promised. Each taka spent returns Tk 1.25 of present value.

TWO CAUTIONS ABOUT THE IRR

The IRR is popular, but it needs two cautions. The first concerns size. A high rate on a small sum can add less wealth than a lower rate on a large sum. The next section shows an example. When projects differ in size, the NPV should decide.

The second caution concerns the pattern of cash flows. Most projects pay out first and receive afterwards, so the sign of the cash flow changes once, from negative to positive. A project with a large payment in the middle of its life, such as the cost of a major repair, changes sign more than once. Such a project can have more than one IRR, and the rates have no clear meaning. The NPV has no such difficulty. It gives a single answer for a given discount rate.

The next figure draws the NPV of the oven and of the van at many discount rates, from 0% to 25%. Each curve falls as the rate rises, because a higher rate shrinks the later sums more. Each curve meets the zero line at its IRR. The picture will help in the next section.

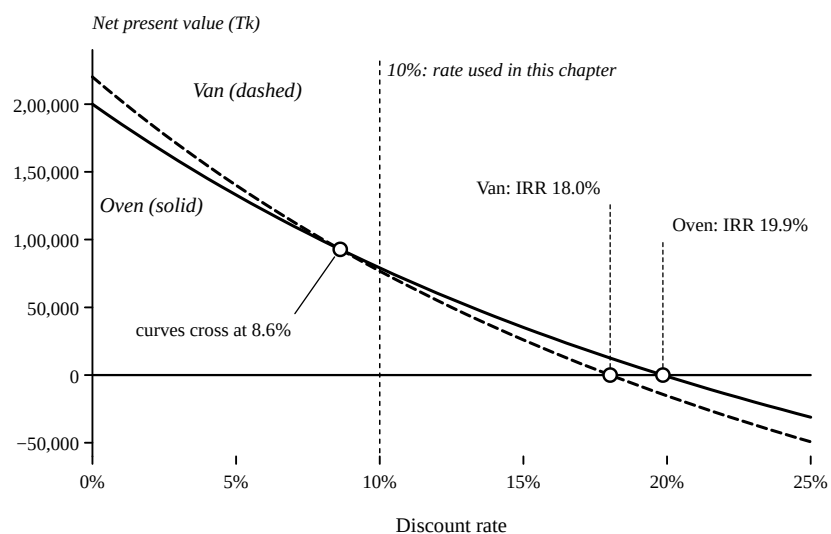


FIG. 20. The NPV of the oven (solid line) and the van (dashed line) at different discount rates. Each curve meets the zero line at the project's internal rate of return.

WHEN THE METHODS DISAGREE

PAYBACK AGAINST NPV

The tests do not always give the same answer. A student must know why this happens and which test to trust. We study three kinds of disagreement, each with a small example. In each case the projects are mutually exclusive, and the discount rate is 10%.

First, payback against NPV. Projects P and Q each cost Tk 2,00,000. Project P returns its money fast and then earns little. Project Q returns its money more slowly and then earns a great deal.

PROJECTS P AND Q AT A DISCOUNT RATE OF 10% (TK)

	PROJECT P	PROJECT Q
Year 1	1,20,000	40,000
Year 2	1,00,000	60,000
Year 3	10,000	80,000
Year 4	10,000	1,00,000
Year 5	10,000	1,20,000
Payback	1.8 years	3.2 years
Net present value	12,288	88,868
Internal rate of return	14.0%	23.3%

Payback prefers project P, because it returns the money in 1.8 years. NPV prefers project Q, by a wide margin: Tk 88,868 against Tk 12,288. The IRR also prefers Q. Project P is quick, but it adds little wealth. Project Q is slow, but it adds a great deal. The decision should follow the wealth added, so the firm chooses Q.

IRR AGAINST NPV

Second, IRR against NPV. Project A costs Tk 1,00,000 and returns Tk 60,000 in each of three years. Project B costs Tk 2,00,000 and returns Tk 1,00,000 in each of four years. At 10%, project A has an NPV of Tk 49,211 and an IRR of 36.3%. Project B has an NPV of Tk 1,16,987 and an IRR of 34.9%.

The two tests now disagree. IRR ranks A first, because its rate is higher. NPV ranks B first, because it adds more taka of wealth. The profitability index agrees with NPV: it is 1.49 for A and 1.58 for B. The IRR hides the fact that B uses Tk 1,00,000 more of the owner's money.

We can test the choice directly. Choosing B instead of A means spending an extra Tk 1,00,000 and receiving the extra inflows. The extra NPV is Tk 1,16,987 – Tk 49,211 = Tk 67,776, which is positive. The extra Tk 1,00,000 therefore earns more than 10%. The owner should take B, unless the money has a better use elsewhere.

A CHANGE IN THE DISCOUNT RATE

Third, a change of rate. Return to the figure of NPV against the discount rate. At a rate of 0%, the van has the higher NPV: Tk 2,20,000 against Tk 2,00,000. This is just the plain total of the opening puzzle, less the outlay. As the rate rises, the van's curve falls faster,

Different tests can give different answers, so we need a way to choose.

Projects P and Q show payback and NPV giving opposite answers.

Payback prefers P, while NPV and IRR both prefer Q.

Projects of different size can be ranked differently by IRR and NPV.

IRR prefers the smaller project A, but NPV prefers the larger project B.

The extra money in B earns more than the discount rate, so B is better.

The best project can change when the discount rate changes.

Above 8.6% the oven has the higher NPV, and below it the van does.

NPV is trusted most because it is measured in taka and uses all the cash flows.

The other tests still have a place as second opinions.

A table sets the four tests side by side.

A fixed habit of testing guards against hasty choices.

The case file records the two options and the provisional discount rate.

because its larger sums come later. The two curves cross at a rate of about 8.6%.³

Below 8.6% the van is better. Above 8.6% the oven is better. Our discount rate is 10%, which is above the crossing point, so the oven wins. The ranking depends on the rate, and so the rate must be chosen with care. Chapter XV will find a more careful rate for the bakery and will test the choice again. Because the oven wins at every rate above 8.6%, a modest rise in the rate will not change the choice.

WHICH TEST TO TRUST

When the tests disagree, the firm follows NPV. There are three reasons. NPV counts every cash flow, and not only those before a date. It gives each flow its true value today. And its answer is in taka, so it states how much wealth the project adds, which is what the goal of the firm requires.

The other tests are not useless. Payback is a quick first filter, and it shows how long the firm waits. IRR is a rate that managers and bankers understand. PI helps when money is limited. A careful firm may compute several tests, but when they disagree, it follows the NPV.

The four tests can be set side by side, so that their questions, rules and weaknesses can be compared. The table below states what each test asks, how it decides, and its main weakness.

THE FOUR TESTS COMPARED

TEST	QUESTION IT ANSWERS	ACCEPT IF	MAIN WEAKNESS
Payback period	How soon is the outlay returned?	Payback is within the cut-off	Ignores time value and later flows
Net present value	How much wealth is added, in taka today?	NPV is above zero	Needs a discount rate
Internal rate of return	What yearly rate does the project earn?	IRR is above the hurdle rate	Can mislead when sizes differ
Profitability index	How much present value for each taka spent?	PI is above 1	Ignores the size of the gain

A good habit is to compute the NPV first, because it decides. Then compute the others as a check. If every test gives the same answer, as with the oven, the owner can decide with confidence. If they differ, the owner looks for the reason, using the three kinds of disagreement in this section as a guide, and follows the NPV.

THE WORKED CASE: THE OVEN AND THE VAN

Case file, illustrative. Sundial Bakery, Dhaka. On 1 July 2026 there are two mutually exclusive options, each with an initial outlay of Tk 3,00,000 and a life of five years. Oven: inflows of Tk 1,00,000 in each year. Van: inflows of Tk 50,000, Tk 80,000, Tk 1,10,000, Tk 1,40,000 and Tk 1,40,000 in years 1 to 5. Provisional discount

³ This crossing rate is often called the crossover rate. Some books call it the Fisher rate, after Irving Fisher, whom the section In History names.

rate: 10% a year. The inflows are the yearly cash flows found in Chapter VI.

Given. The two options above, and a discount rate of 10%. *Find.* The payback, the NPV, the IRR and the PI of each option, and which option to choose. *Formula.* Payback is the time to recover the outlay. NPV is the present value of the inflows minus the outlay. The IRR makes NPV zero. PI is the present value of inflows divided by the outlay.

We state what is given and what the owner must decide.

PAYBACK

The oven's inflow is equal in every year, so its payback is $\text{Tk } 3,00,000 \div \text{Tk } 1,00,000 = 3.0$ years. The van's inflows are uneven, so we keep a running total. After year 3 the van has returned $\text{Tk } 50,000 + \text{Tk } 80,000 + \text{Tk } 1,10,000 = \text{Tk } 2,40,000$. $\text{Tk } 60,000$ of the outlay is still unpaid. Year 4 brings $\text{Tk } 1,40,000$, so the van needs $60,000 \div 1,40,000 = 0.43$ of that year. The van's payback is about 3.4 years. The figure below shows both.

The oven pays back in three years, and the van takes a little longer.

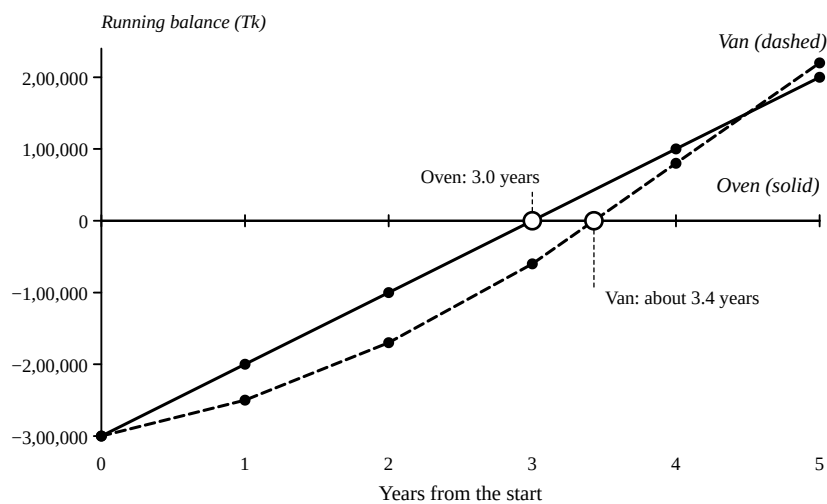


FIG. 21. The running balance of the oven (solid line) and the van (dashed line). Each project pays back where its line crosses zero.

NET PRESENT VALUE

For the oven, the annuity factor for 10% and 5 years is 3.790787. The present value of the inflows is $\text{Tk } 1,00,000 \times 3.790787 = \text{Tk } 3,79,079$. The NPV is $\text{Tk } 3,79,079 - \text{Tk } 3,00,000 = \text{Tk } 79,079$. The van's inflows are uneven, so each is discounted by itself.

The oven's equal inflows allow the annuity factor, and the van needs a table.

NET PRESENT VALUE OF THE VAN AT 10% (TK)

YEAR	CASH INFLOW	DIVIDE BY	PRESENT VALUE
1	50,000	1.10	45,455
2	80,000	1.21	66,116
3	1,10,000	1.331	82,645
4	1,40,000	1.4641	95,622
5	1,40,000	1.61051	86,929
Total present value of inflows			3,76,766
Less initial outlay			3,00,000
Net present value			76,766

The van's NPV is Tk 76,766, which is below the oven's Tk 79,079.

The IRR and the PI both favour the oven, though only by a little.

The choice depends on the discount rate, so we test several rates.

The oven leads at every rate from 10% upward, so the choice is firm.

The owner chooses the oven, and the opening puzzle is solved.

The five tests and the decision rule in a few lines.

The van's total present value is Tk 3,76,766 and its NPV is Tk 76,766. The lines in the table are rounded to the nearest taka, so they add to Tk 3,76,767. The total is found before rounding. The oven's NPV of Tk 79,079 is higher than the van's NPV of Tk 76,766 by Tk 2,313. Both are positive, so each project would be accepted if it stood alone.

THE OTHER TWO TESTS

The oven's IRR is about 19.9% and the van's is about 18.0%. Both are above the hurdle rate of 10%. The oven's PI is $3,79,079 \div 3,00,000 = 1.264$, and the van's PI is $3,76,766 \div 3,00,000 = 1.256$. The oven ranks first on every test.

THE OVEN AND THE VAN ON FOUR TESTS

TEST	OVEN	VAN
Payback	3.0 years	3.4 years
Net present value at 10%	Tk 79,079	Tk 76,766
Internal rate of return	19.9%	18.0%
Profitability index	1.264	1.256

HOW FIRM IS THE CHOICE?

The discount rate of 10% is provisional, so the owner should ask how firm the choice is. We compute the NPV of both options at several rates.

NET PRESENT VALUE AT SEVERAL DISCOUNT RATES (TK)

DISCOUNT RATE	OVEN	VAN
0%	2,00,000	2,20,000
5%	1,32,948	1,40,076
10%	79,079	76,766
15%	35,216	25,947
20%	-939	-15,342

At 0% and 5% the van leads, as the plain total suggested. From 10% upward the oven leads, and its lead widens as the rate rises. At 20% both NPVs are below zero, which agrees with the IRRs of 19.9% and 18.0%. The choice of the oven holds for any rate above 8.6%, and the provisional rate of 10% lies above that point.

Answer. The firm chooses the oven. It pays back sooner, it has the higher NPV and IRR and PI, and it does so at every discount rate above 8.6%. The opening puzzle is solved: the van brings in Tk 20,000 more in total, but its larger sums come later, and discounting takes that into account. *Check.* The oven's NPV can also be found from the line-by-line values in the first figure of this chapter: $90,909 + 82,645 + 75,131 + 68,301 + 62,092 = \text{Tk } 3,79,078$, which agrees with Tk 3,79,079 to within one taka of rounding.

SUMMARY

Payback is the time a project takes to return its outlay. It is quick, but it ignores time value and ignores what follows the payback date. Net present value is the present value of the inflows minus the outlay, and the

accept-reject rule says to accept a project when it is above zero. For mutually exclusive projects the firm chooses the highest NPV, and under limited money it chooses the group with the largest total NPV. The internal rate of return is the rate that makes NPV zero, and it is compared with a hurdle rate. The profitability index is the present value of inflows per taka of outlay. When the tests disagree, the firm follows the NPV.

You should now be able to say: payback tells how soon, NPV tells how much wealth is added, IRR tells the yearly rate earned, and PI tells the return for each taka spent. You should also be able to say why NPV is trusted when the tests disagree.

You should now be able to say what each test does and which to trust.

THINK IT THROUGH

Case file, illustrative. Sundial Bakery is offered two machines for a new counter. A coffee machine costs Tk 2,00,000 and brings in Tk 80,000 at the end of each of four years. A juice press costs Tk 2,00,000 and brings in Tk 20,000, Tk 60,000, Tk 1,00,000 and Tk 1,40,000 in years 1 to 4. The discount rate is 10%, and the owner's cut-off for payback is three years.

A new pair of machines gives the reader a case to decide.

Find the payback and the NPV of each machine. Say which machine you would choose, and which test you used to decide. Then say whether the owner's cut-off of three years changes your answer, and explain why. Finally, say what would have to happen to the juice press's year 4 inflow for its NPV to equal the coffee machine's.

The questions ask for payback, NPV, a choice, and a reason.

WHERE PEOPLE GO WRONG

Mistake: choosing the project that pays back first. *Correction:* payback ignores the time value of money and every sum after the payback date. A fast project can add less wealth than a slow one, as projects P and Q showed.

Mistake: choosing the project with the shortest payback.

Mistake: discounting the initial outlay. *Correction:* the outlay is paid today, at time zero, so it is already measured in today's taka. Only the later inflows are divided by $(1 + r)$.

Mistake: discounting the outlay along with the inflows.

Mistake: calling the present value of the inflows the NPV. *Correction:* NPV is the present value of the inflows minus the outlay. Forgetting the subtraction makes every project look good.

Mistake: confusing present value with net present value.

Mistake: believing that a project with an NPV of zero is a bad project. *Correction:* it earns exactly the discount rate. It adds nothing beyond that, and it is not a loss.

Mistake: treating a zero NPV as a failure.

Mistake: choosing the project with the highest IRR when the projects differ in size. *Correction:* a high rate on a small sum can add less wealth than a lower rate on a larger sum. Compare the NPV.

Mistake: ranking projects of different size by IRR alone.

Mistake: discounting profit figures. *Correction:* the tests of this chapter use cash flows, as found in Chapter VI. Depreciation is not a cash flow, and it enters the cash flows only through its effect on tax.

Mistake: discounting accounting profit in place of cash flow.

Mistake: using a yearly rate with half-yearly cash flows. *Correction:* the rate and the period must match. Convert both to half-years, or both to years, as Chapter IV showed.

Mistake: using a rate and a period that do not match.

Irving Fisher set out the idea of valuing future money in 1930.

A short note on where the ideas of this chapter come from.

Payback.

Payback with uneven inflows.

NPV with equal inflows.

NPV with uneven inflows and a choice.

The accept-reject rule.

Internal rate of return.

Profitability index.

Conflicting tests.

IN HISTORY

In 1930 the American economist Irving Fisher published *The Theory of Interest*. In it he explained how the value of a stream of future payments can be found by discounting each payment to the present, and he studied the rate of return that a choice of investment offers. These ideas lie behind the NPV and IRR tests used by firms today. Fisher was working on a question of economics, and not on the budgets of firms, but the connection was soon seen by others.

SOURCES

Irving Fisher, *The Theory of Interest* (1930), is the landmark source for the idea of discounting a stream of payments. The accept-reject rule and the other tests of this chapter are standard topics in textbooks of financial management.

REVIEW QUESTIONS

1. Define the payback period, and state one strength and two weaknesses of the payback test.

2. A project costs Tk 4,00,000 and brings in Tk 1,00,000, Tk 1,50,000, Tk 1,50,000 and Tk 1,50,000 at the ends of years 1 to 4. Find its payback period.

3. A project costs Tk 1,20,000 and brings in Tk 50,000 at the end of each of three years. At a discount rate of 8%, find its NPV and say whether to accept it.

4. A second project costs Tk 1,20,000 and brings in Tk 30,000, Tk 50,000 and Tk 70,000 at the ends of years 1 to 3. At 8%, find its NPV. Which of the two projects is better?

5. Explain the accept-reject rule, and explain why an NPV of zero means that a project earns exactly the discount rate.

6. A firm pays Tk 1,00,000 today and receives Tk 1,33,100 after three years. Find the internal rate of return.

7. A project's inflows have a present value of Tk 2,70,000 and its outlay is Tk 2,00,000. Find its profitability index, and say what it tells the owner.

8. Two mutually exclusive projects have these results: project X pays back in 2 years with an NPV of Tk 20,000, and project Y pays back in 4 years with an NPV of Tk 70,000. Which should the firm choose, and why?

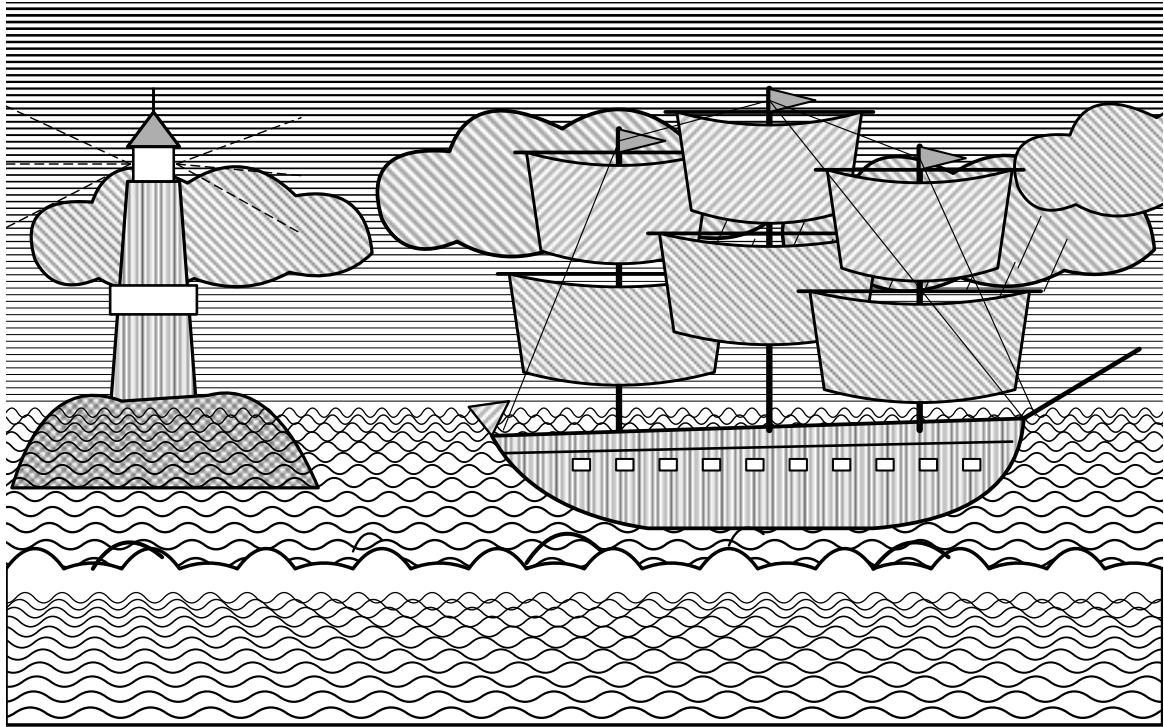


Plate VIII. A Ship in Foul Weather.

CHAPTER VIII

Risk and Return

Two roads can have the same average length and still not be equally safe.



This chapter defines risk and return and teaches how to measure both with numbers. It moves from the plain idea of a result that may differ from our hopes, through chances and averages, to the standard deviation and the coefficient of variation. The bakery's two product ideas serve as the running example.

Two bus routes have the same average journey time but not the same safety.

Most people choose the steady route, although the averages are equal.

Investments behave like the two routes, with results that may wander.

The chapter teaches the numbers in a fixed order, from risk to the coefficient of variation.

In daily speech, risk means danger, but finance gives it a wider meaning.

Risk. The chance that the actual result will differ from the result we hoped for.

Risk is the chance that the result differs from the hoped-for result.

A safe result is one that can be known in advance.

Risk is about the future, because the past is already known.

Suppose that you must catch a train at nine o'clock, and that two bus routes lead to the station. On the first route the journey takes 40 minutes every day, with only a small change from day to day. On the second route the journey takes 20 minutes on some days and 60 minutes on others. Over many days the average is 40 minutes on both routes. Which route would you choose?

Most people choose the first route. They do so even though the averages are the same. The reason is that the second route sometimes takes 60 minutes, and on those days you miss the train. An average hides this danger. It tells you where the results lie in the middle, but it does not tell you how far they may wander from the middle.

Investments behave in the same way. Two investments can offer the same average gain, yet one of them may give results that wander much further from the average. Sundial Bakery now has two product ideas of this kind. Their average returns are equal, but their results are not equally certain. A careful owner needs a way to put a number on this difference.

This chapter gives that number. It first explains what risk is and what kinds of risk exist. It then shows how to calculate a return, and how to describe uncertain returns with chances. Three measures follow: the expected value, the standard deviation, and the coefficient of variation. The bakery's two ideas are compared at the end. You will need the percentages and square roots of Chapter III, and nothing else.

WHAT RISK IS

THE EVERYDAY WORD AND THE FINANCE WORD

In daily speech, the word risk means danger. A person who crosses a busy road takes a risk of injury. Finance uses the word in a wider sense. The danger of a bad result is part of it, but so is the chance of a result that is better than hoped.

In finance, **risk** is the chance that the actual result will differ from the result we hoped for. A result that is better than hoped is a difference, and so is a result that is worse. Both are risk. People usually worry more about the worse result, and we shall see in Chapter X that this worry has a price. But the measure of risk that we use counts differences in both directions.

A result with no risk is one that can be known in advance. A bank may promise to pay Tk 8,000 interest on Tk 1,00,000 after one year. If the bank keeps its promise, the owner knows the result today. A new line of cakes is different. The sales may be low, fair or high, and the owner cannot know which will occur. The cake line has risk, and the bank promise has almost none.

Risk always concerns the future. The sales of last month are known, and they carry no risk. The sales of next month are not known, and they do. This is why risk matters to every decision in this book. The oven of Chapter VII brings in cash over five future years, and each of those yearly sums is an estimate that may prove wrong.

KINDS OF RISK

Risk can be sorted in more than one way. We use two ways in this book. The first sorts risk by where it comes from: from the whole economy, or from one firm alone. The second sorts risk by its cause inside the firm: the firm's work, or the firm's borrowing. The figure below shows both ways.

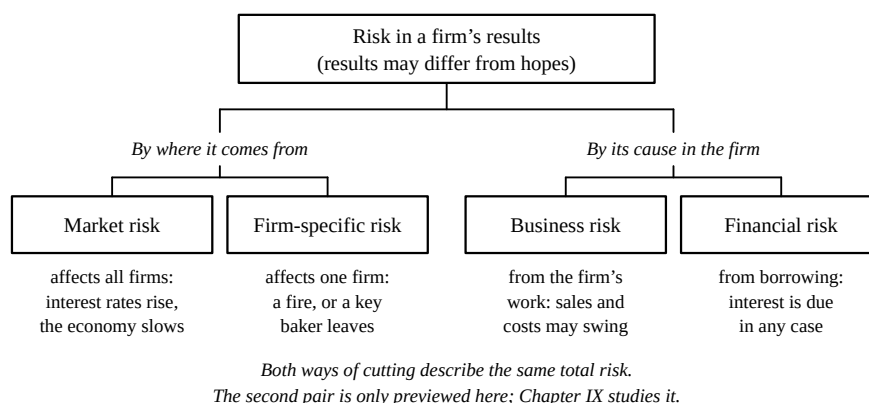


FIG. 22. Two ways of sorting the risk in a firm's results. The second pair of kinds is only previewed here.

The first kind is **market risk**. It is risk that comes from events that affect all firms together. A rise in interest rates is an example. So is a fall in demand across the whole economy. No single firm causes these events, and no single firm can escape them. For this reason market risk is also called risk that cannot be avoided by choosing a different firm.

The second kind is **firm-specific risk**. It is risk that comes from events that affect one firm only. A fire in one bakery is an example. So is the departure of the baker whose recipes the shop depends on. Other bakeries are not harmed by these events, and some may even gain from them. An investor who holds shares in many firms is much less hurt by a fire in one of them.

MARKET RISK AND FIRM-SPECIFIC RISK COMPARED

	MARKET RISK	FIRM-SPECIFIC RISK
Whom it affects	All firms	One firm
Example	Interest rates rise	A fire in one shop
Can an investor avoid it by choosing another firm?	No	Yes, mostly

The table above sets the two kinds side by side. We have introduced it in words first: the difference lies in how many firms an event touches. Chapter X will show that this difference matters greatly for the return that investors require.

The second way of sorting is a preview of Chapter IX. **Business risk** is the risk that comes from the firm's own work. A bakery's sales may rise or fall, and the price of flour may change. These changes make the firm's profit uncertain before any loan is considered. A bakery that has no debt still has business risk.

Risk can be sorted by where it comes from and by what causes it.

Market risk. Risk that comes from events affecting all firms, such as a change in interest rates.

Market risk comes from events that affect every firm at once.

Firm-specific risk. Risk that comes from events affecting one firm only, such as a fire in its shop.

Firm-specific risk comes from events that affect one firm alone.

The table sets the two kinds side by side.

Business risk. Risk that comes from the firm's own work, because sales and costs may change.

Business risk is the risk that the firm's own work brings.

Financial risk. The extra risk that comes from borrowing money, because interest must be paid in any case.

Financial risk is the extra risk that borrowing adds.

Other names for risk are special cases of the two ways of sorting.

A careful owner asks for more reward when the risk is greater.

Financial risk is the extra risk that comes from borrowing. A firm that has borrowed must pay interest whether its sales are good or bad. This fixed duty makes the owners' profit swing more widely. Chapter IX measures both kinds of risk. For now it is enough to know that they exist, and that they are different from market risk and firm-specific risk. They describe the same total risk, cut in a different way.

Books and newspapers use many other names for risk. Interest-rate risk is the risk that a change in interest rates will alter the value of an investment. Inflation risk is the risk that rising prices will reduce what money can buy. Default risk is the risk that a borrower will not repay a loan. Liquidity risk is the risk that an investment cannot be sold quickly without a loss. Each of these fits into the two ways of sorting. The first two usually come from the whole economy, and so they are market risks. Default risk and liquidity risk often belong to one borrower or one investment, and so they are firm-specific.

WHY RISK MATTERS

Why should an owner care about risk? Most people prefer a steady result to an uncertain one when the averages are equal. If an owner is to accept a risky choice, the choice must offer more reward than a safe one. How much more is a question for Chapter X. In this chapter we learn first to measure the amount of risk, because nothing can be priced until it has been measured.

RETURN AND HOW TO FIND IT

WHAT RETURN MEANS

An owner who places money in an investment wants to know what it gave back. The gain in taka is one answer, but it is hard to compare. A gain of Tk 5,000 is large on an investment of Tk 20,000 and small on an investment of Tk 5,00,000. We therefore compare the gain with the cost.

The **return** on an investment is the gain that it brings, written as a share of what it cost. The gain has two parts. One part is the income received while the investment is held, such as a dividend or interest. The other part is the change in the value of the investment, which may be a rise or a fall.

When we state the return as a percentage for one period, usually one year, we call it the **rate of return**. In this chapter we write return for both, and we state it as a percentage. The formula in words is: return equals the money received back, plus any income, minus the money paid, all divided by the money paid.

In symbols, let P_0 be the price paid at the start, P_1 the value at the end, and I the income received during the period. Then $\text{return} = (P_1 + I - P_0) \div P_0$. The top of the fraction is the gain, which may be negative. The bottom of the fraction is the cost.

A FIRST EXAMPLE WITH CLEAN NUMBERS

Given. A person buys a share for Tk 200. During the year the share pays a dividend of Tk 10. At the end of the year the person sells the share for Tk 214. *Find.* The return for the year. *Formula.* Return equals the closing value plus the income, minus the price paid, divided by the price paid.

An investment is judged by what it gives back compared with what it cost.

Return. The gain on an investment, shown as a share of its cost.

The return is the gain written as a share of the cost.

Rate of return. The return stated as a percentage for one period, usually one year.

The rate of return states the return as a percentage for one period.

The formula is the gain divided by the cost.

A share bought for Tk 200 shows the method with easy numbers.

Steps. The gain is $\text{Tk } 214 + \text{Tk } 10 - \text{Tk } 200 = \text{Tk } 24$. $\text{Return} = \text{Tk } 24 \div \text{Tk } 200 = 0.12$, which is 12%. *Answer.* The return for the year is 12%. *Check.* The person paid Tk 200 and has Tk 224 in value and cash. Tk 200 grown by 12% is $\text{Tk } 200 \times 1.12 = \text{Tk } 224$, which agrees.

The gain is Tk 24, and Tk 24 on Tk 200 is 12%.

A SECOND EXAMPLE WITH MESSIER NUMBERS

Given. A person buys a share for Tk 85.50. The share pays a dividend of Tk 4.20. The person then sells the share for Tk 80.30. *Find.* The return. *Steps.* The gain is $\text{Tk } 80.30 + \text{Tk } 4.20 - \text{Tk } 85.50 = -\text{Tk } 1.00$. $\text{Return} = -\text{Tk } 1.00 \div \text{Tk } 85.50 = -0.0117$, which is -1.17% .

A loss gives a negative return, and the method is the same.

Answer. The return is -1.17% , which is a small loss. The share fell by Tk 5.20 in price, and the dividend of Tk 4.20 made up most of the fall but not all of it. *Check.* The person has Tk 80.30 in value and Tk 4.20 in cash, which is Tk 84.50. This is less than the Tk 85.50 paid, so the return must be negative.

The share fell in price, and the dividend only partly made up the fall.

A bank deposit shows the simplest case. A person deposits Tk 50,000 and receives Tk 3,500 as interest after one year. The value at the end is Tk 50,000, so the closing value equals the price paid, and the only gain is the income. $\text{Return} = \text{Tk } 3,500 \div \text{Tk } 50,000 = 0.07$, which is 7%. Interest is a return in the same way as a dividend is.

An investment with no price change, such as a deposit, has a return equal to its interest.

PAST RETURN AND FUTURE RETURN

The two examples measured returns that are already known. A return that lies in the past can be found exactly, and it has no risk. A return that lies in the future is another matter. The owner can only estimate it, and the estimate may prove wrong. The rest of this chapter teaches how to describe such an uncertain future return with numbers. The first tool for this is the idea of chance.

We can measure a past return exactly, but a future return is uncertain.

Three slips are common with returns. The first is to leave out the income, and to count only the change in price. The second is to divide the gain by the closing value instead of the price paid. The third is to forget the period: a return of 12% earned in six months is not the same as a return of 12% earned in a year.

Three slips are common when finding a return.

PROBABILITY AND PROBABILITY DISTRIBUTIONS

CHANCE IN NUMBERS

An owner cannot know next month's sales, but may be able to say how likely each result is. We need a way of writing down likelihood as a number. A coin and a die are the best places to learn it, because their results are easy to count.

GO SLOWLY HERE. THE IDEAS OF CHANCE AND OF A LIST OF OUTCOMES ARE NEW.

A coin and a die let us learn chance with simple cases.

The **probability** of an event is a number from 0 to 1 that shows how likely the event is. A probability of 0 means that the event cannot happen. A probability of 1 means that it is certain. A fair coin falls heads with a probability of 0.5, because the two sides are equally likely and only one of them is heads. A fair die shows a six with a probability of $1 \div 6$, which is about 0.17.

Probability. A number from 0 to 1 that shows how likely an event is.

A probability is a number from 0 to 1 that shows how likely an event is.

One rule governs all probabilities. If we list every possible outcome of an event, the probabilities of the outcomes must add up to exactly 1, because one of them must happen. For the die, six outcomes each have a probability of $1 \div 6$, and $6 \times (1 \div 6) = 1$. The

The probabilities of all possible outcomes must add up to exactly one.

Probabilities for a business come from past records or from judgement.

Business probabilities are estimates, so the results must be used with care.

Probability distribution. A list of all possible outcomes, each shown with its chance.
A probability distribution lists every outcome with its chance.

Both ideas have the same chances, but different possible returns.

A missing chance is found by subtracting the known chances from one.

The plain average ignores chances, so we weight each outcome by its chance.

probability of an even number is the probability of 2, 4 or 6, which is $3 \div 6 = 0.5$.

WHERE THE CHANCES COME FROM

A coin has a known probability, but a bakery's sales do not. Where do the chances come from? There are two sources. The first is past records. If the cake line had been tried on 40 working days, and sales were weak on 10 days, fair on 20 days and strong on 10 days, the chances would be $10 \div 40 = 0.25$, $20 \div 40 = 0.50$ and $10 \div 40 = 0.25$. The second source is judgement, when no records exist.

A business probability is always an estimate. The future need not copy the past, and two owners may judge the same facts differently. The calculations that follow are exact, but their inputs are not. This is why the results must be used with care, and why a careful owner tests how much the answer changes when an estimate changes.

THE PROBABILITY DISTRIBUTION

We now join the outcomes and their chances in one list. A **probability distribution** is a list of all possible outcomes, each shown with its chance. The chances must add up to 1. For the bakery, each outcome is a possible return on the product idea.

THE BAKERY'S TWO PRODUCT IDEAS: POSSIBLE RETURNS AND THEIR CHANCES

OUTCOME	CAKE LINE: RETURN	CAKE LINE: CHANCE	CAFÉ CORNER: RETURN	CAFÉ CORNER: CHANCE
Weak	8%	0.25	0%	0.25
Fair	12%	0.50	12%	0.50
Strong	16%	0.25	24%	0.25
Total of chances		1.00		1.00

Both ideas have the same chances: a quarter for the weak outcome, a half for the fair outcome, and a quarter for the strong outcome. The possible returns differ. The cake line's returns lie close together, from 8% to 16%. The café corner's returns lie far apart, from 0% to 24%. The next two sections turn these lists into single numbers.

If one chance is missing from a list, we find it by subtraction, because the chances must add up to 1. Suppose that a list has three outcomes, and that two of the chances are 0.2 and 0.5. Their total is 0.7, so the third chance is $1 - 0.7 = 0.3$. The same step is a good check on any list that someone else has given us.

EXPECTED VALUE

THE AVERAGE THAT TAKES CHANCES INTO ACCOUNT

An ordinary average gives every outcome the same weight. But an outcome that is likely should count for more than one that is not. We therefore weight each outcome by its probability, and add the results. This gives the average that we should expect over many repeated trials.

The **expected value**, or EV, is the average of all possible outcomes, each weighted by its probability. In words: multiply each outcome by its chance, and add the results. In symbols, $EV = p_1 \times x_1 + p_2 \times x_2 + \dots + p_n \times x_n$, where x is an outcome and p is its probability. We write the same quantity as the expected return when the outcomes are returns.

The die shows the idea. Its outcomes are 1 to 6, each with the chance $1 \div 6$. The expected value is $(1 + 2 + 3 + 4 + 5 + 6) \div 6 = 3.5$. No throw of a die ever shows 3.5. The expected value need not be a possible outcome. It is the average over many throws, and not a prediction of the next throw.

A FIRST EXAMPLE WITH CLEAN NUMBERS

Given. An investment may return 20% with a probability of 0.6, or 5% with a probability of 0.4. *Find.* The expected return. *Formula.* Multiply each return by its chance, and add.

Steps. $0.6 \times 20\% = 12\%$. $0.4 \times 5\% = 2\%$. $EV = 12\% + 2\% = 14\%$. *Answer.* The expected return is 14%. *Check.* The answer lies between the two returns, 5% and 20%. It lies nearer to 20%, which has the larger chance, and the chances add up to $0.6 + 0.4 = 1$.

The figure below shows a useful picture. Imagine the outcomes as weights on a beam, each placed at its return, and each weight proportional to its chance. The expected value is the point at which the beam balances. The larger weight at 20% is balanced by the smaller weight at 5%, which is farther from the balance point.

Two outcomes: 20% with chance 0.6, 5% with chance 0.4

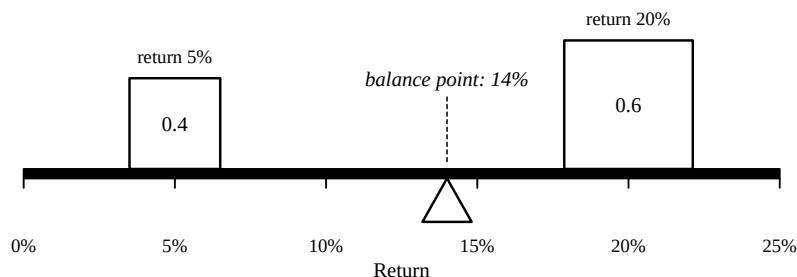


FIG. 23. The expected value as a balance point. The weight of 0.6 at 20% and the weight of 0.4 at 5% balance at 14%.

A SECOND EXAMPLE WITH MESSIER NUMBERS

Now the bakery's ideas, with three outcomes each. *Given.* The distribution of the cake line is in the table of the last section. *Find.* The expected return of each idea. *Steps.* For the cake line: $0.25 \times 8\% = 2\%$, then $0.50 \times 12\% = 6\%$, then $0.25 \times 16\% = 4\%$. The total is 12%. For the café corner: $0.25 \times 0\% = 0\%$, then $0.50 \times 12\% = 6\%$, then $0.25 \times 24\% = 6\%$. The total is 12%.

Answer. The expected return is 12% for both the cake line and the café corner. *Check.* In each list the outcomes are spread evenly on both sides of 12%, with equal chances. The balance point must therefore be 12%, and it is.

Here lies the difficulty that began the chapter. The two ideas have the same expected value, so the expected value cannot separate them. Yet the

Expected value. The average of all possible outcomes, each weighted by its probability.

The expected value is the average with each outcome weighted by its chance.

The expected value of a die is 3.5, a value that never occurs.

A two-outcome case shows the method with the fewest numbers.

The weighted returns are 12% and 2%, which give 14%.

The figure shows the expected value as the balance point of the outcomes.

The cake line has three outcomes, and the method is the same.

Both ideas have the same expected return of 12%.

The expected value cannot tell the two ideas apart.

An outcome may be negative, and the method does not change.

Three slips are common when finding an expected value.

GO SLOWLY HERE. THIS IS THE HARDEST CALCULATION OF THE CHAPTER, AND IT HAS FIVE STEPS.

We want one number for how far outcomes lie from the expected value.

Deviations cannot be averaged as they stand, because they cancel one another.

Variance. The average of the squared deviations from the expected value, each weighted by its chance.

The variance is the weighted average of the squared deviations.

Taking the square root brings the measure back to the original unit.

Standard deviation. The square root of the variance; the usual distance of outcomes from the expected value.

The standard deviation is the square root of the variance.

The calculation follows five fixed steps, in the same order every time.

café corner may return nothing and the cake line cannot. We need a second number, one that measures how far the outcomes wander from the expected value.

A THIRD EXAMPLE WITH A LOSS IN THE LIST

Given. A project may return -10% with a chance of 0.2 , 10% with a chance of 0.5 , or 30% with a chance of 0.3 . *Find.* The expected return. *Steps.* $0.2 \times (-10\%) = -2\%$. $0.5 \times 10\% = 5\%$. $0.3 \times 30\% = 9\%$. $EV = -2\% + 5\% + 9\% = 12\%$. *Answer.* The expected return is 12% , although one of the outcomes is a loss. *Check.* The chances add up to $0.2 + 0.5 + 0.3 = 1$, and the answer lies between the lowest and the highest outcome.

Three slips are common. The first is to take the plain average of the outcomes and ignore the chances. The second is to use chances that do not add up to 1 . The third is to treat the expected value as a promise. It is an average over many trials, and a single trial can give any of the outcomes.

STANDARD DEVIATION

MEASURING THE WANDERING

We want one number that shows how far the outcomes wander from the expected value. A natural first step is to find, for each outcome, its distance from the expected value. This distance is called the deviation. A deviation is positive when the outcome lies above the expected value, and negative when it lies below.

If we weight the deviations by their chances and add them, the positive and negative ones cancel, and the total is zero. For the two-outcome case, $0.6 \times 6\% + 0.4 \times (-9\%) = 3.6\% - 3.6\% = 0$. The total is zero for every distribution, so it tells us nothing. The remedy is to square each deviation before the weighting, because a square is never negative.

The **variance** is the average of the squared deviations from the expected value, each weighted by its probability. A larger variance means that the outcomes lie farther from the expected value. The variance has one inconvenience. If the outcomes are in per cent, the variance is in per cent squared, which is hard to picture.¹

We return to the original unit by taking the square root of the variance. The result is the standard measure of spread.

The **standard deviation**, or SD, is the square root of the variance. It is the usual distance of the outcomes from the expected value, stated in the same unit as the outcomes. A large standard deviation means that the results often lie far from the expected value, and so the risk is large. A small one means that they lie close, and the risk is small.

THE FIVE STEPS

The calculation always follows five steps. First, find the expected value. Second, find each deviation, which is the outcome minus the expected value. Third, square each deviation. Fourth, multiply each squared deviation by its chance, and add the results. This sum is the variance. Fifth,

¹ When a standard deviation is found from a list of past observations and not from chances, books divide the sum of squares by the number of observations, or by one fewer. This chapter uses chances only, so no such division arises.

take the square root of the variance. Use the calculator for the square root, as Chapter III showed.

A FIRST EXAMPLE WITH CLEAN NUMBERS

Given. The investment of the last section: 20% with a probability of 0.6, and 5% with a probability of 0.4. The expected return is 14%. *Find.* The standard deviation.

The two-outcome investment gives a standard deviation of about 7.35%.

STANDARD DEVIATION OF THE TWO-OUTCOME INVESTMENT

RETURN	CHANCE	DEVIATION	SQUARED DEVIATION	CHANCE × SQUARED DEVIATION
20%	0.6	6	36	21.6
5%	0.4	−9	81	32.4
Variance				54.0

The variance is $21.6 + 32.4 = 54.0$, in per cent squared. The standard deviation is the square root of 54, which is 7.35%. *Answer.* The standard deviation is about 7.35%. *Check.* The deviations are 6 and −9, so a usual distance between them, about 7, is sensible. A result of 7.35% lies between 6 and 9, as it should.

The variance is 54, and its square root is the standard deviation.

A SECOND EXAMPLE WITH MESSIER NUMBERS

Now the two ideas of the bakery, each with an expected return of 12%. For the cake line, the deviations are $8 - 12 = -4$, $12 - 12 = 0$ and $16 - 12 = 4$. For the café corner, they are $0 - 12 = -12$, 0 and $24 - 12 = 12$.

The bakery's two ideas have very different standard deviations.

STANDARD DEVIATION OF THE TWO PRODUCT IDEAS

	CAKE LINE	CAFÉ CORNER
Deviations (weak, fair, strong)	−4, 0, 4	−12, 0, 12
Squared deviations	16, 0, 16	144, 0, 144
Each times its chance (0.25, 0.50, 0.25)	4, 0, 4	36, 0, 36
Variance	8	72
Standard deviation	2.83%	8.49%

The cake line has a variance of 8 and a standard deviation of $\sqrt{8} = 2.83\%$. The café corner has a variance of 72 and a standard deviation of $\sqrt{72} = 8.49\%$. *Answer.* The standard deviations are 2.83% and 8.49%. *Check.* Every deviation of the café corner is three times the matching deviation of the cake line, so its standard deviation must be three times as large. $2.83 \times 3 = 8.49$, which agrees.

The café corner's standard deviation is three times the cake line's.

The figure below shows the two distributions as bars. They have the same centre, 12%, and the same chances. But the café corner's bars lie much farther apart. The standard deviation measures exactly this difference.

The figure shows the same two ideas as pictures.

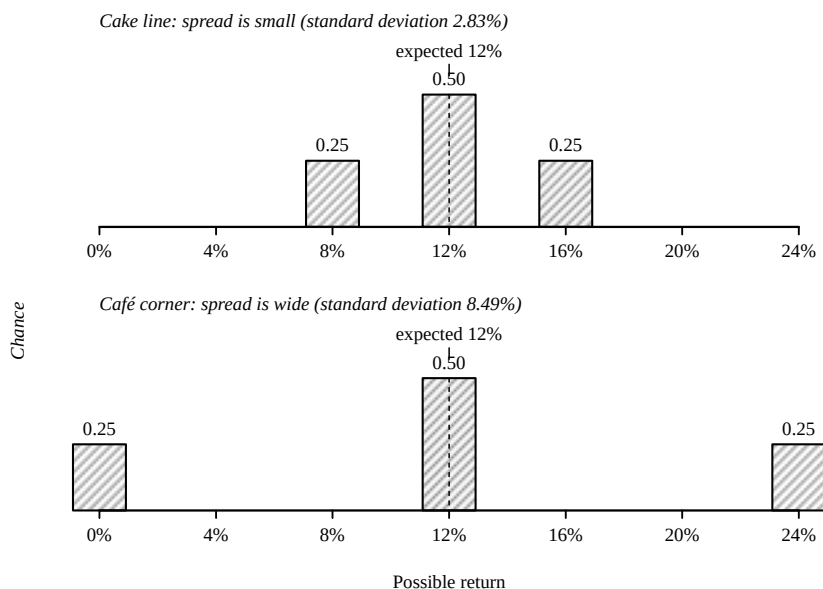


FIG. 24. The two product ideas. Both have an expected return of 12%, but the café corner's outcomes are spread much more widely.

An uneven distribution shows that the five steps still hold.

The variance is 196 and the standard deviation is 14%.

A standard deviation is a usual distance, and not a limit.

Four slips are common when finding a standard deviation.

When the expected values are equal, the smaller standard deviation is the safer choice.

A THIRD EXAMPLE WITH UNEQUAL CHANCES

The five steps work for any distribution, including one whose chances are unequal. Use the project of the last section, with returns of -10% , 10% and 30% , chances of 0.2 , 0.5 and 0.3 , and an expected return of 12% . The deviations are $-10 - 12 = -22$, $10 - 12 = -2$ and $30 - 12 = 18$. The squared deviations are 484 , 4 and 324 . Weighted by the chances, they are $0.2 \times 484 = 96.8$, $0.5 \times 4 = 2.0$ and $0.3 \times 324 = 97.2$.

The variance is $96.8 + 2.0 + 97.2 = 196.0$. The standard deviation is the square root of 196 , which is 14% . *Answer.* The standard deviation is 14% . *Check.* The outcomes lie 22 , 2 and 18 points from the expected value, and the largest chance, 0.5 , belongs to the outcome that lies closest. A usual distance of 14 points is therefore sensible.

READING THE STANDARD DEVIATION

How should we read a standard deviation? The cake line's 2.83% means that its returns usually lie within about three points of 12% . It is a usual distance, and not a limit. A result farther from the expected value is possible, and for some distributions it is quite likely. We should not read the standard deviation as the greatest loss that can occur.

Four slips are common. The first is to forget to square the deviations. The second is to forget to multiply by the chances. The third is to forget the final square root, and to report the variance as if it were the standard deviation. The fourth is to mix units, by using 0.12 for one outcome and 12 for another. A check that the answer lies near the size of the deviations will find most of these slips.

THE COEFFICIENT OF VARIATION

COMPARING CHOICES THAT DO NOT HAVE EQUAL AVERAGES

When two choices have equal expected returns, the comparison is easy. The choice with the smaller standard deviation has less risk. This is the case for the cake line and the café corner, and the cake line has the

smaller standard deviation of 2.83%. But the expected returns of two choices are often unequal, and then the standard deviation alone can mislead.

Suppose that investment X has an expected return of 10% and a standard deviation of 4%. Investment Y has an expected return of 20% and a standard deviation of 6%. Y has the larger standard deviation, but it also offers a much larger expected return. To compare them fairly we need to know how much risk each one carries for each unit of expected return.

The **coefficient of variation**, or CV, is the standard deviation divided by the expected value. It states the risk taken for each unit of expected return. In symbols, $CV = SD \div EV$. A smaller CV means less risk for each unit of return. The CV has no unit, because the unit of the numerator and that of the denominator cancel.

A larger standard deviation may be fair if the expected return is larger as well.

Coefficient of variation. The standard deviation divided by the expected value; the risk taken for each unit of expected return.

The coefficient of variation divides the risk by the expected return.

AN EXAMPLE WITH UNEQUAL AVERAGES

Given. Investment X: expected return 10%, standard deviation 4%. Investment Y: expected return 20%, standard deviation 6%. *Find.* The CV of each, and which carries less risk for each unit of return. *Steps.* CV of X = $4 \div 10 = 0.40$. CV of Y = $6 \div 20 = 0.30$. *Answer.* Y has the smaller CV, so it carries less risk for each unit of expected return, although its standard deviation is larger. *Check.* Y's expected return is twice X's, and its standard deviation is only one and a half times X's, so Y's ratio must be smaller.

The CV of X is 0.40 and the CV of Y is 0.30, so Y carries less risk per unit.

THE BAKERY'S TWO IDEAS

Now apply the CV to the bakery. For the cake line, $CV = 2.83 \div 12 = 0.24$. For the café corner, $CV = 8.49 \div 12 = 0.71$. The expected returns are equal, so the CV ranks the ideas in the same order as the standard deviation does. The café corner carries about three times as much risk for each unit of return as the cake line.

The cake line's CV is 0.24, and the café corner's is 0.71.

The project of the last section has an expected return of 12% and a standard deviation of 14%, so its CV is $14 \div 12 = 1.17$. Its spread is larger than its expected return. Compare this with the cake line's 0.24. A CV above 1 shows that the risk for each unit of return is high, and a careful owner would ask for a strong reason before accepting such a project.

A third project shows what a high coefficient of variation looks like.

The CV has one limit. If the expected value is close to zero, the CV becomes very large, and if it is negative, the CV has no clear meaning. In such cases the standard deviation should be examined by itself. Another point is that the CV measures the amount of risk, but it does not tell the owner how much risk to accept. That depends on how much extra reward the owner asks. Chapter X deals with it.

The CV has limits that a careful student should know.

THE WORKED CASE: CAKE LINE OR CAFÉ CORNER

Case file, illustrative. Sundial Bakery, Dhaka. The owner considers two product ideas for the shop. The cake line is estimated to return 8%, 12% or 16%, with chances of 0.25, 0.50 and 0.25. The café corner is estimated to return 0%, 12% or 24%, with the same chances. The chances come from the owner's judgement, helped by the sales records of the first months.

The case file records the two ideas, their returns and their chances.

We state what is given and what the owner must find.

The steps give the same expected return and different spreads.

The café corner is riskier by every measure, though the average return is equal.

Which idea to choose depends on the reward the owner asks for risk.

The measures also bear on the oven and the van of Chapter VII.

The measures of the chapter in a few lines.

You should now be able to say how each measure is found and used.

Given. The two distributions above. *Find.* The expected return, the standard deviation and the coefficient of variation of each idea, and a statement of which is riskier. *Formula.* EV is the sum of each outcome times its chance. Variance is the sum of each squared deviation times its chance. SD is the square root of the variance. CV is SD divided by EV.

Steps. For the cake line, $EV = 0.25 \times 8 + 0.50 \times 12 + 0.25 \times 16 = 2 + 6 + 4 = 12\%$. The variance is $0.25 \times 16 + 0.50 \times 0 + 0.25 \times 16 = 8$, and $SD = 2.83\%$. $CV = 2.83 \div 12 = 0.24$. For the café corner, $EV = 0 + 6 + 6 = 12\%$. The variance is $0.25 \times 144 + 0 + 0.25 \times 144 = 72$, and $SD = 8.49\%$. $CV = 8.49 \div 12 = 0.71$.

THE TWO PRODUCT IDEAS ON THREE MEASURES

MEASURE	CAKE LINE	CAFÉ CORNER
Expected return	12%	12%
Standard deviation	2.83%	8.49%
Coefficient of variation	0.24	0.71

Answer. The two ideas have the same expected return of 12%. The café corner is riskier on both measures: its standard deviation is 8.49% against 2.83%, and its coefficient of variation is 0.71 against 0.24. *Check.* The probabilities of each idea add up to 1. Each expected value lies in the middle of its outcomes, as the symmetric chances require. The café corner's spread is exactly three times the cake line's, as its deviations are.

The measures do not tell the owner what to choose. They tell the owner what each choice means. With equal expected returns, a cautious owner prefers the cake line, because it offers the same average with less wandering. The owner would choose the café corner only if it promised a higher expected return, enough to pay for its extra risk. Chapter X shows how to put a price on that difference.

These measures return in Chapter X. They also bear on the choices of Chapter VII. The yearly cash flows of the oven were single estimates. In truth each could turn out higher or lower, so the oven's NPV also has a spread. An owner who wished to know how safe the oven's NPV of Tk 79,079 is would describe the possible sales with chances, as we did for the product ideas. We do not carry out that work here.

SUMMARY

Risk is the chance that the actual result differs from the hoped-for result. Market risk affects all firms, and firm-specific risk affects one. Business risk and financial risk sort the same total risk by its cause inside the firm. Return is the gain on an investment as a share of its cost. Probability is a number from 0 to 1, and a probability distribution lists the outcomes with chances that add up to 1. The expected value is the weighted average of the outcomes. The standard deviation is the square root of the weighted average of the squared deviations, and it measures the spread. The coefficient of variation divides the standard deviation by the expected value, and it compares risk per unit of return.

You should now be able to say: the expected value shows where the outcomes centre, the standard deviation shows how far they wander, and the coefficient of variation shows the risk for each unit of return.

You should also be able to find all three from a list of outcomes and chances.

THINK IT THROUGH

Case file, illustrative. The owner of Sundial Bakery is offered two investments for spare cash. Investment M may return 4%, 10% or 16%, with chances of 0.3, 0.4 and 0.3. Investment N may return -5%, 15% or 35%, with chances of 0.2, 0.6 and 0.2.

Find the expected return, the standard deviation and the coefficient of variation of each investment. Say which investment is riskier by the standard deviation, and which carries more risk for each unit of expected return. Explain why the two answers may or may not be the same. Then find the return on a share that is bought for Tk 150, pays a dividend of Tk 6, and is sold for Tk 141.

Two investments with unequal averages give the reader a case to analyse.

The questions ask for the three measures and a comparison.

WHERE PEOPLE GO WRONG

Mistake: believing that risk is only the chance of a loss. *Correction:* in finance, risk is the chance of any difference from the hoped-for result, in either direction. The standard deviation counts both.

Mistake: believing that risk means only the chance of loss.

Mistake: using probabilities that total more or less than 1. *Correction:* a complete list of outcomes must have chances that add up to exactly 1. Add them before any other step.

Mistake: using chances that do not add up to one.

Mistake: averaging the outcomes without weights. *Correction:* each outcome must be multiplied by its chance, so that likely outcomes count for more.

Mistake: taking the plain average of the outcomes.

Mistake: reporting the variance as the standard deviation. *Correction:* the standard deviation is the square root of the variance. The variance is in squared units, and the standard deviation is in the original unit.

Mistake: stopping at the variance.

Mistake: choosing the investment with the smaller standard deviation when the expected returns are not equal. *Correction:* compare the coefficients of variation, which state the risk for each unit of expected return.

Mistake: comparing standard deviations when the expected returns differ.

Mistake: expecting the expected return to occur. *Correction:* the expected value is an average over many trials. A single trial gives one of the outcomes, and it may be far from the average.

Mistake: reading the expected value as a promise.

IN HISTORY

In 1952 the American economist Harry Markowitz published a paper called *Portfolio Selection* in the *Journal of Finance*. In it he proposed that an investor should describe each investment by two numbers: its expected return and the variance of its return. He argued that investors should look at the two together, and that holding several investments can reduce risk without lowering the expected return. Later writers developed these ideas into the theory that Chapter X will introduce.

Harry Markowitz put the expected return and the variance to work in 1952.

SOURCES

Harry Markowitz, *Portfolio Selection*, *Journal of Finance* (1952), is the landmark source for describing an investment by its expected return and the variance of its return. The measures of probability, expected value and standard deviation

A short note on the origin of the ideas in this chapter.

are standard topics in textbooks of statistics and of financial management.

REVIEW QUESTIONS

Risk and its kinds.

1. Define risk as the word is used in finance. Then explain the difference between market risk and firm-specific risk, with one example of each.

Classifying events.

2. Say whether each event is mainly a market risk or a firm-specific risk: a fire at one bakery, a rise in interest rates across the country, a rival shop that opens next door.

Return.

3. A share is bought for Tk 250, pays a dividend of Tk 10, and is sold for Tk 235. Find the return for the period, and say whether it is a gain or a loss.

Probability.

4. A project may return 4%, 9% or 17%. The chances of the first two outcomes are 0.30 and 0.45. Find the chance of the third outcome, and then find the expected return.

Standard deviation.

5. A project may return 2%, 8% or 14%, with chances of 0.25, 0.50 and 0.25. Find its expected return, its variance and its standard deviation.

Coefficient of variation.

6. Asset G has an expected return of 15% and a standard deviation of 6%. Asset H has an expected return of 9% and a standard deviation of 3%. Find the coefficient of variation of each, and say which has less risk for each unit of return.

Why the average is not enough.

7. Explain why the expected value alone is not enough to compare two investments. Use the idea of the two bus routes, or an example of your own.

Reading the measures.

8. Two investments have equal expected returns. One has a standard deviation of 3% and the other has a standard deviation of 9%. Which is riskier, and what extra thing would a cautious owner ask of the riskier one?

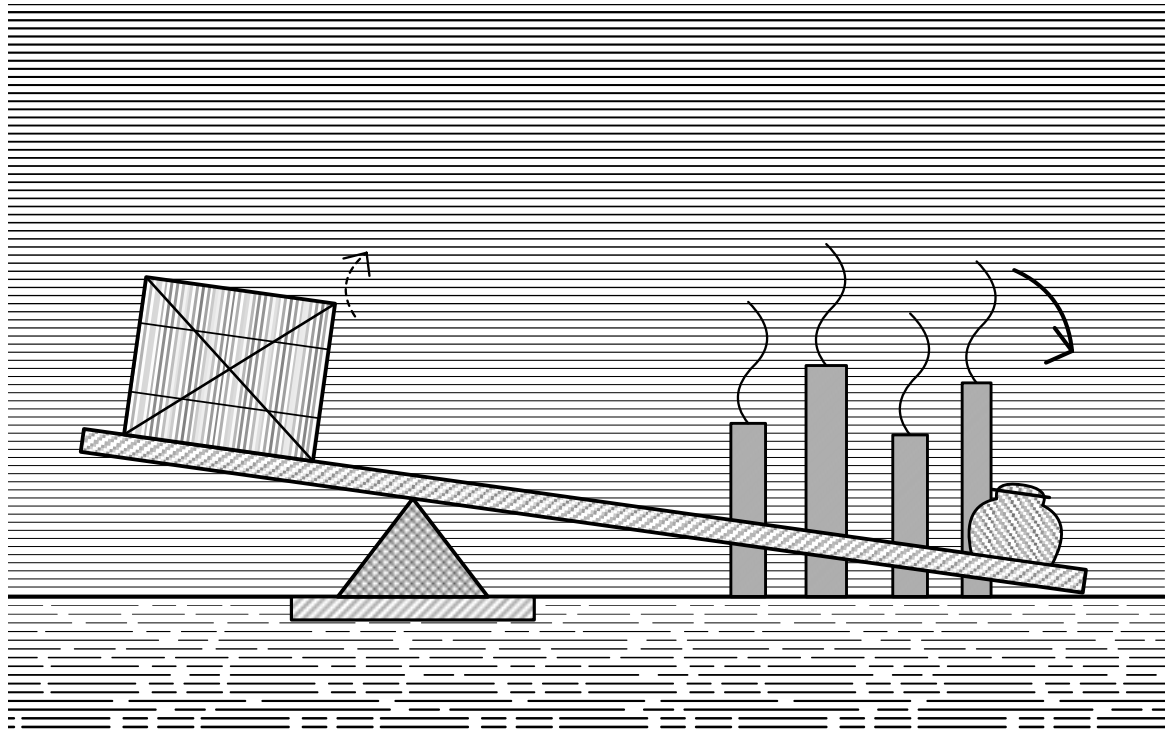


Plate IX. The Lever and the Load.

CHAPTER IX

Leverage and Risk

Why does a small push at one end sometimes move the other end a great deal?



This chapter shows how fixed costs, and fixed payments to lenders, magnify the swings in a firm's profit. It teaches how to find the sales at which a firm earns no profit, and how to measure the magnifying effect in three ways. It ends by asking how much of this effect a careful firm should accept.

Two shops sell the same goods; both sales rise by one-tenth, but only one profit leaps.

The answer lies in how each shop pays its costs: by the cup or by the month.

The same fixed payments that help in a good month hurt in a bad month.

This chapter measures that effect, so that a firm can decide how much of it to accept.

Before this chapter, you should know fixed and variable cost, EBIT and the profit layout, and the idea of risk.

A cart seller faces two uncertainties: how many people buy, and what she must repay.

Business risk is the risk that comes from the firm's own work.

Business risk. The risk that a firm's EBIT differs from what was hoped, because of its own work.

Leverage. In daily life, the extra force a lever gives. In finance, the magnifying effect of fixed payments on profit.

Two small shops stand side by side in the same market. Both sell tea and biscuits. Last month, the sales of each shop rose by one-tenth. The owner of the first shop found that her profit rose by one-tenth as well. The owner of the second shop found that his profit rose by almost one half. Nobody worked harder. No price changed. Why did the second shop gain so much more?

The answer lies in the way each shop pays for its work. The first shop pays for almost everything by the amount it sells. Its helper is paid for each cup served, and its landlord takes a share of sales. When sales rise, its costs rise in step, and so does its profit. The second shop pays a fixed rent and a fixed monthly wage, whatever it sells. When sales rise, those costs stay where they are. Almost every extra taka of sales becomes an extra taka of profit.

The second shop has a hard side, too. Suppose sales fall by one-tenth. The first shop's profit falls in step. The second shop's profit falls by almost one half, because the rent and the wages must still be paid in full. A fixed payment behaves like a long plank resting on a stone. A small push at one end moves the other end a great deal, in either direction. Finance takes a word from this picture: the word is *lever*.

In this chapter we study two kinds of fixed payment. The first kind is the fixed cost of running the business. The second kind is the fixed payment to people who lent money. We study each one alone, and then both together. By the end you will be able to find the sales at which a firm earns no profit. You will also be able to say, in numbers, how many times a change in sales is magnified in profit.

You will need: from Chapter 3, the difference between fixed cost and variable cost, and the profit layout from sales down to net profit (including EBIT, which you have met and which means profit before interest and tax); from Chapter 8, the idea of risk as the chance that results differ from hopes, and the coefficient of variation.



9.1 BUSINESS RISK AND FINANCIAL RISK

Begin with a seller of snacks who has a cart near a bus station. On some days many people buy, and on other days few do. She cannot know in advance. Now suppose she borrowed Tk 20,000 to buy the cart, and she must repay Tk 1,000 every week, whatever she sells. She has two separate worries. The first is about her customers. The second is about her debt. Firms have the same two worries, and finance gives each one a name.

The first name is **business risk**. Business risk means the chance that a firm's EBIT will be different from what was hoped, because of the firm's own work. It is also called operating risk, but this book keeps the name business risk. The word **leverage** needs a note here. In daily life it means the extra force that a lever gives. In finance it means the magnifying effect that fixed payments have on profit, and we study it from section 9.3 onward.¹

¹ In British usage, financial leverage is often called *gearing*. This book uses the word leverage throughout.

What causes business risk? Look at the bakery. The price of flour may rise. Fewer customers may come in the rainy season. A new rival may open in the next street. The electricity may fail for a day. Each of these changes sales or cost, and so each changes EBIT. One more cause matters greatly for this chapter: the share of the firm's costs that are fixed. The more cost is fixed, the more a change in sales is magnified in EBIT. A firm that sells a similar product in a similar market can still have more business risk than its neighbour, only because it has more fixed cost.

The second name is **financial risk**. Financial risk means the extra risk that the owners carry because the firm has promised fixed payments to lenders. The bakery shows how. It owes its bank interest of Tk 7,000 every month. If sales are poor and EBIT is small, the bank still expects Tk 7,000. The owners receive only what is left after the bank is paid. A firm with no loans has no financial risk of this kind.

Chapter 3 gave the profit layout, which starts at sales and ends at net profit. Business risk acts in the upper half of that layout, between sales and EBIT. Financial risk acts in the lower half, between EBIT and net profit, where interest and similar payments are taken away. A figure in section 9.5 draws this idea in one picture.

The table below sets the two risks side by side. Read each row from left to right: the first column names the question, and the next two columns give the answer for each risk.

BUSINESS RISK AND FINANCIAL RISK COMPARED

QUESTION	BUSINESS RISK	FINANCIAL RISK
Where does it come from?	The firm's own work: customers, costs, rivals	Fixed payments promised to lenders
Where in the profit layout?	Between sales and EBIT	Between EBIT and net profit
Can the firm choose it?	Only in part; much belongs to the kind of work	Yes; the firm chooses how much to borrow
Who bears it?	Everyone who has money in the firm	Mostly the owners, after lenders are paid

The two risks look different from different seats. The owner sees both, because the owner is paid last. The lender cares most about financial risk, because the lender wants to know that the fixed payment will be met. The manager must live with business risk and must also decide how much financial risk to add.

The two risks are not separate boxes. They combine, and one can make the other worse. A firm with an uncertain EBIT that also owes a large fixed payment may fail in a bad year. For this reason, firms with high business risk usually borrow less. Business risk can be reduced a little, for example by selling several products, but most of it comes with the kind of work. Financial risk is a choice.

Before moving on. Business risk comes from the work of the firm and shows in EBIT. Financial risk comes from fixed payments to lenders and shows below EBIT. The firm can choose its financial risk more freely than its business risk.

Business risk has many causes; the share of fixed cost is one that the firm controls.

Financial risk is the extra risk that comes from borrowing.

Financial risk. The extra risk owners carry because the firm has promised fixed payments to lenders.

The two risks sit in different places in the profit layout.

A table sets the two risks side by side.

The owner, the lender and the manager each see the two risks from a different seat.

The two risks combine, so a firm with high business risk should think twice before borrowing heavily.

Before moving on: business risk is about EBIT; financial risk is about borrowing.



9.2 THE BREAK-EVEN POINT

A first question: how much must we sell before we earn any profit?

The break-even point is the sales level at which profit is exactly zero.

Break-even point. The level of sales at which profit is exactly zero.

Every unit sold brings a contribution; the break-even point is where contributions cover the fixed cost.

The formula in words, then in symbols.

The symbols used in the formula.

First example: a tea stall, with small and clean numbers.

Steps, answer and check for the tea stall.

Now the bakery, whose figures we carry through the chapter.

Second example: the bakery's break-even point, with a check on both sides.

Steps, answer and check for the bakery.

Break-even sales can also be found from the share of each taka of sales that is contribution.

How many cups must a tea seller sell before the first taka of profit? Every seller asks this question before the first day of trade, because the rent and the pot are already paid for. The answer is a number, and finding it is the first technique of this chapter.

The **break-even point** means the level of sales at which profit is exactly zero. Below it, the firm makes a loss. Above it, the firm makes a profit. In this chapter, "profit" at the break-even point means EBIT, and "fixed cost" means the fixed cost of running the business. Interest is a separate payment, which we study in section 9.4.

Chapter 3 defined contribution as the selling price less the variable cost for each unit. Each unit sold adds its contribution towards paying the fixed cost. At first the contributions are used to cover the fixed cost. When they have covered it all, every further unit adds to profit. The break-even point is the number of units at which the contributions exactly equal the fixed cost. In words:

break-even units = fixed cost $\div$ contribution per unit.

In symbols, let F be the fixed cost, P the selling price of one unit, and V the variable cost of one unit. Then the break-even units are $Q = F \div (P - V)$. We use Q for a number of units.

Worked example 1: a tea stall. Given. A tea stall charges Tk 10 for a cup. The tea, milk and sugar in one cup cost Tk 6. The stall pays Tk 2,000 each day for rent and the helper, whatever it sells. *Find.* The number of cups the stall must sell each day to break even. *Formula.* Break-even units = fixed cost $\div$ (price – variable cost).

Steps. The contribution per cup is Tk 10 – Tk 6 = Tk 4. The break-even units are Tk 2,000 $\div$ Tk 4 = 500 cups. *Answer.* The stall must sell 500 cups a day to break even. *Check.* At 500 cups the stall earns 500 $\times$ Tk 4 = Tk 2,000 of contribution, which exactly equals the fixed cost, so profit is zero. At 501 cups, profit is Tk 4.

Case file, illustrative. Sundial Bakery bought its second oven with a further bank loan of Tk 3,00,000 at 12% a year. Its total debt is now Tk 7,00,000, and its interest is Tk 7,000 a month. A box of buns sells for Tk 40, and the variable cost of one box is Tk 24. The fixed cost of running the bakery is Tk 1,20,000 a month. The bakery plans to sell 10,000 boxes a month.

Worked example 2: the bakery. Given. $P = \text{Tk } 40$; $V = \text{Tk } 24$; $F = \text{Tk } 1,20,000$ a month. *Find.* The break-even boxes each month, and the sales in taka at that level. *Formula.* $Q = F \div (P - V)$.

Steps. The contribution of one box is Tk 40 – Tk 24 = Tk 16. The break-even boxes are Tk 1,20,000 $\div$ Tk 16 = 7,500 boxes. The sales at that level are 7,500 $\times$ Tk 40 = Tk 3,00,000. *Answer.* The bakery breaks even at 7,500 boxes a month, which is Tk 3,00,000 of sales. *Check.* At 7,000 boxes, contribution is 7,000 $\times$ Tk 16 = Tk 1,12,000, which is Tk 8,000 less than the fixed cost, so EBIT is a loss of Tk 8,000. At 8,000 boxes, contribution is Tk 1,28,000, so EBIT is a profit of Tk 8,000. Zero lies between them, at 7,500.

There is a second way to state the answer in taka. Each box sells for Tk 40 and contributes Tk 16, so 40 per cent of every taka of sales is contribution. The fixed cost divided by that share gives the break-even sales: Tk

$1,20,000 \div 0.40 = \text{Tk } 3,00,000$. The two ways agree, which is a useful check.

The chart below shows the same answer as a picture. To read it, look at the foot of the picture first: it shows the boxes sold in a month. The upright side shows money in taka. The rising line from the corner is sales. The dashed line is the fixed cost, which does not move. The second rising line is total cost, which is the fixed cost plus the variable cost of the boxes sold. Where sales meet total cost, profit is zero. To the left of that point the total cost line is above the sales line, which means loss. To the right the sales line is above, which means profit.

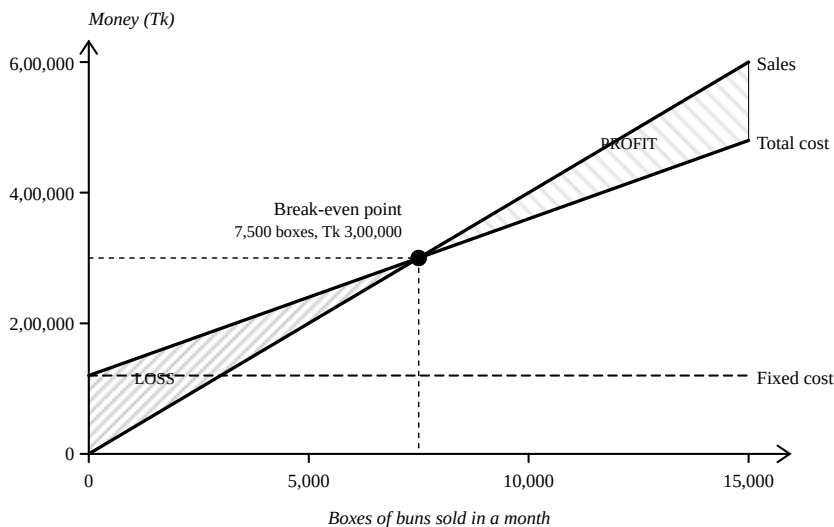


FIG. 25. Break-even chart for the bakery: sales and total cost cross at 7,500 boxes.

The same idea answers a second question. How many boxes must the bakery sell to earn a chosen EBIT? Each box contributes Tk 16. The contributions must first cover the fixed cost of Tk 1,20,000, and then supply the target. In words: units needed = (fixed cost + target EBIT) $\div$ contribution per unit.

Worked example. Given. $F = \text{Tk } 1,20,000$; contribution per box = Tk 16; target EBIT = Tk 56,000 a month. Find. The boxes needed. Steps. Fixed cost plus target is $\text{Tk } 1,20,000 + \text{Tk } 56,000 = \text{Tk } 1,76,000$. Divide by Tk 16: $1,76,000 \div 16 = 11,000$ boxes. Answer. The bakery must sell 11,000 boxes a month. Check. At 11,000 boxes, contribution is $11,000 \times \text{Tk } 16 = \text{Tk } 1,76,000$. Less the fixed cost of Tk 1,20,000, EBIT is Tk 56,000, which is the target.

The break-even point rests on assumptions, and a careful reader should know them. The price and the variable cost of a unit are taken as the same at every level of output. The fixed cost is taken as fixed over the range of sales studied, but a firm that grows far enough must buy another oven, and its fixed cost then jumps. The method as shown suits one product; a firm with many products needs an average contribution. These limits do not make the method wrong. They tell us to use it for planning within a sensible range.

Common slips. Some students divide the fixed cost by the price instead of by the contribution; this gives a break-even point that is far too low. Others mix periods, using a yearly fixed cost with a monthly sales figure. A third slip is to include interest in the fixed cost and then to count it

The chart draws the break-even point as the place where two lines cross.

The same idea finds the sales needed for a target profit.

Worked example for a target EBIT of Tk 56,000.

Break-even results depend on the assumptions behind them.

Three common slips: price instead of contribution, wrong period, and interest inside the fixed cost.

Before moving on: break-even units equal fixed cost divided by contribution per unit.

GO SLOWLY HERE.

Two ways of paying a worker show why fixed costs magnify results.

Operating leverage is the magnifying effect of fixed operating costs on EBIT.

Operating leverage. How fixed costs of running the business magnify changes in EBIT.

The table gives the bakery's EBIT at three levels of sales.

A rise of one-quarter in sales doubles EBIT; a fall of one-quarter removes it.

The degree of operating leverage tells by how many times EBIT changes when sales change.

Degree of operating leverage (DOL). The per cent change in EBIT caused by a one per cent change in sales.

The formula for DOL, in two equal forms.

again in section 9.4. In this chapter, fixed cost means fixed cost of running the business only.

Before moving on. The break-even point is where contributions exactly cover the fixed cost. You find it by dividing the fixed cost by the contribution of one unit. You should now be able to say: "If the fixed cost is Tk 60,000 and a unit contributes Tk 12, the firm must sell 5,000 units to break even."



9.3 OPERATING LEVERAGE

A tailor needs one more worker. She can pay the worker Tk 20 for each shirt sewn, or Tk 6,000 a month whatever is sewn. With the first plan, a month with twice as many shirts costs twice as much in wages. With the second plan, a month with twice as many shirts costs the same wages. The second plan leaves her more profit in a busy month, and less in a quiet month. The more of her costs are fixed, the more her profit swings. We now measure that swing.

Operating leverage means the way that fixed costs of running the business magnify the effect of a change in sales on EBIT. A firm with large fixed costs and small variable costs has high operating leverage. A firm with no fixed cost has none. We begin with a table, because the table shows the effect before any formula does.

The table below shows the bakery's monthly results at three levels of sales. The middle row is the plan of 10,000 boxes. The first row is a fall of one-quarter, and the last row is a rise of one-quarter. Read each row across: sales, less variable cost, equals contribution, less the fixed cost, equals EBIT.

THE BAKERY'S MONTHLY EBIT AT THREE LEVELS OF SALES (TK)

BOXES SOLD	SALES	VARIABLE COST	CONTRIBUTION	FIXED COST	EBIT
7,500	3,00,000	1,80,000	1,20,000	1,20,000	0
10,000	4,00,000	2,40,000	1,60,000	1,20,000	40,000
12,500	5,00,000	3,00,000	2,00,000	1,20,000	80,000

Now compare the rows. From the middle row to the last row, sales rise from Tk 4,00,000 to Tk 5,00,000, which is a rise of 25 per cent. EBIT rises from Tk 40,000 to Tk 80,000, which is a rise of 100 per cent. From the middle row to the first row, sales fall by 25 per cent, and EBIT falls from Tk 40,000 to zero, which is a fall of 100 per cent. A change of 25 per cent in sales became a change of 100 per cent in EBIT. The fixed cost of Tk 1,20,000 did not move, and so it magnified the change.

We give this magnifying factor a name. The **degree of operating leverage (DOL)** means the per cent change in EBIT for each one per cent change in sales. In words and then in symbols:

$$\text{DOL} = \text{per cent change in EBIT} \div \text{per cent change in sales.}$$

For the bakery, $DOL = 100 \div 25 = 4.0$. There is a second form, which needs only one level of sales. $DOL = \text{contribution} \div \text{EBIT}$. Contribution is the total contribution of all units sold, and EBIT is measured at the same level of sales. Both forms give the same answer.

Worked example 3. Given. At 10,000 boxes the bakery has a contribution of Tk 1,60,000 and an EBIT of Tk 40,000. *Find.* The DOL. *Formula.* $DOL = \text{contribution} \div \text{EBIT}$.

Steps. $DOL = \text{Tk } 1,60,000 \div \text{Tk } 40,000 = 4.0$. *Answer.* The DOL at 10,000 boxes is 4.0: each one per cent change in sales changes EBIT by four per cent. *Check.* A rise of 25 per cent in sales should raise EBIT by $4.0 \times 25 = 100$ per cent, which is what the table of EBIT at three levels of sales shows. A second check: 10,000 boxes are 2,500 boxes above break-even, which is one-quarter of 10,000; and $1 \div 0.25 = 4.0$.

The DOL is not one fixed number for a firm. Use the same formula at 12,500 boxes: contribution is Tk 2,00,000 and EBIT is Tk 80,000, so $DOL = 2.5$. The further sales rise above break-even, the smaller the DOL becomes. Near break-even, EBIT is small and a small change in sales is a large change in EBIT, so the DOL is very large. At exactly 7,500 boxes EBIT is zero, and the DOL cannot be found at all.

Now test the idea at its lowest limit. Suppose a firm sells 10,000 units at Tk 40, with a variable cost of Tk 28 and no fixed cost. Its contribution is Tk 1,20,000 and its EBIT is also Tk 1,20,000. $DOL = 1,20,000 \div 1,20,000 = 1.0$. Its EBIT changes by exactly the same per cent as its sales. A DOL of 1.0 is the least that any firm can have, and it means no magnifying at all.

Operating leverage is partly a choice, as an exercise shows. Suppose the bakery could instead buy its buns from a contract baker. It would then have a fixed cost of only Tk 40,000 a month, but a variable cost of Tk 32 for each box. At 10,000 boxes the contribution is $10,000 \times (\text{Tk } 40 - \text{Tk } 32) = \text{Tk } 80,000$, and EBIT is $\text{Tk } 80,000 - \text{Tk } 40,000 = \text{Tk } 40,000$. This is the same EBIT as the bakery's own plan. The table below compares the two plans at three levels of sales.

TWO WAYS OF ORGANISING COSTS: THE BAKERY'S OWN PLAN AND THE CONTRACT-BAKER PLAN (EBIT IN TK A MONTH)

BOXES SOLD	OWN PLAN (FIXED TK 1,20,000, VARIABLE TK 24)	CONTRACT PLAN (FIXED TK 40,000, VARIABLE TK 32)
7,500	0	20,000
10,000	40,000	40,000
12,500	80,000	60,000

At 10,000 boxes the two plans earn the same EBIT, but their risks differ. The own plan has a DOL of 4.0. The contract plan has a DOL of $\text{Tk } 80,000 \div \text{Tk } 40,000 = 2.0$. If sales fall to 7,500 boxes, the contract plan still earns Tk 20,000, and the own plan earns nothing. If sales rise to 12,500 boxes, the own plan earns Tk 80,000 and the contract plan earns Tk 60,000. The plan with more fixed cost does better when sales are high and worse when they are low. A manager who fears weak sales may prefer the plan with less fixed cost, even though it pays less in a good month.

Operating leverage works in both directions. A high DOL is welcome when sales rise, and it is dangerous when sales fall. This is why we speak of it as a source of risk and not only as a source of gain. A bakery with a

For the bakery, $DOL = 100 \div 25 = 4.0$, and a second form needs only one level of sales.

Worked example 3: DOL for the bakery at 10,000 boxes.

Steps, answer and check for DOL.

The DOL is different at each level of sales; it is largest near break-even.

A firm with no fixed cost has a DOL of 1, so profit moves in step with sales.

An exercise compares two ways of organising costs that give the same EBIT at the planned sales.

The plan with more fixed cost has a larger DOL: it does better in good months and worse in bad months.

Operating leverage magnifies gains and losses in the same way.

Slips with DOL: using it far from its base, and forgetting that the fixed cost may jump.

Before moving on: DOL = contribution ÷ EBIT, and it measures how much EBIT is magnified.

DOL of 4.0 and a bad season of 20 per cent fewer sales loses 80 per cent of its EBIT.

Common slips. The DOL is found at one level of sales, and it applies to changes from that level. Using the DOL found at 10,000 boxes to judge a change from 12,500 boxes gives a wrong answer. A second slip is to forget the limit set in section 9.2: if sales grow so much that a new oven must be bought, the fixed cost rises, and the old DOL no longer holds.

Before moving on. Fixed operating costs magnify changes in EBIT. The DOL measures the magnification at one level of sales, and it equals contribution divided by EBIT. You should now be able to say: "A DOL of 3.0 means that a fall of 10 per cent in sales causes a fall of 30 per cent in EBIT."



9.4 FINANCIAL LEVERAGE

GO SLOWLY HERE.

A family with a shop loan shows the same magnifying effect, now below EBIT.

Financial leverage is the magnifying effect of fixed financing payments on the owners' profit.

Financial leverage. How fixed payments to lenders magnify changes in the owners' profit.

The bakery's monthly profit layout, from EBIT to net profit, shows the effect in numbers.

The next table follows EBIT of Tk 40,000 and Tk 80,000 down to net profit.

A family buys a small shop with a loan. After paying the shop's running costs, the shop brings in Tk 10,000 each month. The loan costs Tk 4,000 a month, whatever happens. The family keeps Tk 6,000. Suppose the Tk 10,000 rises by 10 per cent to Tk 11,000. The family now keeps Tk 7,000, which is a rise of 16.7 per cent. Suppose it falls by 10 per cent to Tk 9,000. The family keeps Tk 5,000, a fall of 16.7 per cent. The fixed loan payment magnified the change, in the same way that fixed costs did in the last section.

Financial leverage means the way that fixed payments to lenders magnify the effect of a change in EBIT on the profit left for the owners. The difference from operating leverage is where it acts. Operating leverage links sales to EBIT. Financial leverage links EBIT to net profit.

Case file, illustrative. At 10,000 boxes the bakery has an EBIT of Tk 40,000 a month. Its debt of Tk 7,00,000 at 12% a year costs Tk 7,00,000 × 0.12 ÷ 12 = Tk 7,000 a month in interest. The illustrative tax rate is 25 per cent of profit before tax.

The table below follows the bakery's EBIT down the profit layout, once at the planned level and once at the level of 12,500 boxes. Read each column from top to bottom: EBIT, less interest, equals profit before tax, less tax, equals net profit.

FROM EBIT TO NET PROFIT AT TWO LEVELS OF SALES (TK A MONTH)

ITEM	AT 10,000 BOXES	AT 12,500 BOXES
EBIT	40,000	80,000
Less interest	7,000	7,000
Profit before tax	33,000	73,000
Less tax at 25%	8,250	18,250
Net profit	24,750	54,750

EBIT doubled, but net profit rose by more than double.

EBIT rose by 100 per cent, from Tk 40,000 to Tk 80,000. Net profit rose from Tk 24,750 to Tk 54,750. That is a rise of Tk 30,000 on Tk 24,750, or 121.2 per cent. The interest of Tk 7,000 did not

change, and so the whole of the extra EBIT reached the owners, apart from tax.

The **degree of financial leverage (DFL)** means the per cent change in net profit for each one per cent change in EBIT. For a firm with interest only, a shorter form is available:

$$\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest}).$$

Worked example 4. Given. EBIT is Tk 40,000 a month and interest is Tk 7,000 a month. *Find.* The DFL. *Formula.* $\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest})$.

Steps. $\text{EBIT} - \text{interest} = \text{Tk } 40,000 - \text{Tk } 7,000 = \text{Tk } 33,000$. $\text{DFL} = \text{Tk } 40,000 \div \text{Tk } 33,000 = 1.2121$, which we write as 1.21. *Answer.* The DFL is 1.21: each one per cent change in EBIT changes net profit by 1.21 per cent. *Check.* A rise of 100 per cent in EBIT should raise net profit by $1.2121 \times 100 = 121.2$ per cent, which agrees with the table above. Tax does not change the answer, because tax is the same share of profit before tax at both levels.

Some firms owe a second kind of fixed payment. A preferred dividend is a fixed payment to a special kind of owner, who is paid before the ordinary owners. Chapter 14 teaches this kind of owner fully; here we need only one fact. A preferred dividend is paid out of profit after tax, whereas interest is paid before tax. To compare the two fairly with EBIT, we must raise the preferred dividend to the amount of EBIT that is needed to pay it. We do this by dividing by $(1 - t)$, where t is the tax rate.

The full formula for DFL is therefore: $\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest} - \text{preferred dividend} \div (1 - t))$.

Worked example 5. Given. Suppose, as an exercise only, that the bakery also owed a preferred dividend of Tk 3,000 a month. EBIT is Tk 40,000, interest is Tk 7,000, and $t = 0.25$. *Find.* The DFL. *Formula.* $\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest} - \text{preferred dividend} \div (1 - t))$.

Steps. The preferred dividend divided by $(1 - t)$ is $\text{Tk } 3,000 \div 0.75 = \text{Tk } 4,000$. The bottom line is $\text{Tk } 40,000 - \text{Tk } 7,000 - \text{Tk } 4,000 = \text{Tk } 29,000$. $\text{DFL} = \text{Tk } 40,000 \div \text{Tk } 29,000 = 1.3793$, which we write as 1.38. *Answer.* The DFL rises from 1.21 to 1.38. *Check.* If EBIT doubles to Tk 80,000, the bottom line becomes Tk 69,000, which is 2.379 times Tk 29,000: a rise of 137.9 per cent, as 1.3793×100 predicts. The extra fixed payment raised the magnifying effect.

Financial leverage also cuts the other way. If EBIT falls to Tk 7,000, the bakery's interest uses all of it and profit before tax is zero. If EBIT falls further, the bakery makes a loss, and it may be unable to pay its lender on time. A lender who is not paid can take action against the firm. This is the real danger of financial risk, and it is why the DFL grows very large as EBIT approaches the fixed payments.

A firm with no loans and no preferred shares has a DFL of 1.0, because its net profit moves in the same per cent as its EBIT. Why would any firm borrow, then? A firm borrows because lenders supply money that the owners cannot supply, and because the firm may earn more on that money than the interest it pays. Chapters 13 and 14 treat these

The degree of financial leverage tells by how many times net profit changes when EBIT changes.

Degree of financial leverage (DFL). The per cent change in net profit caused by a one per cent change in EBIT.

The formula for DFL when the firm pays interest only.

Worked example 4: DFL for the bakery with interest only.

Steps, answer and check for DFL.

Preferred dividends are another fixed payment, but they are paid from profit after tax.

The full DFL formula includes interest, preferred dividends and tax.

Worked example 5: a what-if exercise with a preferred dividend of Tk 3,000.

Steps, answer and check for DFL with a preferred dividend.

If EBIT falls to the level of the fixed payments, profit disappears.

Without any interest or preferred dividend, the DFL is 1.0.

Before moving on: $DFL = EBIT \div (EBIT - \text{fixed financing payments}), \text{ after adjusting dividends for tax.}$

The two effects work in a chain, so we can find the combined effect.

Total leverage is the two effects together, and the DTL measures it.

Total leverage. The combined effect of operating and financial leverage on net profit.

Degree of total leverage (DTL). The per cent change in net profit caused by a one per cent change in sales.

The formula for DTL: multiply, never add.

A direct form of the DTL formula also exists.

Worked example 6: the bakery's DTL, found two ways.

Steps, answer and check for DTL.

The figure shows where each leverage acts in the profit layout.

reasons in detail. Here we only note that the gain and the danger arrive together.

Before moving on. Financial leverage magnifies changes in EBIT into larger changes in net profit, because interest and preferred dividends are fixed. The DFL measures the magnification. You should now be able to say: "A DFL of 1.5 means that a fall of 10 per cent in EBIT causes a fall of 15 per cent in net profit."



9.5 TOTAL LEVERAGE

A change in sales passes through two stages before it reaches the owners. At the first stage, operating leverage turns the change in sales into a larger change in EBIT. At the second stage, financial leverage turns the change in EBIT into a still larger change in net profit. The combined effect is the product of the two effects.

Total leverage means the combined effect of operating leverage and financial leverage. The **degree of total leverage (DTL)** means the per cent change in net profit for each one per cent change in sales. It is found by multiplying the two earlier measures:

$$DTL = DOL \times DFL.$$

There is a direct form as well: $DTL = \text{contribution} \div (EBIT - \text{interest} - \text{preferred dividend} \div (1 - t))$.

Worked example 6. Given. At 10,000 boxes, contribution is Tk 1,60,000; DOL is 4.0; DFL is 1.2121 (interest only). Find. The DTL. Formula. $DTL = DOL \times DFL$.

Steps. $DTL = 4.0 \times 1.2121 = 4.848$, which we write as 4.85. By the direct form, $DTL = \text{Tk } 1,60,000 \div \text{Tk } 33,000 = 4.848$. Answer. The DTL is 4.85: each one per cent change in sales changes net profit by 4.85 per cent. Check. A rise of 25 per cent in sales should raise net profit by $4.848 \times 25 = 121.2$ per cent, and the table of profit layouts above shows exactly that. If the bakery also owed the preferred dividend of Worked example 5, the direct form gives $\text{Tk } 1,60,000 \div \text{Tk } 29,000 = 5.52$.

The figure below draws the chain. Read it from top to bottom, in the same order as the profit layout of Chapter 3. The bracket on the right of the upper half is operating leverage. The bracket on the right of the lower half is financial leverage. The long bracket on the left covers the whole chain, and it is total leverage.

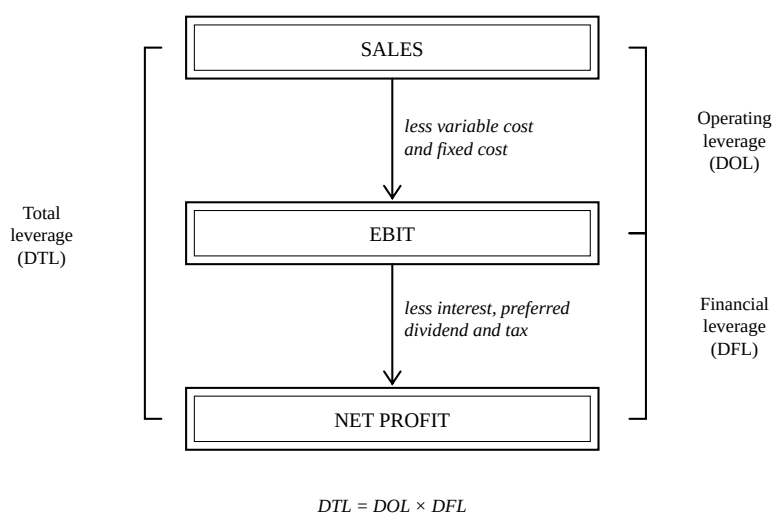


FIG. 26. Where operating, financial and total leverage act in the profit layout.

The next figure shows the effect in a different way. The first bar is the rise of 25 per cent in sales. The second bar is the rise in EBIT of 100 per cent, which is 4.0 times as large. The third bar is the rise in net profit of 121.2 per cent, which is 1.21 times as large again.

The next figure shows the same chain as three bars.

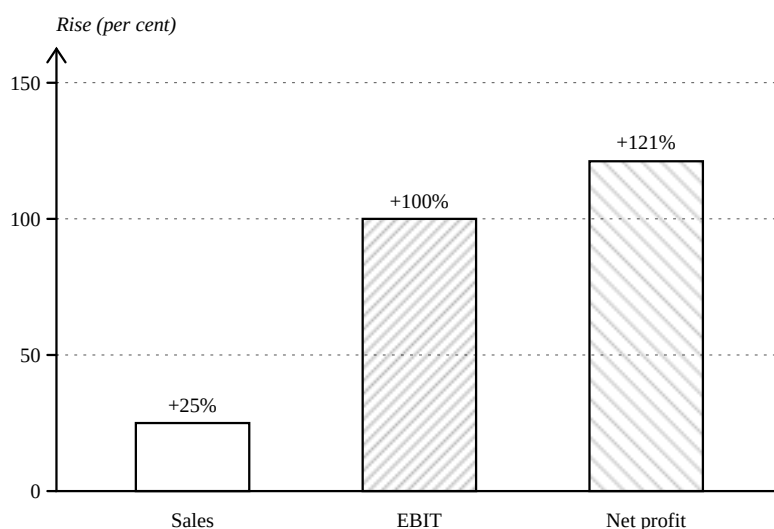


FIG. 27. A rise of 25 per cent in sales becomes a rise of 100 per cent in EBIT and 121 per cent in net profit.

The table below gathers the three measures. Use it as a summary and not as a substitute for the worked examples above.

A summary table gathers the three measures in one place.

THE THREE MEASURES OF LEVERAGE FOR THE BAKERY AT 10,000 BOXES

MEASURE	THE QUESTION IT ANSWERS	FORMULA	BAKERY
DOL	How much does a change in sales change EBIT?	Contribution ÷ EBIT	4.00
DFL	How much does a change in EBIT change net profit?	EBIT ÷ (EBIT – interest)	1.21
DTL	How much does a change in sales change net profit?	DOL × DFL	4.85

The most common error is to add the two measures instead of multiplying.

Before moving on: the DTL multiplies the two leverages and shows how a change in sales reaches the owners.

Leverage widens the range of results, which is the same as raising risk.

Worked case: sales may rise or fall by one-quarter with equal chance.

The next table gives the two outcomes, the expected value, the standard deviation and the coefficient of variation.

The coefficient of variation grows by the DOL, and then by the DFL.

Common slips. The most common error is to add the DOL and the DFL. The bakery's 4.0 plus 1.21 gives 5.21, which is wrong; the right answer is 4.85. The reason is that the second effect acts on an already enlarged change, so the two effects multiply. Another slip is to take the DTL from different levels of sales for the two parts: both measures must be found at the same starting level.

Before moving on. The DTL equals the DOL multiplied by the DFL. It tells by how many times a change in sales is magnified in net profit. You should now be able to say: "A DOL of 2.0 and a DFL of 1.5 give a DTL of 3.0, so a rise of 10 per cent in sales raises net profit by 30 per cent."



9.6 WHY LEVERAGE MATTERS TO A FIRM

Chapter 8 taught that risk means the chance that results differ from hopes, and that the standard deviation and the coefficient of variation measure it. Leverage connects directly to that idea. Leverage does not change the average result by itself. It widens the spread of results around the average. A wider spread is a higher risk. We now show this in numbers.

Case file, illustrative. Suppose the sales of the bakery have an equal chance, 0.5 each, of being 25 per cent below plan (7,500 boxes) or 25 per cent above plan (12,500 boxes). We compare the spread of sales, of EBIT, and of profit before tax. Interest is Tk 7,000. We use profit before tax so that the effect of tax does not hide the effect of leverage.

The table below gives the results. Each line shows the low outcome, the high outcome, the expected value (the average, weighted by chance), the standard deviation (the usual spread around the average) and the coefficient of variation (spread per unit of expected result). The numbers for the low and high outcomes come from the two earlier tables.

SPREAD OF RESULTS WHEN SALES ARE 25 PER CENT BELOW OR ABOVE PLAN (TK)

ITEM	LOW	HIGH	EXPECTED VALUE	STANDARD DEVIATION	COEFFICIENT OF VARIATION
Sales	3,00,000	5,00,000	4,00,000	1,00,000	0.25
EBIT	0	80,000	40,000	40,000	1.00
Profit before tax	-7,000	73,000	33,000	40,000	1.21

Reasoning. The expected value of sales is half of 3,00,000 plus half of 5,00,000, which is Tk 4,00,000. Each outcome lies Tk 1,00,000 from it, so the standard deviation is Tk 1,00,000 and the coefficient of variation is $1,00,000 \div 4,00,000 = 0.25$. For EBIT, the outcomes are 0 and 80,000; the expected value is Tk 40,000; each outcome lies Tk 40,000 from it; and the coefficient is 1.00. For profit before tax, the outcomes are -7,000 and 73,000; the expected value is Tk 33,000; each outcome lies Tk 40,000 from it; and the coefficient is $40,000 \div 33,000 = 1.21$.

Conclusion. The coefficient of variation rose from 0.25 for sales to 1.00 for EBIT, which is 4.0 times as large, the DOL. It rose again to 1.21, which is 1.21 times as large, the DFL. Overall, it is 4.85 times its starting value. *Check.* $0.25 \times 4.848 = 1.21$. The coefficient of variation of profit before tax equals the coefficient of variation of sales multiplied by the DTL. Leverage did not change the average outcome in the way that a better product would. It made the same swing in sales a far greater swing in profit.

What should a firm do with this knowledge? Business risk is mostly fixed by the kind of work the firm does, and by the machines it needs. Financial risk is mostly chosen, by the decision on how much to borrow. A sensible firm looks at the one when deciding the other. A firm with high fixed costs, such as a factory or an airline, should usually keep its debts modest, because the two leverages multiply. A firm with steady sales and low fixed costs can carry more debt more safely.

The same facts look different to different people. An owner sees the chance of a large gain and the chance of a large loss. A lender sees the chance that EBIT will fall below the interest and the loan will not be paid. A manager must balance both views, and must also remember that the people who work in the firm lose their jobs if the firm fails.

Leverage is not a fault. It is a tool that raises gains in good years and losses in bad years. The DOL, the DFL and the DTL do not say whether the risk is worth taking. They say how large the risk is. The decision belongs to the firm, and it uses a further idea, which we study in Chapter 10: the return that owners and lenders require for the risk they carry. Chapters 13 and 14 describe the sources of long-term money, and Chapter 15 puts a price on that money.

Before moving on. Leverage widens the spread of profit around its average. The coefficient of variation of profit is the coefficient of variation of sales multiplied by the DTL. Business risk is mostly given; financial risk is mostly chosen.

Conclusion and check: risk per taka rose by exactly the DOL and then the DFL.

A firm cannot remove its business risk, but it can choose its financial risk.

Different people care about leverage for different reasons.

Leverage is neither good nor bad; it is a tool that raises both gains and dangers.

Before moving on: leverage widens the spread of profit, and the firm chooses how much of it to add.



SUMMARY

Business risk comes from the firm's own work and shows in EBIT. Financial risk comes from borrowing and shows below EBIT. The break-even point is the sales at which profit is zero; it equals fixed cost divided by contribution per unit. Operating leverage, measured by $\text{DOL} = \text{contribution} \div \text{EBIT}$, magnifies changes in sales into larger changes in EBIT. Financial leverage, measured by $\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest} - \text{preferred dividend} \div (1 - t))$, magnifies changes in EBIT into larger changes in net profit. Total leverage, measured by $\text{DTL} = \text{DOL} \times \text{DFL}$, joins the two. For the bakery at 10,000 boxes, the break-even point is 7,500 boxes, DOL is 4.0, DFL is 1.21 and DTL is 4.85.

The chapter's main ideas in a few lines.

THINK IT THROUGH

Suppose, as an exercise only, that the bakery considers borrowing a further Tk 2,00,000 at 12% a year. Its interest would rise by Tk 2,000 a month to Tk 9,000. Use the figures of this chapter: EBIT Tk 40,000 and DOL 4.0

A problem: what happens if the bakery borrows a little more?

at 10,000 boxes. Find the new DFL and the new DTL. Then find what happens to profit before tax if sales fall by 25 per cent. Finally, say in two or three sentences whether you would advise the owner to borrow, and why. (Hint: the new DFL is $40,000 \div 31,000 = 1.29$, and the DTL is $4.0 \times 1.29 = 5.16$. If sales fall by 25 per cent, EBIT falls to zero and profit before tax is a loss of Tk 9,000.)

WHERE PEOPLE GO WRONG

Five common misunderstandings and their corrections.

The DOL changes with the level of sales.

The two leverages multiply.

High operating leverage magnifies gains as well as losses.

Divide fixed cost by contribution.

"Leverage means debt." Leverage is any magnifying effect of fixed payments. Operating leverage comes from fixed costs and needs no loan. Financial leverage comes from fixed payments to lenders.

"The DOL is a fixed number for a firm." It depends on the level of sales at which it is found, and it falls as sales rise above break-even.

"The DTL is the DOL plus the DFL." The two effects multiply. The bakery's DTL is $4.0 \times 1.21 = 4.85$, and not 5.21.

"A high DOL is always bad." It magnifies gains as much as losses. It is a risk, and the firm may accept it for a good reason, for example when the fixed cost buys a cheaper cost per unit.

"Break-even sales equal fixed cost." Fixed cost must be divided by the contribution per unit, or by the share of each taka of sales that is contribution.

IN HISTORY

A fund that borrowed more than twenty times its own money.

Long-Term Capital Management was an American investment fund that began trading in 1994. By early 1998 it had borrowed about US\$125 billion against about US\$4.7 billion of its own money, according to one study of the case.² With so little of its own money compared with its debt, a fall of less than four per cent in the value of what it held could erase its own money. That is our own arithmetic, using the same idea as this chapter: a fixed claim by lenders magnifies every change for the owners. In 1998 markets moved against the fund. In September 1998 the Federal Reserve Bank of New York brought the fund's main lenders together, and they put in about US\$3.6 billion of new money to prevent a disorderly collapse. The Federal Reserve did not lend its own money.³ The fund's history is a large-scale picture of financial leverage working in the wrong direction.

REVIEW QUESTIONS

Practice questions, with short answers at the back of the book.

Practice question 1.

Practice question 2.

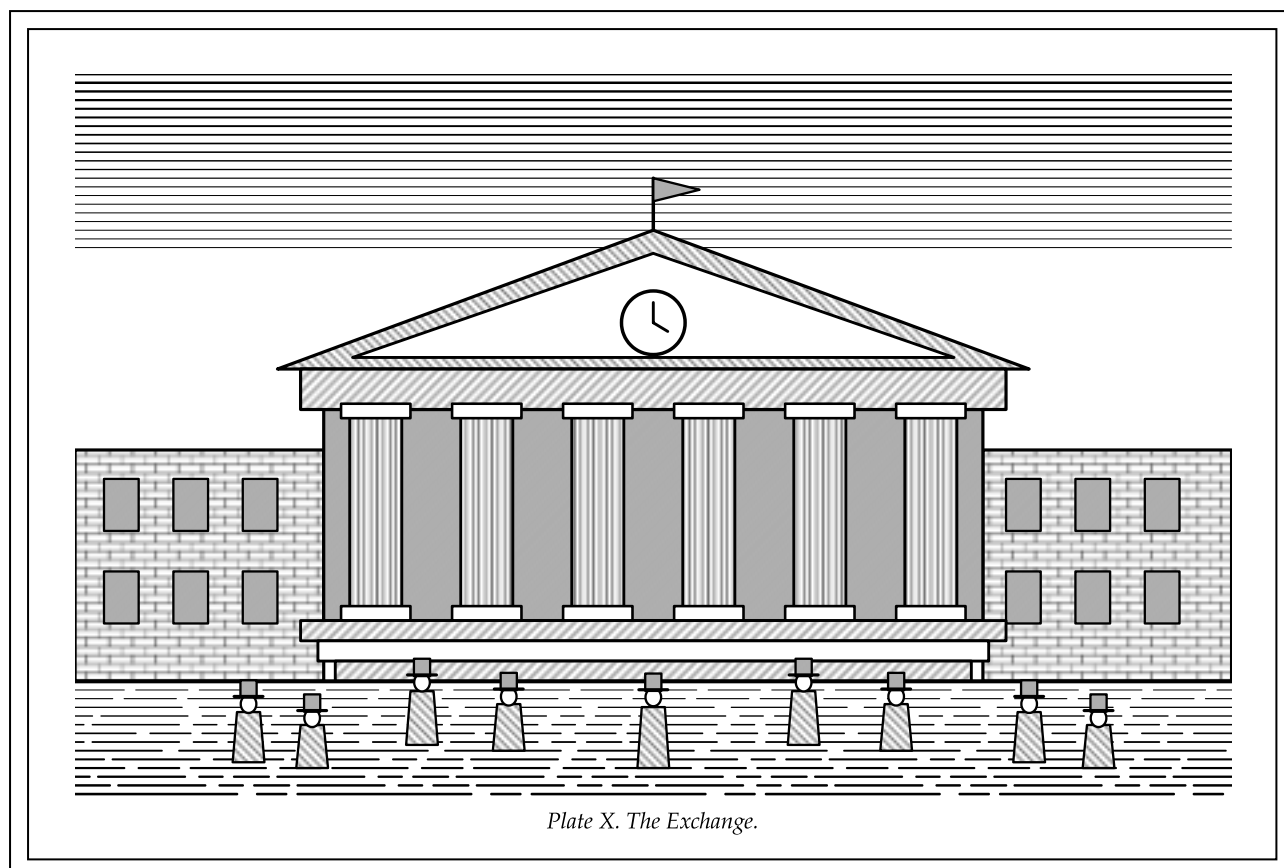
1. Explain business risk and financial risk. Say how they differ, and name one cause of each.

2. A printing shop sells a job for Tk 50. The variable cost of a job is Tk 30 and the fixed cost is Tk 1,00,000 a month. Find the break-even number of jobs and the break-even sales in taka.

² The figures for debt and equity are from Jacque, *Global Derivative Debacles* (2015), as summarised by the publisher; they are approximate and need verification before printing.

³ The date and the role of the Federal Reserve Bank of New York are described by the Federal Reserve's own history pages; reports differ on the number of lenders (13 to 15) and the sum (US\$3.5 to 3.6 billion), so only a rounded figure is used here.

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- | | |
|---|------------------------------|
| 3. For the shop in question 2, find EBIT at 4,000, 5,000 and 8,000 jobs a month. Then find the DOL at 8,000 jobs. | <i>Practice question 3.</i> |
| 4. The shop's fixed cost rises to Tk 1,20,000 a month. Find the new break-even number of jobs and say why it rose. | <i>Practice question 4.</i> |
| 5. Explain what a DOL of 3.0 tells a manager. What is the least value a DOL can take, and when does it take that value? | <i>Practice question 5.</i> |
| 6. A firm has a contribution of Tk 3,00,000 and an EBIT of Tk 1,20,000. It pays interest of Tk 30,000 a month. Find its DOL, DFL and DTL. | <i>Practice question 6.</i> |
| 7. The firm in question 6 also pays a preferred dividend of Tk 15,000 a month, and the tax rate is 25 per cent. Find its new DFL and DTL. | <i>Practice question 7.</i> |
| 8. Using your DTL from question 7, by how much does net profit change if sales rise by 10 per cent? | <i>Practice question 8.</i> |
| 9. A firm has a DFL of 1.0. What does this tell you about the way it is financed? | <i>Practice question 9.</i> |
| 10. Discuss why a firm with high fixed costs may decide to borrow little. | <i>Practice question 10.</i> |



CHAPTER X

Risk Premium, Required Return and the Market

What is the fair price of a chance?



This chapter links risk to the return that investors ask for. It shows how a safe return and an extra reward for risk make up the return a firm must earn, how one number called beta measures the part of risk that is rewarded, and how markets treat news. The ideas prepare the way for valuing securities and for pricing a firm's funds.

Two offers of money are placed side by side: one is sure, and one is a gamble.

Most people ask for more than the sure return before they accept a risk.

The firm faces the same question as the investor, because investors supply its money.

The chapter has four parts: the two ingredients, the rule, the number called beta, and how markets treat news.

Before this chapter, you should know risk, expected value and the standard deviation.

Lending to a trusted borrower earns little, lending to a stranger needs a bigger reward.

The risk-free return is what you earn when there is no real chance of loss.

Risk-free return. The return on an investment where there is no real chance of loss.

The risk-free return is risk-free in taka, but inflation can still reduce what the taka will buy.

Imagine that you have Tk 1,00,000 to invest for one year. A government offers you a certificate that pays 6 per cent, and the payment is certain. A friend offers you a share in a new business. The friend says the share could pay nothing, or it could pay 24 per cent, and on average it will pay 12 per cent. You must choose. Which would you take?

Most people would not accept the share if it offered only 6 per cent on average. The certificate gives the same average, and it gives it without worry. Most people would accept the share only if it promised a good deal more. But how much more? Would 7 per cent be enough? Would 12 per cent? The answer is a number, and it is not chosen by feeling. It is built from two parts, each of which this chapter teaches you to find.

The firm faces this same question, from the other side. The investors who supply a firm's money are people like you, with a certificate always available as an alternative. If a firm wants their money, it must promise a return that makes the risk worth taking. A project that cannot earn that return should not be started. The return that investors ask for is therefore the standard against which a firm judges its projects.

First we meet the two ingredients of the return that investors ask for. Next we see how risk changes it. Then we meet one number, called beta, and a rule that uses it. Last, we ask a question about markets themselves: do prices already know the news?

You will need: from Chapter 8, the idea of risk as the spread of possible results, the expected value, the standard deviation, and the two kinds of risk named there, market risk (which means risk that touches all firms) and firm-specific risk (which means risk that touches one firm).



10.1 THE RISK-FREE RETURN AND THE RISK PREMIUM

Think about lending money to two people. The first is your bank, which will return your money with a small, known payment. The second is a shop owner you hardly know, who says he will pay you back next year. You would not lend to the shop owner for the bank's small payment. You would ask for more, to pay you for the chance that he may not repay. The extra payment is the heart of this section.

The first part of any required return is the **risk-free return**. It means the return on an investment where there is no real chance of loss. No investment is free of every risk, so in practice people take as the risk-free return the rate on short-term loans to a government that borrows in its own money, such as treasury bills. The government can always create the money it owes. In this book, the risk-free return is the base on which every other return is built.

There is an exception. The government will repay the taka, but prices may rise in the meantime, so each taka may buy less. A return that is certain in taka is not certain in what it buys. For the purposes of this chapter we set that problem aside and treat the risk-free return as the starting point.

The second part is the **risk premium**. It means the extra return that investors ask for carrying extra risk. A safe investment needs no premium. A risky one needs a large premium. Putting the two parts together gives the rule of this section.

required return = risk-free return + risk premium.

In symbols, let R be the return that investors ask for, R_f the risk-free return, and RP the risk premium. Then $R = R_f + RP$. We give the name *required return* to the left-hand side in section 10.2.

Worked example 1. Given. The risk-free return is 5 per cent a year. Investors ask a risk premium of 3 per cent to hold shares in a shop. *Find.* The return they ask. *Formula.* $R = R_f + RP$.

Steps. $R = 5\% + 3\% = 8\%$. *Answer.* Investors ask 8 per cent a year. *Check.* The answer is more than the risk-free return of 5 per cent, which it must be if the premium is positive, and the extra part is exactly the 3 per cent premium.

Where does a premium come from in practice? Look at the market as a whole. The **market return** means the average return earned by all the shares in a market taken together, which is often measured by an index of many shares. The market is risky, so its return is higher than the risk-free return. The difference between them is the premium for holding the market as a whole: market premium = market return – risk-free return.

Case file, illustrative. In the bakery's story we take, for teaching, a risk-free return of 6 per cent a year and a market return of 12 per cent a year. These are not real market figures. They keep the arithmetic clean.

Worked example 2. Given. $R_f = 6\%$; market return = 12%. A new product for the bakery needs a return of 13.2%. *Find.* The market premium, and the premium that the new product must offer. *Formula.* Premium = required return – risk-free return.

Steps. The market premium is $12\% - 6\% = 6\%$. The premium for the new product is $13.2\% - 6\% = 7.2\%$. *Answer.* The market premium is 6 per cent, and the new product must offer a premium of 7.2 per cent. *Check.* The new product's premium is larger than the market's, which is reasonable only if the new product is riskier than the market. Section 10.3 shows that it is, by a factor of 1.2.

The three seats give three views of the premium. To an investor, the premium is the reward for waiting while the result is uncertain. To a manager, it is a cost that the firm must be able to pay, because it is the price of attracting the investor's money. To a lender, it is the reason that a loan to a risky firm carries a higher rate of interest than a loan to a safe firm. In every case the premium is the price of risk, and the price grows with the amount of risk.

The rule also shows what happens when the safe return changes. Suppose the risk-free return rises from 6 per cent to 8 per cent, and the new product's premium stays at 7.2 per cent. Its required return rises from 13.2 per cent to 15.2 per cent. If the safe return falls to 4 per cent, the required return falls to 11.2 per cent. Because the safe return is the base of every required return, a change in it moves the whole ladder up or down together.

The risk premium is the extra return asked for carrying extra risk.

Risk premium. The extra return that investors ask for carrying extra risk.

The rule in words and in symbols.

The symbols used in the rule.

First example: small numbers, a clean rule.

Steps, answer and check.

The market return gives a premium for the market as a whole.

Market return. The average return earned by all the shares in a market taken together.

The case file gives illustrative figures for the risk-free return and the market return.

Second example: the market premium and a larger premium for a riskier use of money.

Steps, answer and check.

The investor, the manager and the lender each use the premium in their own way.

A change in the safe return moves every required return by the same amount.

The figure shows the required return as a stack: the safe return, and then the premium.

The figure below shows the idea as a stack. Read it from left to right. The first column is a safe government loan: it has only the risk-free part. The second column is the market as a whole. The third is the new product. In each column, the plain block is the risk-free return and the hatched block is the risk premium. The riskier the use of money, the taller the hatched block.

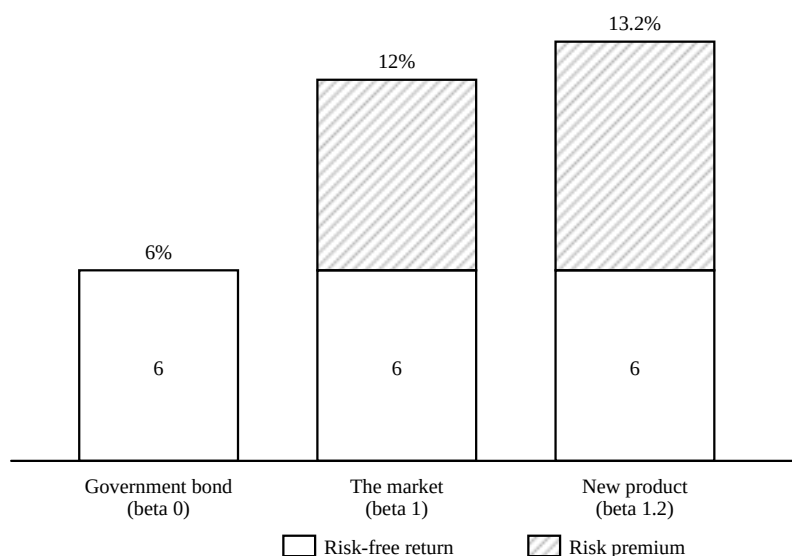


FIG. 28. Required return as a stack: the risk-free return, and a premium that grows with risk.

Figures for the risk-free return and the premium change with time and place.

Before moving on: required return = risk-free return + risk premium.

Both parts of the rule change. The risk-free return rises and falls with the economy, with the government's needs, and with the country. The premium changes with how worried investors are. A figure for either one is an estimate for a certain time and place, and must be dated when it is used. The figures in this book are chosen for teaching.

Before moving on. The return that investors ask has two parts: a safe return for waiting, and a premium for carrying risk. You should now be able to say: "If the safe return is 5 per cent and the premium is 4 per cent, investors ask for 9 per cent."



10.2 RISK AND THE REQUIRED RETURN

GO SLOWLY HERE.

The return that investors ask for has a name, and it is a name for an idea you already know.

Required return. The least return an investor will accept to hold an asset, given its risk.

The required return is the same idea as the discount rate of Chapters 4 to 7.

We now give a name to the sum of the two parts. The **required return** means the least return an investor will accept to hold an asset, given its risk. If an asset's expected return is above its required return, investors want it. If it is below, they do not.

You have met this idea before under another name. In Chapters 4 to 7 we used a *discount rate* to bring later money back to today. That rate is the return that an investor could earn on another use of money of equal risk. It is therefore the required return. From this chapter on, this book uses the name required return, and the two names stand for one idea. In Chapter 7, the least return that a project must earn was called the *hurdle rate*. The hurdle rate is the required return, applied to a project.

The rule for decisions follows. Compare the return a project is expected to earn with the required return. If the expected return is at least as large, the project is acceptable. If it is smaller, the project would leave investors worse off than their alternative, and it should be rejected. For the bakery's new product, with a required return of 13.2 per cent, an expected return of 15 per cent passes the test and an expected return of 11 per cent fails it.

Recall the two product ideas of Chapter 8. The cake line and the café corner both had an expected return of 12 per cent, but the standard deviation of the cake line was 2.83 per cent and that of the café corner was 8.49 per cent. The expected returns are equal and the risks are not, so investors would not ask the same return from each. Suppose, as a thought exercise only, that investors required 11 per cent for the cake line and 14 per cent for the café corner. Then the cake line, which is expected to earn 12 per cent, passes its test by one point. The café corner, which is also expected to earn 12 per cent, fails its test by two points. The lesson is that an expected return is judged against a required return that depends on risk, and not against one number for every project. (Section 10.3 will refine which kind of risk counts.)

The link between risk and required return can be shown as a ladder. The table below goes from the safest use of money at the top to the riskiest at the bottom. Read each row across: the first column names the use of money, the second its risk, and the third the return that investors ask for.

THE LADDER OF RISK AND REQUIRED RETURN

USE OF MONEY	RISK	RETURN ASKED
Short-term loan to a government	Lowest	Lowest: the risk-free return
Loan to a large, steady firm	Low	A little above the risk-free return
Shares in a large, steady firm	Medium	Risk-free return plus a moderate premium
Shares in a young firm with uncertain sales	High	Risk-free return plus a large premium

There is a puzzle in this ladder. In Chapter 8, risk was measured by the standard deviation, which counts all the spread in a result. But a large part of that spread can be removed by an investor, and an investor is not paid a premium for risk that can be removed for free. We show how in the next paragraphs.

Take two shares. The first belongs to a firm that sells umbrellas, and the second to a firm that sells ice cream. In a rainy season the umbrella share returns 16 per cent and the ice-cream share returns 4 per cent. In a sunny season the returns are the other way round. Each season has an equal chance, 0.5. The table below gives the returns for each share, and for an investor who puts half of her money in each.

A project passes the test if its expected return is at least the required return.

Two ideas with equal expected returns can still deserve different required returns.

Higher risk means a higher required return, as a ladder shows.

The risk that matters is not all risk; spreading money across many firms removes some of it.

A small table shows two shares whose swings cancel.

RETURNS ON TWO SHARES, AND ON HALF OF EACH (PER CENT)

SEASON	UMBRELLA SHARE	ICE-CREAM SHARE	HALF IN EACH
Rainy	16	4	10
Sunny	4	16	10
Expected value	10	10	10
Standard deviation	6	6	0

Holding both gives a steady 10 per cent although each share alone swings.

Diversification. Spreading money over many investments, so that the ups of some cancel the downs of others.

Diversification removes firm-specific risk, but not market risk.

Before moving on: the required return is the discount rate; the premium pays for the risk that cannot be spread away.

Each share alone has an expected value of 10 per cent and a standard deviation of 6 per cent. The investor who holds half of each earns 10 per cent in either season, and her standard deviation is zero. This spreading of money over many investments is called **diversification**. Diversification means spreading money over many investments so that the ups of some cancel the downs of others. In the real world the cancelling is never complete, but the effect is large.

Diversification can remove firm-specific risk, which is the risk that touches one firm, such as a fire in its factory or a lost contract. It cannot remove market risk, which touches all firms together, such as a general fall in demand. An investor who holds many shares can dilute the first kind of risk, and she keeps the second. She is therefore paid a premium for market risk only. This is a deep idea, and it prepares the next section: we need a number that measures only the market risk of a share.

Before moving on. The required return is the least return investors accept, given the risk. It is the same idea as the discount rate. Investors are paid for market risk, which diversification cannot remove, and not for firm-specific risk. You should now be able to say: "A project whose expected return is below the required return should be rejected."



10.3 BETA AND THE CAPITAL ASSET PRICING MODEL

GO SLOWLY HERE.

One number tells how strongly a share moves with the market.

Beta measures how strongly a security moves with the market.

Beta. A number showing how strongly a security's return moves with the market's return.

An example: what a beta of 1.2 means in a falling market.

Suppose the market as a whole rises by 10 per cent in a year. Some shares will rise by more than 10 per cent, and some by less. Some may fall. A share that tends to move with the market, and more strongly, carries more market risk than a share that moves weakly. We need a single number that records this tendency.

That number is **beta**. Beta means a number that shows how strongly the return on a security moves with the return on the market. A beta of 1.0 means that the security moves, on average, as the market does. A beta of 1.5 means that for each 1 per cent move in the market, the security tends to move by 1.5 per cent in the same direction. A beta of 0.5 means that it moves by one-half per cent. A beta of zero means that it does not follow the market at all, as with a safe government loan.

For example, a share with a beta of 1.2 would be expected, on average, to rise by 12 per cent when the market rises by 10 per cent, and to fall by 12 per cent when the market falls by 10 per cent. The words "on average" and "expected" matter. Beta describes a tendency found in past data, and it does not forecast any single year.

The table below shows what several betas suggest when the market falls by 10 per cent. Read each row across: the beta, then the average move that a share with that beta tends to make. Betas below 1.0 fall by less than the market, and betas above 1.0 fall by more.

WHAT EACH BETA SUGGESTS WHEN THE MARKET FALLS BY 10 PER CENT

BETA	AVERAGE MOVE OF THE SHARE (%)
0	0
0.5	–5
1.0	–10
1.5	–15

Where does beta come from? It is found by statistics, from past returns of the security and of the market. This book does not ask you to compute it. A share's beta is published by many data services, and a manager usually takes it from there. A small firm such as the bakery has no listed shares, so it has no beta of its own. A manager then uses the beta of similar listed firms as an estimate.

Beta tells us how much of the market premium a security should earn. The **capital asset pricing model (CAPM)** is a rule that finds the required return from the risk-free return, the market premium and beta. In words:

required return = risk-free return + beta $\times$ (market return – risk-free return).

In symbols, $R = R_f + \beta \times (R_m - R_f)$, where R is the required return, R_f is the risk-free return, β (the Greek letter beta) is the security's beta, and R_m is the market return. The part in brackets is the market premium of section 10.1. The rule says: take the market premium, multiply by beta, and add the risk-free return.

Worked example 3. Given. $R_f = 5\%$; $R_m = 10\%$; $\beta = 1.5$. *Find.* The required return. *Formula.* $R = R_f + \beta \times (R_m - R_f)$.

Steps. The market premium is $10\% - 5\% = 5\%$. Multiply by beta: $1.5 \times 5\% = 7.5\%$. Add the risk-free return: $5\% + 7.5\% = 12.5\%$. *Answer.* The required return is 12.5 per cent. *Check.* A beta above 1.0 should give a required return above the market return of 10 per cent, and 12.5 is above 10.

Case file, illustrative. Sundial Bakery considers a new product. For teaching, take a risk-free return of 6 per cent, a market return of 12 per cent, and, for similar listed firms, a beta of 1.2. These figures are illustrative.

Worked example 4. Given. $R_f = 6\%$; $R_m = 12\%$; $\beta = 1.2$. *Find.* The required return of the new product. *Formula.* $R = R_f + \beta \times (R_m - R_f)$.

Steps. The market premium is $12\% - 6\% = 6\%$. Multiply by beta: $1.2 \times 6\% = 7.2\%$. Add the risk-free return: $6\% + 7.2\% = 13.2\%$. *Answer.* The required return is 13.2 per cent. *Check.* With a beta of 1.0, the answer would be the market return of 12 per cent; the beta of 1.2 adds $1.2 - 1.0 = 0.2$ of the 6 per cent premium, which is 1.2 per cent, and $12\% + 1.2\% = 13.2\%$. The two methods agree.

A small table shows what each beta suggests when the market falls by 10 per cent.

Beta is estimated from past data, and its value is usually taken from a data service.

The capital asset pricing model turns beta into a required return.

Capital asset pricing model (CAPM). A rule that finds the required return from the risk-free return, the market premium and beta.

The CAPM formula in words and symbols.

The symbols used in the CAPM formula.

First CAPM example, with clean numbers.

Steps, answer and check for the first CAPM example.

The bakery's new product, the case file figures.

Second CAPM example, with the bakery's figures.

Steps, answer and check for the second CAPM example.

A messier example, with a beta below one.

The decision: compare the expected return of the new product with 13.2 per cent.

If the market return changes, the required return changes by beta times the change.

The security market line shows the required return for every beta.

Security market line. A line that shows the required return for each value of beta.

The figure draws the line; to read it, find a beta at the foot and go up to the line.

Worked example 5. A second security has a beta of 0.8, with the same figures. The market premium is 6%. $0.8 \times 6\% = 4.8\%$. The required return is $6\% + 4.8\% = 10.8\%$. *Check.* A beta below 1.0 should give a required return between the risk-free return of 6% and the market return of 12%, and 10.8 lies between them.

The decision for the bakery follows from section 10.2. If the new product is expected to earn 15 per cent, it passes the test, because 15 is above 13.2. If it is expected to earn only 11 per cent, it fails, because 11 is below 13.2. The result is the discount rate for the new product, and it is the figure that Chapter 15 will use when the bakery retests its projects.

The required return depends on the market premium, and the market premium changes with the market return. Keep the risk-free return at 6 per cent and the beta at 1.2. If the market return is 10 per cent, the premium is 4 per cent and the required return is $6\% + 1.2 \times 4\% = 10.8\%$. If the market return is 14 per cent, the premium is 8 per cent and the required return is $6\% + 1.2 \times 8\% = 15.6\%$. The new product's required return moves by 1.2 times the change in the market premium, which is what a beta of 1.2 says it should do.

If we calculate the required return for several betas, a pattern appears. The table below does this with the case file figures. The **security market line** means a line that shows the required return for each value of beta. It starts at the risk-free return, where beta is zero, and rises in a straight line.

REQUIRED RETURN FOR DIFFERENT BETAS (RF = 6%, RM = 12%)

BETA	CALCULATION	REQUIRED RETURN (%)
0	$6 + 0 \times 6$	6.0
0.5	$6 + 0.5 \times 6$	9.0
1.0	$6 + 1.0 \times 6$	12.0
1.5	$6 + 1.5 \times 6$	15.0
2.0	$6 + 2.0 \times 6$	18.0

The figure below draws the line. To read it, find a beta on the bottom edge, go straight up to the line, and read across to the left edge. The first dot is the risk-free return, at beta zero. The second dot is the market, at beta 1.0. The open circle is the bakery's new product, at beta 1.2. The upright arrow shows its risk premium, which is the distance between the safe return and the line.

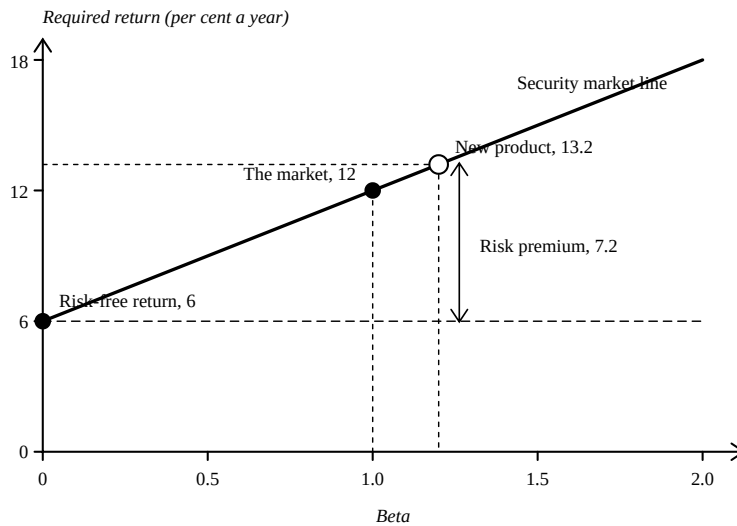


FIG. 29. The security market line: the required return rises in a straight line as beta rises.

Limits and common confusions. Beta measures only market risk. The standard deviation of Chapter 8 measures all risk. A share with a high standard deviation may have a low beta if its swings do not follow the market, and the CAPM would then ask for a low premium. Beta is also an estimate from the past, and different data services give different values. The risk-free return and the market return are estimates too. For these reasons the answer from the CAPM is a guide and not a fact.

Before moving on. Beta measures how strongly a security moves with the market. The CAPM turns beta into a required return by multiplying the market premium by beta and adding the risk-free return. You should now be able to say: "With a risk-free return of 6 per cent, a market return of 12 per cent and a beta of 1.2, the required return is 13.2 per cent."



Beta is not the same as the standard deviation, and it has limits.

Before moving on: required return = risk-free return + beta × market premium.

10.4 EFFICIENT MARKETS

Suppose a large firm announces at nine in the morning that it has won a very big contract. By ten, its share price has risen. By eleven, it has stopped rising. Can you, reading the news at noon, still buy the share at a bargain? Most people who study markets would say you cannot, because the price already knows the news. This idea has a name.

An **efficient market** means a market in which prices already reflect the information that is available. In such a market, a share price is a fair summary of what is known. Nobody can find a bargain by using information that the price has already absorbed. Because information is of different kinds, three forms of efficiency are described.

The *weak form* says that prices already reflect all past prices and trading figures. If it is true, studying charts of old prices cannot earn extra profit. The *semi-strong form* says that prices reflect all public information, which includes company reports, news and government announcements. If it is true, reading public news after it appears cannot earn extra profit. The *strong form* says that prices reflect all information, including private

A puzzle about news: why do prices move so fast?

An efficient market is one where prices already reflect the information available.

Efficient market. A market in which prices already reflect the information that is available.

There are three forms of efficiency, which differ in the information the price has absorbed.

The figure shows the three forms as circles inside one another.

information that only insiders hold. If it is true, even insiders cannot profit.

The figure below shows the three forms as circles, one inside the other. Read it from the middle. The inner circle is the weak form, which covers the least information. The middle ring adds all public information. The outer ring adds private information. A market that is efficient in a wider form is also efficient in the narrower forms inside it.

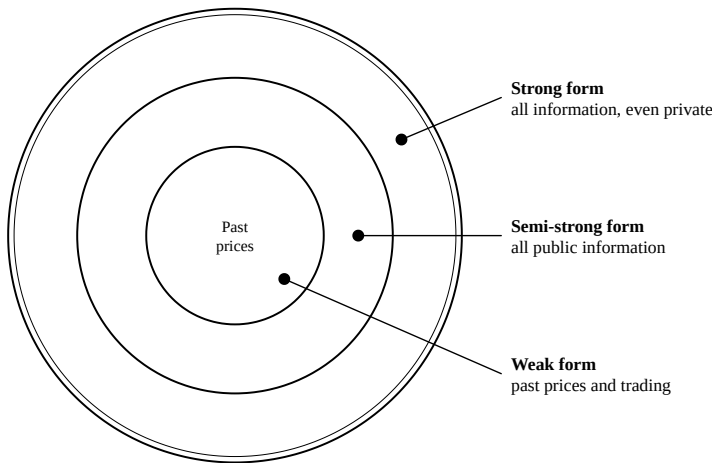


FIG. 30. The three forms of market efficiency, each containing the one inside it.

A table sets out what each form says and what it would mean for an investor.

The table below sets out the three forms. Read each row across: the form, the information that the price has absorbed, and what the form means for an investor who hopes to earn an extra profit.

THE THREE FORMS OF AN EFFICIENT MARKET

FORM	PRICES REFLECT	MEANING FOR AN INVESTOR
Weak	Past prices and trading	Studying old price charts gives no extra profit
Semi-strong	All public information	Reading published news, after it appears, gives no extra profit
Strong	All information, even private	Not even insiders earn an extra profit

Most people doubt the strong form, and the use of private information is controlled by law.

The strong form is the hardest to believe. Insiders often do seem to know things that outsiders do not, and using such secret information to trade is forbidden in many countries. Most writers therefore treat the weak and the semi-strong forms as the useful ones. Even these are tested against evidence, and the results differ between markets and between periods.

Efficiency matters to the manager because the share price is a fair judge of what the firm does.

Why does a financial manager need this idea? If the market is efficient, the share price reflects what the market knows about the firm. A manager cannot raise the price by hiding bad news, because the market will uncover it, and cannot gain by guessing the best time to sell shares, because the price already allows for what is known. What raises the price is a better firm. This agrees with the goal studied in Chapter 2.

If markets are efficient, a share's price will stay close to the value that its future payments justify.

The idea of efficiency prepares the next chapter. In Chapter 11 we value a security as the present value of what it will pay, discounted at its required return. If the market is efficient, the price in the market and the value found in that way will be close to each other, because the market has already absorbed what is known about the future payments. A large

gap between the two would be a puzzle that needs an explanation, such as new information or an error in the estimates.

Common confusions. An efficient market does not mean that prices never fall, or that the price is always right when looked at later. It means that no easy extra profit can be found by using information the price has already absorbed. A market with few traders and little public information is more likely to fall short of efficiency than a large and closely watched one. The efficient-market idea and the CAPM are separate ideas, and a student may accept one without the other.

Before moving on. In an efficient market, prices already reflect available information. The weak, semi-strong and strong forms differ in the kind of information. You should now be able to say: "If the semi-strong form is true, I cannot earn an extra profit by reading a company's published report after it appears."

Efficient does not mean perfect: prices can be wrong in hindsight, and some markets are less efficient.

Before moving on: an efficient market's prices already reflect the information at the level of its form.



SUMMARY

The return that investors ask for has two parts: the risk-free return and a risk premium. This required return is the same idea as the discount rate, and applied to a project it is the hurdle rate. Investors are paid for market risk, because diversification can remove firm-specific risk. Beta measures how strongly a security moves with the market. The CAPM gives the required return as $R = R_f + \beta \times (R_m - R_f)$, and the security market line draws the rule. With $R_f = 6\%$, $R_m = 12\%$ and $\beta = 1.2$, the bakery's new product has a required return of 13.2%. An efficient market is one where prices already reflect available information, in the weak, semi-strong or strong form.

The chapter's main ideas in a few lines.

THINK IT THROUGH

A firm considers a project with a beta of 0.9. The risk-free return is 7 per cent and the market return is 13 per cent. The project is expected to earn 12.5 per cent. Find the required return and say whether to accept. Then suppose that a different data service says the beta is 1.0, and decide again. In two or three sentences, say what this teaches about using the CAPM. (Hint: the first required return is $7\% + 0.9 \times 6\% = 12.4\%$, so the project passes by a very small margin. With a beta of 1.0 the required return is 13.0%, and the project fails.)

A problem: a project that only just passes or fails, and what that teaches.

WHERE PEOPLE GO WRONG

"A risk-free return has no risk at all." It has no real chance of loss in taka, but prices may rise and reduce what the taka buy.

"A risk premium is a guaranteed extra." It is the extra return investors ask for carrying risk. It is an expected reward, and some years it is not earned.

"Beta measures all of a share's risk." Beta measures only the risk that moves with the market. The standard deviation measures the whole spread.

Five common misunderstandings and their corrections.

A premium is asked for, not promised.

Beta is a different measure from the standard deviation.

A high beta means high market sensitivity, not a poor firm.

Efficiency does not mean that prices are never wrong.

A rule for the price of risk, and a way of sorting the evidence on markets.

The two landmark papers behind this chapter.

The second landmark paper.

Practice questions, with short answers at the back of the book.

Practice question 1.

Practice question 2.

Practice question 3.

Practice question 4.

Practice question 5.

Practice question 6.

Practice question 7.

"A high beta means a bad firm." A high beta means that the share moves strongly with the market. Investors ask a higher return for it. The firm may be sound.

"An efficient market means prices are always right." It means that no easy extra profit can be found from information the prices already reflect.

IN HISTORY

Before the 1960s, investors knew that both risk and return mattered, but they had no accepted rule for how much extra return a given risk deserved. In 1964 William Sharpe published a paper in the *Journal of Finance* that supplied such a rule. It is the model we call the CAPM.¹ In 1970 Eugene Fama published a review of the evidence on how prices respond to information, in the same journal, and he sorted the question into the three forms of efficiency that this chapter describes.² Both men later received the Nobel memorial prize in economic sciences: Sharpe in 1990, and Fama in 2013.³

SOURCES

Sharpe, W. F. (1964). *Capital asset prices: a theory of market equilibrium under conditions of risk*. *Journal of Finance*, 19(3). It proposed the model of required return that rests on beta.

Fama, E. F. (1970). *Efficient capital markets: a review of theory and empirical work*. *Journal of Finance*, 25(2). It set out the weak, semi-strong and strong forms.

REVIEW QUESTIONS

1. Define the risk-free return and the risk premium, and say how they combine into the return that investors ask for.

2. The risk-free return is 5 per cent. Investors ask a premium of 4.5 per cent to hold the shares of a small firm. Find the required return.

3. The risk-free return is 7 per cent and the market return is 13 per cent. Find the market premium. Then find the required return for shares with a beta of 0.5, of 1.4, and of 0.

4. A share has a beta of 0.8. The risk-free return is 6 per cent and the market return is 12 per cent. Find the required return and the risk premium of this share.

5. A firm with a required return of 13.4 per cent is offered a project that is expected to earn 13 per cent. Should it accept? Give your reason.

6. Explain what a beta of 1.5 tells an investor. If the market falls by 8 per cent, what fall would you expect in the share, on average?

7. Explain why an investor who holds many shares asks a premium for market risk and not for firm-specific risk.

¹ Sharpe's paper is cited under Sources. The facts of year, journal and title need verification before printing.

² Fama's review is cited under Sources. The year and the three-form classification should be checked against the paper.

³ The years of the Nobel memorial prizes in economic sciences (1990 for Sharpe; 2013 for Fama) should be verified against the prize committee's records.

8. State the three forms of an efficient market and the information each one covers. Which form would be broken by an investor who earns an extra profit from secret news before it is announced?

Practice question 8.

9. Explain how the discount rate of Chapters 4 to 7 relates to the required return of this chapter.

Practice question 9.

10. Discuss why a manager should use a required return that depends on risk, and not one rate for every project.

Practice question 10.

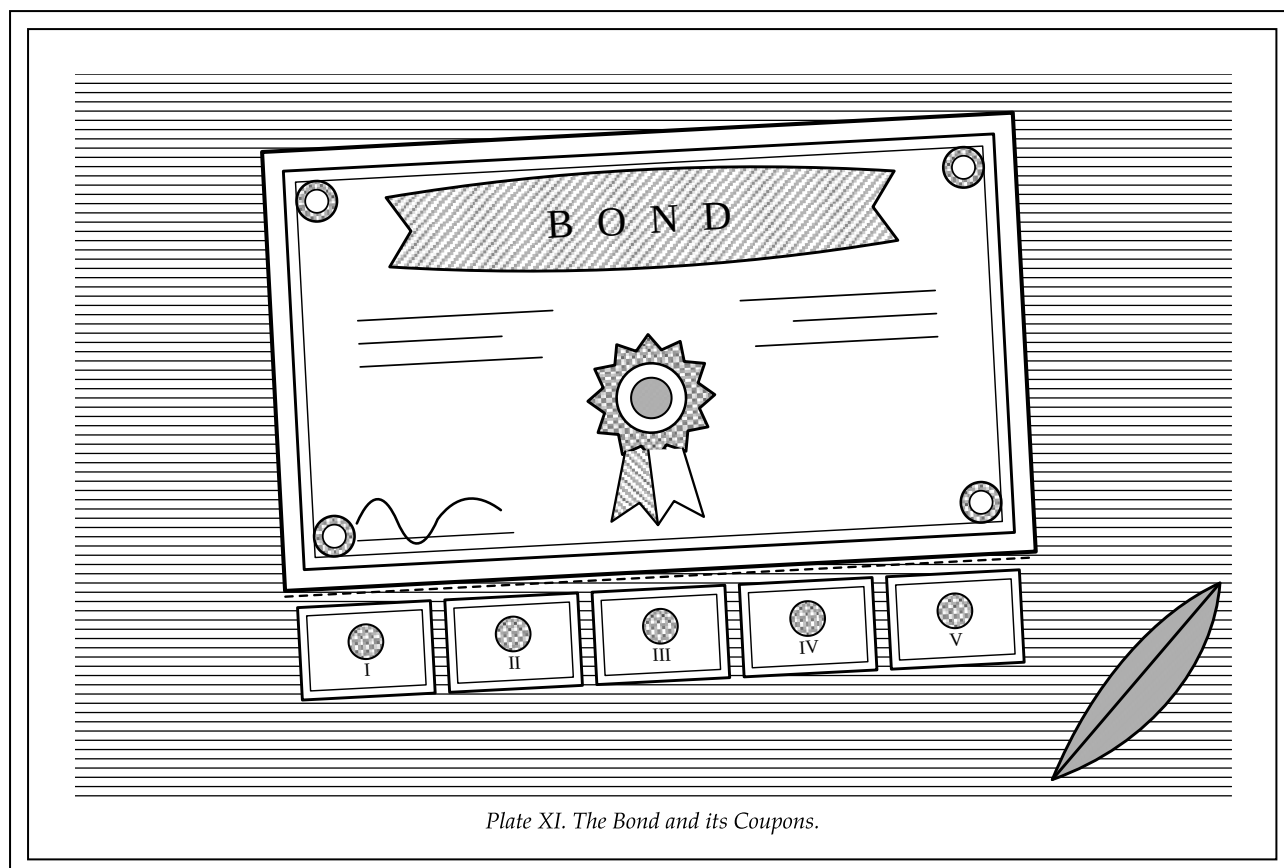


Plate XI. The Bond and its Coupons.

CHAPTER XI

Valuing Bonds and Shares

What is a promise worth today?



This chapter shows how to put a price on a security by adding up the present value of everything it will pay. It teaches the features of a bond, the value of a bond with yearly and half-yearly interest, and why a bond sells at a discount, at a premium or at par. It then values a share from its dividends and compares bonds with shares.

A neighbour offers to sell you a promise of future money, and you must name a fair price.

The answer needs two things you already have: the timing of money and the return that you could earn elsewhere.

This chapter turns those two tools into a method for pricing any promise of future money.

Before this chapter, you should know present value, annuity factors, and the required return.

A neighbour who owns a small shop offers you a piece of paper. On it he has written that he will pay the holder Tk 100 at the end of each year for five years, and Tk 1,000 at the end of the fifth year as well. He asks Tk 1,000 for the paper. You stare at it. The total of the payments is Tk 1,500, and you are asked to pay Tk 1,000 now. Is that a good price, a poor price, or a fair one?

The answer cannot come from adding the payments. A taka paid in five years is worth less than a taka today, as Chapter 4 showed, and the paper's risk matters too, as Chapter 10 showed. To price the paper you must know when each payment arrives and what return you could earn on another use of your money of equal risk. You have both tools. The first is present value. The second is the required return.

This chapter joins the two tools. We start with a single idea that prices every promise of money: its value today is the present value of all that it will pay. We then use the idea on two kinds of paper. The first is the **bond**, which pays a fixed sum each year and repays a fixed sum at the end. The second is the share, which pays a dividend that can rise over time. At the end we put them side by side.

You will need: from Chapter 5, the present value of an annuity and of a single sum, together with the factor tables; from Chapter 10, the required return, which is the same idea as the discount rate.



11.1 VALUE AND PRICE: THE FUNDAMENTAL IDEA

GO SLOWLY HERE.

A farmer's fruit tree gives a plain picture of how an asset is valued.

The value of anything that pays money later is the present value of its payments.

Intrinsic value. What an asset is worth, found by adding the present value of everything it is expected to pay.

The rule has three parts: the payments, their timing, and the discount rate.

Begin with a fruit tree. A farmer owns a mango tree that will give fruit worth Tk 5,000 every year for the next ten years, and then it will die. What is the tree worth to him today? He could sell it for a price, and a buyer would think carefully. The buyer would not pay Tk 50,000, the total of the ten harvests, because the later harvests arrive later. He would pay the present value of ten harvests of Tk 5,000, discounted at the return that he could earn on another use of the money. The tree is worth what it will give, brought back to today.

This idea has a name. The **intrinsic value** of an asset means what the asset is worth, found by adding up the present value of everything it is expected to pay. It is a value that comes from the asset's own future payments. It does not come from what other people happen to be offering today. The rule has three parts, and each one is something you have already learned.

First, list the payments the asset will make, and the year of each. Second, bring each payment back to today by dividing by $(1 + r)$ for each year of waiting, where r is the discount rate. Third, add the results. In symbols, the intrinsic value is $V = CF_1 \div (1 + r) + CF_2 \div (1 + r)^2 + \dots + CF_n \div (1 + r)^n$, where CF_1 is the payment at the end of year 1, and so on. The discount rate r is the required return of Chapter 10: the least return that investors will accept for an asset of this risk.

We use the rule for **securities**. A security is a tradable claim to future payments, such as a bond or a share. People buy and sell securities every day, and the price paid for one is its market price. The intrinsic value is the value that the rule gives. The price and the value are different things, and the next paragraph shows why the difference matters.

If the price is below the intrinsic value, the security is cheap, because you pay less than its payments are worth. If the price is above it, the security is dear. Chapter 10 told us that in an efficient market the two stay close, because prices already absorb what is known. We use the intrinsic value as the standard, and we treat any gap between it and the price as a signal to look more closely.

An everyday bridge may help. You lend Tk 10,000 to a friend, who promises to repay Tk 11,000 in one year. If your required return is 10 per cent, the value of the promise is $Tk\ 11,000 \div 1.10 = Tk\ 10,000$, which is exactly what you lent. If your required return were 15 per cent, the value would be $Tk\ 11,000 \div 1.15 = Tk\ 9,565$ (to the nearest taka). The promise is the same in both cases, but its value to you differs, because your required return differs. We shall see the same thing with bonds.

Before moving on. The intrinsic value of a security is the present value of all its expected payments, found with the required return. You should now be able to say: "A promise of Tk 1,100 in one year is worth Tk 1,000 today if my required return is 10 per cent."



11.2 BOND FEATURES

Chapter 1 explained that a lender gives money and expects it back with interest. Suppose a firm needs Tk 10 crore. Few lenders have that much. The firm can split the loan into 1,00,000 units of Tk 1,000 each and sell the units one by one to many lenders. A unit of such a loan is a bond. A bond is a loan split into units that can be traded. The person who buys one has lent Tk 1,000 to the firm, and may later sell the unit to someone else.

The firm or government that borrows by selling bonds is the **issuer**, and the person who holds a bond is the bondholder. The bondholder is a lender and not an owner. This matters, because lenders are paid before owners, and they have no vote in the firm. Chapter 13 describes the kinds of long-term debt, including debentures and mortgages, and Chapter 14 describes owners' shares. In this chapter we take only the features that are needed to value a bond.

Four features fix everything that a bond pays. The **face value** means the sum that the bond repays at the end. It is also called *par*, and it is the amount on which interest is calculated. The **coupon rate** means the yearly interest rate on the face value. The coupon rate times the face value gives the **coupon**, which is the yearly interest payment in taka. The **maturity** means the date on which the bond is repaid, so the time to maturity is the number of years left until that date. For example, a bond with a face value of Tk

A security is a tradable claim to future payments, such as a bond or a share.

Security. A tradable claim to future payments, such as a bond or a share.

Price is what people pay, and value is what the payments justify.

An everyday bridge: a loan to a friend is a security with one payment.

Before moving on: the value of a security is the present value of what it will pay, discounted at the required return.

A bond is a loan split into small units, so that many lenders can each take a part.

The firm that issues a bond is called the issuer, and the buyer is the bondholder.

Four features fix a bond's payments: face value, coupon rate, interest payment, and maturity.

Face value. The sum that a bond repays at its maturity; also called *par*.

Coupon rate. The yearly interest rate on a bond's face value.

Maturity. The date on which a bond is repaid.

A bond's coupon is fixed when the bond is issued, and it does not change when the market changes.

A bond's payments can be drawn on a timeline.

1,000 and a coupon rate of 10 per cent pays a coupon of Tk 100 each year.¹

The coupon is fixed on the day that the bond is issued. If market rates later rise, the coupon stays the same. This single fact explains the main surprise of the chapter, which is that the value of a bond rises and falls with the market even though the bond's payments do not. We return to it in section 11.4.

The figure below shows a five-year bond on a timeline. Read it from left to right. Year 0 is today. The short arrows are the coupons of Tk 100 at the end of years 1 to 4. At the end of year 5 there are two payments: the last coupon of Tk 100 and the face value of Tk 1,000, which the tallest arrow stands for. The box underneath says what we shall do with the timeline.

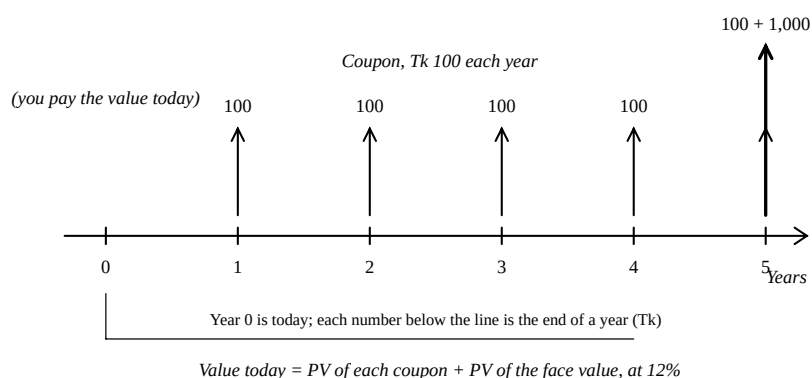


FIG. 31. The payments of a five-year bond with a face value of Tk 1,000 and a coupon rate of 10 per cent.

The same bond is an annuity plus a single sum, which the reader already knows how to value.

The bond-value formula in words and symbols.

A bond's risk depends on the issuer, and the required return reflects it.

The timeline shows a pattern that you know. The coupons are equal payments at equal gaps: that is an annuity, from Chapter 5. The face value is a single sum, paid once, from Chapter 4. So the value of a bond is the present value of an annuity plus the present value of a single sum. No new arithmetic is needed. In words and in symbols:

value of a bond = coupon $\times$ annuity factor + face value $\times$ present value factor.

In symbols, $V = C \times AF(r, n) + F \times PVF(r, n)$. Here C is the yearly coupon in taka, F is the face value, r is the required return for one year, n is the number of years to maturity, AF is the present value annuity factor, and PVF is the present value factor of a single sum.

Different bonds have different risk. A bond from a government that borrows in its own money is nearly risk-free. A bond from a young firm is not. The required return in the formula is chosen to match the bond's risk, using the ideas of Chapter 10: the risk-free return plus a premium that grows with risk. Some bonds are also backed by an asset that the lender may take if the issuer does not pay, and others are not. Chapter 13 describes these differences.

¹ In British usage, a bond that is issued by a firm and backed only by the firm's general credit is often called a *debenture*. Chapter 13 explains the term. In this chapter we use the word bond for all such loans.

Before moving on. A bond is a loan split into tradable units. Its face value, coupon rate and maturity fix every payment that it will make. You should now be able to say: "A bond with a face value of Tk 5,000 and a coupon rate of 8 per cent pays Tk 400 each year."



Before moving on: a bond's face value, coupon rate and maturity fix its payments.

11.3 VALUING A BOND

We value a bond by following the same steps each time. We shall start with a very short bond so that you can see every line of arithmetic, before we use the factor tables for longer ones. The method is the idea of section 11.1: write down the payments, bring each back to today, and add.

Worked example 1. Given. A bond has a face value of Tk 1,000, a coupon rate of 10 per cent, and three years to maturity. Interest is paid at the end of each year. The required return is 10 per cent. *Find.* The value of the bond. *Formula.* $V = \text{the sum of each payment divided by } (1 + r) \text{ for each year of waiting.}$

Steps. The coupon is $10\% \times \text{Tk } 1,000 = \text{Tk } 100$. Year 1: $\text{Tk } 100 \div 1.10 = \text{Tk } 90.91$. Year 2: $\text{Tk } 100 \div 1.21 = \text{Tk } 82.64$. Year 3: the coupon and the face value arrive together, $\text{Tk } 1,100 \div 1.331 = \text{Tk } 826.45$. The sum is $\text{Tk } 90.91 + \text{Tk } 82.64 + \text{Tk } 826.45 = \text{Tk } 1,000.00$. *Answer.* The bond is worth Tk 1,000. *Check.* When the required return equals the coupon rate, a bond should be worth its face value, because it pays exactly the return that investors want. Tk 1,000 is the face value.

Worked example 2. Given. The same bond, but the required return is 12 per cent. *Find.* The value. *Formula.* The same.

Steps. Year 1: $\text{Tk } 100 \div 1.12 = \text{Tk } 89.29$. Year 2: $\text{Tk } 100 \div 1.2544 = \text{Tk } 79.72$. Year 3: $\text{Tk } 1,100 \div 1.404928 = \text{Tk } 782.96$. The sum of these rounded lines is Tk 951.97. The exact value, kept to more places, is Tk 951.96, and a difference of one paisa from rounding is normal. *Answer.* The bond is worth about Tk 952, which is less than its face value. *Check.* The required return is above the coupon rate, so the bond must be worth less than Tk 1,000. It is.

Year-by-year work becomes slow for a bond of five or ten years, so we use the factor method of Chapter 5, which needs only two multiplications. The factors for bonds are given here to six places, because a bond's face value is large and small rounding differences would show.

Case file, illustrative. In 2026 Sundial Bakery has spare cash and considers a bond with a face value of Tk 1,000, a coupon rate of 10 per cent, and five years to maturity. Interest is paid at the end of each year. The bakery's required return for a bond of this risk is 12 per cent a year, which is the illustrative discount rate carried through earlier chapters.

Worked example 3: the bakery's bond. Given. $F = \text{Tk } 1,000$; coupon rate 10%, so $C = \text{Tk } 100$; $n = 5$ years; $r = 12\%$. *Find.* The value of the bond. *Formula.* $V = C \times AF(12\%, 5) + F \times PVF(12\%, 5)$.

Steps. From the factor tables, $AF(12\%, 5) = 3.604776$ and $PVF(12\%, 5) = 0.567427$. The coupons are worth $\text{Tk } 100 \times 3.604776 = \text{Tk } 360.4776$. The face value is worth $\text{Tk } 1,000 \times 0.567427 = \text{Tk } 567.4270$. The sum is $\text{Tk } 360.4776 + \text{Tk } 567.4270 = \text{Tk } 927.9046$, which we write as Tk 927.90. *Answer.* The bond is worth Tk 927.90. *Check.* Year by year, the lines are Tk 89.29, Tk 79.72, Tk 71.18, Tk 63.55 and Tk 56.74 for the coupons (sum Tk

GO SLOWLY HERE.

A first example uses the smallest numbers, so the method is the only thing to learn.

First example: a three-year bond whose coupon rate equals the required return.

Steps, answer and check for the first example.

Second example: the same bond when the required return is higher.

Steps, answer and check for the second example.

For longer bonds, the factor method is faster than year-by-year work.

The bakery's case: spare cash into a bond.

Third example: the bakery's bond, with the factor method.

Steps, answer and check for the bakery's bond.

Fourth example: half-yearly interest changes the periods, the rate and the coupon.

Steps, answer and check for the half-yearly bond.

A table shows how the value depends on the required return.

The value falls as the required return rises, and the figure shows the curve.

360.48), and Tk 567.43 for the face value, which gives the same total to within a paisa. The value is below Tk 1,000, as it must be when the required return of 12 per cent is above the coupon rate of 10 per cent.

Many bonds pay interest twice a year. In this case three things change together. The number of periods is doubled, the rate for one period is half of the yearly rate, and each coupon is half of the yearly coupon. If the numbers in one place change and the others do not, the answer is wrong, which is the most common slip.

Worked example 4. Given. The bakery's bond, but interest is paid every six months. $F = \text{Tk } 1,000$; yearly coupon rate 10%, so each coupon is Tk 50; $n = 5$ years, which is 10 half-years; the required return is 12% a year, which is 6% for each half-year. *Find.* The value. *Formula.* $V = C \times AF(i, N) + F \times PVF(i, N)$, where i is the rate for one period and N is the number of periods.

Steps. $AF(6\%, 10) = 7.360087$ and $PVF(6\%, 10) = 0.558395$. The coupons are worth $\text{Tk } 50 \times 7.360087 = \text{Tk } 368.0044$. The face value is worth $\text{Tk } 1,000 \times 0.558395 = \text{Tk } 558.3950$. The sum is Tk 926.3994, which we write as Tk 926.40. *Answer.* The bond is worth Tk 926.40. *Check.* The value is slightly lower than the Tk 927.90 of the yearly bond. This is right: the half-yearly bond pays its coupons earlier, which should raise its value, but a required return of 12 per cent a year is treated here as 6 per cent for each half-year, which is a little less than 12 per cent compounded yearly, and so the discounting is stronger. The two answers are close, which is what we expect.²

The coupon and the face value of the bond do not change, but its value does change when the required return changes. The table below shows the bakery's bond at four required returns. Read each row across: the required return, the factors, the value, and a word for the bond's status, which section 11.4 explains.

VALUE OF THE BAKERY'S BOND (FACE VALUE TK 1,000, COUPON RATE 10%, FIVE YEARS) AT FOUR REQUIRED RETURNS

REQUIRED RETURN	AF(R, 5)	PVF(R, 5)	VALUE (TK)	STATUS
8%	3.992710	0.680583	1,079.85	Premium
10%	3.790787	0.620921	1,000.00	Par
12%	3.604776	0.567427	927.90	Discount
14%	3.433081	0.519369	862.68	Discount

The figure below draws these values and the values between them. To read it, find a required return on the bottom edge, go straight up to the curve, and read across to the left edge to find the value. The curve slopes downwards: the higher the required return, the lower the value. The dashed line is the face value of Tk 1,000. The curve crosses it at 10 per cent, which is the coupon rate.

² The yearly equivalent of 6 per cent for each half-year is 12.36 per cent. Some books give the bond's required return as an effective yearly rate and convert it. This book states the rate for each half-year directly, as above.

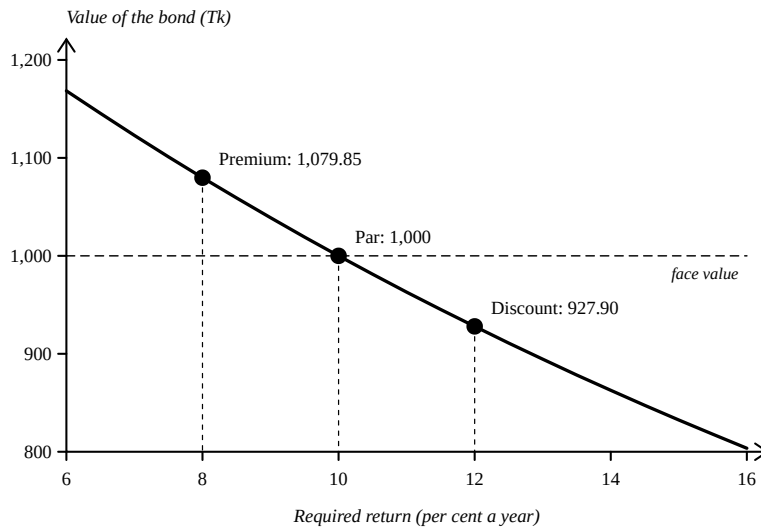


FIG. 32. The value of the bakery's bond falls as the required return rises; it equals the face value when the required return equals the coupon rate.

Common slips. A frequent slip is to discount the coupons and forget the face value, which gives a value far too low. A second is to use the yearly rate with half-yearly periods, as in section 11.3 above. A third is to use the annuity factor for the face value, or the single-sum factor for the coupons. The coupons are an annuity, and the face value is a single sum. Always draw the timeline before you use the factors.

Before moving on. The value of a bond is the present value of its coupons, found with the annuity factor, plus the present value of its face value, found with the single-sum factor. For half-yearly interest, double the periods and halve both the rate and the coupon. You should now be able to say: "A bond that pays Tk 80 a year and Tk 1,000 at the end is worth less than Tk 1,000 if the required return is above 8 per cent."



11.4 DISCOUNT, PREMIUM AND PAR; YIELD TO MATURITY

Suppose the bakery buys its bond for Tk 927.90, which is less than the Tk 1,000 face value. A student might think that the bakery was cheated, or that the seller made a gift. Neither is true. The price is fair, because the bond pays a coupon of 10 per cent while the bakery requires 12 per cent. The lower price makes up the difference. The bakery will receive Tk 1,000 at the end though it paid only Tk 927.90, and that extra gain lifts its return from 10 per cent to 12 per cent.

Slips with bond values: mixing periods, forgetting the face value, and using the wrong factor.

Before moving on: the value of a bond is its coupons as an annuity plus its face value as a single sum.

A puzzle: how can a bond be sold for less than its face value and still be a fair deal?

A bond sells at a discount, a premium or par, according to how its coupon rate compares with the required return.

Selling at a discount. Selling for less than the face value, which happens when the coupon rate is below the required return.

Selling at a premium. Selling for more than the face value, which happens when the coupon rate is above the required return.

The rule has three cases. A bond is **selling at a discount** when its value is less than its face value, and this happens when the coupon rate is below the required return. A bond is **selling at a premium** when its value is more than its face value, and this happens when the coupon rate is above the required return. A bond is *selling at par* when its value equals its face value, and this happens when the two rates are equal. The table below sets out the three cases, with the bakery's bond as the example.

BOND STATUS AND THE REQUIRED RETURN (BAKERY'S BOND, COUPON RATE 10%)

IF THE REQUIRED RETURN IS...	COMPARED WITH THE COUPON RATE	THE BOND SELLS AT...	VALUE (TK)
12%	Above	A discount	927.90
10%	Equal	Par	1,000.00
8%	Below	A premium	1,079.85

The reason is that the coupon is fixed, while the required return moves with the market.

Why does the status change? The coupon is fixed when the bond is issued. The required return is not fixed, because it follows the market. Suppose market rates rise, so that new bonds of the same risk pay 12 per cent. Nobody will pay Tk 1,000 for a bond that pays only 10 per cent when a new one pays 12 per cent. The price of the old bond must fall until a buyer earns 12 per cent on the lower price. If market rates fall to 8 per cent, the opposite happens. The old bond's coupon of 10 per cent is attractive, and the price rises.

A bond's value moves in the opposite direction to the required return.

This reasoning gives the most important rule about bonds: when required returns rise, bond values fall, and when required returns fall, bond values rise. The figure of the previous section showed it. A student who has learned the rule can say, on hearing that market interest rates have gone up, that holders of existing fixed-coupon bonds have lost value on paper.

Time also matters: a bond's value moves toward its face value as maturity nears.

Time to maturity changes the value as well. Keep the required return at 12 per cent and the coupon rate at 10 per cent, and vary the years left. The bond is worth Tk 982.14 with one year left, Tk 966.20 with two, Tk 951.96 with three, Tk 927.90 with five, and Tk 887.00 with ten. The shorter the time, the closer the value is to Tk 1,000. On the day of maturity, the bond pays exactly its face value, and the discount has disappeared.

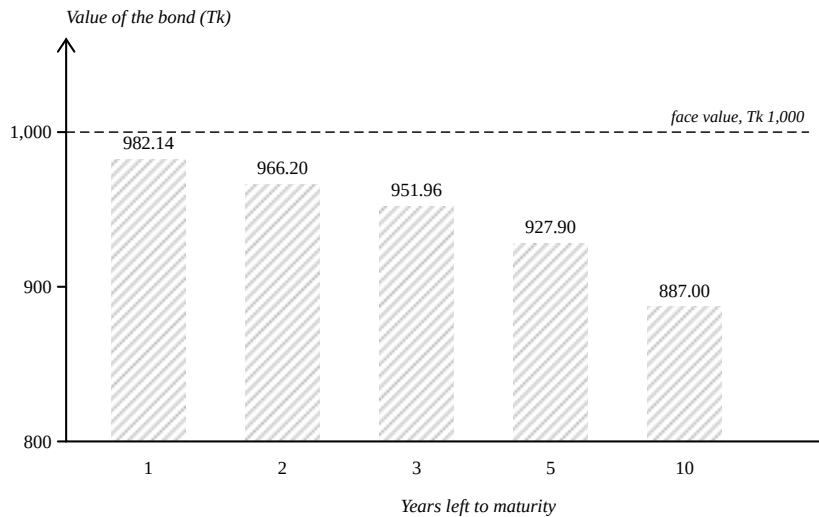


FIG. 33. How the value of a 10 per cent bond moves toward Tk 1,000 as maturity nears, when the required return stays at 12 per cent.

The figure above shows the five values as bars. Read it from the right: with ten years left the bar is lowest, and each step to the left brings the bar closer to the dashed face-value line. The discount of the bond is a gain that arrives, little by little, as the years pass.

A buyer often knows the price and wants the return. The **yield to maturity** means the yearly return that a buyer earns by paying today's price and holding the bond until it is repaid. In the bakery's bond, the price of Tk 927.90 gives a yield to maturity of exactly 12 per cent, because 12 per cent is the discount rate that makes the present value of the payments equal to the price. Yield to maturity is therefore the required return seen from the other side: given the value, we find the rate; before, given the rate, we found the value.

Worked example 5. Given. The bakery's bond (face value Tk 1,000; coupon rate 10%; five years) is offered at Tk 900. *Find.* The yield to maturity, to the nearest tenth of a per cent. *Method.* Try rates until the value of the bond is close to Tk 900.

Steps. At 12% the value is Tk 927.90, which is above Tk 900, so the yield is above 12%. At 13%, $AF = 3.517231$ and $PVF = 0.542760$, so the value is $Tk\ 351.72 + Tk\ 542.76 = Tk\ 894.48$, which is below Tk 900, so the yield is below 13%. The price of Tk 900 lies between Tk 927.90 and Tk 894.48. The fall from 12% to 13% is Tk 33.42, and the price needs a fall of Tk 27.90 from the value at 12%. That is $27.90 \div 33.42 = 0.83$ of the way, so the yield is about 12.8%. *Answer.* The yield to maturity is about 12.8 per cent. *Check.* At 12.8% the value is Tk 901.04, and at 12.9% it is Tk 897.75, so Tk 900 lies between them. The yield is higher than the coupon rate of 10 per cent, which it must be for a bond selling at a discount.

Limits. The yield to maturity assumes that the issuer pays every coupon and the face value in full, and on time. A bond whose issuer may not pay should be valued with a higher required return. The yield also assumes that the buyer holds to maturity. A buyer who sells early earns a return that depends on the price at the time of sale. The word yield is also used for other measures in some books; in this book it always means yield to maturity.

The figure shows the bars rising toward the face value as the years left fall.

Yield to maturity is the yearly return from buying at today's price and holding to maturity.

Yield to maturity. The yearly return a buyer earns by paying today's price and holding the bond until it is repaid.

Finding a yield needs trial and error, as a worked example shows.

Steps, answer and check for the yield.

Yield to maturity assumes the issuer pays in full and that coupons can be invested at the same rate.

Before moving on: a bond sells at a discount if its coupon rate is below the required return, and its yield is the rate that makes value equal to price.

Before moving on. A bond sells at a discount when the coupon rate is below the required return, at a premium when it is above, and at par when the two are equal. The value falls when the required return rises, and it moves toward the face value as maturity nears. You should now be able to say: "A bond with a 9 per cent coupon sells at a discount when buyers require 11 per cent."



GO SLOWLY HERE.

A share gives the owner a claim on the firm, and its payments are dividends, not coupons.

The ordinary kind of share has a word in use: common stock.

Common stock. The ordinary ownership shares of a company. Word in use: in daily life, stock means goods kept in a shop.

First case: a share that pays the same dividend every year for ever.

The formula for a share with a constant dividend.

Worked example 6: a constant dividend of Tk 6 and a required return of 12 per cent.

Steps, answer and check for the constant-dividend share.

A different check: a buyer who holds the share for one year only.

Dividends often grow, and growth raises the value of a share.

11.5 VALUING SHARES

Chapter 2 introduced the shareholder, who owns a part of a company, and the dividend, which is a part of profit paid out to owners. We now value a share in the same way that we valued a bond: by the present value of what it will pay. The share pays dividends. But there are three differences from a bond. A dividend is not fixed in advance, and the firm may raise it, cut it, or skip it. A share has no maturity, because the firm may continue for many years. And a share has no face value to be repaid, because the owner may only sell it to another buyer.

The ordinary kind of share is called **common stock**. It means the ordinary ownership shares of a company, and it carries the right to dividends after others are paid, and the right to vote. In this chapter we need only the dividends. Chapter 14 describes the rights of the owner in full, together with preferred shares, which are a different kind.

Begin with the simplest case. A firm is expected to pay the same dividend, year after year, for ever. The share is then a perpetuity, which Chapter 5 described briefly: an annuity that never ends. The present value of such a stream is found by a simple rule. In words, and in symbols:

value of a share = yearly dividend $\div$ required return.

In symbols, $V = D \div r$, where D is the dividend each year and r is the required return, written as a decimal.

Worked example 6. Given. A share pays a dividend of Tk 6 at the end of each year for ever. The required return is 12 per cent. *Find.* The value of the share. *Formula.* $V = D \div r$.

Steps. $V = \text{Tk } 6 \div 0.12 = \text{Tk } 50$. *Answer.* The share is worth Tk 50. *Check.* A buyer who pays Tk 50 and receives Tk 6 every year earns $6 \div 50 = 12$ per cent a year, which is the required return.

Another way to see the answer is to suppose that a buyer holds the share for one year and then sells it. She receives the dividend of Tk 6 and the selling price of Tk 56, and she wants 12 per cent. The value today is $(\text{Tk } 6 + \text{Tk } 56) \div 1.12 = \text{Tk } 55.36$. Here the selling price of Tk 56 was assumed and was not found. If the buyer in turn expected to sell for the value found by the rule, the one-year view would give back Tk 50. Every holder, in the end, is paid by the dividends that follow.

Most firms hope to raise their dividends as they grow. The bakery's profit, for example, may rise as it opens more shops. A share whose dividend grows is worth more than one with the same first dividend that does not grow. The simplest case is a dividend that grows at the same per cent each year for ever. This case is called **dividend growth**: the dividend

of each year is the last year's dividend times $(1 + g)$, where g is the yearly growth rate.

The value is found by a rule that is a small change in the one before. In words: the value of a share is the dividend expected next year, divided by the required return less the growth rate. In symbols, $V = D_1 \div (r - g)$, where D_1 is the dividend expected at the end of the coming year, r is the required return, and g is the yearly growth rate of the dividend. The rule needs r to be larger than g . If g were as large as r or larger, the formula would give a meaningless or negative answer, and a growth rate that high could not continue for ever.

Worked example 7. Given. The first dividend, due at the end of the coming year, is $D_1 = \text{Tk } 6$. It then grows by 4 per cent each year. The required return is 12 per cent. *Find.* The value of the share. *Formula.* $V = D_1 \div (r - g)$.

Steps. $r - g = 0.12 - 0.04 = 0.08$. $V = \text{Tk } 6 \div 0.08 = \text{Tk } 75$. *Answer.* The share is worth Tk 75. *Check.* Without growth, the same first dividend gives Tk 50. Growth raises the value to Tk 75, which is more, as it should be. The dividends in the first years would be Tk 6.00, Tk 6.24, Tk 6.49, and so on.

Worked example 8. Given. A firm has just paid a dividend of Tk 5 (call it D_0). Dividends are expected to grow at 6 per cent each year. The required return is 14 per cent. *Find.* The value of the share. *Formula.* First find $D_1 = D_0 \times (1 + g)$. Then $V = D_1 \div (r - g)$.

Steps. $D_1 = \text{Tk } 5 \times 1.06 = \text{Tk } 5.30$. $r - g = 0.14 - 0.06 = 0.08$. $V = \text{Tk } 5.30 \div 0.08 = \text{Tk } 66.25$. *Answer.* The share is worth Tk 66.25. *Check.* A buyer pays Tk 66.25 and expects Tk 5.30 as the first dividend, which is a yield of 8 per cent, and growth of 6 per cent supplies the rest of the 14 per cent. $8\% + 6\% = 14\%$.

The table below uses the first dividend of Tk 6 and the required return of 12 per cent, and changes only the growth rate. Read each row across: the growth rate, the gap between the required return and the growth, and the value. Notice how the value rises faster and faster as the growth rate approaches the required return.

VALUE OF A SHARE WITH A FIRST DIVIDEND OF TK 6 AND A REQUIRED RETURN OF 12 PER CENT, AT DIFFERENT GROWTH RATES

GROWTH RATE	$R - G$	VALUE (TK)
0%	0.12	50.00
2%	0.10	60.00
4%	0.08	75.00
6%	0.06	100.00
8%	0.04	150.00

Some firms are known to pay a fixed dividend for a few years, and then to grow. The bakery's shop network, for example, may take two years to open before dividends start to rise. The share is then valued in two steps: first find what the share will be worth at the end of the fixed years, and then bring everything back to today. This is a more advanced idea, and a beginner may return to it after the rest of the chapter has been mastered.

The growing-dividend formula is the constant-dividend formula with growth taken away from the required return.

Worked example 7: a dividend of Tk 6 growing at 4 per cent a year.

Steps, answer and check for the growing share.

Worked example 8: starting from the dividend just paid, which must be grown by one year.

Steps, answer and check for the second growing share.

The table shows how sensitive the value is to the growth rate.

A share with a dividend that is fixed for some years and then grows is valued in two parts.

Worked example 9: a fixed dividend for two years, then growth.

Steps, answer and check for the two-stage share.

The limits of share valuation are the estimates of dividends and the required return.

Slips with share values: using the wrong dividend, and letting growth equal the required return.

Before moving on: a share is worth its expected dividends brought to today; growth raises the value.

A bondholder is a lender, a shareholder is an owner, and this difference runs through everything else.

Worked example 9. Given. The dividend is Tk 6 at the end of year 1 and Tk 6 at the end of year 2. From year 3 it grows at 4 per cent a year for ever, so $D_3 = \text{Tk } 6 \times 1.04 = \text{Tk } 6.24$. The required return is 12 per cent. *Find.* The value of the share today. *Formula.* The value at the end of year 2 is $D_3 \div (r - g)$. Then the value today is the present value of the two dividends and of that end-of-year-2 value.

Steps. The value at the end of year 2 is $\text{Tk } 6.24 \div 0.08 = \text{Tk } 78$. Now bring everything back. Dividend 1: $\text{Tk } 6 \div 1.12 = \text{Tk } 5.36$. Dividend 2: $\text{Tk } 6 \div 1.2544 = \text{Tk } 4.78$. The value of Tk 78 at the end of year 2: $\text{Tk } 78 \div 1.2544 = \text{Tk } 62.18$. The sum is $\text{Tk } 5.36 + \text{Tk } 4.78 + \text{Tk } 62.18 = \text{Tk } 72.32$. *Answer.* The share is worth Tk 72.32 today. *Check.* The answer lies between Tk 50 (no growth ever) and Tk 75 (growth from the first year), which is where we expect it, because growth begins late.

Limits. Share valuation is far less certain than bond valuation. A bond's payments are fixed in a contract, but a share's dividends are only estimates. A small change in the growth rate changes the value a great deal, as the table showed. The required return is also an estimate, and Chapter 10 showed how it is found. The result is a guide to the value and not a fact. A firm that pays no dividend today can still have a value, because investors expect dividends in the future, or a sale of the firm; the rule is then used with more care.

Common slips. The most frequent slip is to use the dividend just paid in place of the one expected next year. The formula needs D_1 , the dividend of the coming year. If you are given D_0 , you must multiply by $(1 + g)$ first. Another slip is to enter the required return as a whole number (12) rather than a decimal (0.12), which gives an answer a hundred times too small.

Before moving on. A share is valued as the present value of its dividends. With a constant dividend the value is $D \div r$. With a dividend growing at g each year, the value is $D_1 \div (r - g)$. You should now be able to say: "A share with a first dividend of Tk 10 and growth of 5 per cent is worth Tk 100 when the required return is 15 per cent."



11.6 BONDS AND SHARES COMPARED

We now set the two securities beside each other. The table below does this. Read each row across: the question, the answer for a bond, and the answer for a share. Every row follows from the one difference at the top: the bondholder lent money and the shareholder owns a part of the firm.

BONDS AND SHARES COMPARED

QUESTION	BOND	SHARE (COMMON STOCK)
What does the holder have?	A loan to the firm	A part of the firm
What does it pay?	A fixed coupon, and the face value at maturity	A dividend, which may rise, fall or stop
Is there a maturity?	Yes	No
Who is paid first?	The bondholder, before owners	The shareholder, after lenders
Is there a vote?	No	Yes
What is the risk to the holder?	Lower; payments are fixed by contract	Higher; dividends and price are uncertain
How is its value found?	Coupons as an annuity plus face value as a single sum	Dividends, with or without growth, brought to today
What return is required?	Lower	Higher

The last row connects to Chapter 10. Because bondholders are paid first, and their payments are fixed, a bond is usually less risky than a share of the same firm. A lower risk means a lower premium, and so a lower required return. A share must offer a higher expected return, or no investor would hold it. The firm, in turn, pays more for the owners' money than for the lenders' money, and Chapter 15 returns to this fact when it prices the firm's funds.

The two securities also differ from the firm's side. A firm must pay a bond's coupons and its face value, even in a bad year, or the lenders can take action against it. It need not pay dividends in a bad year. A firm that borrows by selling bonds therefore takes on a fixed payment, which is the financial risk of Chapter 9. A firm that raises money by selling shares takes on no fixed payment, but it shares its profits and its control with a wider group of owners.

The valuation method is the same for both securities: list the payments, bring each back to today at the required return, and add. For the bond the payments are known, and the difficulty is the rate. For the share the rate is also an estimate, and so are the payments. A student who has understood this has understood the whole chapter. Chapters 13 and 14 describe how firms raise long-term money from lenders and from owners, and Chapter 15 puts a single price on all of it.

Before moving on. Bondholders are lenders, with fixed payments and priority; shareholders are owners, with uncertain dividends and more risk. Both are valued by present value at the required return. You should now be able to say: "A share needs a higher expected return than a bond of the same firm because it carries more risk."

The required return is lower for a bond than for a share, because its risk to the holder is lower.

The two securities differ in what the firm must do.

The valuation method is the same for both, and only the payments differ.

Before moving on: a bond and a share are valued in the same way, but the bond has fixed payments and lower risk.



The chapter's main ideas in a few lines.

A problem: pricing a bond at two required returns, and judging an offer.

Five common misunderstandings and their corrections.

The coupon rate and the required return are different ideas.

A discount bond is not a bad bond.

A dividend is not the same as a profit.

The growth formula has a limit.

The first public offer of shares to many ordinary buyers.

SUMMARY

The intrinsic value of a security is the present value of all its expected payments, found with the required return. A bond has a face value, a coupon rate and a maturity. Its value is the coupons, as an annuity, plus the face value, as a single sum: $V = C \times AF + F \times PVF$. For half-yearly interest, double the periods and halve the rate and the coupon. A bond sells at a discount when its coupon rate is below the required return, at a premium when it is above, and at par when they are equal. Its value falls when the required return rises, and moves toward the face value as maturity nears. The yield to maturity is the rate that makes the value equal to the price. A share with a constant dividend is worth $D \div r$, and a share with a dividend growing at g is worth $D1 \div (r - g)$. The bakery's bond, with a coupon of 10 per cent and a required return of 12 per cent, is worth Tk 927.90, and it sells at a discount.

THINK IT THROUGH

The bakery is offered a bond with a face value of Tk 1,000, a coupon rate of 9 per cent, and four years to maturity. Interest is paid at the end of each year. First, find the value of the bond if the required return is 12 per cent. Then find its value if the required return falls to 9 per cent. State in each case whether the bond sells at a discount, a premium or par. Finally, the bond is offered at Tk 950. Should the bakery buy it, if the required return is 12 per cent? Give a reason in two sentences. (Hint: at 12%, $AF(12\%, 4) = 3.037349$ and $PVF(12\%, 4) = 0.635518$, so the value is $Tk\ 90 \times 3.037349 + Tk\ 1,000 \times 0.635518 = Tk\ 273.36 + Tk\ 635.52 = Tk\ 908.88$, a discount. At 9% the bond is worth Tk 1,000, at par. At Tk 950 the price is above the value of Tk 908.88, so the bond is dear and should not be bought.)

WHERE PEOPLE GO WRONG

"A bond's price is always its face value." The face value is repaid at the end. The value in between changes with the required return.

"The coupon rate is the return the buyer earns." It is only the rate of interest on the face value. The buyer's return depends on the price paid, and the yield to maturity measures it.

"A bond selling at a discount is a poor bond." It sells at a discount because its coupon rate is below the required return. The price is fair for that rate.

"A share is worth its profit." A share is worth the dividends it will pay, brought to today. A firm may keep part of its profit, as Chapter 14 shows, and growth then comes from the profit that is kept.

"The share formula works for any growth rate." It needs the growth rate to be less than the required return. A growth rate equal to or above the required return cannot last for ever.

IN HISTORY

In 1602 the Dutch East India Company, a trading firm of the Netherlands, was formed to send ships to Asia. A single voyage was costly and dangerous, and a few rich merchants could not carry the risk alone. The company's charter allowed members of the public to buy shares, and

more than a thousand investors did so in the first offering.³ Because buyers wished to sell when they needed cash, a market for the shares grew in Amsterdam, and a dedicated exchange building was later built there. The company's shares did not repay a fixed sum or pay a fixed coupon. Their value rested on the dividends that the voyages were expected to bring, which is the idea of this chapter.

REVIEW QUESTIONS

1. Define face value, coupon rate, coupon and maturity. A bond has a face value of Tk 5,000 and a coupon rate of 8 per cent. Find its yearly coupon.

Practice questions, with short answers at the back of the book.

Practice question 1.

2. A bond has a face value of Tk 1,000, a coupon rate of 12 per cent and four years to maturity. Interest is paid yearly. Find its value if the required return is 10 per cent. (Use $AF(10\%, 4) = 3.169865$ and $PVF(10\%, 4) = 0.683013$.) Does it sell at a discount or a premium?

Practice question 2.

3. A bond has a face value of Tk 1,000, a coupon rate of 8 per cent and four years to maturity. Find its value if the required return is 10 per cent, and say whether it sells at a discount or a premium.

Practice question 3.

4. Explain, in your own words, why the value of a bond falls when the required return rises, even though the bond's payments do not change.

Practice question 4.

5. A bond has a coupon rate of 9 per cent. State whether it sells at a discount, at a premium or at par when the required return is 7 per cent, 9 per cent and 11 per cent.

Practice question 5.

6. Value the bond of question 2 if it pays interest half-yearly. (Use $AF(5\%, 8) = 6.463213$ and $PVF(5\%, 8) = 0.676839$.)

Practice question 6.

7. A three-year bond with a 10 per cent coupon sells at its value of Tk 951.96 when the required return is 12 per cent. Without calculation, state its yield to maturity, and explain.

Practice question 7.

8. A share pays a dividend of Tk 8 each year for ever. The required return is 16 per cent. Find its value.

Practice question 8.

9. A firm has just paid a dividend of Tk 10, and the dividend is expected to grow by 4 per cent each year. The required return is 12 per cent. Find the value of the share.

Practice question 9.

10. Compare a bond and a share of the same firm in a short table, covering what the holder has, what is paid, who is paid first, and the required return. Explain why the required returns differ.

Practice question 10.

³ The figures for the number of investors and the amount raised differ between sources (about 1,100 investors and about 6.4 to 6.5 million guilders), and they need verification before printing. Only a rounded figure is used here.

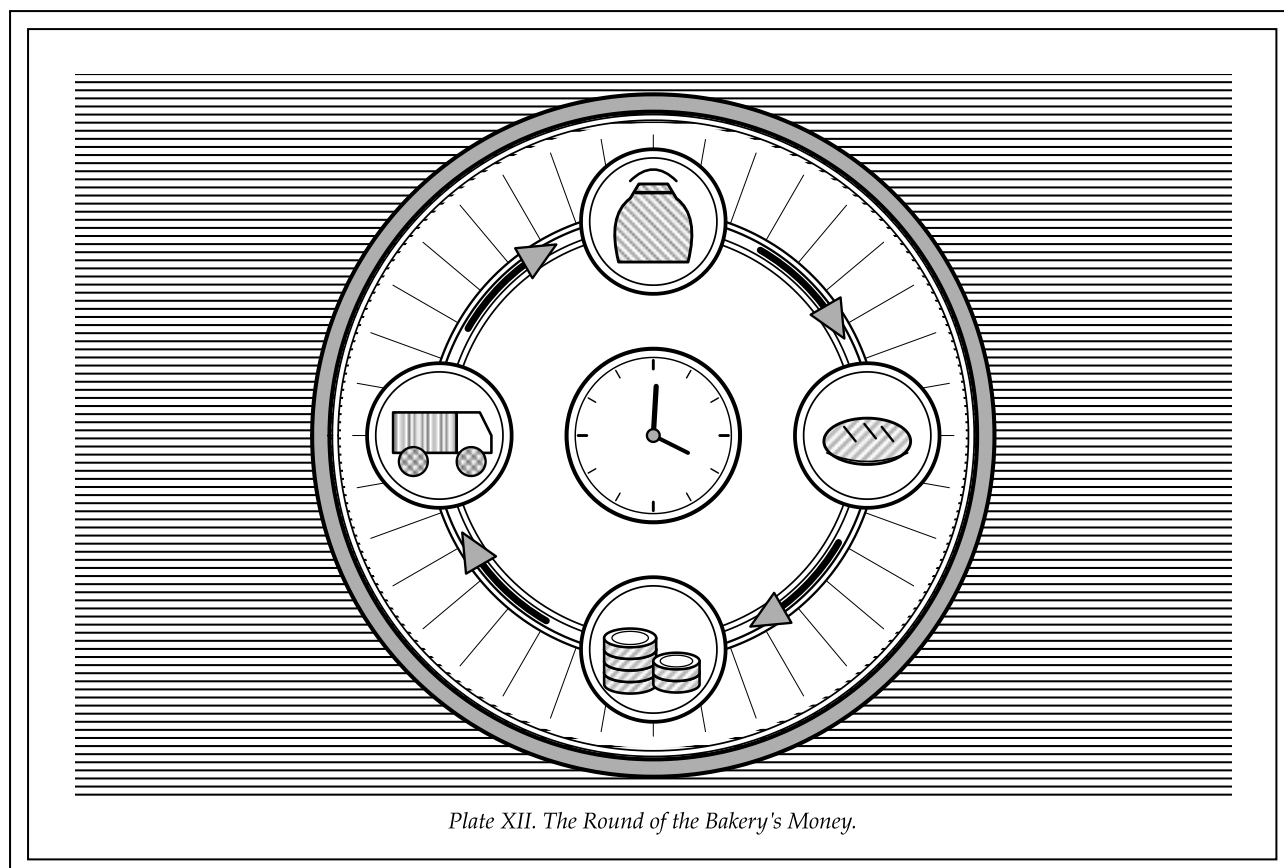


Plate XII. The Round of the Bakery's Money.

CHAPTER XII

Working Capital and Short-Term Financing

A profitable bakery can still run short of cash on a Thursday.



This chapter explains the money that moves through a firm in its daily work: cash, goods waiting to be sold, and bills not yet paid. It shows how to measure the days that money stays tied up, how to match short needs with short funds, and how to judge whether paying a supplier early is worth the cost.

A firm can earn profit and still lack the cash to pay one bill.

The answer is timing: profit and cash arrive on different days.

This chapter studies the daily movement of money through a firm.

A case-file snapshot of the bakery's day-to-day money.

The chapter relies on percentages and on interest on interest.

GO SLOWLY HERE.

A household's monthly routine is a small model of a firm's daily money.

A current asset is something owned that will be used or turned into cash within a year.

Current asset. Something the firm owns that it expects to use or turn into cash within one year.

Receivable: money a customer owes for goods already delivered.

Receivable. Money a customer owes the firm for goods or services already delivered.

Inventory: goods held for use or sale.

Inventory. Goods that a firm holds for use or for sale.

Consider a puzzle that troubles many young firms. A bakery sells well all month. Its accounts, shown in the layout of Chapter 3, show a profit. Yet on Thursday morning the owner cannot pay the flour supplier. How can a firm that earns profit lack the money to pay one bill?

The answer lies in timing. In Chapter 1 we learned that profit is what remains of sales after costs, while cash flow is money actually moving in or out. Profit counts a sale on the day it is made. Cash counts the sale on the day the customer pays. If a customer takes thirty days to pay, the bakery has already paid for flour, wages and gas before the money returns. The profit exists in the accounts, but the money is still with the customer.

This chapter studies that daily movement of money. We ask what a firm must hold to keep working, how long its money stays tied up, where short-term money can be found, and what it costs. We begin with the bakery's position on one day.

Case file, illustrative. On 10 September 2026 Sundial Bakery holds cash of Tk 18,000. Catering customers owe it Tk 1,20,000 for orders already delivered. It holds flour and other ingredients worth Tk 40,000. It owes the flour supplier Tk 50,000 and owes a further Tk 30,000 for wages and gas bills. The supplier's invoice carries the terms 2/10, net 40, explained in section 12.6. The bakery cannot pay the Tk 50,000 from its cash, although its customers owe it more than twice that sum.

You will need percentages from Chapter 3 and the idea of interest on interest from Chapter 4. Chapter 2 named working capital in one line. Here we study it fully.

12.1 WHAT WORKING CAPITAL IS

Think first of a household. A family is paid on the first day of the month. It buys food, pays the electricity bill on the tenth day and the school fee on the twentieth. Money in the wallet, food in the kitchen and bills not yet paid all belong to this monthly routine. The family does not plan to keep any of them for years. They are used and replaced, again and again. A firm has the same routine, with larger sums.

A **current asset** means something the firm owns that it expects to use or turn into cash within one year. Cash in the till is one example. Flour in the store room is another. A bakery has three main kinds of current asset, and we use each one throughout the chapter.

The first kind is the **receivable**. A receivable means money a customer owes the firm for goods already delivered. When an office orders cakes for a meeting and promises to pay in thirty days, the bakery holds a receivable until the money arrives. The bakery has done its work, but it does not yet hold the cash.

The second kind is **inventory**. Inventory means goods that a firm holds for use or for sale. For the bakery it is flour, sugar, butter and the buns waiting on the shelf. Some writers say stock for inventory. This book uses inventory, because Chapter 11 gave the word stock another meaning, the shares of a company.¹

¹ In Chapter 11 the word *common stock* named ordinary ownership shares. In this chapter, inventory always means goods held for use or sale.

The third kind is cash itself, in the till and in the bank. Cash is the only current asset that can pay a bill today. A receivable must first be collected, and inventory must first be sold and then collected.

The firm also owes money in the short term. A **current liability** means an amount the firm owes that it must pay within one year. The most common is the **payable**, which means money the firm owes a supplier for goods already received. Wages not yet paid and the gas bill are also current liabilities.

We can now name the whole idea. **Working capital** means the money tied up in a firm's current assets: cash, receivables and inventory. It is called working capital because the money works every day. It is bought as flour, turned into buns, sold, and collected again as cash. Then the circle begins once more.

Most firms owe money in the short term as well, so one more measure is useful. **Net working capital** means current assets less current liabilities. It shows how much of the firm's daily money belongs to the firm after the short-term debts are paid. The next table shows the bakery on 10 September, with each item explained in turn.

THE BAKERY'S DAY-TO-DAY MONEY, 10 SEPTEMBER 2026

ITEM	TK
Cash	18,000
Receivables (catering)	1,20,000
Inventory	40,000
Current assets	1,78,000
Payable to the flour supplier	50,000
Wages and gas bills due	30,000
Current liabilities	80,000
Net working capital	98,000

Read the table from the top. The three current assets add to Tk 1,78,000. The two current liabilities add to Tk 80,000. Net working capital is therefore Tk 1,78,000 less Tk 80,000, which is Tk 98,000. The figure is positive, so the bakery owns more in the short term than it owes. Yet the puzzle remains: the bakery cannot pay the Tk 50,000 invoice today.

The puzzle has a plain answer. Of the Tk 98,000, only Tk 18,000 is cash. The rest is money in customers' hands and flour on the shelf. A positive net working capital tells us the firm is sound in the short term. It does not tell us whether the firm can pay a bill this morning. For that, we must also know how quickly the other assets turn into cash.

There are limits on both sides. A firm with too little working capital may fail to pay wages or suppliers on time. A firm with too much has money lying idle in unsold goods and slow customers. That money could have bought a second oven. The owner and the supplier see the matter differently. The owner wants little money tied up. The supplier wants a customer who always has cash to pay.

Cash is the third and most useful current asset.

A current liability is a debt due within one year.

Current liability. An amount the firm owes that it must pay within one year.

Working capital is the money tied up in cash, receivables and inventory.

Working capital. Money tied up in a firm's daily work, held as cash, receivables and inventory.

Capital. In daily life, a city or a large letter. In finance, money put into a business.

Net working capital is current assets less current liabilities.

Net working capital. Current assets less current liabilities.

Net working capital is positive, yet cash is short.

Most of the net working capital is not cash.

Too little working capital risks unpaid bills; too much leaves money idle.

Recap: working capital is cash, receivables and inventory, less short-term debts.

People borrow for different lengths of time for different purposes.

Short, medium and long are fixed by the number of years.

Short-term. Lasting up to one year.

Medium-term. Lasting more than one year and up to five years.

Long-term. Lasting more than five years.

Each period has its own uses and its own sources of money.

The length of the need is the guide to the length of the funding.

Recap: short, medium and long are periods of up to one, five and more than five years.

GO SLOWLY HERE.
A short loan for a long need, or a long loan for a short need, causes trouble.

Before moving on. Working capital is the money tied up in cash, receivables and inventory. Net working capital subtracts the short-term debts. You should now be able to say why a firm with positive net working capital can still be unable to pay a bill today.

12.2 FINANCING BY TIME

Return to the household. A family borrows a little money from a neighbour for a week to buy medicine. It borrows a large sum from a bank for twenty years to buy a house. Nobody would borrow for twenty years to buy lunch. The length of the loan follows the length of the need. Firms reason in the same way.

Finance uses three periods. **Short-term** means lasting up to one year. **Medium-term** means lasting more than one year and up to five years. **Long-term** means lasting more than five years. Writers and banks draw these lines in slightly different places, so a reader meeting another book should check its lines.² This book keeps the three lines above throughout.

The next table shows what each period is usually used for, and where the money usually comes from. We will study the sources one at a time, the short-term sources in this chapter and the others in Chapters 13 and 14. Read each row from left to right: the use is a need, and the source is the way the need is met.

FINANCING BY TIME		
PERIOD	TYPICAL USE	TYPICAL SOURCES
Short-term	Flour, wages, goods waiting for customers	Supplier credit, overdraft, short-term bank loan
Medium-term	A delivery van, a small oven	Term loan of one to five years
Long-term	Premises, large ovens, expansion	Long-term loans, debentures, shares, retained profit

The table shows a pattern, and the pattern leads to the next section. Each use has a natural length. A bag of flour is used in days. An oven is used for ten years. The funds should have a length that fits the need.

Before moving on. Short-term means up to one year, medium-term up to five years, and long-term more than five years. Each period has its own uses and sources. You should now be able to name one use and one source for each period.

12.3 THE MATCHING PRINCIPLE

A student wants a motorcycle that will last ten years. A friend offers a loan, but demands full repayment in one month. The student cannot repay, because the motorcycle has earned nothing yet. Now reverse the case. The student borrows for ten years to buy lunch for one week. For nine years and fifty-one weeks the student pays interest on

² Some banks treat up to three years as medium-term; others treat up to ten years. The lines in this book are a teaching choice and not a law.

money that was needed for seven days. Both loans are poorly fitted to the need.

The **matching principle** means fund short-term needs with short-term money, and long-term needs with long-term money. A firm that follows it repays each loan from the money the asset earns, at about the time the asset earns it. A firm that ignores it must find new money to repay old money.

Working capital shows the principle well. A firm's need for working capital has two parts. The **permanent part** is the level the firm needs all year, even in a quiet month. The **temporary part** rises in busy weeks and disappears afterwards. The next figure shows the two parts. The horizontal axis shows time through one year, and the vertical axis shows the money needed. The wavy line is the firm's total need. The flat line is the permanent part. The distance between the two lines, in busy weeks, is the temporary part.

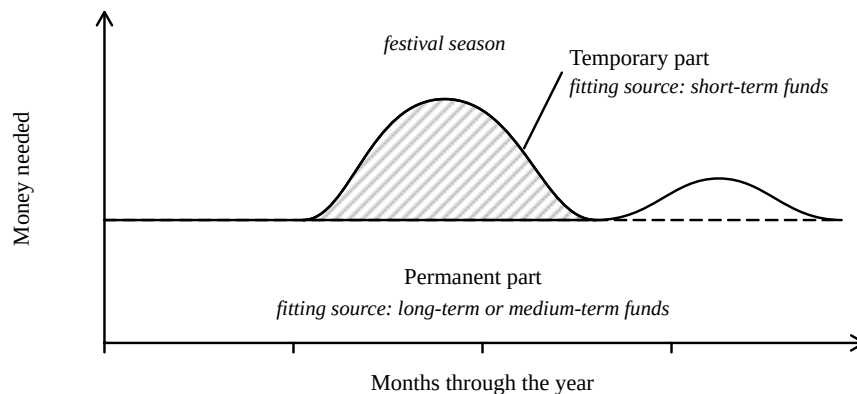


FIG. 34. The bakery's need for working capital through one year: a flat permanent part and a temporary part that rises in busy weeks.

By the matching principle, the permanent part should come from long-term or medium-term funds, and the temporary part from short-term funds that the firm repays when the busy weeks end. The figure labels each part with its fitting source.

Now we test the principle with numbers. Suppose the bakery needs Tk 60,000 more than usual for the two months of a festival season. It borrows Tk 60,000 for five years at 12 per cent a year. The interest for one year is Tk 60,000 multiplied by 0.12, which is Tk 7,200. The bakery needs the money for only two months, so the interest it should pay is two-twelfths of Tk 7,200, which is Tk 1,200. The remaining Tk 6,000 pays for money that sits idle. The loan was too long for the need.

The reverse mismatch is more dangerous. Suppose the bakery buys a Tk 3,00,000 oven with an overdraft that the bank may call back at short notice. The oven will earn money for years, but the bank may ask for repayment in months. The bakery may then have to sell the oven or borrow again on worse terms. Finding new money to repay old money is called refinancing, and it is a risk whenever the lender has the stronger position.

There are exceptions. Short-term money is often cheaper than long-term money, so a firm that wants low interest may accept some mismatch, and some firms do. They accept more risk in return for lower cost.

The matching principle: fund short needs with short money, long needs with long money.

Matching principle. Fund short-term needs with short-term money and long-term needs with long-term money.

A firm's working capital need has a permanent part and a temporary part.

Matching sends the permanent part to long money and the peaks to short money.

A worked example with small numbers shows what a mismatch costs.

The reverse mismatch is more dangerous, because the lender can demand repayment.

Matching is a guide, not a law; firms sometimes accept a mismatch on purpose.

Recap and a self-test sentence.

GO SLOWLY HERE.

Money leaves the firm as ingredients and returns as customers' cash, after a delay.

The operating cycle runs from buying ingredients to collecting cash.

Operating cycle. The days from receiving ingredients to collecting cash from the customer.

Supplier credit shortens the time the firm itself must fund.

Cash conversion cycle. The days between paying the supplier and collecting from the customer.

Words first: $\text{cycle} = \text{inventory days} + \text{receivable days} - \text{payable days}$.

A first worked example with clean numbers.

The steps are shown in a table, one line of arithmetic each.

The bakery funds 28 days by itself; the check is a timeline.

The matching principle does not forbid this. It warns the firm to decide knowingly and to know which risk it carries.

Before moving on. Fund short needs with short money and long needs with long money. A mismatch costs idle interest in one direction and risks a forced repayment in the other. You should now be able to say, in one sentence, why a firm should not pay for an oven with an overdraft.

12.4 THE CASH CYCLE, FLOAT AND ESTIMATING NEEDS

THE OPERATING CYCLE AND THE CASH CONVERSION CYCLE

Follow one sack of flour. The bakery pays the supplier for it, keeps it in the store for some days, bakes it into cakes, delivers the cakes to an office, and waits for the office to pay. Money left the firm when the supplier was paid, and returns only when the office pays. The time between the two is the time the firm must find money from somewhere else.

The **operating cycle** means the days from receiving ingredients to collecting cash from the customer. It has two parts: the days the goods wait in inventory, and the days the customer takes to pay. If the bakery holds ingredients for 8 days and customers take 30 days to pay, the operating cycle is 8 plus 30, which is 38 days.

The supplier helps the bakery. If the supplier allows 10 days before payment, the bakery does not pay on the day it receives the flour. The **cash conversion cycle** means the days between paying the supplier and collecting from the customer. It is the operating cycle less the days the bakery takes to pay.

In words: cash conversion cycle equals inventory days, plus receivable days, less payable days. In symbols: $\text{CCC} = \text{ID} + \text{RD} - \text{PD}$. Here ID is the days goods wait in inventory, RD is the days customers take to pay, and PD is the days the firm takes to pay its supplier. The following example uses the catering line of the bakery on its own.

Case file, illustrative. Sundial Bakery's catering line sells to offices on credit. Ingredients stay in the store for 8 days. Offices pay 30 days after delivery. The bakery takes the supplier's cash discount and pays on day 10. *Given.* ID is 8 days, RD is 30 days, PD is 10 days. *Find.* the cash conversion cycle.

CASH CONVERSION CYCLE OF THE CATERING LINE, DISCOUNT TAKEN

STEP	WORKING	RESULT
Operating cycle	$8 + 30$	38 days
Less days to pay the supplier	$38 - 10$	28 days
Cash conversion cycle	28 days	28 days

Answer. the cash conversion cycle is 28 days. The bakery pays out on day 10 and collects on day 38, so it must find money for 28 days. *Check.* day 38 less day 10 is 28 days. The figure below draws the same cycle as a timeline. Each bar starts on the day shown on the line below it, and its length shows the number of days.

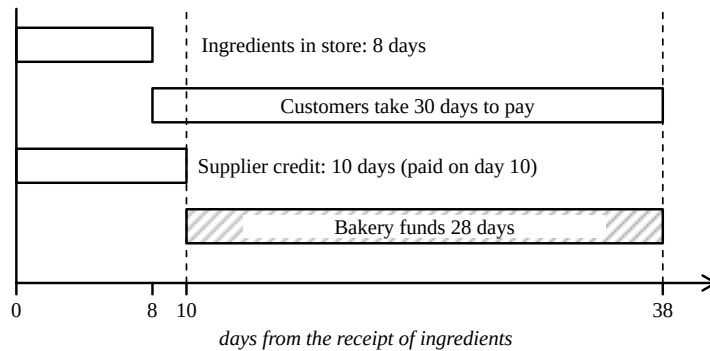


FIG. 35. The catering line's cycle: ingredients wait 8 days, customers pay after 30 more, and the supplier is paid on day 10, leaving 28 days to fund.

Now a second case. The bakery does not take the discount. It pays the supplier on day 40, as the terms allow. Then PD is 40. The cycle is 8 plus 30 less 40, which is minus 2 days. A negative cycle means the customer's cash arrives two days before the bakery pays the supplier. The supplier then funds the whole cycle. This looks attractive, but it has a price, which section 12.6 measures.

ESTIMATING THE MONEY NEEDED

How much money does the cycle tie up? The answer follows from the days and from the daily cost. Money tied up is the money the bakery has actually spent, so we count it at cost and not at the selling price. In words: money needed equals the daily cost, multiplied by the cash conversion cycle.

The catering line sells Tk 4,000 a day on credit. Ingredients are 60 per cent of sales, as in Chapter 3, which is Tk 2,400 a day. We count only ingredients, to keep the example small. With the discount taken, the cycle is 28 days. The money tied up is Tk 2,400 multiplied by 28, which is Tk 67,200. Without the discount, the cycle is minus 2 days. The money tied up is Tk 2,400 multiplied by minus 2, which is minus Tk 4,800: the supplier holds the bakery's money, not the reverse.

Check. the difference between the two cases is Tk 67,200 less minus Tk 4,800, which is Tk 72,000. That equals 30 days of ingredients at Tk 2,400 a day. By paying 30 days sooner, the bakery funds 30 days of ingredients itself. A firm also keeps a small cash balance above this figure, for surprises, and the owner should add it.

There is one more point. The accounts record a receivable at the selling price. Thirty days of sales at Tk 4,000 is Tk 1,20,000, the figure in the bakery's snapshot. The bakery has spent only the cost, Tk 72,000, on those orders. The rest, Tk 48,000, is profit not yet collected. Managers who want a cautious estimate count receivables at the selling price, and they must say so.

LOAT

A cheque from a customer does not become usable money on the day it is written. It must be posted or carried, recorded, deposited, and cleared by the banks. The money in this gap is called **float**: money that has been paid but cannot yet be used by the receiver. It is not lost, but while it is in the gap, nobody can spend it.

A second example shows what happens if the bakery pays the supplier late.

Estimating the money needed multiplies days by the daily cost.

The catering line spends Tk 2,400 a day on ingredients, so 28 days ties up Tk 67,200.

Taking the discount raises the money the bakery must fund by Tk 72,000.

The accounts record a receivable at the selling price, which is higher.

Float is money paid but not yet usable.

Float. Money that has been paid but cannot yet be used by the receiver.

Collection float delays the firm's receipts; disbursement float delays its payments.

A worked example finds the average delay and the money held in the gap.

Multiply each amount by its days, add, and divide.

Average delay is 4.8 days; average float is Tk 19,200.

Disbursement float works in the bakery's favour, but by less.

Reducing float releases money, but the method must cost less than it saves.

Common slips in cycle and float arithmetic.

Recap and self-test sentence.

Float has two directions. **Collection float** is money the firm has been paid, but its bank has not yet credited. It hurts the firm. **Disbursement float** is money the firm has paid by cheque, but its bank has not yet deducted. It helps the firm, because the bank balance stays higher for a few days. Net float is disbursement float less collection float. A positive net float helps; a negative one hurts.

The bakery receives catering cheques of Tk 1,20,000 in one 30-day month. *Given.* Tk 72,000 comes from offices in the city, and each cheque takes 4 days to become usable. Tk 48,000 comes from firms outside the city, and each takes 6 days. *Find.* the average delay and the average collection float.

The average delay is a weighted average, because the larger amount should count for more. In words: average delay equals the sum of each amount multiplied by its days, divided by the total amount. The average float equals the same sum divided by the number of days in the month.

COLLECTION FLOAT, ONE MONTH

GROUP	AMOUNT (TK)	DAYS	AMOUNT × DAYS
City offices	72,000	4	2,88,000
Outside the city	48,000	6	2,88,000
Total	1,20,000		5,76,000

Steps. the average delay is 5,76,000 divided by 1,20,000, which is 4.8 days. The average float is 5,76,000 divided by 30, which is Tk 19,200. *Answer.* cheques wait 4.8 days on average, and Tk 19,200 of the bakery's money is in the gap on an average day. *Check.* daily collections are Tk 4,000, and Tk 4,000 multiplied by 4.8 days is Tk 19,200.

Suppose the bakery pays the supplier Tk 2,400 a day on average by cheque, and each cheque takes 3 days to clear. The disbursement float is Tk 2,400 multiplied by 3, which is Tk 7,200. The net float is Tk 7,200 less Tk 19,200, which is minus Tk 12,000. On an average day, the bank shows Tk 12,000 less than the bakery's own records.

A firm can reduce collection float by depositing cheques on the day received, or by asking customers to pay by bank transfer. Suppose one day is cut from the average delay. The bakery releases one day of collections, Tk 4,000. At the 12 per cent bank rate, that money earns or saves Tk 4,000 multiplied by 0.12, which is Tk 480 a year. A method that costs more than Tk 480 a year would not be worth using.

Common slips. Students add the payable days instead of subtracting them. They use selling price when the question concerns money spent. They divide by the number of cheques instead of the amount. They forget that a negative cycle is possible and meaningful. Check the sign of each answer before moving on.

Before moving on. The cash conversion cycle is inventory days plus receivable days less payable days. Money needed is daily cost multiplied by the cycle. Float is money in the gap between payment and use. You should now be able to say how a 10-day delay in paying a supplier changes the cash a firm must find.

12.5 SHORT-TERM SOURCES: TRADE CREDIT, BANK AND OTHERS

A firm that needs money for a few weeks has three families of source. It may ask its suppliers for time to pay. It may ask its bank. Or it may use sources that arise from the firm's own routine. This section describes each; section 12.6 measures the cost of the first.

The first source is **trade credit**, which means buying goods now and paying the supplier at a later agreed date. Every payable is trade credit. For the bakery, ten days of credit on flour is worth the price of ten days of flour. Trade credit is the largest source of short-term money for most small firms, because suppliers give it as part of the sale.

The second source is the bank. An **overdraft** means bank permission to spend more than the balance in the account, up to a stated limit. The bank charges interest only on the amount used, and only for the days it is used. A short-term bank loan is different: the bank lends a fixed sum for a fixed period and charges interest on the whole sum.

Compare the two with numbers. The bakery needs Tk 60,000 for 60 days. At 12 per cent a year, an overdraft of Tk 60,000 costs Tk 60,000 multiplied by 0.12 multiplied by 60 and divided by 365, which is Tk 1,184. If the bank instead lends a fixed Tk 1,00,000 for the same 60 days, the interest is Tk 1,973. The larger loan costs more because the bakery pays for money it does not need.

The third family needs no bank and no supplier. A firm pays wages at the month's end and not daily, so for part of the month it owes its workers money: an accrued expense. A customer may pay in advance for a large order, which gives the firm money before it spends. Some firms also sell their receivables to a finance company for cash now, at a fee. Each of these is short-term money, and each has limits.

The next table compares the main sources. Read each row for one source. The cost column tells what the firm pays. The limit column tells what holds the source back.

SHORT-TERM SOURCES COMPARED

SOURCE	WHAT IT IS	COST	MAIN LIMIT
Trade credit	Time to pay a supplier	Free if paid in time; high if a discount is lost	Supplier may refuse or stop supplying
Overdraft	Spending beyond the balance	Interest on the amount used	Bank sets a limit and may ask for repayment at short notice
Short-term bank loan	Fixed sum for a fixed period	Interest on the whole sum	Bank may ask for collateral
Accrued expenses	Wages and bills not yet due	None	Limited by pay dates
Customer advances	Payment before delivery	None	Only if customers agree

The pattern in the table is common. The cheapest sources are limited in size, while the more flexible sources cost more. A firm usually uses the cheap sources first and the bank for the rest. From the supplier's seat,

Short-term money has three families of source.

Trade credit is buying now and paying the supplier later.

Trade credit. Buying goods now and paying the supplier at a later agreed date.

An overdraft lets the firm spend beyond its balance, up to a limit.

Overdraft. Bank permission to spend more than the balance in the account, up to a stated limit.

Overdraft interest is paid only on the money used.

The third family arises from the firm's own routine.

A table sets the main sources side by side.

Cheap sources are limited; costly sources are plentiful.

Recap: three families of source, with different costs and limits.

GO SLOWLY HERE.

A two per cent discount sounds small, but the period matters.

The terms 2/10, net 40 are a short code for a whole offer.

Credit terms. The days allowed for payment and any discount offered for early payment.

Cash discount. A price cut for paying within a stated short period.

Skipping the discount is borrowing from the supplier.

Words first, then symbols, for the yearly cost.

The formula for the yearly cost of giving up a discount.

First worked example: the bakery's invoice.

The cost is 24.8 per cent, twice the bank rate.

trade credit is a cost: the supplier waits for money it could use itself, and gives the credit only because it wants the sale.

Before moving on. Short-term money comes from suppliers, banks and the firm's own routine. An overdraft charges interest only on money used. You should now be able to name two sources that cost nothing and say what limits each.

12.6 THE COST OF TRADE CREDIT AGAINST A BANK LOAN

A supplier offers the bakery a two per cent discount for early payment. Two per cent sounds small. Is it worth taking? The answer depends on a number the offer does not show: the number of days the bakery gives up in return.

Credit terms are the days allowed for payment and any discount offered for early payment. The supplier writes them as a short code. The terms 2/10, net 40 mean this: take a **cash discount** of 2 per cent, a price cut for paying within a stated short period, if you pay within 10 days. Otherwise the full bill is due on day 40.

Skipping the discount is borrowing. Suppose the invoice is Tk 50,000. The discount is 2 per cent, which is Tk 1,000. If the bakery pays on day 10, it pays Tk 49,000. If it waits until day 40, it pays Tk 50,000. The extra Tk 1,000 buys 30 more days of use of Tk 49,000. The supplier is lending the bakery Tk 49,000 for 30 days, and charging Tk 1,000 for the loan.

We turn this into a yearly rate in two steps. First, find the cost for one period: the discount divided by the money actually used. Second, scale it up to a year by multiplying by the number of such periods in a year. In words: annual cost equals discount per taka paid, times 365, divided by the extra days.

In symbols: $\text{annual cost} = d / (100 - d) \times 365 / (N - D)$. Here d is the discount in per cent, N is the day the full bill is due, and D is the last day for the discount. Writing $100 - d$ in the first fraction reflects the money used: for each Tk 100 of price, the bakery uses only Tk 100 less the discount.³

Given. invoice Tk 50,000, terms 2/10, net 40, so d is 2, N is 40, D is 10. The bank charges 12 per cent a year. *Find.* the annual cost of skipping the discount, and whether the bakery should take it.

ANNUAL COST OF SKIPPING THE 2/10, NET 40 DISCOUNT

STEP	WORKING	RESULT
Cost for one period	$2 / 98$	2.04 per cent
Periods in a year	$365 / (40 - 10)$	12.17
Annual cost	$2.04 \text{ per cent} \times 12.17$	24.8 per cent

Answer. skipping the discount costs about 24.8 per cent a year, which is twice the bank rate of 12 per cent. The bakery should take the discount

³ The first fraction is the cost for one period: the discount per taka actually paid. The second fraction counts the periods in a year. This is the simple method. The compound method, which allows for interest on interest, gives a higher figure: about 27.9 per cent for the bakery's terms.

and borrow the money from the bank. *Check.* pay Tk 49,000 for 30 days at 12 per cent, which costs Tk 49,000 multiplied by 0.12 multiplied by 30 and divided by 365, or Tk 483. The discount saves Tk 1,000. The net gain is Tk 1,000 less Tk 483, which is Tk 517 on one invoice.

The rule follows. If the annual cost of giving up the discount is higher than the cost of other short-term money, take the discount and borrow. If it is lower, skip the discount and use the credit. In the bakery's case, 24.8 per cent is above 12 per cent, so the choice is easy.

Now a second, more awkward example. A second supplier offers 1/15, net 45 on the same Tk 50,000. Here d is 1, N is 45, D is 15. The cost for one period is 1 divided by 99, which is 1.01 per cent. The periods in a year are 365 divided by 30, which is 12.17. The annual cost is 1.01 per cent multiplied by 12.17, which is 12.3 per cent. The bank rate is 12 per cent, so the two are almost equal.

Test the gain. The discount is Tk 500, so the bakery would pay Tk 49,500. Borrowing Tk 49,500 for 30 days at 12 per cent costs Tk 488. The net gain is Tk 500 less Tk 488, which is Tk 12. Such a small gain may not repay the effort of arranging a loan, and a rise in the bank rate would remove it. When the two rates are this close, other matters, such as the bank's reliability, can decide.

There is a further exception. Suppose the bakery pays the first supplier on day 60 instead of day 40. The extra days then number 50, and the annual cost falls to 2 divided by 98 multiplied by 365 divided by 50, which is 14.9 per cent. The cost falls, because the same Tk 1,000 buys more days. Even so, late payment can harm the relationship with the supplier, who may stop giving credit. The arithmetic gives only one part of the decision.

Common slips. Students divide the discount by 100 instead of by 100 less the discount. They use the full credit period N and not the extra days N less D . They forget that the discount is a saving to be set against the loan's interest. They mix years of 360 and 365 days.⁴ Check each answer by asking: if I borrowed this money from the bank instead, what would it cost?

Before moving on. Giving up a discount is borrowing from the supplier at a yearly cost of $d / (100 - d) \times 365 / (N - D)$. Take the discount when this cost exceeds the bank rate. You should now be able to say why 2/10, net 40 costs far more than 2 per cent.



THE WORKED CASE: ONE INVOICE, ONE DECISION

Case file, illustrative. Situation. On 10 September 2026 Sundial Bakery holds a flour invoice of Tk 50,000 with terms 2/10, net 40, dated that day. Cash is Tk 18,000. Customers owe the bakery Tk 1,20,000, but not before the end of the month. The bank would give an overdraft at 12 per cent a year.

Question. Should the bakery pay on day 10 with an overdraft, or wait until day 40?

The rule: take the discount when the credit costs more than the loan.

A second example has awkward numbers and a close decision.

When rates are close, the gain is too small to matter.

Paying late is an exception, and it carries risks that arithmetic does not show.

Common slips in the trade credit formula.

Recap: a discount lost is a loan taken from the supplier.

The situation: a Tk 50,000 invoice and Tk 18,000 in cash.

The question: pay early with borrowed money, or wait?

⁴ Some books use a year of 360 days to keep the arithmetic short. This book uses 365 days throughout. The answers differ by a small amount, and the method is the same.

The reasoning uses the cost formula and the matching principle.

Conclusion: take the discount and borrow.

The check repeats the gain by the second method.

The chapter's main ideas in one paragraph.

A new problem with new data, to test the whole chapter.

Profit is not cash.

Positive net working capital is not the same as cash in hand.

A small discount percentage hides a large yearly cost.

Long-term money is not always the safer choice.

Reasoning. Paying on day 10 costs Tk 49,000. The cash is Tk 18,000, so the bakery needs an overdraft. Skipping the discount costs 24.8 per cent a year, as section 12.6 showed. The overdraft costs 12 per cent. The bakery needs the money for 30 days, which is a short need, and an overdraft is a short-term source, so the funding fits the need by the matching principle.

Conclusion. Take the discount and pay Tk 49,000 on day 10, using an overdraft. The overdraft interest for 30 days is Tk 483, and the discount is Tk 1,000. The bakery gains Tk 517.

Check. Cost of the discount lost is 24.8 per cent, against 12 per cent: the difference favours borrowing. Net gain by the money method is Tk 1,000 less Tk 483, which is Tk 517. Both methods agree.

SUMMARY

Working capital is the money tied up in cash, receivables and inventory, and net working capital subtracts the short-term debts. A firm can have positive net working capital and still lack cash, because most of it is not cash. Financing is short-term, medium-term or long-term, and the matching principle says to fund short needs with short money and long needs with long money. The cash conversion cycle is inventory days plus receivable days less payable days, and money needed is daily cost multiplied by that cycle. Float is money paid but not yet usable, and net float is disbursement float less collection float. Short-term sources are trade credit, bank overdraft and loans, and the firm's own routine. Giving up a discount is borrowing from the supplier at a yearly cost that is often far above the bank rate.

THINK IT THROUGH

Case file, illustrative. A different supplier offers the bakery terms of 3/15, net 60 on an invoice of Tk 80,000. The bank rate is 14 per cent a year. The owner says: "Three per cent is a small gain, and I prefer to keep my cash." Find the annual cost of skipping the discount, state whether the owner is right, and name one matter, apart from the arithmetic, that the owner should weigh. Then say what happens to the bakery's cash conversion cycle if it takes the discount.

WHERE PEOPLE GO WRONG

A profitable firm always has cash. It does not. Profit counts a sale on the day it is made; cash arrives when the customer pays. A firm can earn profit and fail to pay its bills in the same month.

Positive net working capital means the firm can pay now. It does not. Net working capital includes receivables and inventory, which are not cash. Only cash pays a bill today.

Two per cent is too small to matter. The discount is earned over a short period. Scaled to a year, 2 per cent for 30 days is 24.8 per cent, which is more than the bank charges.

Borrowing for the longest period is always safer. It is safer against refinancing, but it costs interest on idle money when the need is temporary. Matching, not length alone, is the guide.

Float is lost money. Float is money in transit. It is not lost, but while it is in the gap, nobody can use it. The task is to shorten the gap when the saving exceeds the cost.

Float is delay, not loss.

IN HISTORY

After the Second World War, Japanese manufacturers had little money to spare for large stocks of parts. At Toyota Motor Corporation, engineers led by Taiichi Ohno developed a system in which parts arrive at the production line shortly before they are used. The idea is widely known as the just-in-time method, and writers connect it with Kiichiro Toyoda, the company's founder. The system lowered inventory, and so it lowered the money tied up in daily work. The lesson for this chapter is plain: when a firm shortens the time goods wait, it releases money for other uses.

A manufacturer cut the money tied up in inventory by changing how goods arrived.

REVIEW QUESTIONS

REVIEW QUESTIONS

Ten questions to test understanding of the chapter.

NO.	QUESTION
1	Define working capital, and explain in two sentences why net working capital can be positive while cash is short.
2	Name the three kinds of current asset. Give one example of each from a bakery.
3	State the periods that this book calls short-term, medium-term and long-term.
4	State the matching principle. Explain what goes wrong when a firm uses a short loan for a long need.
5	A firm holds inventory for 12 days, customers pay in 45 days, and the firm pays its supplier in 20 days. Find the cash conversion cycle.
6	A firm spends Tk 3,000 a day on ingredients and has a cash conversion cycle of 25 days. How much money is tied up?
7	Explain collection float and disbursement float. State which one helps the firm, and why.
8	Cheques of Tk 60,000 take 3 days and cheques of Tk 40,000 take 5 days. Find the average delay and the average float over a 30-day month.
9	Compare an overdraft with a short-term bank loan. Which charges interest on money not used?
10	Terms are 1/10, net 30. Find the annual cost of skipping the discount. Should a firm that borrows at 10 per cent take the discount?

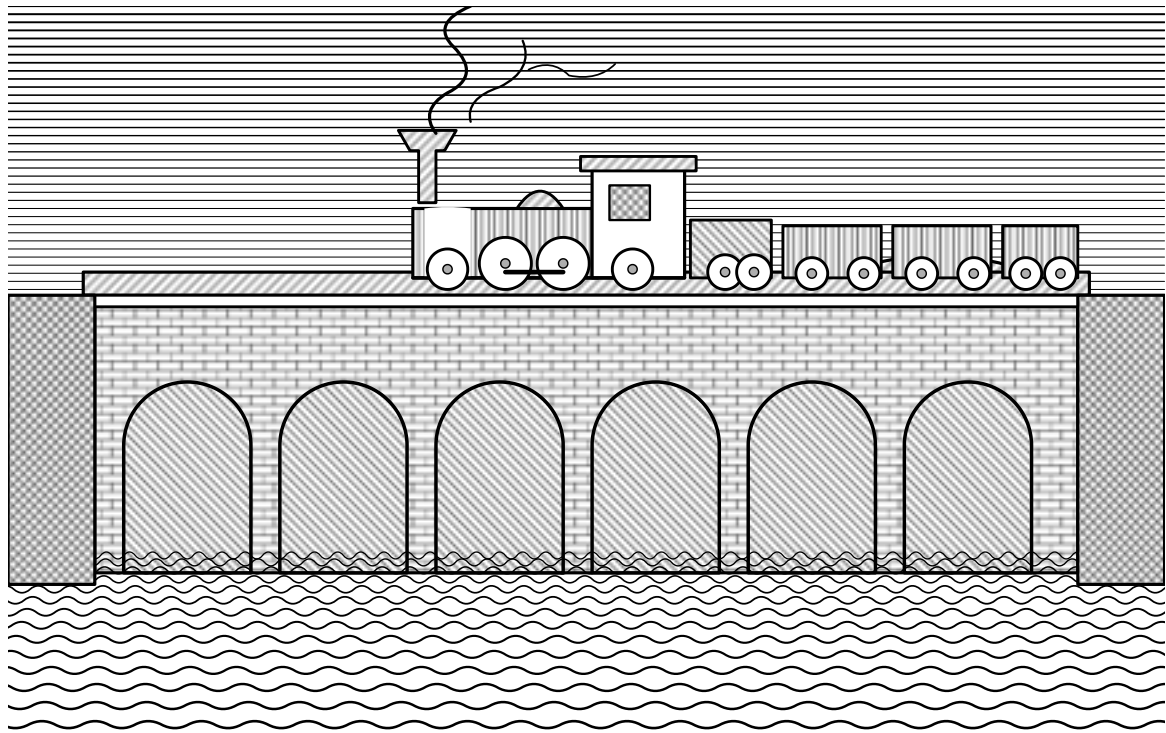


Plate XIII. The Viaduct.

CHAPTER XIII

Long – Term Debt Financing

A loan is a promise about the future, and the future keeps its own accounts.



This chapter explains how a firm borrows money for years and not for weeks. It shows why a new firm needs long-term funds, what every source of money asks in return, how term loans, debentures and mortgages differ, and how a repayment schedule and the cost of debt are worked out.

A longer loan can cost more interest and still be the better loan.

The answer depends on when the money arrives, not only on how much interest is paid.

The chapter relies on annuities, financial leverage and bond features.

A firm should hold long-lasting assets with long-lasting money.

A new firm earns slowly, so it needs time before money returns.

A new firm has fewer options than an old one.

Long-term funds also keep short-term money free for daily work.

A test with numbers compares a one-year loan with five yearly instalments.

Consider a puzzle. A bank offers the baker Tk 3,00,000 at 12 per cent a year. The baker may repay all of it after one year, or repay it in five equal yearly instalments. The rate is the same in both cases. The five-year loan costs far more interest in total. Why would a sensible baker choose it?

The answer depends on a question that the interest figure does not ask: when does the money to repay arrive? A loan is a promise about the future. The borrower promises to hold, on certain days, the money to repay. If the business earns that money later than the lender demands it, the promise cannot be kept, however low the interest. This chapter studies loans of long length, and the way a borrower and a lender fit the promise to the business.

You will need the annuity tables of Chapter 5, the idea of financial leverage from Chapter 9, and the bond words of Chapter 11: face value, coupon rate and maturity. We apply them to the oven loan of Sundial Bakery.

13.1 WHY A NEW FIRM NEEDS LONG-TERM FUNDS

In Chapter 12 we learned the matching principle: fund long needs with long money. An oven lasts for years, so the money that buys it should be available for years. This is the first reason a firm needs long-term funds. It buys assets that last, and it should not have to repay their cost before they have earned it.

The second reason is speed. A new firm earns slowly. Customers must learn the shop exists, a baker must learn how much to bake, and the first months often bring losses. A loan that is due in one year may fall due before the firm has learned to earn. A loan of five years gives the business time.

The third reason concerns a new firm in particular. It has no record of paying debts, and its profit is uncertain, so lenders see it as risky. Short loans must be renewed again and again, and each renewal is a new decision for the lender. If one lender refuses, the firm may be forced to repay at once. A long loan, agreed once, protects the firm from that risk.

The fourth reason follows from Chapter 12. Working capital is needed every day. If a firm uses its cash and overdraft to pay for ovens, it has less money for flour and wages. Long-term funds pay for the long-term assets, and short-term money stays free for daily work.

We now test the puzzle with numbers. The oven costs Tk 3,00,000 and brings in Tk 1,00,000 a year, as in Chapter 7. For this test, we ignore tax and treat the oven's yearly inflow as cash available to repay. The next table compares what the bakery owes at the end of the first year under each plan. Read each row from left to right. The last column is the difference between what comes in and what must be paid out.

CASH DUE AT THE END OF YEAR 1 AGAINST THE OVEN'S INFLOW

PLAN	DUE (TK)	OVEN INFLOW (TK)	DIFFERENCE (TK)
Repay all after one year	3,36,000	1,00,000	minus 2,36,000
Five equal yearly instalments	83,223	1,00,000	plus 16,777

The one-year loan falls due as Tk 3,00,000 plus Tk 36,000 of interest, which is Tk 3,36,000. The oven has brought in only Tk 1,00,000, so the bakery is Tk 2,36,000 short. The five-year plan asks for Tk 83,223 in the first year, which is less than the oven's inflow, so the bakery keeps Tk 16,777. Section 13.5 shows how the instalment of Tk 83,223 is found.

The five-year plan costs more interest, Tk 1,16,114 against Tk 36,000. This is the price of time: the borrower keeps the lender's money for longer and pays for each year of keeping it. The baker accepts the larger sum because the smaller sum would have to be paid before the oven could earn it.

Before moving on. A firm needs long-term funds to match long-lived assets, to give a new business time to earn, to avoid repeated renewal risk, and to keep short-term money free. You should now be able to say why a larger total of interest can still be the better choice.

The one-year loan leaves a gap of Tk 2,36,000; the five-year loan leaves a surplus.

The longer loan pays more interest in total, which is the price of more time.

Recap: long-term funds fit long-lived assets and give a new firm time and safety.

13.2 WHAT EVERY SOURCE OF MONEY ASKS IN RETURN

Whoever gives money to a firm expects something in return. Before we study individual loans, we need four questions that can be asked of any source of money. We call them claim, cost, control and risk. They apply to a bank, to a supplier and to an owner. Each question has a simple everyday form.

The first question is about the **claim**, which means a right to be paid a stated sum, or to share in what is left. A friend who lends Tk 1,000 has a claim for exactly Tk 1,000, plus any agreed interest. A friend who becomes a partner in a business has a claim to a share of the profit, whatever it turns out to be. The first claim is fixed. The second is not.

The second question is about cost: what does the firm pay for using the money? For a loan, the cost is interest and any fees. The third question is about control: what say does the giver of money have in running the business? A bank that lends does not vote on what to bake. An owner usually does. The fourth question is about risk, and it needs more care.

Risk to the firm arises from the **fixed charge**, which means a payment the firm must make on a set date, whatever its profit. In Chapter 9 the bakery's interest of Tk 7,000 a month was a fixed charge. It was due in a quiet month as well as a busy one, and it made profit swing more widely when sales moved. If the firm fails to pay, the lender may take legal action and may take the collateral, which section 13.4 explains.

The next table compares the two main kinds of money on the four questions. The first kind is debt, which is money borrowed from lenders. The second is owners' money, which Chapter 14 studies fully. Read down each column to see the character of each kind.

GO SLOWLY HERE.

Every source of money asks for four things in return; four questions reveal them.

A claim is the right to be paid, or to share what is left.

Claim. A right to be paid a stated sum, or to share in what is left.

Cost is the price the firm pays for using the money.

A fixed charge must be paid whether or not the firm earns a profit.

Fixed charge. A payment the firm must make on a set date, whatever its profit.

A table compares borrowed money with owners' money on the four questions.

DEBT AND OWNERS' MONEY ON FOUR QUESTIONS

QUESTION	DEBT (LENDER'S MONEY)	OWNERS' MONEY (SEE CHAPTER 14)
Claim	A fixed sum, repaid on agreed dates	A share of what is left after all debts
Cost	Interest, and fees	A share of profit, which is not fixed
Control	None over daily decisions, but promises may limit the firm	Votes and a say in decisions
Risk to the firm	High: fixed charges must be paid	Lower: no fixed payment

Debt is cheaper and fixed; owners' money is flexible and carries a share of control.

Recap: four questions for any source: claim, cost, control and risk.

A term loan is a bank loan repaid over several years.

Term loan. A bank loan that is repaid over several years.

Every term loan has a short list of features.

The lender asks three questions before lending.

The loan moves through a sequence from request to final payment.

The table shows a balance. Debt gives the firm money without sharing control, and the interest it costs is fixed in advance. But the firm must pay it. Owners' money asks for no fixed payment, but it gives the owners a say and a share. Neither is better for every firm. A wise firm uses both and watches the balance, as Chapters 9 and 15 explain.

Before moving on. Ask four questions of any source of money: what is the claim, what is the cost, who has control, and what is the risk? Debt has a fixed claim and a fixed charge. You should now be able to say why debt adds risk even when its interest is low.

13.3 TERM LOANS

A **term loan** means a bank loan that is repaid over several years. The word term here means the period of the loan. Term loans are the most common way for a small firm to borrow for an oven, a van or a shop fitting. Chapter 12 named the medium-term and long-term periods; term loans usually run from one year to ten years or more, so they cover both.

A term loan has features that are agreed before the money is paid. The first is the amount. The second is the term, which is the number of years until the loan is fully repaid, and the date of full repayment is its maturity, a word we met in Chapter 11. The third is the interest rate, which may stay fixed or may change with market rates. The fourth is the pattern of repayment: regular instalments, or one payment at the end. The fifth is any collateral, and the sixth is any promise the borrower must keep. We study these in section 13.4.

Why does the bank lend, and on what conditions? A bank wants three things. It wants a clear purpose, such as an oven that will earn income. It wants a clear source of repayment, which is usually the cash the business earns. And it wants a second source in case the first fails, which is usually collateral. A firm that shows all three has a better chance of a loan on good terms.

A term loan moves through a sequence. The firm asks for the loan and shows its plans. The bank examines the plans and the firm's accounts. If the bank agrees, both sign a written agreement that states the terms. The bank then pays out the money, and the firm uses it for the stated purpose. The firm repays on the agreed dates until the last payment is made. The details differ between banks and countries, and the borrower should read each document before signing.

In Bangladesh, term loans are given by commercial banks and by institutions set up to finance long-term projects. Chapter 14 describes these institutions. The loan terms, rates and rules change over time, so any figure in this book is illustrative and should be checked against a lender's current offer.

Before moving on. A term loan is a bank loan repaid over several years, with an agreed amount, term, rate, repayment pattern, collateral and promises. A bank lends when it sees a purpose, a source of repayment and a second source. You should now be able to list the features of a term loan.

Lenders in Bangladesh include commercial banks and development institutions.

Recap: a term loan has an amount, term, rate, repayment pattern, collateral and promises.

13.4 METHODS OF DEBT FINANCING

A firm has more than one way to borrow for the long term. It may borrow from one bank, which is the term loan. It may borrow from many lenders at once, by selling written promises to repay. Or it may borrow against land or a building. This section studies the second and third ways, and then the terms on which any loan is made.

A **debenture** means a firm's written promise to repay a loan with interest, backed by the firm's general credit and not by one named asset. A debenture is a kind of bond, so it has the features of Chapter 11: a face value, a coupon rate and a maturity. A firm sells debentures to many lenders, each of whom lends a small part of the total. Debentures suit large, well-known firms. A small bakery would not issue them.¹

A firm may borrow from one bank, from many lenders, or against property.

A debenture is a written promise to repay, backed by the firm's general credit.

Debenture. A firm's written promise to repay a loan with interest, backed by the firm's general credit.

A **mortgage** means a loan secured on land or a building, which the lender may sell if the loan is not repaid. A baker who wishes to buy the shop in which the bakery works might take a mortgage from a bank. The bank holds a claim on the shop until the last instalment is paid. When the loan is repaid, the claim ends.

A mortgage is a loan secured on land or a building.

Mortgage. A loan secured on property, which the lender may sell if the loan is not repaid.

The mortgage is one case of a wider idea. **Collateral** means an asset pledged to the lender, to be sold if the loan is not repaid. A **secured loan** is a loan backed by collateral. An **unsecured loan** is a loan not backed by collateral, and its lender relies on the borrower's general credit. Lenders charge less for a secured loan, because their risk is smaller. Some writers say security for collateral, but this book keeps collateral, because Chapter 11 gave security another meaning.²

Collateral is an asset pledged to back a loan.

Collateral. An asset pledged to the lender, to be sold if the loan is not repaid.

Secured loan. A loan backed by collateral.

Unsecured loan. A loan not backed by collateral.

Lenders also protect themselves by promises. A **covenant** means a promise in a loan agreement that limits what the borrower may do, or requires it to do. A covenant may require the firm to send its accounts to the bank each year, or to insure the oven. It may forbid the firm to borrow more without the bank's agreement. The firm gives up some freedom, and in return the lender lends on better terms.

A covenant is a promise by the borrower that limits what the firm may do.

Covenant. A promise in a loan agreement that limits what the borrower may do or requires it to do.

The last word in this group is the one every borrower wishes to avoid. **Default** means failure to pay a loan as agreed. A default allows the lender to use the rights in the agreement: to demand repayment of the whole balance, and to sell the collateral. Covenants are often

Default is failure to pay as agreed, and the lender's rights then begin.

Default. Failure to pay a loan as agreed.

¹ In some countries a debenture is a loan backed by named assets, and in others it is an unsecured loan. This book follows the second meaning. A reader should check how the word is used in the law of the country concerned.

² In daily speech, lenders say they take security for a loan, meaning collateral. In Chapter 11, security meant a bond or a share that can be traded. The two meanings are different.

A table sets the three methods side by side.

written so that breaking one counts as a default, even if all payments are on time.

The next table sets the three methods side by side. Read each row for one method. The last column applies the method to Sundial Bakery.

THREE WAYS OF BORROWING FOR THE LONG TERM

METHOD	WHO LENDS	BACKING	USUAL USE	FOR SUNDIAL BAKERY
Term loan	One bank	Often collateral	Equipment, expansion	Suitable: the oven
Debenture	Many lenders	General credit	Large sums for large firms	Not suitable: too small
Mortgage	A bank or similar lender	Land or building	Premises	Suitable if the bakery buys its shop

Recap: methods differ by lender, backing and use.

Before moving on. Methods of long-term debt are the term loan, the debenture and the mortgage. Collateral backs a secured loan, covenants limit the borrower, and default lets the lender act. You should now be able to say which method suits a small firm that buys an oven.

13.5 REPAYMENT SCHEDULES AND THE COST OF DEBT

GO SLOWLY HERE.

Most term loans are repaid in regular instalments, each with two parts.

An instalment loan is repaid by regular payments of interest and principal.

Instalment loan. A loan repaid by regular payments, each including interest and part of the amount borrowed.

Principal (of a loan). The amount borrowed that is still owed. In Chapter 2, principal meant the one who hires an agent.

A repayment schedule lists every instalment and splits it.

Repayment schedule. A table listing each instalment, how much of it is interest, and how much repays the principal.

The instalment is the loan divided by the annuity factor.

A student who borrows money for fees might repay a little each month. Each payment does two jobs. It pays the interest for the month, and it reduces the amount still owed. Most term loans work in the same way.

An **instalment loan** means a loan repaid by regular payments, each including interest and part of the amount borrowed. The amount borrowed that is still owed is called the **principal**. Interest is charged on the principal that remains, so as the principal falls, the interest in each instalment falls too.

A **repayment schedule** means a table that lists each instalment, how much of it is interest, and how much repays the principal. A borrower who understands a schedule can see exactly what is owed on any date.

How is the instalment found? The instalments form an ordinary annuity, which we studied in Chapter 5. The loan is the present value of those instalments, at the loan's interest rate. In words: instalment equals the loan divided by the present value annuity factor. In symbols: $P = L / A$, where P is the instalment, L is the loan, and A is the factor for the rate and the number of years.

Given. The loan L is Tk 3,00,000. The rate is 12 per cent a year, and the term is 5 years, with one instalment at the end of each year. The factor table of Chapter 5 gives A as 3.6048. *Find.* The yearly instalment.

A first worked example finds the oven loan's instalment.

Steps. P equals 3,00,000 divided by 3.6048, which is Tk 83,222.4. A bank rounds an instalment up to the next whole taka, so that the loan is surely repaid, and the instalment is Tk 83,223. *Answer.* The bakery pays Tk 83,223 at the end of each of the five years. *Check.* The schedule below must end at zero.

The next table is the repayment schedule. Read each row from left to right. The interest is 12 per cent of the opening balance. The principal repaid is the instalment less the interest. The closing balance is the opening balance less the principal repaid, and it becomes the next year's opening balance. In the last year, the instalment is Tk 1 smaller, because of rounding.

REPAYMENT SCHEDULE: TK 3,00,000 AT 12 PER CENT, FIVE EQUAL YEARLY INSTALMENTS

YEAR	OPENING BALANCE	INTEREST	INSTALMENT	PRINCIPAL REPAID	CLOSING BALANCE
1	3,00,000	36,000	83,223	47,223	2,52,777
2	2,52,777	30,333	83,223	52,890	1,99,887
3	1,99,887	23,986	83,223	59,237	1,40,650
4	1,40,650	16,878	83,223	66,345	74,305
5	74,305	8,917	83,222	74,305	0
Total		1,16,114	4,16,114	3,00,000	

The check passes. The closing balance in year 5 is zero, and the principal repaid adds to Tk 3,00,000. Total interest is Tk 1,16,114, which is Tk 4,16,114 paid less Tk 3,00,000 borrowed. Each instalment is the same size, but its content changes: in year 1, Tk 36,000 of it is interest, and in year 5 only Tk 8,917. The figure below shows this change as bars. Each bar is one year's instalment. The hatched part is interest, and the blank part repays principal.

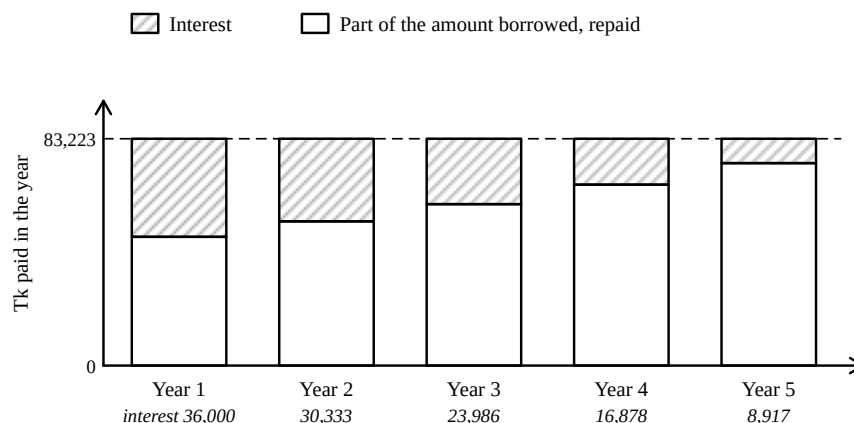


FIG. 36. Each year's instalment of the oven loan: the hatched part is interest, which falls, and the blank part repays the amount borrowed, which rises.

Not every loan has equal instalments. Some banks ask the borrower to repay equal parts of the principal each year, and to add the interest on the balance. Here the principal repaid is Tk 3,00,000 divided by 5, which is Tk 60,000 each year. The interest is 12 per cent of the opening balance, so the yearly payment starts high and falls.

One division gives the instalment.

The schedule shows interest, principal and the balance, year by year.

The check passes: the balance falls to zero and the principal repaid is Tk 3,00,000.

A second worked example repays equal parts of the principal instead.

REPAYMENT SCHEDULE: THE SAME LOAN, EQUAL REPAYMENTS OF PRINCIPAL

YEAR	OPENING BALANCE	INTEREST	PRINCIPAL REPAID	PAYMENT	CLOSING BALANCE
1	3,00,000	36,000	60,000	96,000	2,40,000
2	2,40,000	28,800	60,000	88,800	1,80,000
3	1,80,000	21,600	60,000	81,600	1,20,000
4	1,20,000	14,400	60,000	74,400	60,000
5	60,000	7,200	60,000	67,200	0
Total		1,08,000	3,00,000	4,08,000	

The equal-principal plan pays less interest but asks more in the early years.

The total interest is Tk 1,08,000, which is Tk 8,114 less than under equal instalments, because the balance falls faster. But the first payment is Tk 96,000, against Tk 83,223. The oven brings in Tk 1,00,000 in year 1, so under this plan only Tk 4,000 remains, and a poor month could cause trouble. The next figure draws the balance still owed under the two plans. The horizontal axis shows years after the loan is drawn, and the vertical axis shows the amount still owed.

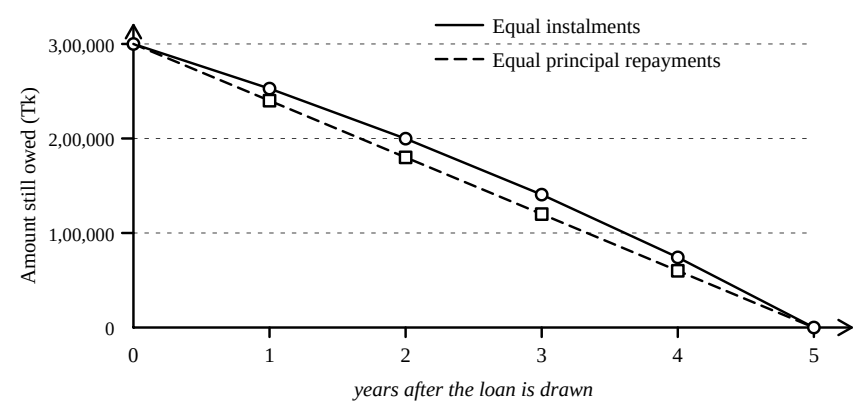


FIG. 37. The amount still owed on the oven loan under two repayment plans: the equal-principal plan falls faster.

The cost of a loan is the interest rate on the balance, before tax.

Interest reduces taxable profit, and so it reduces tax.

Tax shield on interest. The tax saved because interest is subtracted from profit before tax is calculated.

A table compares tax with and without the loan's interest in year 1.

Now the cost of the debt. In a simple loan with no fees, the cost before tax is the interest rate on the balance, which is 12 per cent a year. A fee paid at the start raises the cost, because the borrower receives less than the amount on which interest is charged. A borrower should always ask for the total of interest and fees, and not the rate alone.

A second effect lowers the cost. The firm subtracts interest from its profit before it calculates tax. The tax saved is called the **tax shield on interest**, which means the tax saved because interest is subtracted from profit before tax is calculated. Chapter 3 introduced interest and tax in the profit layout. We use the illustrative tax rate of 25 per cent.³

Given. In year 1 the firm's profit before interest and tax is Tk 1,00,000, an illustrative figure. The loan's interest for the year is Tk 36,000. *Find.* The tax saved, and the real cost of the interest.

³ This rate is a placeholder. The rate of company tax in Bangladesh depends on the type of company, and it changes. Verify the current rate and the firm's type before using it in practice.

YEAR 1: TAX WITH AND WITHOUT THE LOAN'S INTEREST

LINE	WITHOUT THE INTEREST	WITH THE INTEREST
Profit before interest and tax	1,00,000	1,00,000
Interest	0	36,000
Profit before tax	1,00,000	64,000
Tax at 25 per cent	25,000	16,000
Profit after tax	75,000	48,000

Steps. Tax falls from Tk 25,000 to Tk 16,000, a saving of Tk 9,000, which equals 25 per cent of Tk 36,000. Profit after tax falls only from Tk 75,000 to Tk 48,000, a fall of Tk 27,000. *Answer.* Tk 36,000 of interest costs the firm Tk 27,000 after tax. As a rate on Tk 3,00,000, that is 9 per cent. *Check.* In words: after-tax cost equals the rate times one less the tax rate. In symbols: $12 \text{ per cent} \times (1 - 0.25) = 9 \text{ per cent}$. Chapter 15 uses this figure.

The tax shield has limits. It works only when the firm has taxable profit. A firm that makes a loss pays no tax, so interest saves it nothing now. It also does not make debt free. The firm still pays Tk 27,000 for Tk 3,00,000 in this year, a cost of 9 per cent. And tax rules change, so the rate and the rules should be verified for the firm's own type.

Common slips. Students charge interest on the original loan each year instead of on the balance. They forget that the last instalment may differ by rounding. They treat the instalment as all interest or all principal. They apply the tax saving to the whole instalment, when it applies only to the interest. A schedule that does not end at zero has an error.

Before moving on. An instalment loan is repaid by equal instalments, found by dividing the loan by the annuity factor. A schedule shows interest falling and principal rising. The tax shield lowers the cost of 12 per cent to 9 per cent. You should now be able to say why year 5's instalment holds less interest than year 1's.

The firm saves Tk 9,000 of tax, so Tk 36,000 of interest costs Tk 27,000.

The tax shield works only when the firm has profit to tax.

Common slips in schedules and cost.

Recap: a schedule splits each instalment; after-tax cost is lower than the rate.



THE WORKED CASE: THE OVEN LOAN

Case file, illustrative. Situation. On 1 August 2026 Sundial Bakery draws a term loan of Tk 3,00,000 to buy the second oven chosen in Chapter 7. The rate is 12 per cent a year for 5 years. The bakery repays by equal instalments of Tk 83,223 on 1 August of each year from 2027 to 2031, with a final instalment of Tk 83,222. The new oven is the collateral. There are two covenants: the bakery sends its accounts to the bank each year, and does not borrow more without the bank's agreement.

Question. Can the oven's inflow of Tk 1,00,000 a year meet the instalment? What is the loan's cost after tax?

Reasoning. The instalment of Tk 83,223 is less than the inflow of Tk 1,00,000, so the oven covers it, with Tk 16,777 to spare in each of the first four years and Tk 16,778 in the fifth. The loan's term of five years equals the oven's life of five years, so the long money fits the long asset. Interest in year 1 is Tk 36,000, and the tax shield at 25 per cent is Tk 9,000.

The situation: a loan of Tk 3,00,000 on agreed terms.

The question: can the oven repay the loan, and what does the loan cost?

The reasoning uses the schedule and the matching principle.

Conclusion: the loan can be repaid from the oven's earnings.

The check confirms the schedule closes at zero.

The chapter's main ideas in one paragraph.

A new problem with new data, to test the whole chapter.

A longer loan is not necessarily a worse loan.

Instalments are equal, but their parts are not.

The tax shield does not make debt free.

A debenture is a loan, not a share.

Collateral is not a payment.

Railway companies showed how long-term debt can finance very large projects.

Conclusion. The loan can be repaid from the oven's earnings. Its cost before tax is 12 per cent, and after tax it is 9 per cent. The total interest is Tk 1,16,114.

Check. The schedule ends at zero. The principal repaid adds to Tk 3,00,000. After-tax cost is 12 per cent $\times$ 0.75, which is 9 per cent, as in the table of year 1.

SUMMARY

A new firm needs long-term funds to match long-lived assets, to give the business time to earn, and to avoid repeated renewal risk. Any source of money can be examined by four questions: claim, cost, control and risk. Debt gives a fixed claim and a fixed charge. A term loan is a bank loan repaid over several years. Debentures are sold to many lenders, and a mortgage is secured on property. Collateral backs a secured loan, covenants limit the borrower, and default lets the lender act. An instalment is found by dividing the loan by the annuity factor, and a repayment schedule splits each instalment between interest and principal. Interest reduces tax, so the cost of debt after tax is the rate times one less the tax rate.

THINK IT THROUGH

Case file, illustrative. A café borrows Tk 2,00,000 at 10 per cent a year for 4 years, with equal yearly instalments. The annuity factor for 10 per cent and 4 years is 3.1699, and the tax rate is 25 per cent. Find the yearly instalment, the interest in year 1, the principal repaid in year 1, and the balance at the end of year 1. Find the tax saved in year 1 and the cost after tax. Then say which would be more dangerous for a new café, a loan of one year or a loan of four years, and why.

WHERE PEOPLE GO WRONG

The loan with the most interest is the worst loan. Not so. A longer loan pays more interest because it uses the money for longer. If the shorter loan falls due before the asset earns, the shorter loan is the worse choice.

Equal instalments mean equal interest each year. They do not. Interest is charged on the balance, which falls each year, so the interest part falls and the principal part rises.

The tax shield makes borrowing free. It does not. The firm saves tax only on the interest, and it still pays the rest. At 25 per cent tax, 12 per cent debt costs 9 per cent.

The holder of a debenture is an owner. No. A debenture holder is a lender. The holder has a fixed claim and no vote, unlike an owner.

Collateral is paid to the lender. No. Collateral remains the borrower's, and the lender takes it only if the borrower defaults.

IN HISTORY

In the nineteenth century, railway companies in Britain and elsewhere needed sums far larger than their founders could supply. Alongside shares, many raised money by issuing debentures, which promised fixed interest and were often backed by the company's property and earnings.

The method allowed thousands of lenders each to lend a small sum to one firm. It made large, long-lived assets possible. It also showed the risk of fixed charges: companies that did not earn enough from their traffic could not pay the interest, and some fell into financial difficulty. The lesson for this chapter is that long-term debt can build what short-term money cannot, but its promise must be kept.

REVIEW QUESTIONS

REVIEW QUESTIONS

*Ten questions to test
understanding of the chapter.*

NO.	QUESTION
1	Give two reasons why a new firm needs long-term funds.
2	Name the four questions that can be asked of any source of money, and apply each to a bank loan.
3	Define fixed charge. Explain why a fixed charge increases risk.
4	List the features of a term loan.
5	State two differences between a debenture and a mortgage.
6	Define collateral and covenant. Give one example of each.
7	A loan of Tk 1,00,000 at 10 per cent for 3 years has a factor of 2.4869. Find the yearly instalment.
8	For the loan in question 7, find the interest and the principal repaid in year 1, and the closing balance after year 1.
9	Explain the tax shield on interest. A firm pays interest of Tk 20,000 and the tax rate is 25 per cent. How much tax does it save?
10	Compare equal instalments with equal repayments of principal. State one advantage of each.

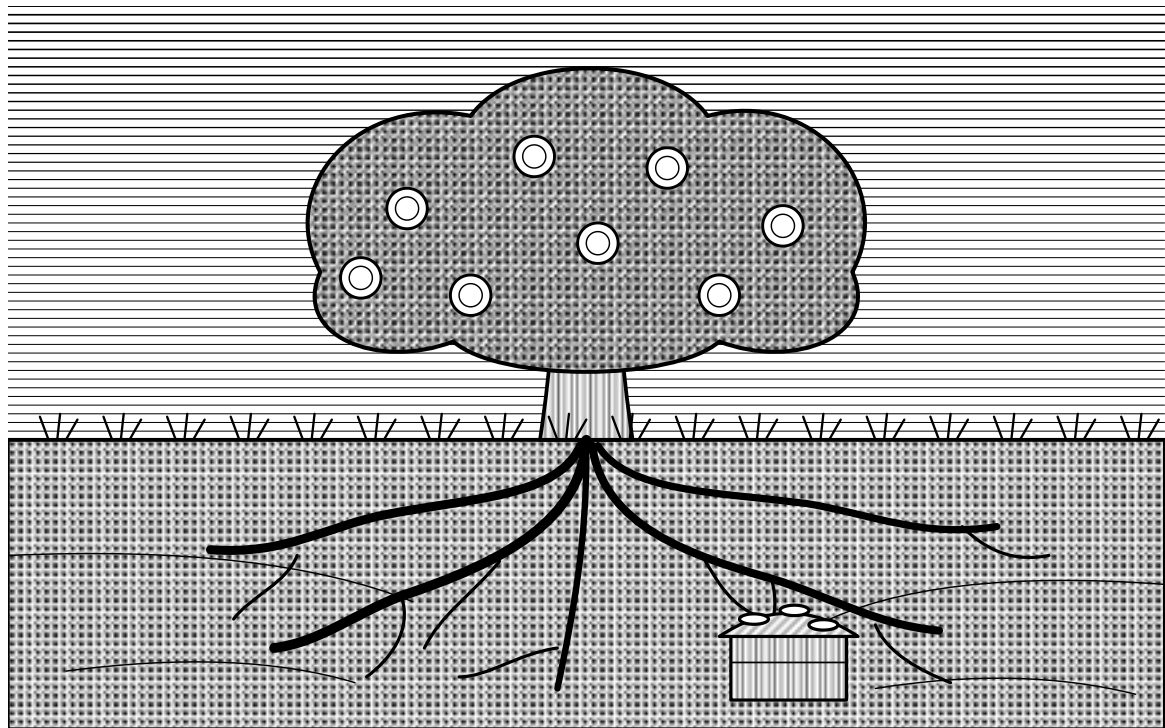


Plate XIV. The Tree and its Roots.

CHAPTER XIV

Equity, Retained Earnings and the Capital Market

An owner's money asks for no promise, and so it asks for a share of everything.



This chapter explains the money that owners supply: common shares, preferred shares and profit kept inside the firm. It shows how a share is priced when a new investor joins, how the claims of lenders, preferred owners and common owners are ranked, how a firm raises funds in the capital market, and which institutions in Bangladesh supply long-term finance.

A friend's money buys less of the firm than the friend expects.

The bakery is worth more than the money first put in, because it has earned and kept profit.

The chapter has five parts: common shares, preferred shares, kept profit, the capital market and Bangladeshi institutions.

The chapter relies on Chapters 9, 11 and 13.

GO SLOWLY HERE.

Ownership of a firm can be divided into many small equal parts.

A share is one equal part of a company's ownership.

Share. One of the equal parts into which the ownership of a company is divided.

Common stock is the ordinary share, held by the owners who carry the final risk.

Common stock. The ordinary shares of a company, held by the owners who are paid last and who control the company.

A share has a printed value, called its face value, which is not its worth.

Consider a puzzle. The owner of Sundial Bakery put in Tk 6,00,000 in January. An investor now offers Tk 2,80,000 to join the business. The investor reasons: "Together we hold Tk 8,80,000, and I supply Tk 2,80,000, so I should own 31.8 per cent." The owner replies that 25 per cent is fair. Which of them is right, and what does each person overlook?

The answer lies in what the bakery has done since January. It has earned profit, and the owner has kept most of that profit inside the business. The bakery is therefore worth more than the Tk 6,00,000 that the owner first paid in. The investor counted the money put in, and the owner counted what the business is now worth. This chapter explains the second way of counting, and the other ways in which owners supply money to a firm.

We study five matters. First, the common share, which is the basic form of ownership. Second, the preferred share, which is a form of ownership with a fixed payment. Third, retained earnings, which is profit that owners leave in the firm. Fourth, the capital market, where firms raise long-term money from many people. Fifth, the institutions in Bangladesh that supply long-term finance.

You will need the leverage ideas of Chapter 9, the share and bond words of Chapter 11, and the debt ideas of Chapter 13: claim, cost, control, risk, and the tax shield on interest.

14.1 COMMON STOCK AS OWNERSHIP

Imagine that three friends buy a field together and agree to divide it into ten equal plots. Each friend owns some plots, and a friend who wants to leave can sell plots to someone else. The field is not cut up; the ownership is. A company works in the same way.

A **share** means one of the equal parts into which the ownership of a company is divided. If a company has 1,000 shares and you hold 10, you own one hundredth of the company. Chapter 2 named the person who holds shares a shareholder. A company may have one shareholder, as the bakery has today, or many thousands.

Chapter 11 valued shares as a kind of security. Here we study them as ownership. **Common stock** means the ordinary shares of a company, held by the owners who are paid last and who control the company. Chapter 11 used the words common stock for a security that can be valued. Now the word names the owner's position in the firm. We keep to the word common stock for these ordinary shares, as in Chapter 11.

Each share has a **face value**, which is the value printed on the share. The bakery's shares have a face value of Tk 10. Chapter 11 used the same words for the sum that a bond repays at maturity. A share is different: its face value is a label used in the company's records, and it does not say what the share is worth today.¹

¹ Some writers say par value or nominal value. This book says face value, as in Chapter 11. Tk 10 is a common figure for the face value of shares in Bangladesh, but the figure here is illustrative.

A shareholder holds a safe position in one respect. **Limited liability** means a shareholder can lose no more than the amount paid for the shares. Suppose the bakery fails and owes lenders Tk 20,00,000, but its assets sell for only Tk 12,00,000. The shareholders lose their shares, but the lenders cannot demand the missing Tk 8,00,000 from the shareholders' own homes or savings. This protection explains why many people are willing to buy shares in firms they do not manage.

Common stock gives its holders three main rights. The first is a vote on important matters, such as who will be the directors. The second is a share of any profit that the directors decide to pay out as dividend. The third is a claim on what is left if the firm closes, after every debt has been paid. The vote has force only in proportion to the shares held: an owner of 75 per cent has three times the votes of an owner of 25 per cent.

The last two rights have a common feature. They are the **residual claim**, which means the right to what remains after every other claim has been paid. A lender is promised a fixed sum. A common owner is promised nothing, and receives whatever is left. When the firm does well, what is left is large, and the owners gain. When the firm does badly, what is left may be nothing.

The next table applies the four questions of Chapter 13 to a common owner. Read each row for one question, and compare the two columns.

A LENDER AND A COMMON OWNER COMPARED

QUESTION	LENDER	COMMON OWNER
Claim	A fixed sum on agreed dates	The residual: whatever is left
Cost to the firm	Interest, fixed in advance	Dividends, which the directors decide
Control	None over daily decisions	A vote in proportion to shares held
Risk to the firm	High: fixed charges must be paid	Low: no payment is compulsory

The table shows a balance between the firm and the owner. For the firm, common stock is the safest money, because no payment is compulsory. For the owner, it is the riskiest money, because the owner is paid last. For this reason, owners expect a higher return than lenders, a point that Chapter 15 will use.

Now we return to the puzzle. When a new investor joins, the firm must decide how many shares to give for the money. The decision depends on what each existing share is worth. A simple starting point uses the owners' equity: the owners' total claim, which is the money they put in plus profit kept in the business. In words: **equity per share** equals total owners' equity, divided by the number of shares. We use it as a fair starting price, and the two sides may bargain from there.²

Given. The owner holds 60,000 shares of face value Tk 10, which is Tk 6,00,000. Profit kept in the business is Tk 2,40,000 on 30 September 2026. The investor offers Tk 2,80,000. *Find.* The price of a

Limited liability means a shareholder can lose only the money paid for the shares.

Limited liability. A shareholder can lose no more than the amount paid for the shares.

Common owners have three rights: to vote, to share profit and to share what remains.

The common owner's claim is the last in line, so it is called the residual claim.

Residual claim. The right to what remains after every other claim has been paid.

A table compares a lender and a common owner on the four questions of Chapter 13.

Common stock is the safest money for the firm, and the riskiest for the owner.

A new investor must be given the right number of shares for the money paid.

Equity per share. Total owners' equity divided by the number of shares.

First worked example: the bakery's share price and the investor's shares.

² The equity per share is only a starting point. It looks backward, at what the firm has put in and kept. Chapter 11 valued shares by the dividends they are expected to pay. A seller and a buyer usually bargain between the two.

share, the number of new shares, and the investor's share of the company.

THE INVESTOR'S SHARES, WORKED IN STEPS

STEP	WORKING	RESULT
Owners' equity before	6,00,000 + 2,40,000	Tk 8,40,000
Equity per share	8,40,000 / 60,000	Tk 14
New shares	2,80,000 / 14	20,000
Shares after the issue	60,000 + 20,000	80,000
Investor's part	20,000 / 80,000	25 per cent

The investor owns 25 per cent, as the owner said.

The investor's first count failed because the old owner's equity includes kept profit.

A second example shows what happens when new shares are sold too cheaply.

The owner loses what the investor gains, and this loss is called dilution.

Dilution. The fall in the value of existing shares when new shares are sold for less than they are worth.

Common slips in share arithmetic.

Recap: common stock is ownership with votes and a residual claim; price decides fairness.

Answer. The price is Tk 14 a share. The investor receives 20,000 shares and owns 25 per cent of the bakery. *Check.* Owners' equity after the issue is Tk 8,40,000 plus Tk 2,80,000, which is Tk 11,20,000. Divided by 80,000 shares, it is Tk 14 again. Neither side gained or lost, because the new money entered at the price of the old shares.

The investor counted only the money put in, which gives 2,80,000 divided by 8,80,000, or 31.8 per cent. That count ignores Tk 2,40,000 of profit that belongs to the owner and stays in the firm. The investor must pay for a share of that profit as well. Each of the two persons held part of the truth: the owner knew what the business is worth, and the investor knew the money offered.

Now a harder test. Suppose the firm sells the 20,000 new shares at their face value of Tk 10, so that the investor pays Tk 2,00,000. The investor still owns 25 per cent. But owners' equity after the issue is Tk 8,40,000 plus Tk 2,00,000, which is Tk 10,40,000, and 25 per cent of that is Tk 2,60,000. The investor pays Tk 2,00,000 and receives a share worth Tk 2,60,000, a gain of Tk 60,000.

The gain comes from the old owner. Before the issue, the owner's claim was Tk 8,40,000. After it, the owner holds 75 per cent of Tk 10,40,000, which is Tk 7,80,000, a loss of Tk 60,000. This loss is called **dilution**, which means the fall in the value of existing shares when new shares are sold for less than they are worth. The owner should always ask what price the new shares are sold at, and not only how many.

Common slips. Students divide the new money by the money already put in, and not by the owners' equity. They use face value as the price. They forget that the number of shares after the issue is the old number plus the new number. After every issue, check that the equity per share is unchanged when the price equals the old equity per share.

Before moving on. A share is one equal part of ownership. Common owners vote, share profit and hold the residual claim, and limited liability protects them. The issue price decides whether old owners are treated fairly. You should now be able to say why an investor who pays 31.8 per cent of the total money receives only 25 per cent of the shares.

14.2 PREFERRED STOCK

Some investors want a steady payment and little control.

Not every investor wants the risks of a common owner. A retired teacher with savings may wish for a steady yearly payment, and may not care to vote on the price of buns. A firm may also wish to raise money without adding to its fixed

charges or to the number of votes. Both wishes lead to a second kind of share.

Preferred stock means shares that pay a fixed dividend, which must be paid before any common dividend. The dividend is a fixed percentage of the face value. A preferred share of face value Tk 100 with a rate of 10 per cent pays Tk 10 a year. Chapter 3 gave a one-line preview of the preferred dividend; now we study it fully.

The word preferred names a position in a line. When the firm earns profit, it pays in a fixed order: lenders first, then preferred owners, then common owners. The same order holds if the firm closes and sells its assets. Preferred owners therefore carry less risk than common owners, and more than lenders. The next figure draws this ranking for one year's profit. Read it from the top to the bottom: each box is a claim, and the arrow shows the order in which the claims are met.

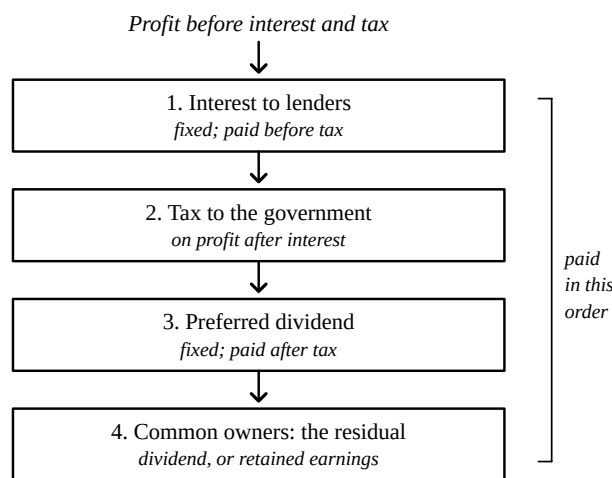


FIG. 38. The order in which a firm's profit is shared: interest to lenders, then tax, then the preferred dividend, and last the common owners, who may retain what is left.

The fixed dividend has a price for the owner. A preferred owner usually has no vote, except in special cases, for example when dividends have not been paid for some time. Preferred owners accept a limited reward and little control. In return they have priority.³

Suppose the firm cannot pay the preferred dividend in a poor year. Unlike unpaid interest, this is not a default, and the preferred owners cannot force the firm to close. But the firm cannot pay any dividend to common owners while the preferred dividend is unpaid. A preferred share is **cumulative** if unpaid dividends are carried forward and must be paid before any common dividend. A share that does not carry them forward is non-cumulative.

Given. Suppose the investor of section 14.1 took preferred shares instead: 2,800 shares of face value Tk 100, which is Tk 2,80,000, at 10 per cent a year. The firm paid no dividend last year. This year Tk 50,000 is available for dividends. *Find.* How the Tk 50,000 is shared if the shares are cumulative, and if they are non-cumulative.

Preferred stock pays a fixed dividend before common owners are paid.

Preferred stock. Shares that pay a fixed dividend, which must be paid before any common dividend.

A preferred owner is ranked after lenders and before common owners.

The preferred owner usually gives up the vote.

A missed preferred dividend is not a default, but it has consequences.

Cumulative. A preferred share is cumulative if unpaid dividends are carried forward and paid before any common dividend.

First worked example: how a cumulative dividend takes profit from common owners.

³ The rules for preferred stock differ between companies and between countries. Whether a firm may issue them, what rights they give, and whether they may be redeemed, must be checked in the company's own documents and in the law.

SHARING TK 50,000 BETWEEN PREFERRED AND COMMON OWNERS

STEP	CUMULATIVE	NON-CUMULATIVE
Yearly preferred dividend: 10% of 2,80,000	28,000	28,000
Unpaid from last year	28,000	Lost
Preferred claim this year	56,000	28,000
Paid to preferred owners	50,000	28,000
Paid to common owners	0	22,000
Still unpaid to preferred owners	6,000	0

The cumulative holders take all Tk 50,000; the others take Tk 28,000.

A second example compares a preferred dividend with interest on the same sum.

Answer. If the shares are cumulative, the preferred owners receive the whole Tk 50,000, the common owners receive nothing, and Tk 6,000 is still owed. If they are non-cumulative, the preferred owners receive Tk 28,000 and the common owners receive Tk 22,000. *Check.* In each column, the two payments add to Tk 50,000. The extra protection costs the common owners Tk 22,000 this year.

Now the main difference between preferred stock and debt. Interest is subtracted from profit before tax is calculated, as in Chapter 13. A preferred dividend is paid out of profit after tax. The next table gives the bakery a profit before interest and tax of Tk 1,00,000 and a tax rate of 25 per cent. In the first column the bakery pays Tk 28,000 as interest. In the second it pays Tk 28,000 as preferred dividend. Read down each column.

THE SAME TK 28,000 PAID AS INTEREST AND AS A PREFERRED DIVIDEND

LINE	INTEREST OF 28,000	PREFERRED DIVIDEND OF 28,000
Profit before interest and tax	1,00,000	1,00,000
Interest	28,000	0
Profit before tax	72,000	1,00,000
Tax at 25 per cent	18,000	25,000
Profit after tax	54,000	75,000
Preferred dividend	0	28,000
Left for common owners	54,000	47,000

The preferred dividend costs Tk 7,000 more, because it has no tax shield.

A table sets the three kinds of money side by side.

Answer. The common owners keep Tk 54,000 in the first case and Tk 47,000 in the second. The difference is Tk 7,000, which is 25 per cent of Tk 28,000: the tax shield that interest gains and a preferred dividend does not. *Check.* Tax in the second column is Tk 7,000 higher: Tk 25,000 less Tk 18,000. The cost of a fixed payment to the firm is therefore higher for preferred stock than for debt, a point that Chapter 15 will use.

The next table brings together the three kinds of money. Read each row for one feature, and compare the three columns.

DEBT, PREFERRED STOCK AND COMMON STOCK COMPARED

FEATURE	DEBT	PREFERRED STOCK	COMMON STOCK
Holder is	A lender	An owner	An owner
Payment	Fixed interest	Fixed dividend	Dividend decided by directors
Paid	First	Second	Last
Missed payment	Default	Carried forward if cumulative	No claim
Vote	None	Usually none	Yes
Tax shield	Yes	No	No

There are limits to preferred stock. It adds a fixed payment, so it raises the firm's financial risk, as Chapter 9 showed with leverage. It has no tax shield. And its rules vary between companies, so each issue must be read carefully. For these reasons, many firms prefer a mix of debt and common stock.

Before moving on. Preferred stock pays a fixed dividend before common owners, usually with no vote. Cumulative shares carry unpaid dividends forward. The dividend has no tax shield. You should now be able to say why a preferred dividend costs a firm more than interest of the same size.

Many small firms never issue preferred stock, because the arrangement is complex.

Recap: preferred stock is owners' money with a fixed dividend and priority.

14.3 RETAINED EARNINGS

A family business earns Tk 3,20,000 in a year after tax. The family may take all of it for household needs, or leave some in the business to buy a new oven. The second choice is a source of money that needs no bank, no new owner and no form to sign.

Retained earnings means the part of profit after tax that is kept in the firm and not paid out as dividend. Retained earnings belong to the common owners, so they are part of owners' equity. When a firm retains profit, the owners agree, in effect, to lend their share of the profit to their own business and take the reward later, as a larger firm.

The directors decide how to divide the profit between dividend and retained earnings. We measure the division with a ratio. In words: the **payout ratio** equals dividends divided by net profit. The retention ratio equals retained earnings divided by net profit, which is one less the payout ratio.

Given. Net profit for 2026 is Tk 3,20,000: Tk 2,40,000 for the first nine months and Tk 80,000 for the last three. The directors pay a dividend of Tk 0.80 on each of 80,000 shares. For simplicity, all shares receive the same amount. *Find.* The dividend, the payout and retention ratios, the retained earnings, and owners' equity at 31 December.

The simplest source of owners' money is profit left in the firm.

Retained earnings are the profit after tax that is kept in the firm.

Retained earnings. The part of profit after tax that is kept in the firm and not paid out as dividend.

The dividend decision divides profit into a part paid and a part kept.

Payout ratio. Dividends divided by net profit; the part of profit paid to owners.

Worked example: the bakery's profit in 2026.

THE BAKERY'S PROFIT SHARED, 2026

LINE	WORKING	TK
Net profit	2,40,000 + 80,000	3,20,000
Dividend	0.80 × 80,000	64,000
Retained earnings	3,20,000 - 64,000	2,56,000
Payout ratio	64,000 / 3,20,000	20 per cent
Retention ratio	2,56,000 / 3,20,000	80 per cent

Owners' equity is the owner's money, the investor's money and kept profit, held in three parts.

The bakery began the year with no profit kept, so the kept profit at 31 December is Tk 2,56,000. The case file keeps the owner's money and the investor's money apart. The owner's Tk 6,00,000 buys 60,000 shares. The investor's Tk 2,80,000 buys 20,000 shares: Tk 2,00,000 at face value, and Tk 80,000 more because each share was sold at Tk 14.⁴ Owners' equity is the sum of the three parts. The figure below shows them as one bar. Each block is one part, and its length is its size in taka.

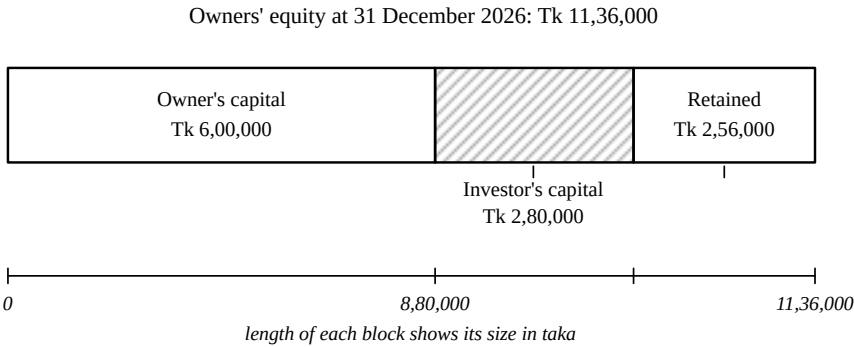


FIG. 39. Owners' equity of Sundial Bakery at 31 December 2026: the owner's capital, the investor's capital and retained earnings.

Owners' equity at 31 December is Tk 11,36,000, or Tk 14.20 a share.

Answer. Owners' equity at 31 December is 6,00,000 plus 2,80,000 plus 2,56,000, which is Tk 11,36,000. Divided by 80,000 shares, it is Tk 14.20 a share, which is Tk 0.20 higher than at the issue, because of the profit kept since then. Check. 3,20,000 less 64,000 is 2,56,000, and the kept profit in the figure matches.

Retained earnings are not a pile of cash.

A common error needs attention. Retained earnings are a claim of the owners, not a sum of money in a drawer. The firm has already used the kept profit: it buys flour, fills the shop with stock, and waits for customers to pay. Suppose that at 31 December only Tk 40,000 of the Tk 2,56,000 is held as cash. The rest is in goods, equipment and amounts owed by customers. A firm can have large retained earnings and little cash, as Chapter 12 showed.

Retained earnings avoid issue costs and new owners, but they are not free.

Retained earnings have advantages. They cost nothing to arrange, they add no new owner who must share control, and they create no fixed payment. They also have limits. They are no larger than the profit the firm has made, and a young firm has little. And they are not free of cost: the owners could have taken the money as dividend and used it elsewhere. Chapter 15 treats this lost use as the cost of retained earnings.

Owners may differ about the best dividend.

There is also a difference of view. An owner who needs income wants a high dividend. An owner who wishes the firm to grow wants a low

⁴ Accountants call the amount paid above the face value a share premium. In this book we keep the word premium for a bond that sells above its face value, as in Chapter 11, and we say that the shares were sold at more than their face value.

one. A firm with a good use for the money, such as a profitable oven, can reasonably keep more. A firm with no good use for it should return it to owners. This is the dividend decision that Chapter 2 named.

Before moving on. Retained earnings are profit after tax kept in the firm. The payout ratio plus the retention ratio is one. Kept profit is a claim of the owners and is not itself cash. You should now be able to find retained earnings from profit and dividend.

Recap: retained earnings are kept profit; the payout ratio shows the division.

14.4 RAISING FUNDS FROM THE CAPITAL MARKET

The bakery found its investor through an introduction. A firm that wishes to raise Tk 10,00,00,000 cannot depend on introductions. It needs a way to reach many people with savings, and those people need a way to sell their shares if they later need money. The system that serves both needs is called the capital market.

The bakery found its investor by asking; larger firms need many investors.

The **capital market** means the financial market in which long-term funds are raised and securities are traded. It is one part of the financial markets that Chapter 1 described. The market for short-term money, such as an overdraft, is a different part. It has two sides, which we must keep apart.

The capital market is where long-term money is raised and traded.

Capital market. The financial market in which long-term funds are raised and securities are traded.

The **primary market** means the market in which a firm sells new securities and receives the money. The **secondary market** means the market in which investors trade securities that already exist. When the bakery's investor paid Tk 2,80,000 to the bakery, that was a primary sale. If the investor later sells the shares to a friend, the sale is secondary, and the bakery receives nothing. The figure below shows the two flows. Follow the arrows: in the primary market, money goes to the firm; in the secondary market, money moves between investors.

The primary market sells new securities, and the secondary market trades existing ones.

Primary market. The market in which a firm sells new securities and receives the money.

Secondary market. The market in which investors trade securities that already exist.

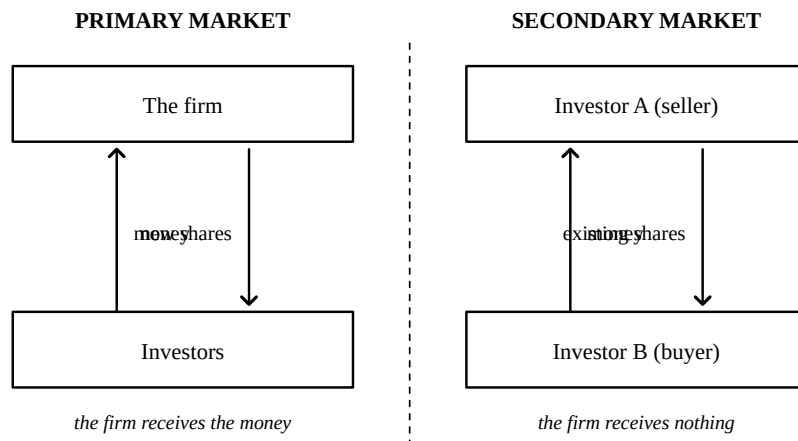


FIG. 40. The two sides of the capital market: in the primary market the firm receives the money, and in the secondary market one investor pays another.

Why should a firm care about the secondary market, when it receives nothing there? The answer is confidence. An investor who knows the shares can be sold later is more willing to buy them now. A firm whose shares trade easily can usually raise new money more cheaply.

The secondary market matters to the firm, although the firm receives no money from it.

A share issue is the sale of new shares to raise money.

Share issue. The sale of new shares by a company to raise money.

Three methods of share issue differ in who buys and how much they cost.

Public offering. The sale of new shares to the public at large.

Rights issue. An offer of new shares to existing shareholders, in proportion to the shares they hold.

A **share issue** means the sale of new shares by a company to raise money. A firm has three common ways to make an issue. The first is a private placement, in which the firm sells to one investor or a few chosen ones, as the bakery did. The second is a public offering, in which the firm sells to the public at large. The third is a rights issue, in which the firm offers new shares first to its existing shareholders.

In a **public offering**, the firm publishes a document, called a prospectus, that describes the business, its risks and its accounts. Investors read it and apply for shares. A **rights issue** is an offer of new shares to existing shareholders, in proportion to the shares they hold. The next table sets the three methods side by side. Read each row for one method.

THREE WAYS TO ISSUE SHARES

METHOD	WHO BUYS	COST AND EFFORT	FITS THE BAKERY WHEN
Private placement	One investor or a few chosen ones	Low: a private agreement	The firm needs Tk 2,80,000 from one person
Public offering	The public at large	High: prospectus, advisers, approval	The firm is large and wants many owners
Rights issue	Existing shareholders	Medium: an offer to each holder	The owners wish to keep their shares in proportion

A worked example shows how a rights issue protects each owner's share.

The offer raises Tk 2,80,000 and leaves each holder's part unchanged.

Every issue has costs, so the firm receives less than investors pay.

Given. Suppose that, after the issue of 14.1, the bakery offers one new share for every four held, at Tk 14. The bakery has 80,000 shares: 60,000 held by the owner, 20,000 by the investor. *Find.* The number of new shares, the money raised, and each holder's cost and part of the company if each takes up the offer in full.

Steps. New shares are 80,000 divided by 4, which is 20,000. Money raised is 20,000 times Tk 14, which is Tk 2,80,000. The owner may buy 60,000 divided by 4, or 15,000 shares, costing Tk 2,10,000. The investor may buy 5,000 shares, costing Tk 70,000. *Answer.* After the offer the owner holds 75,000 of 1,00,000 shares, which is 75 per cent, and the investor holds 25,000, which is 25 per cent. *Check.* The two purchases add to 20,000 shares and Tk 2,80,000. A holder who declines the offer ends with a smaller part of the company.

An issue is not free. The firm pays lawyers, auditors, advisers and fees. Suppose the costs of a Tk 2,80,000 issue are 2 per cent. They are Tk 5,600, and the firm receives Tk 2,74,400. The cost of the issue is part of the cost of the money, and it is larger for small issues than for large ones.

A **stock exchange** means an organised market in which securities are bought and sold under fixed rules. A company that wishes its shares to be traded there seeks a **listing**, which means the admission of its shares to trading on the exchange. Only listed shares are traded. In Bangladesh the main exchanges are the Dhaka Stock Exchange and the Chittagong Stock Exchange.

Why would a firm seek a listing? It gains access to many investors, its shares can be sold easily, and its name becomes better known. But it accepts duties. It must publish accounts, follow the exchange's rules, and tell investors promptly about important news. Its owners lose some privacy, and its managers must answer to many shareholders. Chapter 2 named this relation between owners and managers as the agency problem.

A market in which ordinary people trade needs a referee. A **regulator** means a public body that makes rules for a market and checks that they are followed. In Bangladesh, the Bangladesh Securities and Exchange Commission, known as BSEC, regulates the securities market, and a company that offers shares to the public must follow its rules. The central bank, Bangladesh Bank, regulates banks and certain other financial institutions. The rules, fees and conditions change, so any student who plans an issue must read the current rules.

Before moving on. The primary market sells new securities, and the secondary market trades existing ones. A firm may issue shares by private placement, public offering or rights issue. A listing brings access and resale but also duties. You should now be able to say who receives the money when shares are sold on a stock exchange.

14.5 INSTITUTIONS SUPPLYING LONG-TERM FINANCE IN BANGLADESH

A baker who needs a large sum for many years has more than one door to try. Each kind of institution has its own purpose, size and rules. This section describes the main kinds, with a few examples. All facts here are illustrative of the kinds of institution, and the current details must be checked before any use.

The first kind is the commercial bank. Banks lend for the term loans of Chapter 13, and they are the usual choice for a small firm. A bank also holds the firm's accounts, so it can see how the firm's money moves. For the bakery's oven loan, a bank is the natural lender.

A **development finance institution** means a body that supplies long-term funds to build industry or infrastructure, often with public backing. Its aim is not only profit but also growth of the economy. Such institutions lend for longer periods than many commercial banks, and often for projects that private lenders find too large or too slow to repay.

One example is the Infrastructure Development Company Limited, known as IDCOL. The Government of Bangladesh established it on 14 May 1997, and the Bangladesh Bank licensed it as a non-bank financial

A stock exchange is an organised market where listed shares are traded.

Stock exchange. An organised market in which securities are bought and sold under fixed rules.

Listing. The admission of a company's shares to trading on a stock exchange.

Listing brings money and easy resale, but also duties.

A regulator makes rules for the market and watches that they are kept.

Regulator. A public body that makes rules for a market and checks that they are followed.

Recap: the primary market raises money, the secondary market trades; issue methods differ.

A firm in Bangladesh has several kinds of institution to approach.

Commercial banks are the most common source of long-term loans for small firms.

A development finance institution lends for the long term to build industry and infrastructure.

Development finance institution. A body that supplies long-term funds to build industry or infrastructure, often with public backing.

IDCOL is an example that finances large infrastructure and renewable energy projects.

ICB is an investment bank that supports industry and the securities market.

Non-bank financial institutions lend and lease without taking deposits from the public.

Non-bank financial institution. A financial firm that lends, leases or invests but is not a bank.

A table sets the institutions side by side and tests each against the bakery's needs.

institution in January 1998. It finances medium and large projects in infrastructure and renewable energy. These projects are far larger than the bakery's oven, so IDCOL is not a source for the bakery.

A second example is the Investment Corporation of Bangladesh, known as ICB. It was established on 1 October 1976 as an investment bank, to speed up industrial growth and to build a sound securities market. Investment banks help firms to issue shares and bonds, and may themselves invest in them. A firm planning a public offering usually seeks advisers of this kind.

A third kind is the **non-bank financial institution**, which means a financial firm that lends, leases or invests but is not a bank. Bangladesh Bank regulates them under the Financial Institutions Act of 1993. Leasing companies and investment companies belong here. Chapter 16 returns to leasing companies when it compares leasing with buying.

The next table sets the kinds of institution side by side. Read each row for one kind. The last column asks whether it suits a small bakery needing Tk 3,00,000.

INSTITUTIONS SUPPLYING LONG-TERM FINANCE

KIND	EXAMPLE	WHAT IT SUPPLIES	SUITS THE BAKERY?
Commercial bank	A bank holding the bakery's account	Term loans, mortgages	Yes: the usual choice
Development finance institution	IDCOL	Loans for large infrastructure projects	No: the projects are too large
Investment bank	ICB	Advice on issues, investment in securities	Later, if the bakery issues shares to the public
Non-bank financial institution	A leasing company	Loans and leases	Possible: see Chapter 16
Stock exchange and investors	Dhaka Stock Exchange	Share capital from many investors	Not yet: too small to list

Institutions change, so the facts here must be verified.

Recap: institutions differ in purpose and size, and each suits a different firm.

Institutions are created, merged and closed as laws and needs change. A student who reads about them in a later year should look for the current names, rules and sizes. The pattern is more lasting than the names: each kind of institution suits a particular size and purpose of firm.

Before moving on. Long-term funds in Bangladesh come from commercial banks, development finance institutions, investment banks, non-bank financial institutions, and investors through the stock exchanges. Each suits a different size of firm. You should now be able to name two institutions and say which of them suits a small firm.



THE WORKED CASE: THE INVESTOR JOINS

Case file, illustrative. Situation. On 1 October 2026 the bakery's ownership is divided into 60,000 shares of face value Tk 10, all held by the owner, who paid Tk 6,00,000. Profit kept in the firm is Tk 2,40,000. The bakery owes Tk 7,00,000 to banks, as in Chapter 9. An investor offers Tk 2,80,000.

Question. How many shares should the investor receive, at what price, and how does the new money change the bakery's debt in relation to owners' money?

Reasoning. Equity per share is Tk 8,40,000 divided by 60,000, which is Tk 14. The investor's Tk 2,80,000 buys 20,000 shares. The bakery's debt is Tk 7,00,000. Before the issue, that is Tk 0.83 of debt for each taka of owners' equity. After the issue, equity is Tk 11,20,000, and the figure is Tk 0.63.

Conclusion. The investor receives 20,000 shares at Tk 14 and owns 25 per cent of the bakery. The new equity lowers the debt carried by each taka of owners' money, so financial leverage, as Chapter 9 defined it, falls. The owner's part falls from 100 per cent to 75 per cent, and the owner shares control and profit.

Check. Equity per share after the issue is 11,20,000 divided by 80,000, which is Tk 14, so neither side gained. Debt over equity is 7,00,000 divided by 8,40,000, which is 0.83, and 7,00,000 divided by 11,20,000, which is 0.63. At 31 December, equity is Tk 11,36,000 and equity per share is Tk 14.20.

The situation: the bakery has one owner, kept profit and a loan.

The question: how many shares, at what price, and with what effect on risk?

The reasoning uses equity per share and the debt ratio.

Conclusion: 20,000 shares at Tk 14; the investor owns 25 per cent; debt weighs less.

The check repeats the arithmetic and tests the price.

SUMMARY

A share is one equal part of a company's ownership. Common owners vote, share profit and hold the residual claim, and limited liability protects them. The issue price decides whether old owners are treated fairly: a price below equity per share dilutes them. Preferred stock pays a fixed dividend before common owners are paid, but it has no tax shield, and cumulative preferred shares carry unpaid dividends forward. Retained earnings are profit after tax that the firm keeps; they are a claim of owners, not a sum of cash, and they are not free. The primary market sells new securities, and the secondary market trades existing ones. A firm may issue shares by private placement, public offering or rights issue, and a listing brings access and duties. In Bangladesh, long-term funds come from commercial banks, development finance institutions, investment banks, non-bank financial institutions and the stock exchanges.

The chapter's main ideas in one paragraph.

THINK IT THROUGH

Case file, illustrative. A café has 50,000 shares of face value Tk 10, held by two owners who paid Tk 5,00,000 in all. The café has kept Tk 1,50,000 of profit. An investor offers Tk 2,60,000 for new shares. Find the equity per share, the number of new shares, and the investor's part of the café. Then suppose the shares are sold at Tk 10, and find the investor's gain and the owners' loss. Last, suppose the investor takes 2,600 preferred shares of face value Tk 100 at 12 per cent, and say how much dividend the café

A new problem with new data, to test the whole chapter.

must pay each year, and whether the café can deduct it from profit before tax.

WHERE PEOPLE GO WRONG

A share's face value is not its worth.

A share of face value Tk 10 is worth Tk 10. It is not. Face value is a printed label. The worth of a share depends on the firm's equity and its earnings. The bakery's shares of face value Tk 10 are priced at Tk 14.

Owners' part follows equity, not only money put in.

An investor who pays 31.8 per cent of the money owns 31.8 per cent of the firm. Not so. The firm is worth more than the money first put in, because it holds kept profit. The investor owns 25 per cent.

Preferred stock is not debt.

A preferred share is a loan. It is not. A preferred owner is an owner with a fixed dividend and priority. A missed dividend is not a default, and the dividend has no tax shield.

Kept profit is not cash.

Retained earnings are money in the bank. They are a claim of the owners. The firm has already used most of that profit in its assets.

Trading on the exchange does not bring money to the firm.

When shares are sold on the stock exchange, the firm receives the money. It does not. In the secondary market, one investor pays another. The firm receives money only when it sells new shares.

Owners' money is not free.

Retained earnings and new shares cost nothing, because there is no interest. They have a cost: owners expect a return, and Chapter 15 shows how to measure it.

IN HISTORY

A refusal in Calcutta led to a stock exchange in Dhaka.

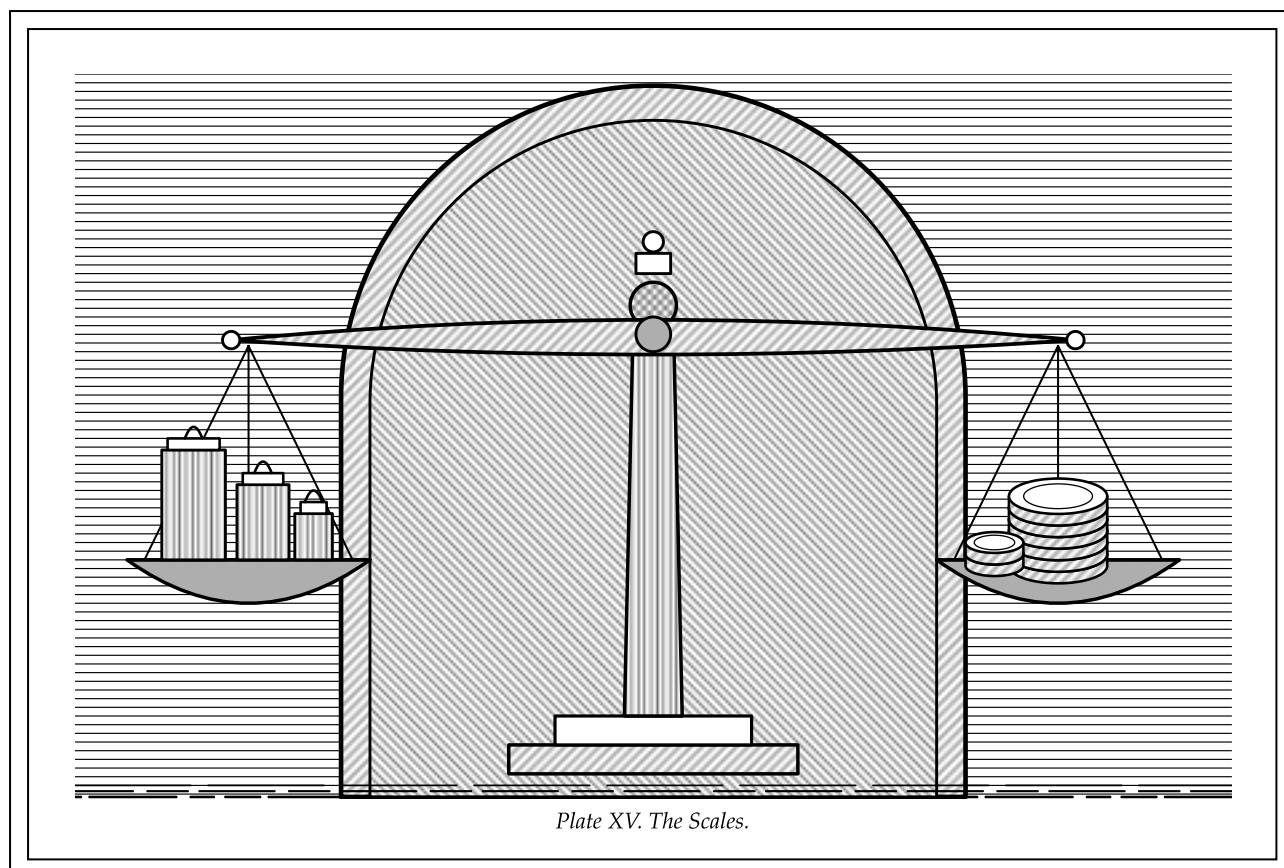
In 1952, the stock exchange in Calcutta stopped trading in the shares of companies in Pakistan. East Pakistan, whose capital was Dhaka, then had no market of its own for the shares of its firms. A group of eight promoters incorporated the East Pakistan Stock Exchange Association Limited on 28 April 1954, and formal trading began in 1956. The exchange was renamed the Dacca Stock Exchange in 1964, and the spelling later became Dhaka. The lesson for this chapter is plain: a firm that wants to raise owners' money from many people needs a place where those people can later sell, and a country that wants such firms must build that place.

REVIEW QUESTIONS

REVIEW QUESTIONS

*Ten questions to test
understanding of the chapter.*

NO.	QUESTION
1	Define share, common stock and limited liability. State the three rights of a common owner.
2	A firm has 40,000 shares and owners' equity of Tk 6,00,000. An investor pays Tk 3,00,000 for new shares at the equity per share. Find the price, the number of new shares and the investor's part of the firm.
3	For question 2, suppose the shares were sold at their face value of Tk 10. Find the investor's gain and the old owners' loss. Name this effect.
4	Explain what preferred stock is. State two ways in which it differs from debt and two ways in which it differs from common stock.
5	A firm has cumulative preferred shares of face value Tk 2,00,000 at 9 per cent. No dividend was paid last year. This year Tk 30,000 is available for dividends. Find what the preferred and the common owners receive, and what is still owed.
6	A firm pays Tk 20,000 as interest in one case and as preferred dividend in another. The tax rate is 25 per cent. Find the difference in what is left for common owners.
7	Net profit is Tk 2,00,000 and the dividend is Tk 50,000. Find the payout ratio, the retention ratio and the retained earnings.
8	Explain why a firm with large retained earnings can have little cash.
9	Distinguish the primary market from the secondary market. A shareholder sells 100 shares on a stock exchange: does the company receive money?
10	A firm has 50,000 shares and offers one new share for every five held, at Tk 12. Find the new shares and the money raised. A holder of 5,000 shares takes up the offer: how many shares does the holder buy, and what do they cost?



CHAPTER XV

Cost of Capital

What price does a firm pay for the money it uses?



Every project needs money, and money has a price. This chapter shows how to find the price of each kind of fund, how to combine the prices into one yearly rate, and how to use that rate to test the projects of earlier chapters again.

A question left open in Chapter 7 now needs an answer.

The guessed rate decided how large each project's value looked.

A fair rate must come from the real price of the firm's funds.

In Chapter 7 the owner of the bakery compared an oven and a van. Both projects looked good, and the oven won by a small margin. Yet one number in that comparison was only a guess. The owner had chosen a provisional rate of 10% to bring the later cash back to today, and had promised to replace it with a better one.

Does the guess matter? Suppose the true rate is 15%. Then the value of the oven falls from Tk 79,079 to Tk 35,216. Suppose it is 20%. Then the oven no longer pays for itself. The same cash flows can give a clear yes or a clear no, and only the rate has changed. A project is never judged alone. It is judged against the price of the money that pays for it.

This chapter finds that price. We begin with an idea from daily life, then price each source of money in turn, then combine the prices into one rate. At the end we return to the oven and the van and test them again.

YOU WILL NEED

From Chapter 7: net present value and the hurdle rate. From Chapter 10: required return and the capital asset pricing model. From Chapter 11: the value of a share that pays a growing dividend. From Chapters 13 and 14: debt, the tax saved on interest, preferred stock, common stock and retained earnings. Chapter 3: percentages and the profit layout.



WHAT THE COST OF CAPITAL IS

GO SLOWLY HERE. THIS IDEA IS NEW.

A taka that is borrowed or invested always has a price.

Each part of the money has its own price in yearly percentage.

A firm does the same sum, with more kinds of money.

Cost of capital. The yearly price, as a percentage, that a firm pays for the money it uses.

The word cost has a different meaning here.

Word in use. In daily life, cost is a price paid once. Here it is a yearly percentage.

Every price has two owners: the one who pays and the one who is paid.

Begin with a student who needs a laptop that costs Tk 60,000. Her uncle lends her Tk 30,000 and asks for Tk 3,000 each year in return. She pays the other Tk 30,000 from her savings. Those savings were earning 6% a year in a bank, so using them means she gives up Tk 1,800 a year. What does the laptop cost her, as a yearly percentage of the money used?

The uncle's part costs 10%, because Tk 3,000 is one tenth of Tk 30,000. The savings part costs 6%, because that is the reward she gives up. The two parts are equal in size, so the average price is half of 10% plus half of 6%, which is 8%. The laptop is a good use of money only if it gives her a benefit worth more than 8% a year. We shall use exactly this method for a firm.

A firm also uses money from several places: lenders, owners who put in money, and profit kept in the business. We now give the idea its name. The **cost of capital** means the price a firm pays for its funds, stated as a yearly percentage of the money used. If a firm uses Tk 1,00,000 of funds and the cost of capital is 11%, the funds must earn Tk 11,000 a year to pay their price.

Notice that the word *cost* has changed its meaning. In earlier chapters, cost meant money spent to make or buy something. Here it is a percentage, a price for using money over time. Keep the two meanings apart. When we say cost of capital, we always mean the yearly percentage.

There is one more point, and it is the most important. The price that the firm pays is the reward that its investors receive. A lender who is paid 12% earns 12%. So the firm calls this number its cost, and the

investor calls it the **required return**, which means the least reward that the investor will accept for lending or investing. It is one number seen from two sides. The next figure shows this.

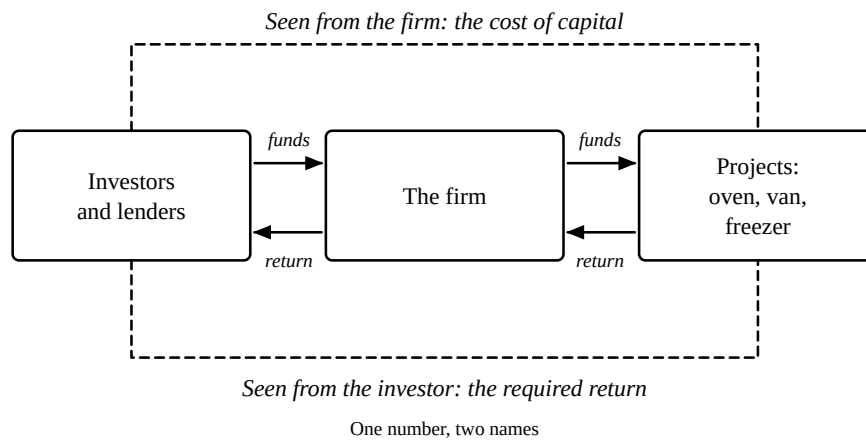


FIG. 41. The firm's cost of capital and the investor's required return are one number seen from two sides.

You have met this rate before. In Chapters 4 to 7 we called it the **discount rate**, the rate used to bring later cash back to today. In Chapter 8 and later we called it the required return. All three names describe one idea. A project is worth doing only if its cash flows, brought back to today at this rate, are worth more than its outlay. That is the net present value rule of Chapter 7. The cost of capital is the rate that the rule needs.

A firm gets its funds from three kinds of source. The first is debt, which is money borrowed from lenders (Chapter 13). The second is preferred stock, which is shares with a fixed dividend (Chapter 14). The third is common equity, which is the money of the ordinary owners, either put in by them or kept from profit (Chapters 11 and 14). Each source has its own price. The mix of debt and equity that a firm uses is its **capital structure**, and it decides how the separate prices are combined.

Here is a common confusion. Debt shows the lowest price, so it seems wise to use only debt. This is not correct. Each new loan raises the risk of the firm (Chapter 9), and owners and lenders ask for more reward when the risk is higher. So the price of the other sources rises as debt grows. The cost of capital is a price for the whole mix of funds, and not for one source alone.

Consider the viewpoints in turn. The owner sees the cost of capital as a limit: a project that earns less than this rate takes value away. The lender sees it as a promise, the interest that must be paid on time. The financial manager sees it as a tool, a single rate that lets many different projects be compared fairly. All three are looking at the same number, and each uses it for a different task.

Pause and check. (1) The student's uncle had lent her all Tk 60,000 instead of half, at 10%. What would the laptop have cost her, as a yearly percentage? (2) Her savings earn 6% in the bank. Why is 6% a cost to her, even though she pays nobody? Short answers are given at the back of the book.

Before moving on. The cost of capital is the yearly price of a firm's funds. For the firm it is a cost, for the investor it is a required return, and

The same rate appears in earlier chapters under other names.

Funds come from three sources, so there are three prices.

Capital structure. The mix of debt and equity that a firm uses to pay for its assets.

Cheap-looking money may be costly in a way that is hard to see.

The idea looks different from each seat.

Two questions to test the idea before moving on.

Recap: the cost of capital is one price with three parts.

in Chapters 4 to 7 it was the discount rate. Debt, preferred stock and common equity each have a price, and the mix decides the total.

A check the reader can use at once.

You should now be able to say: the cost of capital is the yearly percentage that a firm must earn on its funds to satisfy those who supplied them.



COST OF DEBT AFTER TAX

A loan is the simplest fund to price.

Cost of debt. The yearly percentage a firm pays on its borrowed money, after the tax saved is taken into account.

Interest reduces tax, so the real price is lower.

The rule has one multiplication.

First example, with small numbers.

Four steps, each short.

Answer and a quick test.

Sometimes the firm receives less than the face value of a bond.

Issue cost. The fees a firm pays to sell new bonds or shares, which reduce the money it receives.

The method must use what the firm actually receives.

Use a short formula that averages the yearly gain.

The steps show each line.

Debt is the easiest source to price, because the lender states the rate. If the bakery borrows Tk 4,00,000 at 12%, the interest is Tk 48,000 a year. Before tax, the **cost of debt** is therefore 12%. For a bond that is traded, the yearly return to a buyer who holds it until maturity is the yield to maturity of Chapter 11, and that is the before-tax cost of debt for a firm that sells such bonds.

Now recall the tax shield of Chapter 13. Interest is an expense, so it reduces the profit on which the firm pays tax. Every taka of interest therefore saves a part of a taka in tax. If the tax rate is 25%, each taka of interest saves 25 paisa of tax, and the firm really pays only 75 paisa. We call this the cost of debt after tax.

In words: the after-tax cost of debt is the interest rate multiplied by one less the tax rate. In symbols, $k(d) = i \times (1 - T)$, where i is the interest rate before tax and T is the tax rate written as a decimal. Here $k(d)$ stands for the cost of debt.

Given. A firm borrows Tk 1,00,000 at 10% interest. The tax rate is 25%. *Find.* The cost of debt after tax.

Steps. Interest in one year is 10% of Tk 1,00,000, which is Tk 10,000. Tax saved is 25% of Tk 10,000, which is Tk 2,500. The firm really pays Tk 10,000 less Tk 2,500, which is Tk 7,500. As a share of the loan, Tk 7,500 divided by Tk 1,00,000 is 7.5%.

Answer. The cost of debt after tax is 7.5%. *Check.* Use the rule: $10\% \times (1 - 0.25) = 10\% \times 0.75 = 7.5\%$. Both ways agree, and the answer is lower than 10%, as it should be.

The second example is less tidy. When a firm sells new bonds, it often pays fees to the bank that sells them. These fees are the **issue cost**, sometimes called the flotation cost. The firm then receives less money than the face value, and this amount is the net proceeds.

Given. A firm sells debentures with a face value of Tk 1,000, a coupon rate of 11% and a life of 10 years. After issue costs, it receives Tk 950 for each one. The tax rate is 25%. *Find.* The cost of debt after tax.

Formula. A close estimate of the before-tax cost is: (yearly interest + yearly share of the gain on repayment) divided by the average of the face value and the net proceeds. The gain on repayment is the face value less the net proceeds, spread over the years.

Steps. Yearly interest is 11% of Tk 1,000, which is Tk 110. The gain on repayment is Tk 1,000 – Tk 950 = Tk 50, and over 10 years this is Tk 5 a year. The average of Tk 1,000 and Tk 950 is Tk 975. Before tax the cost is $(110 + 5) \div 975 = 115 \div 975 = 11.79\%$. After tax it is $11.79\% \times 0.75 = 8.85\%$.

Answer. The cost of debt after tax is about 8.85%. *Check.* The exact yield to maturity, found by trial as in Chapter 11, is 11.88%. After tax this is 8.91%. The short formula is close, and it is accepted in an answer when the question gives no more.

Common slips. First, the reader forgets the tax step and uses the before-tax rate. Second, the reader uses the coupon rate and forgets issue costs. Third, the reader applies the tax adjustment to dividends, which the tax law does not allow as an expense. We return to this last point in the next section.

There is one exception. The tax saving is real only if the firm earns enough profit to pay tax. A firm that makes a loss pays no tax, so interest saves it nothing, and its cost of debt after tax is the full interest rate. In this chapter we assume the firm earns taxable profit.

Case file, illustrative. Sundial Bakery borrows at 12% a year (loans of Tk 4,00,000 and Tk 3,00,000 in 2026). At the illustrative tax rate of 25%, the cost of debt after tax is $12\% \times (1 - 0.25) = 9\%$.¹

For a term loan repaid in instalments (Chapter 13), the lender states a yearly rate on the balance still owed, and that stated rate is the before-tax cost. The tax step is the same: multiply by one less the tax rate. Only when the firm pays large fees, or receives much less than the face amount, does the cost need the longer method used for bonds. Most exercises that give one interest rate and one tax rate need only the short rule.

Before moving on. The cost of debt after tax is the interest rate multiplied by one less the tax rate. If the firm pays issue costs, use the money it receives, not the face value. The saving holds only if the firm pays tax.



COST OF PREFERRED AND COMMON FUNDS

Preferred stock was described in Chapter 14. It pays a fixed dividend each year, and the holders are paid before the ordinary owners. The **cost of preferred stock** is the yearly dividend divided by the money the firm receives for each share. In symbols, $k(p) = D(p) \div N(p)$, where $D(p)$ is the yearly preferred dividend and $N(p)$ is the net proceeds per share.

Notice that there is no tax adjustment. A dividend is paid out of profit that has already been taxed, so it saves no tax. This is why preferred stock costs more than debt of the same face rate, even though both pay a fixed amount.

Given. A preferred share is sold for Tk 100 and pays a dividend of Tk 10 each year. The firm pays no issue cost. *Find.* The cost of preferred stock. *Steps.* $k(p) = 10 \div 100 = 0.10$. *Answer.* The cost is 10%.

Given. A preferred share has a face value of Tk 100 and pays a 10% dividend. The firm receives Tk 95 for each share after issue costs. *Find.* The cost. *Steps.* The yearly dividend is 10% of Tk 100, which is Tk 10. Then $k(p) = 10 \div 95 = 0.1053$. *Answer.* The cost is 10.5%. *Check.* It is higher than 10%, because the firm receives less than the face value.

Answer and check.

Three slips are common here.

The tax saving has a condition.

The bakery's own debt.

Bank loans with instalments are priced in the same way.

Recap: debt is priced after tax.

Preferred stock is the next cheapest source after debt.

Cost of preferred stock. The yearly fixed dividend divided by the money the firm receives per preferred share.

Dividends are not an expense, so no tax step.

First example, small numbers.

Second example, with issue costs.

¹ The tax rate of 25% is a placeholder for illustration. The rate that applies depends on the type of business and the law of Bangladesh in force, and must be checked before the book is printed.

Common equity is the hardest fund to price.

Cost of equity. The yearly return that the common owners require on the money they have in the firm.

Profit that is kept in the business is not free.

Two ways to estimate the cost of equity are in use.

Method one uses beta.

A small check with round numbers first.

Method two uses the dividend a share is expected to pay.

Worked with two data sets.

New shares cost more than kept profit.

Estimates of this kind can be wrong.

A table of the three costs side by side.

Common equity is harder to price, because the owners are not promised any amount. The firm pays no stated interest and no fixed dividend. Yet owners are not giving their money free. They could have placed it elsewhere, so they expect a reward that matches the risk they accept. The **cost of equity** is the return that the common owners require on the money they have in the firm.

Profit that the firm keeps is part of the owners' money. It is tempting to say that it costs nothing, because nobody is paid for it. This is a mistake. If the firm had paid the profit out as a dividend, the owners could have invested it at the same level of risk. They require the same return on kept profit as on money they put in newly. So the cost of retained earnings equals the cost of equity.

Two methods are used to estimate the cost of equity. The first uses the capital asset pricing model of Chapter 10. The second uses the value of a share that pays a growing dividend, from Chapter 11. Each method has strengths, and a careful person may calculate both and compare.

Method one. In Chapter 10 the required return was the risk-free return plus beta times the market risk premium. In symbols, $k(e) = r(f) + \beta \times (r(m) - r(f))$, where $r(f)$ is the risk-free return, $r(m)$ is the return of the market and β is beta. For the bakery, the illustrative figures were $r(f) = 6\%$, $r(m) = 12\%$ and $\beta = 1.2$.

Example. If $r(f) = 5\%$, $r(m) = 11\%$ and $\beta = 1.0$, then $k(e) = 5\% + 1.0 \times (11\% - 5\%) = 11\%$. For the bakery, $k(e) = 6\% + 1.2 \times (12\% - 6\%) = 6\% + 7.2\% = 13.2\%$. *Check.* A beta above 1 raises the answer above the market return of 12%, and 13.2% does.

Method two. In Chapter 11 a share with a dividend that grows at a steady rate has the value $P(0) = D(1) \div (k(e) - g)$. Here $P(0)$ is the price now, $D(1)$ is the dividend expected next year and g is the yearly growth rate of the dividend. Rearranging gives $k(e) = D(1) \div P(0) + g$. The first part is the dividend yield, and the second is the growth that the owner expects.

Example 1. A share sells for Tk 100. The next dividend is Tk 5, and it grows at 5% a year. Then $k(e) = 5 \div 100 + 0.05 = 0.05 + 0.05 = 10\%$.

Example 2. A share sells for Tk 80. The next dividend is Tk 6, with growth of 4%. Then $k(e) = 6 \div 80 + 0.04 = 0.075 + 0.04 = 11.5\%$.

When a firm sells new common shares, it pays issue costs. Suppose the issue cost in Example 2 is Tk 4 on each share. The firm receives only Tk 76. Then $k(e) = 6 \div 76 + 0.04 = 0.0789 + 0.04 = 11.9\%$. So the cost of new shares is higher than the cost of retained earnings, which is 11.5%.

The two methods rarely give the same answer, and each depends on estimates. Beta, the market return and the growth rate are all uncertain. A private firm such as the bakery has no share price, so only the first method can be used, and with illustrative figures. A person who uses either method should state the numbers, show the arithmetic and accept that the answer is an estimate, not a fact.

The table below compares the three kinds of fund. Read each row from left to right: it gives the kind of fund, what the investor is promised, the usual tax treatment and the way the cost is found.

THREE KINDS OF FUND COMPARED

FUND	REWARD TO THE INVESTOR	TAX EFFECT	HOW THE COST IS FOUND
Debt	Interest, fixed, paid first	Interest saves tax	Interest rate $\times$ (1 – tax rate)
Preferred stock	Fixed dividend, paid before common	None	Dividend $\div$ net proceeds
Common equity	What is left, not promised	None	CAPM, or next dividend $\div$ price + growth

The costs usually follow this order: debt is cheapest, preferred stock is next and common equity is dearest. The reason is the order of claims from Chapters 11 and 14. Lenders are paid first, preferred holders next and common owners last. The one who is paid last carries the most risk, so he asks for the largest reward.

Case file, illustrative. For the bakery, the cost of debt after tax is 9% and the cost of equity by the capital asset pricing model is 13.2%. The order, with debt below equity, is as expected. The bakery has no preferred stock at this stage.

Before moving on. Preferred stock costs its dividend divided by what the firm receives, with no tax step. Common equity costs what the owners require, found by beta or by the dividend method. Kept profit has the same cost as the owners' money. Issue costs raise the price of new funds.



THE WEIGHTED AVERAGE COST OF CAPITAL

A project is paid for by the whole pool of the firm's funds and not by one source. So the firm needs a single rate. Think of a bag of mixed rice, half cheap and half dear. The price of one kilogram of the mix is not the price of the cheap rice. It is the average of the two prices, with each price counted in proportion to its share. The same idea gives the firm one rate.

The **weighted average cost of capital** means the average of the costs of all the firm's funds, with each cost counted in proportion to the share of funds that it supplies. It is often written WACC. The share of each fund is its **weight**.

In words: multiply each cost by its weight and add the products. In symbols, $WACC = w(d) \times k(d) + w(p) \times k(p) + w(e) \times k(e)$. Here w stands for a weight, and the small letters d , p and e stand for debt, preferred stock and equity. The weights are decimals, and together they must add to 1.

Given. A firm has Tk 1,00,000 of debt with an after-tax cost of 8% and Tk 1,00,000 of equity with a cost of 12%. *Find.* The WACC. *Steps.* The total is Tk 2,00,000. Each weight is $Tk\ 1,00,000 \div Tk\ 2,00,000 = 0.5$. Then $WACC = 0.5 \times 8\% + 0.5 \times 12\% = 4\% + 6\% = 10\%$. *Check.* The answer lies between 8% and 12%, in the middle, because the weights are equal.

Given. The funds are debt of Tk 3,00,000 (cost after tax 9%), preferred stock of Tk 1,00,000 (cost 10.5%) and common

Order of cost follows order of risk.

The bakery's two costs, side by side.

Recap: three funds, three methods.

GO SLOWLY HERE. SEVERAL PARTS COMBINE.

The firm has one rate to find, not three.

Each cost is counted by its share of the funds.

Weighted average cost of capital. The average of the costs of all funds, each counted by its share of the total.

The formula is a sum of products.

First example, two funds and clean numbers.

Second example, three funds with unequal weights.

equity of Tk 6,00,000 (cost 13%). *Find*. The WACC. The table sets out the work.

WEIGHTED AVERAGE COST OF CAPITAL, SECOND EXAMPLE

FUND	AMOUNT (TK)	WEIGHT	COST	WEIGHT × COST
Debt	3,00,000	0.30	9.0%	2.70%
Preferred stock	1,00,000	0.10	10.5%	1.05%
Common equity	6,00,000	0.60	13.0%	7.80%
Total	10,00,000	1.00		11.55%

How to read the table.

Read a row at a time. For debt, the weight is Tk 3,00,000 ÷ Tk 10,00,000 = 0.30, and $0.30 \times 9.0\% = 2.70\%$. Do the same for each fund, then add the last column. *Answer*. The WACC is 11.55%. *Check*. The weights add to 1.00, and 11.55% lies between the lowest cost, 9%, and the highest, 13%. It lies nearer to 13%, because equity has the largest weight.

Weights can be taken from books, markets or a plan.

Target weights. The shares of debt and equity that a firm plans to hold in the long run.

Which amounts should give the weights? There are three choices. Book weights use the amounts written in the firm's records, which are easy to find. Market weights use the present market value of each fund, which is more accurate for traded securities. **Target weights** are the shares that the firm plans to keep in the long run, and most careful firms prefer them, because the funds of the future are what new projects will use.

The cost rises as more funds are raised.

There is an exception that every reader should know. The price of funds does not stay fixed as the firm raises more. At first the firm uses retained earnings, which have no issue cost. When those are used up, it must sell new shares, which cost more. Lenders may also ask for higher interest on a large loan. So the cost of the next taka can be higher than the average cost of the earlier ones. In this chapter we assume the rate stays the same for the sums we use.

Does the mix of funds change the average?

A long debate remains open: can a firm lower its WACC by changing the mix of debt and equity? Interest saves tax, and this favours debt. But debt raises risk, which raises the price of equity and may harm the firm if it cannot pay. Most writers agree that, in practice, there is a mix that is best for each firm, but that it is hard to find. We do not try to find it here.

The bakery's weights and its new hurdle rate.

Case file, illustrative. The owner of the bakery plans to hold debt and equity in equal parts as a long-run target, so the weights are 0.5 and 0.5. The costs are 9% for debt after tax and 13.2% for equity. $WACC = 0.5 \times 9\% + 0.5 \times 13.2\% = 4.5\% + 6.6\% = 11.1\%$. This is the new hurdle rate for the bakery.

A short test with new figures.

Pause and check. A firm has debt of Tk 4,00,000 with a cost after tax of 8%, and equity of Tk 6,00,000 with a cost of 12%. Find the weights, and then the WACC. Say why the answer must lie between 8% and 12%. A short answer is given at the back of the book.

Recap: one weighted rate.

Before moving on. The WACC multiplies each cost by its weight and adds the products. The weights must add to 1. Target weights suit new projects best. The rate can rise when the firm must raise more funds.

A check the reader can test.

You should now be able to say: the WACC is one yearly rate that shows what the firm pays, on average, for each taka of its funds.



USING THE RATE TO RETEST PROJECTS

We now return to the question from the start of the chapter. The WACC serves as the hurdle rate of Chapter 7, which means the least return that a project must earn. A project is accepted if its net present value at the WACC is above zero. This is the same as saying that its internal rate of return is greater than the WACC. If the two tests disagree on a ranking, Chapter 7 shows how to settle the matter.

Example. A project costs Tk 1,00,000 and brings Tk 40,000 at the end of each of three years. At a WACC of 10% the annuity factor for 3 years is 2.4869, so the present value of the inflows is $40,000 \times 2.4869 = \text{Tk } 99,476$. The net present value is $99,476 - 1,00,000 = -\text{Tk } 524$. The project is rejected, narrowly.

Check. At a rate of 8% the annuity factor is 2.5771. The present value is Tk 1,03,084, and the net present value is +Tk 3,084. Had the owner guessed 8%, the project would have been accepted. The internal rate of return is about 9.7%, which is below 10%, so the two tests agree on rejection at 10%. A small difference in the rate decided the matter, and this is why the rate must be found with care.

The WACC rests on an assumption. It assumes that the new project has about the same risk as the business as a whole, and that the mix of funds does not change. This is true for another oven. It is not true for a very different venture. A risky new product line should be tested at a higher rate than the WACC, as Chapter 10 explained, because investors ask for more reward for more risk. A very safe project may deserve a lower rate.

Another point: suppose a project is paid for with one specific loan at 12%. Should we use its after-tax cost of 9%? No. Funds are pooled. If the firm uses up its borrowing power on this project, the next project will have to be paid for with more costly equity. So both projects should be judged at the rate of the pool, which is the WACC.

Before moving on. Use the WACC as the discount rate in the net present value rule, or compare it with the internal rate of return. Use a higher rate for riskier projects. Always use the rate of the whole pool, not the rate of the loan that happens to be used.



WORKED CASE: THE BAKERY RETESTS THE OVEN AND THE VAN

Case file, illustrative. In Chapter 7 the owner chose between two projects at a provisional rate of 10%. Each costs Tk 3,00,000 and lasts five years. The oven brings Tk 1,00,000 a year. The van brings Tk 50,000, Tk 80,000, Tk 1,10,000, Tk 1,40,000 and Tk 1,40,000 in years 1 to 5. The owner has now found the hurdle rate, 11.1%.

Situation. The new rate of 11.1% replaces the guess of 10%. *Question.* Does the oven remain the better choice, and are both projects still worth doing? *Reasoning.* The owner finds the present value of the inflows of each project at 11.1% and subtracts the outlay of Tk 3,00,000. Because the rate is not in the printed tables, the owner uses a calculator for the factors, $1 \div 1.111$ raised to the year number, as shown in Chapter 3.

The rate is useful only if it changes decisions.

A project can pass at one rate and fail at another.

The earlier rate would have said yes.

One rate is right only for projects of the same kind of risk.

A project financed by one loan still uses the common rate.

Recap: retest, then decide.

The case file ties the chapter together.

Situation, question, reasoning, conclusion, check.

PRESENT VALUES AT A DISCOUNT RATE OF 11.1%

YEAR	FACTOR	OVEN INFLOW (TK)	PRESENT VALUE (TK)	VAN INFLOW (TK)	PRESENT VALUE (TK)
1	0.9001	1,00,000	90,010	50,000	45,005
2	0.8102	1,00,000	81,020	80,000	64,816
3	0.7292	1,00,000	72,920	1,10,000	80,212
4	0.6564	1,00,000	65,640	1,40,000	91,896
5	0.5908	1,00,000	59,080	1,40,000	82,712
Total			3,68,670		3,64,641

Reading the figures.

Read each row from left to right: the factor turns the inflow of that year into today's taka. The totals are the present values of all the inflows. Because the factors are rounded to four places, the totals differ from a full calculator answer by a few taka. The full calculator totals are Tk 3,68,662 for the oven and Tk 3,64,632 for the van. We use these.

Conclusion and check with a figure.

Conclusion. The net present value of the oven is Tk 3,68,662 – Tk 3,00,000 = Tk 68,662. For the van it is Tk 3,64,632 – Tk 3,00,000 = Tk 64,632. Both are above zero, so both are worth doing, and the oven is still ahead. The next figure shows the net present value of each project at different rates.

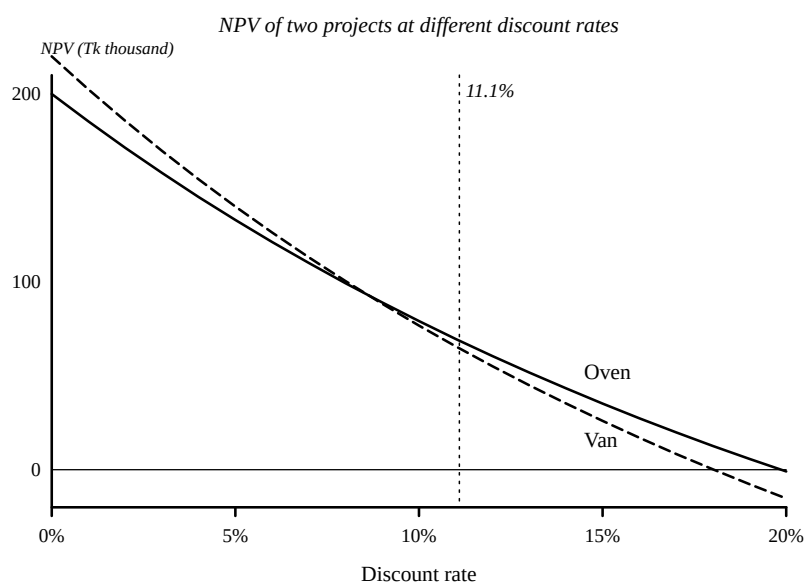


FIG. 42. Net present value of the oven (solid) and the van (dashed) at different discount rates. Each line crosses zero at the project's internal rate of return.

How to read the figure.

The horizontal axis shows the discount rate and the vertical axis shows net present value in thousands of taka. A point on a line gives the value of the project at that rate. The lines fall as the rate rises, because later cash is worth less. The dotted line marks 11.1%. At this rate both lines stand above zero, with the oven above the van.

The check agrees with the figure.

Check. The oven's line crosses zero at an internal rate of return of about 19.9%, and the van's at about 18.0%. Both are well above the hurdle of 11.1%, which agrees with the positive net present values. The two lines cross each other near 8.6%. Below that rate the van would be the better project, and above it the oven. The owner's hurdle is

above 8.6%, so the oven stays the first choice, with a lead of about Tk 4,030.

At the guess of 10% the oven led by Tk 2,313. At 11.1% it leads by Tk 4,030. The ranking has not changed, but the values have fallen, from Tk 79,079 to Tk 68,662 for the oven. If the owner had used the cost of the 12% loan before tax, the hurdle would have been too high, and the projects would have looked less attractive than they really are.

The next figure shows the three rates that the case file uses: the cost of debt after tax, the cost of equity and the WACC that lies between them.

The result also shows how the figures compare with the guess.

Another bar shows where the 11.1% comes from.

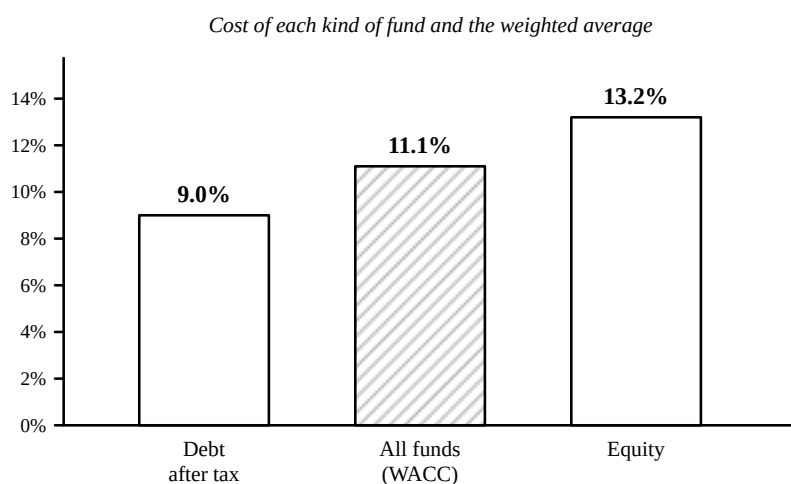


FIG. 43. The bakery's cost of debt after tax, its weighted average cost of capital and its cost of equity.

The bars show that the WACC is above the cost of debt and below the cost of equity. With equal weights, it lies exactly midway between them. Had the owner chosen more debt, the weighted average would fall nearer to 9% on paper. But, as the first section explained, more debt would raise the risk and with it the cost of equity.

The WACC lies between the costs it combines.



SUMMARY

The cost of capital is the yearly price a firm pays for its funds. For the firm it is a cost; for the investor it is the required return; in earlier chapters it was the discount rate. Debt costs its interest rate multiplied by one less the tax rate. Preferred stock costs its dividend divided by what the firm receives. Common equity costs what the owners require, estimated by the capital asset pricing model or by the dividend method. Kept profit has the same cost as the owners' money. The weighted average cost of capital combines the costs by weight. It is the hurdle rate for projects of the firm's usual risk.

The chapter in a few lines.

IN SHORT

The weighted average cost of capital is the sum of each fund's cost multiplied by its weight. It serves as the discount rate for projects that have the same risk as the firm.

THINK IT THROUGH

A reasoning problem with a wrong argument to correct.

The owner of the bakery wishes to buy a second van. It costs Tk 2,00,000, lasts four years and brings Tk 63,100 a year, and its internal rate of return is about 10%.² The owner argues: "I shall pay for it with a new loan at 12%, which costs 9% after tax, so I should test it at 9%." The adviser replies: "Test it at 11.1%." Who is correct, and what happens if the van's risk is higher than the bakery's usual risk?

A hint without giving the answer.

Consider these questions in order. What do the funds of the firm look like after the loan is used? Which projects would then have to be paid with equity? Does the van's return still pass the hurdle if the van is risky and the rate must be raised? A short answer is given at the back of the book.

WHERE PEOPLE GO WRONG

Using the rate of one loan.

Using the rate of the loan that pays for the project. The funds of the firm are pooled. A project is judged at the cost of the pool, which is the WACC, and not at the price of one source.

Forgetting tax.

Using the interest rate without the tax step. The interest saves tax, so the cost of debt after tax is lower than the interest rate. But the step does not apply to dividends.

Thinking debt is always best.

Believing that the cheapest source should be used alone. Debt looks cheapest, but more debt raises the risk and the price of equity. The cost of capital belongs to the whole mix.

Thinking kept profit is free.

Treating retained earnings as costless. The owners could have received that profit and invested it elsewhere. They require the same return as on the money they put in.

Weights that do not add up.

Weights that do not add to 1. If the weights add to more or less than 1, the answer is wrong. Add them first. A second check is that the WACC must lie between the lowest and highest costs.

IN HISTORY

A famous paper changed the way economists think about funds.

In 1958 two American economists, Franco Modigliani and Merton Miller, published a paper called "The Cost of Capital, Corporation Finance and the Theory of Investment". They asked whether a firm could lower its cost of capital merely by changing the mix of debt and equity. Under very simple conditions, with no tax and no cost of failure, they argued that the answer is no. The value of a firm depends on what its assets earn, and not on how the money to buy them was divided. Their later work, which added tax, showed why debt can still have an advantage. The two men received the Nobel Prize in Economics, Modigliani in 1985 and Miller in 1990.³

SOURCES

The one landmark work named in this chapter.

Modigliani, F. and Miller, M. H. (1958), "The Cost of Capital, Corporation Finance and the Theory of Investment", *American Economic*

² Four yearly inflows of Tk 63,100 against an outlay of Tk 2,00,000 give an internal rate of return of about 10.0%. The figure is rounded for the problem.

³ Dates and titles are given from memory of the published record and must be confirmed against the journal and the Nobel Prize website before printing.

Review, volume 48, number 3. The paper is the starting point for all modern discussion of the link between a firm's mix of funds and its value.

REVIEW QUESTIONS

- | | |
|--|------------------------------|
| 1. Define the cost of capital. In two or three sentences, explain why a firm must know it before it judges a project. | <i>Explain in words.</i> |
| 2. Explain why the same number is called a cost by the firm and a required return by the investor. | <i>Explain in words.</i> |
| 3. A firm borrows Tk 2,00,000 at 14% interest. The tax rate is 25%. Find the cost of debt after tax. | <i>Calculate.</i> |
| 4. A firm sells debentures with a face value of Tk 1,000, a coupon rate of 9% and a life of 5 years. After issue costs it receives Tk 960 for each one. The tax rate is 25%. Estimate the cost of debt after tax. | <i>Calculate.</i> |
| 5. A preferred share is sold at Tk 200 and pays a yearly dividend of Tk 24. The issue cost is Tk 10 a share. Find the cost of preferred stock. | <i>Calculate.</i> |
| 6. A share sells for Tk 50. The next dividend is Tk 3, and the dividend grows at 5% a year. (a) Find the cost of equity. (b) Find the cost of new shares if the issue cost is Tk 2.50 a share. | <i>Calculate.</i> |
| 7. The risk-free return is 7%, the market return is 13% and beta is 0.8. Find the cost of equity by the capital asset pricing model. | <i>Calculate.</i> |
| 8. A firm has debt of Tk 4,00,000 with a cost after tax of 8%, preferred stock of Tk 2,00,000 with a cost of 11%, and common equity of Tk 6,00,000 with a cost of 14%. Find the weighted average cost of capital. | <i>Calculate.</i> |
| 9. A project costs Tk 5,00,000 and brings Tk 1,50,000 at the end of each of five years. The firm's WACC is 12%, and the annuity factor for 5 years at 12% is 3.6048. Find the net present value and say whether to accept the project. | <i>Calculate and decide.</i> |
| 10. Discuss why the cost of equity is usually higher than the cost of debt. Give two reasons. | <i>Discuss.</i> |
| 11. A manager says: "Retained earnings cost nothing, because we pay nobody for them." Explain why the manager is wrong. | <i>Discuss.</i> |



CHAPTER XVI

Leasing

Is it wiser to own the machine or to rent it?



A firm often needs an asset but does not need to own it. This chapter explains what a lease is, the two main kinds, why firms choose to lease, and how to compare leasing with buying in today's taka and after tax.

A rent that adds up to more than the price looks like a bad bargain.

The simple sum leaves out time, borrowing and tax.

This chapter gives a method that does so.

The bakery needs a display freezer. It costs Tk 1,50,000 to buy. A leasing company offers the same freezer for Tk 42,000 at the end of each year for five years. Add the five payments and the total is Tk 2,10,000. That is Tk 60,000 more than the price. Many people would stop here and say that buying must be cheaper.

The simple sum is not wrong, but it is not complete. Buying the freezer needs Tk 1,50,000 today, and the bakery does not have it. It must borrow the money, and the loan has interest. Buying also gives tax benefits that renting does not give. And a taka paid in five years is worth less than a taka paid today. A fair comparison must put all these things on one scale.

This chapter builds that scale. We first learn what a lease is and the two kinds of lease. We then examine why firms lease at all. After this we compare leasing with buying by following four steps. The freezer will show whether the extra Tk 60,000 is real.

YOU WILL NEED

From Chapter 5: the present value of a stream of equal payments, the annuity. From Chapter 6: depreciation. From Chapter 3: the tax step in the profit layout. From Chapter 13: term loans, repayment schedules and the tax saved on interest. From Chapter 15: the cost of debt after tax.



WHAT LEASING IS

Using an asset and owning it are two different things.

Lease. An agreement in which the owner of an asset lets another party use it in return for regular payments.

The agreement has a name, and so do its two parties.

Lessor. The owner of the asset, who lets another party use it.

Lessee. The party that uses the asset and pays for the right to do so.

The owner and the user need separate names.

Lease rental. The regular payment that the lessee makes to the lessor for the use of the asset.

The regular payment is the price of using the asset.

The next figure shows who gives what.

Think of a student who rents a room instead of buying a house. She has the use of the room every day, but the owner of the house is another person. She pays a rent for the right to use it, and when the agreement ends she gives the room back. Many firms treat machines in the same way.

A **lease** means an agreement in which the owner of an asset lets another party use it, in return for regular payments, for a stated time. A bakery that agrees to use a freezer for five years and to pay each year has made a lease. The agreement is written in a contract that states the asset, the time and the payments.

The owner of the asset is the **lessor**. The user is the **lessee**. A simple way to remember the two: the lessor lets the asset go to another, and the lessee is the party that has the lease. In our case, the leasing company is the lessor and the bakery is the lessee.

The payment that the lessee makes at fixed times is the **lease rental**. It may be paid each month, each quarter or each year. In this chapter we assume that it is paid at the end of each year, the same timing as in the annuities of Chapter 5.

The next figure shows how the parties are linked. The lessor buys the asset from a

supplier. It then allows the lessee to use the asset, and the lessee pays the rental.

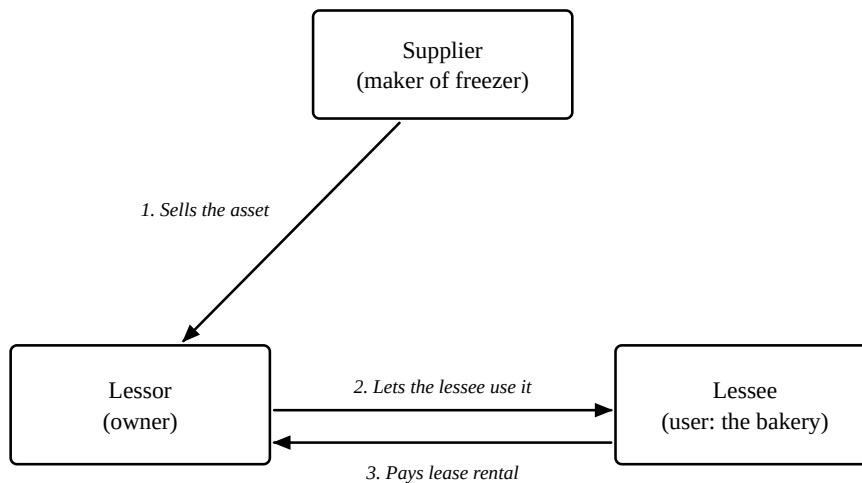


FIG. 44. The parties to a lease. The lessor buys the asset from a supplier and lets the lessee use it; the lessee pays the rental.

Read the figure from the top. The supplier makes or sells the freezer and receives its price from the lessor. The arrow from the lessor to the lessee shows the freezer in use. The arrow below it shows the rental going back. Note that the lessee never pays the supplier. The lessor carries the large payment, and recovers it through the rentals.

The key idea is that the lease separates ownership from use. This has two results. First, the lessee does not need a large sum on the first day. Second, the lessee does not receive what the asset is worth when the lease ends, unless the contract says so. We shall see in the next section that the contracts differ on this point.

In Bangladesh, leases are offered by some finance companies and banks.¹ The ideas in this chapter apply wherever a lease is used, but the tax rules differ between countries, and a student should check the rules for the place and year concerned.

Before moving on. A lease lets one party use an asset owned by another, for payments called lease rentals. The owner is the lessor and the user is the lessee. The lessee gains use without paying the full price at the start.



OPERATING AND FINANCIAL LEASES

A lease for one weekend and a lease for the whole life of a machine are very different agreements. So leases are divided into two main kinds, according to how long they last and who carries the risk. The names are **operating lease** and **financial lease**.

¹ Leasing by financial institutions in Bangladesh is understood to be regulated by Bangladesh Bank under the Financial Institutions Act, 1993. This statement and the tax treatment of lease rentals and depreciation must be verified against current law before printing.

How to read the figure.

Ownership and use are separated, and this has consequences.

Leasing exists in Bangladesh, in several forms.

Recap: a lease divides ownership from use.

Operating lease. A short lease in which the lessor keeps most of the risks of owning the asset.

Financial lease. A long lease that covers most of the life of the asset, and in which the lessee carries most of the risks.

Not all leases are alike.

The first kind is short and flexible.

An operating lease is a short rental in which the owner keeps most of the risk of owning the asset. It lasts for much less than the working life of the asset, and the lessee can often cancel it with notice. The lessor usually pays for repair and insurance, and the rental includes these costs. Renting a car for a month is an example. The lessor expects to lease the asset again to another user, or to sell it.

The second kind is long and firm.

A financial lease is a long rental that covers most of the life of the asset. The lessee usually cannot cancel it without a heavy penalty. The lessee pays for repair and insurance. In effect, the lessor has lent the price of the asset, and the rentals repay it with interest. This kind of lease is much like a loan, and the decision method that we shall learn is built for it.

Residual value. What the asset is worth at the end of the lease or at the end of its useful life.

At the end of a lease, the asset still has some worth. This is its **residual value**, which means the value of the asset when the lease ends. If a freezer that cost Tk 1,50,000 can be sold for Tk 10,000 after five years, its residual value is Tk 10,000. Who receives that Tk 10,000? In an operating lease, the lessor does. In a financial lease, the contract may let the lessee buy the asset for a small sum, or return it.

The end of the lease raises a question about value.

The table lists the main differences.

The table below puts the two kinds side by side. Read each row to compare them on one point.

OPERATING LEASE AND FINANCIAL LEASE COMPARED

POINT	OPERATING LEASE	FINANCIAL LEASE
Length	Short, much less than the asset's life	Long, most of the asset's life
Cancelling	Often allowed with notice	Usually not allowed, or heavy penalty
Repair and insurance	Usually paid by the lessor	Usually paid by the lessee
Residual value	Belongs to the lessor	May go to the lessee by contract
Nature	Rental for use	Like a loan to buy the asset

Sale and leaseback. A firm sells an asset it owns to a lessor and leases the same asset back at once.

One more arrangement is common. In a **sale and leaseback**, a firm that already owns an asset sells it to a lessor and at once leases it back. The firm receives cash and keeps the use of the asset. A firm that needs money for other purposes can use this method, but it then pays rentals for something it once owned.

A third arrangement links to the second.

The name of the lease does not settle the matter.

There is an exception. A contract may be called an operating lease and yet behave like a financial lease, if it lasts for most of the asset's life and cannot be cancelled. The substance counts and not the name. Accounting rules also treat leases in a particular way. Under the international standard that many countries now use, a lessee records most leases as an asset and a liability in its accounts, so the old difference matters less for the lessee's records, but it is still useful for understanding the risks.²

Recap: two kinds, one test.

Before moving on. An operating lease is short and the lessor keeps most of the risk. A financial lease is long and works like a loan. The residual value is the worth of

² The standard referred to is IFRS 16, which took effect for periods beginning on or after 1 January 2019. Whether and from when it applies in Bangladesh, and the exact rules for lessees and lessors, must be verified with the Institute of Chartered Accountants of Bangladesh before printing.

the asset at the end. The substance of the contract decides which kind it is.



WHY FIRMS LEASE

If a firm can borrow and buy, why does it lease? There are several reasons, and each is true in some situations and false in others. We list them with care.

Little cash at the start. A lease needs no large payment on the first day. The bakery can use the freezer immediately and pay from the sales that the freezer helps to earn. A firm that is short of cash, or a young firm that finds borrowing hard, may value this very much. A loan, however, can also pay for the whole price, so this advantage is smaller than it seems when a loan is available.

Flexibility and change. Some assets become old-fashioned quickly, such as computers and phones. A short lease lets a firm return the asset and take a newer one. The firm avoids being stuck with something it no longer wants. The cost of this flexibility is usually a higher rental.

Service and skill. Some lessors specialise in an asset. They know how to maintain it and may include repair in the rental. A bakery does not need to learn to repair freezers.

Tax. An owner of an asset claims depreciation, and this saves tax. A lessor that pays much tax can use the depreciation well, and may pass part of the benefit to the lessee through a lower rental. A firm with little taxable profit cannot use depreciation, so leasing may suit it. The rental is also an expense that reduces taxable profit.

Fewer conditions. A lender may place conditions, called covenants in Chapter 13, on a loan. A lease may carry fewer such conditions. A lessor also holds the asset as its own security, so it may agree when a bank has refused.

There are drawbacks too, and a careful student states them. The total of the rentals can exceed the cost of buying and borrowing. The rentals are fixed obligations, like loan payments, and they add to financial risk as described in Chapter 9. The lessee loses the residual value. And the lessee usually cannot change the asset. A firm should lease for a clear reason, not from habit.

Pause and check. (1) A new café needs an oven for six months to test a recipe. Which kind of lease suits it, and why? (2) Give one reason why a firm with little taxable profit may prefer to lease. A short answer is given at the back of the book.

Before moving on. Firms lease to save cash at the start, to keep choices open, to receive service and skill, to use tax benefits that they cannot use themselves, and to avoid conditions. The drawbacks are higher total cost, fixed obligations and the loss of residual value.



A good decision begins with the reasons.

The first reason is to save cash at the start.

The second reason is to keep choices open.

The third reason is skill and service.

The fourth reason is tax.

The fifth reason is fewer conditions.

The drawbacks are as real as the reasons.

Two questions to test the section.

Recap: reasons and drawbacks.

THE LEASE-OR-BUY DECISION: FOUR STEPS

GO SLOWLY HERE. SEVERAL PARTS COMBINE.

Two ways of getting the same asset must be compared fairly.

We assume that buying is paid for with a loan.

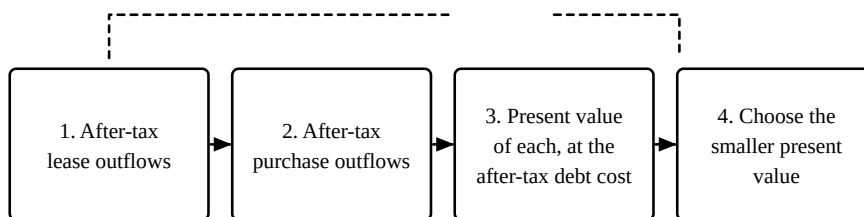
The decision follows four steps.

The bakery can obtain the freezer in two ways. It can lease, or it can borrow and buy. We want to know which costs less. A fair comparison needs three things. The cash flows must be in the same units, in taka that are paid out. They must be after tax, because tax differs between the two ways. And they must be brought to today, because they fall at different times.

We assume that if the bakery buys, it pays by borrowing the whole price. This keeps the comparison clean. Both choices then involve fixed yearly payments, and the real question is which set of payments costs less after tax. If the owner could pay with cash, the same method would apply, with the cash outlay in place of the loan.

The decision is made in four steps. First, find the after-tax cash outflows of the lease. Second, find the after-tax cash outflows of buying. Third, find the present value of each set of outflows. Fourth, choose the choice with the smaller present value, because it costs less in today's taka. The next figure shows the steps.

The lease-or-buy decision in four steps



Same discount rate and same years for both choices

FIG. 45. *The four steps of the lease-or-buy decision.*

The first step is the easier of the two.

The second step has three parts.

Each year's net outflow is built from these parts.

Step 1: the lease. The rental is an expense, so it reduces taxable profit and saves tax. If the rental is Tk 42,000 and the tax rate is 25%, the tax saved is Tk 10,500. The after-tax outflow is Tk 42,000 less Tk 10,500, which is Tk 31,500. In symbols, the after-tax rental equals the rental $\times (1 - T)$. This is the same rule as for interest in Chapter 15.

Step 2: buying. The outflows of buying have three parts. The firm pays the loan instalments, which are interest plus part of the loan. It saves tax on the interest, and it saves tax on the depreciation of the asset. Only interest and depreciation save tax. The repayment of the loan itself is not an expense, so it saves none. At the end, the firm may sell the asset, and the sale brings a cash inflow that reduces the outflow of that year.

For each year, the after-tax outflow of buying equals the loan instalment, less the tax saved on interest, less the tax saved on depreciation, and less any cash received from selling the asset. It may help to write the interest and the depreciation together: tax saved = tax rate $\times$ (interest + depreciation).

Step 3: present values. We must choose a rate to bring each outflow to today. We use the after-tax cost of debt from Chapter 15, and not the weighted average cost of capital. The reason is that lease rentals are fixed, and the lessor has a claim like that of a lender. They carry the same kind of risk as loan payments, so they should be discounted at the price of loan money. In the bakery's case this is 9%.

Step 4: choose. Add the discounted outflows of each choice. The choice with the smaller total is cheaper. If the totals are close, the owner should also weigh matters that the numbers leave out, such as the need for flexibility or service.

Before moving on. Compare the after-tax outflows of the lease and of buying, bring both to today at the after-tax cost of debt, and choose the smaller present value. Interest and depreciation save tax; loan repayments do not.

You should now be able to say: the lease-or-buy decision compares the present values of the after-tax cash outflows of two ways of obtaining the same asset.

The third step needs a discount rate, and the choice matters.

The fourth step is the decision.

Recap: four steps, one discount rate.

A check the reader can test.



AFTER-TAX OUTFLOWS AND THE PRESENT-VALUE COMPARISON

We now carry out the four steps with numbers. The first example uses small, round figures so that the method is the only new thing. The loan in this first example is repaid in equal parts of principal, which makes the interest easy to calculate. The second example, in the case that follows, uses equal instalments.

Given. A machine costs Tk 1,00,000 and lasts 4 years. It will have no value at the end. The firm can borrow the whole price at 12%, repaying Tk 25,000 of the loan at the end of each year, together with the interest on the balance. Depreciation is Tk 25,000 a year. The firm can instead lease the machine for Tk 30,000 at the end of each year. The tax rate is 25%, and the after-tax cost of debt is 9%. *Find.* Whether to lease or buy.

Step 1. The after-tax rental is $30,000 \times (1 - 0.25) = \text{Tk } 22,500$ in each of the 4 years.

Step 2. The interest is 12% of the balance owed at the start of each year: Tk 12,000, Tk 9,000, Tk 6,000 and Tk 3,000. The table sets out the work.

We begin with a small example to learn the layout.

The data of the first example.

The lease side is a single annuity.

The purchase side builds up year by year.

BUYING WITH A LOAN: AFTER-TAX OUTFLOWS (TK)

YEAR	LOAN PAID	INTEREST	INTEREST + DEPRECIATION	TAX SAVED (25%)	AFTER-TAX OUTFLOW
1	37,000	12,000	37,000	9,250	27,750
2	34,000	9,000	34,000	8,500	25,500
3	31,000	6,000	31,000	7,750	23,250
4	28,000	3,000	28,000	7,000	21,000

Read the first row. The firm pays Tk 25,000 of the loan and Tk 12,000 of interest, which is Tk 37,000. The interest plus the depreciation of Tk 25,000 is Tk 37,000, and 25% of this, Tk 9,250, is saved in tax. The outflow after tax is $\text{Tk } 37,000 - \text{Tk } 9,250 = \text{Tk } 27,750$. The other rows follow the same rule.

How to read the table.

Step three finds the present values.

Step 3. The discount factors at 9% from the table are 0.9174, 0.8417, 0.7722 and 0.7084 for years 1 to 4, and the annuity factor for 4 years is 3.2397. For the lease: Tk 22,500 $\times$ 3.2397 = Tk 72,893. For buying: 27,750 $\times$ 0.9174 = 25,458; 25,500 $\times$ 0.8417 = 21,463; 23,250 $\times$ 0.7722 = 17,954; 21,000 $\times$ 0.7084 = 14,876. The total is Tk 79,751.

Step four is the answer and its check.

Answer. The present value of the lease is Tk 72,893 and that of buying is Tk 79,751. Leasing costs about Tk 6,858 less, so the firm should lease. *Check.* Before tax, the lessee pays Tk 1,20,000 in rentals, while buying costs Tk 1,00,000 plus Tk 30,000 of interest, which is Tk 1,30,000. Leasing is cheaper before tax as well, so the direction of the answer is sensible.

Slips that students make.

Common slips. First, the whole loan instalment is treated as an expense, but only the interest saves tax. Second, depreciation is forgotten, although it saves tax in every year. Third, the weighted average cost of capital is used as the discount rate, when the after-tax cost of debt is correct. Fourth, the rental is not reduced by tax. Fifth, the two choices are discounted at different rates or over different numbers of years.

Timing can change the figures.

There are exceptions. Some leases require the rental at the start of each year, and not at the end. Then the payments form an annuity due as in Chapter 5, and each rental is one year closer to today. If the lessee pays for repair and the lessor does not, those costs must be added to the lease side, with their tax effect. If the residual value is uncertain, the owner may wish to test the decision with a lower value to see whether it changes.

Recap: build both sets of outflows, then compare.

Before moving on. After-tax rental is the rental times one less the tax rate. For buying, subtract the tax saved on interest and depreciation from the loan payment, and subtract any after-tax sale value. Discount both at the after-tax cost of debt. The lower present value wins.

A test with new figures.

Pause and check. A lessor asks Tk 20,000 a year for four years, and the after-tax cost of debt is 9%. What is the present value of the after-tax rentals if the tax rate is 25%? A short answer is given at the back of the book.



WORKED CASE: THE FREEZER

The case file now contains the second example, with equal instalments and a residual value.

Case file, illustrative. The bakery needs a display freezer that costs Tk 1,50,000 and is expected to last five years. The lessor offers it for Tk 42,000 at the end of each year for five years. The alternative is to borrow Tk 1,50,000 at 12%, repaid in five equal yearly instalments, and to buy. The owner expects to sell the freezer for Tk 10,000 at the end of year 5. The tax rate is 25%, and the after-tax cost of debt is 9%, as found in Chapter 15.³

The situation, the question and the layout of the reasoning.

Situation. The owner wishes to obtain the freezer at the lowest cost in today's taka. *Question.* Should the owner lease or buy with a loan? *Reasoning.* Follow the four steps. We first find the loan schedule, because the interest in each year is needed for the tax saved.

³ Depreciation is taken as a straight line to nil over five years, Tk 30,000 a year. The instalment is calculated as Tk 1,50,000 divided by the annuity factor for 5 years at 12%, 3.6048, which is about Tk 41,611. The schedule is rounded to the nearest taka, so a small balance of Tk 3 remains after the last instalment. Repair and insurance are assumed to be the same under both choices, and so are left out.

LOAN SCHEDULE: TK 1,50,000 AT 12%, INSTALMENT TK 41,611 (TK)

YEAR	INTEREST	PRINCIPAL REPAYED	BALANCE AT YEAR END
1	18,000	23,611	1,26,389
2	15,167	26,444	99,945
3	11,993	29,618	70,327
4	8,439	33,172	37,155
5	4,459	37,152	3

Interest in year 1 is 12% of Tk 1,50,000, which is Tk 18,000. The principal repaid is the instalment less the interest, Tk 41,611 – Tk 18,000 = Tk 23,611. The balance falls to Tk 1,26,389, and interest in year 2 is 12% of that. Each row repeats the rule.

Now find the tax saved. Depreciation is Tk 30,000 a year. In year 1, interest plus depreciation is Tk 18,000 + Tk 30,000 = Tk 48,000, and 25% of this is Tk 12,000. The after-tax outflow is the instalment less this saving, Tk 41,611 – Tk 12,000 = Tk 29,611. In year 5 the sale also brings Tk 10,000. This is a taxable gain, because the book value is nil, so after 25% tax the firm keeps Tk 7,500.

How the schedule is built.

The tax saved on interest and depreciation is the next step.

BUYING WITH A LOAN: PRESENT VALUE OF AFTER-TAX OUTFLOWS (TK)

YEAR	INSTALMENT	TAX SAVED	SALE AFTER TAX	NET OUTFLOW	FACTOR AT 9%	PRESENT VALUE
1	41,611	12,000	0	29,611	0.9174	27,165
2	41,611	11,292	0	30,319	0.8417	25,520
3	41,611	10,498	0	31,113	0.7722	24,025
4	41,611	9,610	0	32,001	0.7084	22,670
5	41,611	8,615	7,500	25,496	0.6499	16,570
Total						1,15,950

For the lease, the after-tax rental is $42,000 \times (1 - 0.25) = \text{Tk } 31,500$ each year. Its present value is $31,500 \times 3.8897 = \text{Tk } 1,22,526$, using the annuity factor for 5 years at 9%. If each year is discounted separately with the four-place factors, the total is Tk 1,22,523, and we use this figure.

The lease side is a repeated payment.

Conclusion. The present value of buying is Tk 1,15,950 and that of leasing is Tk 1,22,523. Buying with a loan costs Tk 6,573 less in today's taka, so the owner should buy. *Check.* The lease pays slightly more in total, Tk 2,10,000 against Tk 2,08,055 for the instalments, and it gives up the Tk 7,500 residual value. So buying should cost less, as found. The figure below shows the yearly outflows of both choices side by side.

Conclusion and check.

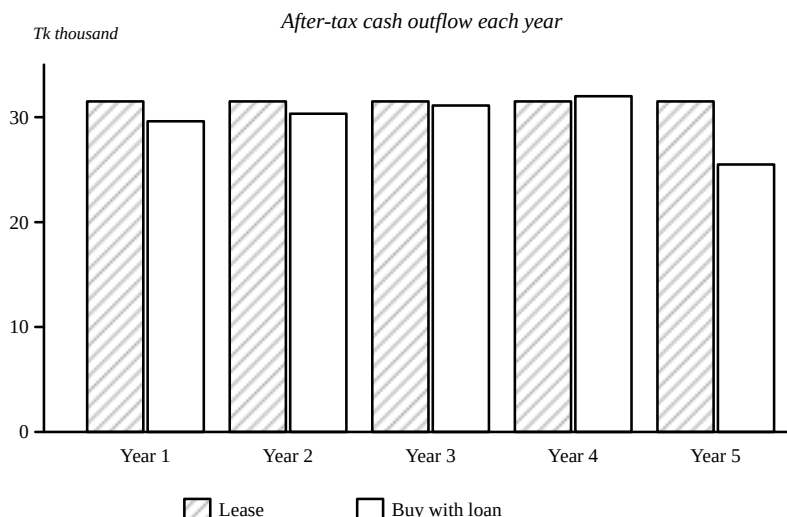


FIG. 46. After-tax cash outflow each year, for leasing and for buying with a loan. Buying is lower in year 5 because of the sale of the freezer.

How to read the figure.

Each pair of bars shows one year. The shaded bar is the lease, which is the same each year at Tk 31,500. The plain bar is buying. Buying costs less than leasing in years 1 to 3, a little more in year 4, and much less in year 5. The lower outflows of the early years are worth more than later ones, because of discounting.

The rental that would make the choice equal.

How high may the rental be before leasing is the better choice? The total present value of buying is Tk 1,15,950. Leasing costs the present value of $0.75 \times \text{rental} \times 3.8897$, which is $2.9173 \times \text{rental}$. Setting these equal gives $\text{rental} = 1,15,950 \div 2.9173$, which is about Tk 39,750. A lessor who offers less than this would make leasing the cheaper choice.

The owner's decision and what else matters.

The lessor asks Tk 42,000, which is above Tk 39,750, so the decision is to buy with the loan. The difference in present value, Tk 6,573, is small against the Tk 1,50,000 price. If the owner values the repair service that a lessor might include, or fears that the freezer may soon be out of date, these matters could outweigh the saving. Numbers guide the choice, but they do not make it alone.



SUMMARY

The chapter in a few lines.

A lease lets a lessee use an asset that a lessor owns, in return for lease rentals. An operating lease is short, and the lessor keeps most of the risk. A financial lease is long, and works like a loan. The residual value is the worth of the asset at the end. Firms lease to save cash, to keep choices open, to receive service and to use tax benefits, but the total cost may be higher. The lease-or-buy decision follows four steps: after-tax outflows of the lease, after-tax outflows of buying, present values at the after-tax cost of debt, and a choice of the smaller one.

IN SHORT

To compare leasing with buying, find the present value of the after-tax cash outflows of each choice at the after-tax cost of debt. Choose the choice with the smaller present value.

THINK IT THROUGH

The lessor of the freezer lowers its offer to Tk 39,000 a year for five years. Everything else in the case stays the same. Find the present value of the lease under the new offer and say whether the decision changes. A friend then says: "Leasing is better, because the freezer will not appear as a debt of the bakery." Is the friend's reason sound? What should the owner consider besides the numbers?

Begin with the break-even rental of about Tk 39,750 found in the case. Then ask whether the lease obligation is a fixed payment that the bakery must make in the same way as a loan instalment. A short answer is given at the back of the book.

A reasoning problem about a changed offer.

A hint without the answer.

WHERE PEOPLE GO WRONG

Treating the whole loan instalment as a tax-saving expense. Only the interest and the depreciation save tax. The part of the instalment that repays the loan does not.

Comparing the sum of the rentals with the price. The sum of rentals is paid over several years, and part of the price would be borrowed. Compare present values after tax.

Discounting at the weighted average cost of capital. Lease rentals and loan payments are fixed claims, so they are discounted at the after-tax cost of debt.

Believing that a lease is not a form of borrowing. A financial lease is a fixed obligation for years. It adds to financial risk in the same way as a loan, even when the records show it differently.

Forgetting the residual value. The lessee gives up the worth of the asset at the end. It belongs in the purchase side of the comparison.

Treating the whole instalment as an expense.

Comparing totals without time.

Using the wrong rate.

Believing a lease is free of debt.

Forgetting what is lost.

IN HISTORY

Ancient records show that rented tools and land are very old. But leasing as the main business of a company is new. In 1952 a company called the United States Leasing Corporation was founded in San Francisco, and it is described as the first company in the United States to treat leasing as its main business. Its founder is named in the records of the equipment finance industry as Henry Schoenfeld. Within a few years other leasing companies appeared, and the idea spread to Europe in the early 1960s.⁴

A new kind of company began in California.

⁴ The founding date and place are supported by a university leasing institute and an academic history of French leasing. The founder's name and the later spread must be confirmed in a second source before printing.

REVIEW QUESTIONS

- Define.* 1. Define lease, lessor, lessee and lease rental. Give an example of each from a business that you know.
- Compare.* 2. Explain the difference between an operating lease and a financial lease. Give three points of difference.
- Explain.* 3. What is residual value? Explain why it matters to the lessee at the end of a lease.
- Discuss.* 4. Discuss three reasons why a firm may lease an asset instead of buying it, and two drawbacks of leasing.
- List.* 5. State the four steps of the lease-or-buy decision.
- Explain.* 6. Explain why the lease-or-buy decision discounts the cash outflows at the after-tax cost of debt and not at the weighted average cost of capital.
- Calculate and decide.* 7. A machine costs Tk 80,000 and lasts 4 years, with no value at the end. The firm can borrow the price at 12%, repaying Tk 20,000 of the loan at the end of each year together with interest on the balance. Depreciation is Tk 20,000 a year. The firm can instead lease the machine for Tk 24,000 at the end of each year. The tax rate is 25% and the after-tax cost of debt is 9%. The discount factors at 9% are 0.9174, 0.8417, 0.7722 and 0.7084, and the annuity factor for 4 years is 3.2397. Should the firm lease or buy?
- Calculate.* 8. The present value of the after-tax outflows of buying an asset is Tk 90,000. The lease would run for 5 years, with the rental paid at the end of each year. The tax rate is 25%, and the after-tax cost of debt is 9%, with an annuity factor for 5 years of 3.8897. Find the highest yearly rental at which leasing is still no more costly than buying.
- Discuss.* 9. A firm that makes no taxable profit is deciding whether to lease or buy. Explain how its position differs from that of a firm that pays tax, and how this may change the decision.
- Explain.* 10. Why do the loan repayments not appear as a tax-saving expense, while interest and depreciation do?

GLOSSARY

Accept-reject rule. Accept a project if its NPV is above zero; reject it if it is below zero. (*Chapter VII*)

Accounting. Recording and reporting what has happened to a firm's money. (*Chapter I*)

Agency cost. The cost of the conflict, or of preventing it. (*Chapter II*)

Agency problem. A situation in which the agent's aims differ from the principal's. (*Chapter II*)

Agent. The one hired to act for another. (*Chapter II*)

Annuity. Equal payments made at equal gaps of time. (*Chapter V*)

Annuity due. An annuity whose payments come at the start of each period. (*Chapter V*)

Annuity factor. A table number that, times the payment, gives an annuity's value. (*Chapter V*)

Asset. Something the business owns that has value. (*Chapter I*)

Beta. A number showing how strongly a security's return moves with the market's return. (*Chapter X*)

Borrowing. Taking money now and agreeing to repay it. (*Chapter I*)

Break-even point. The level of sales at which profit is exactly zero. (*Chapter IX*)

Business. An activity that sells goods or services to earn money. (*Chapter I*)

Business risk. Risk that comes from the firm's own work, because sales and costs may change. (*Chapter VIII*)

Capital. Money put in to start or run a business. In daily life, a capital is also a country's chief city. (*Chapter I*)

Capital asset pricing model (CAPM). A rule that finds the required return from the risk-free return, the market premium and beta. (*Chapter X*)

Capital budgeting. Planning and choosing long-lived investments. (*Chapter VI*)

Capital expenditure. Money spent on assets that serve for more than a year. (*Chapter VI*)

Capital market. The financial market in which long-term funds are raised and securities are traded. (*Chapter XIV*)

Capital rationing. A limit on the money available for projects. (*Chapter VI*)

Capital structure. The mix of debt and equity that a firm uses to pay for its assets. (*Chapter XV*)

Cash conversion cycle. The days between paying the supplier and collecting from the customer. (*Chapter XII*)

Cash discount. A price cut for paying within a stated short period. (*Chapter XII*)

Cash flow. Money actually moving in or out. (*Chapter I*)

Claim. A right to receive money from the firm. (*Chapter I*)

Coefficient of variation. The standard deviation divided by the expected value; the risk taken for each unit of expected return. (*Chapter VIII*)

Collateral. An asset pledged to the lender, to be sold if the loan is not repaid. (*Chapter XIII*)

Column. One line of a table that runs down the page. (*Chapter III*)

- Common stock.** The ordinary ownership shares of a company. Word in use: in daily life, stock means goods kept in a shop. (*Chapter XI*)
- Company.** A business that the law treats as a separate unit, owned through shares. (*Chapter I*)
- Compound interest.** Interest worked on the first sum and on the interest already added. (*Chapter IV*)
- Compounding.** The process of earning interest on interest already added. (*Chapter IV*)
- Contribution.** The selling price of one unit less its variable cost. (*Chapter III*)
- Cost.** What is paid to get or make something. In daily life, a cost can mean a price. (*Chapter I*)
- Cost of capital.** The yearly price, as a percentage, that a firm pays for the money it uses. (*Chapter XV*)
- Cost of debt.** The yearly percentage a firm pays on its borrowed money, after the tax saved is taken into account. (*Chapter XV*)
- Cost of equity.** The yearly return that the common owners require on the money they have in the firm. (*Chapter XV*)
- Cost of preferred stock.** The yearly fixed dividend divided by the money the firm receives per preferred share. (*Chapter XV*)
- Coupon rate.** The yearly interest rate on a bond's face value. (*Chapter XI*)
- Covenant.** A promise in a loan agreement that limits what the borrower may do or requires it to do. (*Chapter XIII*)
- Credit terms.** The days allowed for payment and any discount offered for early payment. (*Chapter XII*)
- Cumulative.** A preferred share is cumulative if unpaid dividends are carried forward and paid before any common dividend. (*Chapter XIV*)
- Current asset.** Something the firm owns that it expects to use or turn into cash within one year. (*Chapter XII*)
- Current liability.** An amount the firm owes that it must pay within one year. (*Chapter XII*)
- Debenture.** A firm's written promise to repay a loan with interest, backed by the firm's general credit. (*Chapter XIII*)
- Debt.** Money owed to a lender. (*Chapter I*)
- Default.** Failure to pay a loan as agreed. (*Chapter XIII*)
- Degree of financial leverage (DFL).** The per cent change in net profit caused by a one per cent change in EBIT. (*Chapter IX*)
- Degree of operating leverage (DOL).** The per cent change in EBIT caused by a one per cent change in sales. (*Chapter IX*)
- Degree of total leverage (DTL).** The per cent change in net profit caused by a one per cent change in sales. (*Chapter IX*)
- Depreciation.** Spreading an asset's cost over the years of its life. (*Chapter VI*)
- Development finance institution.** A body that supplies long-term funds to build industry or infrastructure, often with public backing. (*Chapter XIV*)
- Dilution.** The fall in the value of existing shares when new shares are sold for less than they are worth. (*Chapter XIV*)
- Discount rate.** The yearly rate used to bring a later sum back to today. (In daily life, a discount is a cut in price.) (*Chapter IV*)
- Discounting.** Finding the present value of a later sum. (*Chapter IV*)

- Diversification.** Spreading money over many investments, so that the ups of some cancel the downs of others. (*Chapter X*)
- Dividend.** The part of profit paid to the owners. (*Chapter II*)
- EBIT.** Profit before interest and tax: sales less variable cost and fixed cost. (*Chapter III*)
- Economics.** The study of how people and societies choose when resources are limited. (*Chapter I*)
- Effective annual rate.** The yearly rate that gives the same growth as the stated rate with interest added once a year. (*Chapter IV*)
- Efficiency.** Getting the most result from the money and effort used. (*Chapter II*)
- Efficient market.** A market in which prices already reflect the information that is available. (*Chapter X*)
- Equity.** The owners' share of what the business is worth. (*Chapter I*)
- Equity per share.** Total owners' equity divided by the number of shares. (*Chapter XIV*)
- Expected value.** The average of all possible outcomes, each weighted by its probability. (*Chapter VIII*)
- Expense.** Money going out. (*Chapter I*)
- Exponent.** The small raised figure that says how many times a number is multiplied by itself. (*Chapter III*)
- Face value.** The sum that a bond repays at its maturity; also called par. (*Chapter XI*)
- Factor.** A table number to multiply by. (In daily life, a cause or influence.) (*Chapter IV*)
- Finance.** How money is raised and used wisely. (*Chapter I*)
- Financial lease.** A long lease that covers most of the life of the asset, and in which the lessee carries most of the risks. (*Chapter XVI*)
- Financial leverage.** How fixed payments to lenders magnify changes in the owners' profit. (*Chapter IX*)
- Financial manager.** The person who decides how a firm raises and uses money. (*Chapter II*)
- Financial market.** A place or system where money is lent, borrowed or invested. (*Chapter I*)
- Financial risk.** The extra risk that comes from borrowing money, because interest must be paid in any case. (*Chapter VIII*)
- Financing decision.** Choosing where the money comes from. (*Chapter II*)
- Firm.** A business seen as one money-using unit. (*Chapter I*)
- Firm-specific risk.** Risk that comes from events affecting one firm only, such as a fire in its shop. (*Chapter VIII*)
- Fixed charge.** A payment the firm must make on a set date, whatever its profit. (*Chapter XIII*)
- Fixed cost.** A cost that stays the same however much the firm makes or sells. (*Chapter III*)
- Float.** Money that has been paid but cannot yet be used by the receiver. (*Chapter XII*)
- Future value.** What a sum today grows to at a later date. (*Chapter IV*)
- Hurdle rate.** The least return a project must earn to be accepted. (*Chapter VII*)
- Incentive.** A reward that guides behaviour. (*Chapter II*)
- Income.** Money coming in. (*Chapter I*)

Incremental cash flow. The extra cash flow caused by the project alone. (*Chapter VI*)

Independent project. A project whose acceptance does not affect other projects. (*Chapter VI*)

Initial outlay. All the money spent at the start of a project. (*Chapter VI*)

Instalment. One of several regular payments that together repay a debt. (*Chapter V*)

Instalment loan. A loan repaid by regular payments, each including interest and part of the amount borrowed. (*Chapter XIII*)

Interest. A fee for using someone's money. In daily life, interest also means curiosity. (*Chapter I*)

Interest expense. The money a firm pays lenders for using their money. (*Chapter III*)

Interest rate. The interest for one year, shown as a percentage of the sum lent. (*Chapter IV*)

Internal rate of return. The discount rate at which a project's NPV is exactly zero. (*Chapter VII*)

Intrinsic value. What an asset is worth, found by adding the present value of everything it is expected to pay. (*Chapter XI*)

Inventory. Goods that a firm holds for use or for sale. (*Chapter XII*)

Investment decision. Choosing what long-lasting things to buy. (*Chapter II*)

Investor. A person who puts money in, hoping to earn more. (*Chapter I*)

Issue cost. The fees a firm pays to sell new bonds or shares, which reduce the money it receives. (*Chapter XV*)

Lease. An agreement in which the owner of an asset lets another party use it in return for regular payments. (*Chapter XVI*)

Lease rental. The regular payment that the lessee makes to the lessor for the use of the asset. (*Chapter XVI*)

Lender. One who lends money for repayment plus interest. (*Chapter I*)

Lessee. The party that uses the asset and pays for the right to do so. (*Chapter XVI*)

Lessor. The owner of the asset, who lets another party use it. (*Chapter XVI*)

Leverage. In daily life, the extra force a lever gives. In finance, the magnifying effect of fixed payments on profit. (*Chapter IX*)

Liability. An amount the business owes. (*Chapter I*)

Limited liability. A shareholder can lose no more than the amount paid for the shares. (*Chapter XIV*)

Listing. The admission of a company's shares to trading on a stock exchange. (*Chapter XIV*)

Long-term. Lasting more than five years. (*Chapter XII*)

Management. Directing people and resources to reach a goal. (*Chapter I*)

Market return. The average return earned by all the shares in a market taken together. (*Chapter X*)

Market risk. Risk that comes from events affecting all firms, such as a change in interest rates. (*Chapter VIII*)

Matching principle. Fund short-term needs with short-term money and long-term needs with long-term money. (*Chapter XII*)

Maturity. The date on which a bond is repaid. (*Chapter XI*)

- Medium-term.** Lasting more than one year and up to five years. (*Chapter XII*)
- Mixed stream.** A stream whose payments have different sizes. (*Chapter V*)
- Money.** What people accept in payment for goods and services. (*Chapter I*)
- Mortgage.** A loan secured on property, which the lender may sell if the loan is not repaid. (*Chapter XIII*)
- Mutually exclusive project.** A project that rules out another if it is chosen. (*Chapter VI*)
- Net present value.** The present value of a project's inflows minus its initial outlay. (*Chapter VII*)
- Net profit.** The profit left after interest and tax have been paid. (*Chapter III*)
- Net working capital.** Current assets less current liabilities. (*Chapter XII*)
- Non-bank financial institution.** A financial firm that lends, leases or invests but is not a bank. (*Chapter XIV*)
- Operating cash flow.** The cash a project brings in each year of its life. (*Chapter VI*)
- Operating cycle.** The days from receiving ingredients to collecting cash from the customer. (*Chapter XII*)
- Operating lease.** A short lease in which the lessor keeps most of the risks of owning the asset. (*Chapter XVI*)
- Operating leverage.** How fixed costs of running the business magnify changes in EBIT. (*Chapter IX*)
- Order of operations.** The fixed order in which a calculation is worked: brackets, powers, times and divide, add and subtract. (*Chapter III*)
- Ordinary annuity.** An annuity whose payments come at the end of each period. (*Chapter V*)
- Overdraft.** Bank permission to spend more than the balance in the account, up to a stated limit. (*Chapter XII*)
- Owner.** The person whose money and risk the business carries. (*Chapter I*)
- Payback period.** The time a project takes to bring back its initial outlay. (*Chapter VII*)
- Payout ratio.** Dividends divided by net profit; the part of profit paid to owners. (*Chapter XIV*)
- Percentage.** A part out of 100. So 25% means 25 out of every 100. (*Chapter III*)
- Percentage change.** The change in a number, shown as a percentage of the old number. (*Chapter III*)
- Percentage point.** The gap between two percentages. From 10% to 12% is two percentage points. (*Chapter III*)
- Perpetuity.** An annuity that never ends. (*Chapter V*)
- Power.** A number multiplied by itself a stated number of times. (In daily life, strength or authority.) (*Chapter III*)
- Preferred dividend.** A fixed payment to a special kind of owner, taught in Chapter XIV. (*Chapter III*)
- Preferred stock.** Shares that pay a fixed dividend, which must be paid before any common dividend. (*Chapter XIV*)

Present value. What a later sum is worth today. (In daily life, a present can mean a gift.) (*Chapter IV*)

Primary market. The market in which a firm sells new securities and receives the money. (*Chapter XIV*)

Principal. The one who hires another to act for them. (*Chapter II*)

Principal (of a loan). The amount borrowed that is still owed. In Chapter 2, principal meant the one who hires an agent. (*Chapter XIII*)

Probability. A number from 0 to 1 that shows how likely an event is. (*Chapter VIII*)

Probability distribution. A list of all possible outcomes, each shown with its chance. (*Chapter VIII*)

Profit. What is left of sales after costs. (*Chapter I*)

Profit before tax. Profit left after interest expense and before tax is paid. (*Chapter III*)

Profit maximisation. Aiming for the biggest profit. (*Chapter II*)

Profitability index. The present value of a project's inflows divided by its initial outlay. (*Chapter VII*)

Project. One investment plan, such as buying a single oven. (*Chapter VI*)

Public offering. The sale of new shares to the public at large. (*Chapter XIV*)

Rate of return. The return stated as a percentage for one period, usually one year. (*Chapter VIII*)

Ratio. One number compared with another, such as 3 to 2. (*Chapter III*)

Receivable. Money a customer owes the firm for goods or services already delivered. (*Chapter XII*)

Regulator. A public body that makes rules for a market and checks that they are followed. (*Chapter XIV*)

Repayment schedule. A table listing each instalment, how much of it is interest, and how much repays the principal. (*Chapter XIII*)

Required return. The least return an investor will accept to hold an asset, given its risk. (*Chapter X*)

Residual claim. The right to what remains after every other claim has been paid. (*Chapter XIV*)

Residual value. What the asset is worth at the end of the lease or at the end of its useful life. (*Chapter XVI*)

Retained earnings. The part of profit after tax that is kept in the firm and not paid out as dividend. (*Chapter XIV*)

Return. The gain on an investment, shown as a share of its cost. (*Chapter VIII*)

Rights issue. An offer of new shares to existing shareholders, in proportion to the shares they hold. (*Chapter XIV*)

Risk. The chance that the actual result will differ from the result we hoped for. (*Chapter VIII*)

Risk premium. The extra return that investors ask for carrying extra risk. (*Chapter X*)

Risk-free return. The return on an investment where there is no real chance of loss. (*Chapter X*)

Root. The number that, raised to a stated power, gives a given number. (In daily life, part of a plant.) (*Chapter III*)

Rounding. Replacing a number by a nearby shorter one, such as 7.1428 by 7.14. (*Chapter III*)

- Row.** One line of a table that runs across the page. (*Chapter III*)
- Sale and leaseback.** A firm sells an asset it owns to a lessor and leases the same asset back at once. (*Chapter XVI*)
- Sales.** Money earned from selling. (*Chapter I*)
- Saving.** Keeping money for later use. (*Chapter I*)
- Secondary market.** The market in which investors trade securities that already exist. (*Chapter XIV*)
- Secured loan.** A loan backed by collateral. (*Chapter XIII*)
- Security.** A tradable claim to future payments, such as a bond or a share. (*Chapter XI*)
- Security market line.** A line that shows the required return for each value of beta. (*Chapter X*)
- Selling at a discount.** Selling for less than the face value, which happens when the coupon rate is below the required return. (*Chapter XI*)
- Selling at a premium.** Selling for more than the face value, which happens when the coupon rate is above the required return. (*Chapter XI*)
- Share.** One equal part of the ownership of a company. (*Chapter I*)
- Share issue.** The sale of new shares by a company to raise money. (*Chapter XIV*)
- Share price.** The price at which one share can be bought or sold. (*Chapter II*)
- Shareholder.** The owner of a share of a company. (*Chapter II*)
- Short-term.** Lasting up to one year. (*Chapter XII*)
- Simple interest.** Interest worked on the first sum only. (*Chapter IV*)
- Single sum.** One payment made at one date. (*Chapter IV*)
- Sinking fund payment.** A regular saving that reaches a target sum. (*Chapter V*)
- Standard deviation.** The square root of the variance; the usual distance of outcomes from the expected value. (*Chapter VIII*)
- Stock exchange.** An organised market in which securities are bought and sold under fixed rules. (*Chapter XIV*)
- Stream.** A series of payments made or received over time. (*Chapter V*)
- Sunk cost.** Money already spent that cannot be recovered. (*Chapter VI*)
- Table.** Numbers arranged in rows and columns so that each can be found quickly. (In daily life, a piece of furniture.) (*Chapter III*)
- Target weights.** The shares of debt and equity that a firm plans to hold in the long run. (*Chapter XV*)
- Tax.** Money a firm must pay the government on its profit. (*Chapter III*)
- Tax shield on interest.** The tax saved because interest is subtracted from profit before tax is calculated. (*Chapter XIII*)
- Term loan.** A bank loan that is repaid over several years. (*Chapter XIII*)
- Terminal cash flow.** Cash received or paid at the end of a project's life. (*Chapter VI*)
- Time value of money.** The idea that a taka now is worth more than a taka later. (*Chapter IV*)
- Timeline.** A line that shows the dates at which money moves. (*Chapter IV*)
- Total leverage.** The combined effect of operating and financial leverage on net profit. (*Chapter IX*)
- Trade credit.** Buying goods now and paying the supplier at a later agreed date. (*Chapter XII*)

Unit. One item of what the firm sells, such as one box of buns. (*Chapter III*)

Unsecured loan. A loan not backed by collateral. (*Chapter XIII*)

Variable cost. A cost that rises and falls with the amount the firm makes or sells. (*Chapter III*)

Variance. The average of the squared deviations from the expected value, each weighted by its chance. (*Chapter VIII*)

Wealth. The total value of what a person owns less what the person owes. (*Chapter II*)

Wealth maximisation. Aiming to raise the owners' total value over time. (*Chapter II*)

Weighted average cost of capital. The average of the costs of all funds, each counted by its share of the total. (*Chapter XV*)

Working capital. Money tied up in daily running, such as flour, unpaid bills and cash. (*Chapter II*)

Yield to maturity. The yearly return a buyer earns by paying today's price and holding the bond until it is repaid. (*Chapter XI*)

AN INTRODUCTION TO FINANCE

A first course in finance for readers who begin with no background. Through the story of one small bakery, Sundial Bakery of Dhaka, the book builds each idea from plain words and exact numbers: how money behaves over time, how to judge a project, how risk is priced, and how a firm finds and pays for the money it uses. Every page keeps its summaries and key terms in the margin, every chapter ends with practice, and sixteen engraved plates mark the way.

MONEY AND THE FIRM • TIME AND VALUE • CAPITAL BUDGETING •
RISK AND RETURN • LEVERAGE • BONDS AND SHARES • WORKING
CAPITAL • DEBT AND EQUITY • COST OF CAPITAL • LEASING

KUTUSH CATUSH